

CYBER ELIAS ACADEMY

Computer Repairs

Complete Course Materials — Instructor & Learner Edition

PROFESSIONAL TRAINING PROGRAMME

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Table of Contents

Course Design & Curriculum

CYBER ELIAS ACADEMY

Session Materials

SESSION 1 â€" COMPUTER COMPONENTS: WHAT IS INSIDE THE BOX

SESSION 2 â€" SAFETY AND TOOLS: WORKING WITHOUT HARMING YOURSELF OR THE MACHINE

SESSION 3 â€" DISASSEMBLY AND REASSEMBLY: OPEN IT AND PUT IT BACK CORRECTLY

SESSION 4 â€" THE DIAGNOSTIC WORKFLOW: FINDING THE FAULT SYSTEMATICALLY

SESSION 5 â€" HARDWARE DIAGNOSTICS

SESSION 6 â€" SOFTWARE DIAGNOSTICS

SESSION 7 â€" COMPONENT REPLACEMENT

SESSION 8 â€" CLEANING AND MAINTENANCE

SESSION 9 â€" OPERATING SYSTEM INSTALLATION

SESSION 10 â€" CUSTOMER SERVICE AND COMMUNICATION

SESSION 11 â€" SERVICE REPORTS AND DOCUMENTATION

SESSION 12 â€" FINAL PROJECT (CAPSTONE): DIAGNOSE, REPAIR, DOCUMENT AND PRESENT

Learner Workbook

COMPUTER REPAIRS — LEARNER WORKBOOK

Instructor Manual

COMPUTER REPAIRS — INSTRUCTOR MANUAL

Practical Lab Pack

COMPUTER REPAIRS — PRACTICAL LAB PACK

Capstone Project

CAPSTONE ASSESSMENT RUBRIC

CAPSTONE INSTRUCTOR GUIDE

QA & Audit

FINAL QUALITY ASSURANCE AUDIT REPORT

PART 1

Course Design & Curriculum

CYBER ELIAS ACADEMY

CYBER ELIAS ACADEMY

COMPUTER REPAIRS — COURSE DESIGN DOCUMENT

Institution:
Cyber Elias Academy (CEA) **Location:** 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State **Philosophy:** From Zero to Expert, Together **Delivery Mode:** In-person (primary), with online/hybrid option **Session Length:** 105 minutes (1 hour 45 minutes) **Sessions Per Week:** 3 **Course Duration:** 4 weeks **Total Sessions:** 12 **Course Fee:** ₦50,000 **Certificate Requirement:** A diagnosed and serviced machine with a full service report

PART 1 — COURSE DESIGN DOCUMENT

1. Course Overview

Course Name:
Computer Repairs **Course Category:** Hardware & Security — Technical Diagnostics **Target Duration:** 4 weeks (12 sessions × 105 minutes = 21 contact hours) **Target Learner:** Beginner with basic computer skills, no prior hardware experience

Course Description:
This course transforms a beginner into a competent computer repair technician who can safely diagnose and fix common hardware and software problems. The course covers computer components, safe diagnostic procedures, hardware identification, common faults, repair techniques, customer service and professional documentation.

Every session involves hands-on practice with real computer components and simulated faults. Learners work with actual hardware from the first session — identifying components, running diagnostics, replacing parts and documenting their work.

The final assessment is a complete repair project: the learner receives a computer with a simulated fault, diagnoses the problem, performs the repair, verifies the fix and produces a professional service report.

Why This Course Exists:
In Port Harcourt and across Nigeria, computer repair is a critical skill. Thousands of computers develop faults every day — failed hard drives, faulty RAM, overheating, virus infections, broken screens. Most computer technicians learned through informal apprenticeship. This course provides structured, safe, professional training in four weeks.

Connection to Other CEA Courses:
This course assumes basic computer skills (Typing & Computer Basics). It complements Cybersecurity (software protection) and is complemented by Business & Freelancing (running a repair business).

2. Target Learners

Primary Audience:

- Job seekers who want to enter the computer repair trade
- Small business owners who want to maintain their own computers
- Students and NYSC members looking for a practical skill
- Existing computer users who want to understand how their machines work
- Aspiring technicians who want formal training

What "Beginner" Means Here:

What This Course Does NOT Assume:

- No prior hardware experience
- No familiarity with computer components
- No ownership of a personal laptop
- No internet connection at home

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3. Entry Requirements

Formal Entry Requirements:
None, but Typing & Computer Basics is strongly recommended.

What the Learner Actually Needs:

1. A practice machine — provided at the academy
2. Basic computer skills — file management, typing, internet
3. Willingness to work with their hands
4. Respect for safety procedures

What Is NOT Required:

- Prior hardware experience
- A personal laptop
- Internet access at home
- Any specific educational qualification

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4. Graduate Competency Profile

A graduate of Computer Repairs can demonstrate the following:

Hardware Knowledge:

- Identify all major computer components (motherboard, CPU, RAM, hard drive, PSU, GPU, etc.)
- Explain the function of each component
- Understand how components connect and communicate

Diagnostic Skills:

- Follow a systematic diagnostic workflow
- Identify common hardware faults (failed hard drive, faulty RAM, overheating, etc.)
- Identify common software faults (virus, corrupted OS, driver issues)
- Use diagnostic tools and software

Repair Skills:

- Safely open and reassemble a computer case
- Replace common components (RAM, hard drive, power supply)
- Clean and maintain computer hardware
- Install and configure operating systems

Safety:

- Follow ESD (electrostatic discharge) safety procedures
- Work safely with electrical components
- Handle tools correctly

- Produce a professional service report
- Communicate clearly with customers
- Document all work performed
- Provide honest assessments and recommendations

5. Measurable Learning Outcomes

#	Learning Outcome	Assessment Method
LO1	Identify and name all major computer components	Practical identification
LO2	Explain the function of each component	Knowledge check
LO3	Follow safe ESD and electrical safety procedures	Observed practical
LO4	Diagnose common hardware faults using a systematic workflow	Practical task
LO5	Replace common components (RAM, hard drive, PSU)	Practical task
LO6	Diagnose and resolve common software faults	Practical task
LO7	Use diagnostic tools and software	Practical task
LO8	Clean and maintain computer hardware	Practical task
LO9	Install and configure an operating system	Practical task
LO10	Produce a professional service report	Written report
LO11	Complete a full repair project from diagnosis to service report	Capstone project

6. Tools and Resources

Hardware (Provided by CEA):

- Desktop computers for disassembly/reassembly (at least 2 per class)
- Assorted components: RAM sticks, hard drives, power supplies, cables
- Tool kits: screwdrivers, anti-static wrist strap, tweezers, flashlight
- Cleaning supplies: compressed air, isopropyl alcohol, microfiber cloths

Software:

- Windows 10/11 installation media
- Diagnostic software (e.g., CrystalDiskInfo, MemTest86, HWMonitor)
- Antivirus software

Printed Materials:

- Learner workbook
- Component identification charts
- Diagnostic workflow posters
- Service report templates

Safety Equipment:

- Anti-static wrist straps (one per learner)

7. Full Curriculum Map

```
WEEK 1 — Computer Components and Safety
├── Session 1: Computer Components
│   ├── Major components: motherboard, CPU, RAM, hard drive, PSU, GPU
│   ├── Storage types: HDD, SSD, NVMe
│   ├── Input/output devices
│   └── Component identification practical
│
├── Session 2: Safety and Tools
│   ├── ESD (electrostatic discharge) safety
│   ├── Electrical safety
│   ├── Tool identification and proper use
│   ├── Workspace setup
│   └── Safety procedures practical
│
└── Session 3: Disassembly and Reassembly
    ├── Safe disassembly workflow
    ├── Cable management
    ├── Component removal and installation
    ├── Reassembly and verification
    └── Full disassembly/reassembly practical

WEEK 2 — Diagnostics
├── Session 4: Diagnostic Workflow
│   ├── Systematic diagnostic approach
│   ├── Common symptoms and likely causes
│   ├── Diagnostic questions to ask customers
│   └── Diagnostic scenario exercises
│
├── Session 5: Hardware Diagnostics
│   ├── RAM testing (MemTest86)
│   ├── Hard drive testing (CrystalDiskInfo)
│   ├── Temperature monitoring (HWMonitor)
│   ├── Power supply testing
│   └── Hardware diagnostic practical
│
└── Session 6: Software Diagnostics
    ├── Operating system issues
    ├── Virus and malware identification
    ├── Driver issues
    ├── System restore and recovery
    └── Software diagnostic practical

WEEK 3 — Repair and Maintenance
├── Session 7: Component Replacement
│   ├── RAM replacement
│   ├── Hard drive replacement
│   ├── Power supply replacement
│   ├── GPU replacement
│   └── Component replacement practical
│
├── Session 8: Cleaning and Maintenance
│   ├── Physical cleaning (dust removal)
│   ├── Thermal paste replacement
│   ├── Fan cleaning and replacement
│   ├── Preventive maintenance schedule
│   └── Cleaning and maintenance practical
│
└── Session 9: Operating System Installation
    ├── Pre-installation preparation
    ├── Windows installation
    ├── Driver installation
    ├── Windows updates and activation
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├── Session 10: Customer Service and Communication
│   ├── Customer intake process
│   ├── Explaining technical issues in plain language
│   ├── Providing estimates and timelines
│   ├── Handling difficult customers
│   └── Role-play exercises
│
├── Session 11: Service Reports and Documentation
│   ├── Service report structure
│   ├── Recording diagnostic findings
│   ├── Documenting repairs performed
│   ├── Providing recommendations
│   └── Service report writing practical
│
└── Session 12: Final Project — Diagnose and Repair
    ├── Receiving the faulty machine
    ├── Diagnosing the fault
    ├── Performing the repair
    ├── Verifying the fix
    ├── Producing a service report
    ├── Presenting to the instructor
    └── Certificate sign-off
```

8. Week-by-Week Programme

WEEK 1 — Computer Components and Safety

Week Theme:
Learning what is inside a computer and how to work with it safely.

Sessions Included:
1, 2, 3

Skills Being Developed:

- Component identification
- Safety procedures
- Disassembly and reassembly

Mini-Project:
"My First Teardown" — Learner safely disassembles and reassembles a desktop computer, identifying every component.

Deliverables:

- Completed disassembly/reassembly
- Component identification checklist signed off
- Safety procedures observed

Quality Standards:

- All components correctly identified
- No damage to components during disassembly
- Correct reassembly with all cables connected
- All safety procedures followed

Contribution to Capstone:
The component knowledge and disassembly skills taught in Week 1 are essential for the Session 12 capstone repair project.

WEEK 2 — Diagnostics

Sessions Included:
4, 5, 6

Skills Being Developed:

- Systematic diagnostic workflow
- Hardware diagnostic tools
- Software diagnostic tools
- Symptom-to-cause analysis

Mini-Project:
"My Diagnostic Report" — Learner diagnoses three simulated faults (one hardware, one software, one mixed) and produces a diagnostic report for each.

Deliverables:

- Three diagnostic reports
- Correct identification of all simulated faults
- Appropriate use of diagnostic tools

Quality Standards:

- Systematic diagnostic workflow followed
- Correct tools used for each diagnosis
- Accurate identification of root cause
- Clear documentation of findings

Contribution to Capstone:
The diagnostic skills taught in Week 2 are directly used in the Session 12 capstone to identify the fault.

WEEK 3 — Repair and Maintenance

Week Theme:
Learning to replace components, clean and maintain computers, and install operating systems.

Sessions Included:
7, 8, 9

Skills Being Developed:

- Component replacement
- Cleaning and maintenance
- OS installation
- Driver configuration

Mini-Project:
"My Repair Log" — Learner replaces a component, cleans a computer and installs an operating system, documenting each step.

Deliverables:

- Component replacement completed
- Cleaning and maintenance completed
- OS installation completed
- Repair log documenting all work

Quality Standards:

- Component replaced correctly

- All drivers installed
- Work documented professionally

Contribution to Capstone:
The repair and installation skills taught in Week 3 are directly used in the Session 12 capstone to fix the fault.

WEEK 4 — Professional Practice and Capstone

Week Theme:
Learning professional customer service, documentation and completing the final repair project.

Sessions Included:
10, 11, 12

Skills Being Developed:

- Customer communication
- Service report writing
- Complete repair project management
- Presentation skills

Mini-Project:
"My Service Report" — The final capstone project. Learner receives a faulty computer, diagnoses the problem, performs the repair, verifies the fix and produces a professional service report.

Deliverables:

- Fault correctly diagnosed
- Repair performed successfully
- Fix verified
- Professional service report
- Presentation to the instructor

Quality Standards:

- Fault correctly identified
- Repair performed safely and correctly
- Computer fully functional after repair
- Service report is complete and professional
- Presentation is clear and confident

Contribution to Capstone:
The service report IS the capstone deliverable. It is presented and assessed in Session 12.

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9. Complete Session Schedule

Session	Week	Title	Duration	Key Skills	Assessment
1	1	Computer Components	105 min	Component identification	Identification checklist
2	1	Safety and Tools	105 min	ESD safety, tool use	Safety observation
3	1	Disassembly and Reassembly	105 min	Disassembly, reassembly	Practical task
4	2	Diagnostic Workflow	105 min	Systematic diagnostics	Diagnostic scenarios

5	2	Hardware Diagnostics	105 min	RAM, HDD, PSU testing	Diagnostic practical
6	2	Software Diagnostics	105 min	OS, virus, driver issues	Diagnostic practical
7	3	Component Replacement	105 min	RAM, HDD, PSU, GPU replacement	Replacement practical
8	3	Cleaning and Maintenance	105 min	Cleaning, thermal paste, fans	Maintenance practical
9	3	Operating System Installation	105 min	Windows installation, drivers	Installation practical
10	4	Customer Service	105 min	Communication, intake, estimates	Role-play
11	4	Service Reports	105 min	Documentation, reporting	Report writing
12	4	Final Project (Capstone)	105 min	Full repair + service report	Capstone assessment

10. Assignment and Assessment Schedule

Assessment	Session	Type	Weight	Description
Component Identification	1	Practical identification	5%	Identify all major components
Disassembly/Reassembly	3	Practical task	10%	Full teardown and rebuild
Diagnostic Scenarios	4	Written exercise	5%	Three diagnostic scenarios
Hardware Diagnostics	5	Practical task	10%	RAM, HDD, PSU testing
Software Diagnostics	6	Practical task	10%	OS, virus, driver diagnosis
Component Replacement	7	Practical task	10%	Replace RAM and hard drive
OS Installation	9	Practical task	10%	Install Windows and drivers
Service Report	11	Written report	10%	Professional service report
Capstone: Repair Project	12	Integrated project	25%	Full diagnosis + repair + report
Knowledge Checks	All	Mixed quiz	5%	Short quizzes each session

Total: 100%

Grading Scale:

Grade	Percentage	Descriptor
A — Excellent	80–100%	Can diagnose and repair common faults independently
B — Good	65–79%	Can diagnose and repair with minor guidance
C — Competent	50–64%	Can diagnose and repair with some guidance
D — Developing	40–49%	Requires significant guidance
F — Not Yet Competent	0–39%	Does not meet minimum requirements

Passing Requirement:
Minimum 50% overall AND successful completion of the Session 12 capstone repair project. A learner who scores 90% on knowledge checks but cannot diagnose and repair a fault does NOT pass.

Week	Mini-Project	Skills Practised	Contribution to Capstone
1	"My First Teardown"	Disassembly, reassembly	Foundation for capstone
2	"My Diagnostic Report"	Hardware and software diagnostics	Direct preparation for capstone
3	"My Repair Log"	Component replacement, OS installation	Direct preparation for capstone
4	"My Service Report"	Full repair + documentation	IS the capstone project

12. Capstone Overview

Capstone Title:
Diagnosed and Serviced Machine

Real-World Context:
A customer brings a computer to a repair shop with a fault. The technician must diagnose the problem, perform the repair, verify the fix and provide a service report to the customer.

The Deliverables:

1. **A correctly diagnosed fault** — the learner identifies the root cause of the problem.
2. **A successfully performed repair** — the learner fixes the fault.
3. **A verified fix** — the learner confirms the computer is fully functional.
4. **A professional service report** — documenting the fault, the repair and recommendations.

What the Instructor Observes:

- Does the learner follow a systematic diagnostic workflow?
- Does the learner use the correct tools and safety procedures?
- Does the learner correctly identify the fault?
- Does the learner perform the repair safely and correctly?
- Does the learner verify the fix?
- Can the learner explain their process and decisions?

Passing Standard:
The learner must correctly diagnose the fault, perform the repair, verify the fix and produce a professional service report. A repair that the learner cannot explain must not automatically receive a passing grade.

13. Certification and Competence Requirements

Certificate Name:
CEA Certificate of Computer Repair Competence

Awarded When:

1. The learner has attended a minimum of 10 out of 12 sessions
2. The learner has completed all assignments to a satisfactory standard
3. The learner has successfully completed the Session 12 capstone repair project
4. The learner has achieved a minimum overall score of 50%
5. The learner has produced a professional service report

What the Certificate Means:
The holder has demonstrated the ability to safely diagnose and repair common computer hardware and software faults. They can work as a junior computer technician or start a small repair business.

- It does not guarantee employment
- It does not cover advanced repairs (e.g., motherboard-level repair, data recovery)
- It does not make the learner an expert — it makes them a competent beginner ready for the next level

Verification:
Certificates include a verification code that employers can check at cea.ng/certificates/verify.

SESSION SCHEDULE — DETAILED

SESSION 1 — Computer Components

Week:
1 | **Module:** Components and Safety | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Basic computer skills
Preparation: Prepare component samples, identification charts

SESSION 2 — Safety and Tools

Week:
1 | **Module:** Components and Safety | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Session 1 **Preparation:** Prepare tool kits, safety equipment

SESSION 3 — Disassembly and Reassembly

Week:
1 | **Module:** Components and Safety | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Sessions 1–2 **Preparation:** Prepare practice machines for disassembly

SESSION 4 — Diagnostic Workflow

Week:
2 | **Module:** Diagnostics | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Sessions 1–3 **Preparation:** Prepare diagnostic scenario cards

SESSION 5 — Hardware Diagnostics

Week:
2 | **Module:** Diagnostics | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Session 4 **Preparation:** Prepare diagnostic software, faulty components

SESSION 6 — Software Diagnostics

Week:
2 | **Module:** Diagnostics | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Sessions 4–5 **Preparation:** Prepare software fault simulations

SESSION 7 — Component Replacement

Week:
3 | **Module:** Repair and Maintenance | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Sessions 1–6 **Preparation:** Prepare replacement components

SESSION 8 — Cleaning and Maintenance

Week:

SESSION 9 — Operating System Installation

Week:
3 | **Module:** Repair and Maintenance | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Sessions 1–8 **Preparation:** Prepare Windows installation media

SESSION 10 — Customer Service

Week:
4 | **Module:** Professional Practice | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Sessions 1–9 **Preparation:** Prepare role-play scenarios

SESSION 11 — Service Reports

Week:
4 | **Module:** Professional Practice | **Duration:** 105 minutes **Difficulty:** Beginner | **Prerequisite:** Session 10 **Preparation:** Prepare service report templates

SESSION 12 — Final Project (Capstone)

Week:
4 | **Module:** Professional Practice | **Duration:** 105 minutes **Difficulty:** Beginner (tests all accumulated skills) **Prerequisite:** Sessions 1–11 **Preparation:** Prepare faulty machines, assessment rubrics, certificate sign-off sheets

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COMPETENCY STANDARDS FRAMEWORK

Level 1 — Know

The learner can identify components, name tools and explain safety procedures.

Level 2 — Do

The learner can diagnose and repair common faults independently, following safety procedures.

Level 3 — Create

The learner can run a small repair business, handling customer intake, diagnosis, repair and documentation.

QUALITY ASSURANCE CHECKLIST (Self-Audit)

- [ ] All 12 sessions have complete learning outcomes
- [ ] Every session has a practical component
- [ ] Every session has an instructor demonstration
- [ ] Every session has a guided practical exercise
- [ ] Every session has an independent exercise
- [ ] Every session has an assignment
- [ ] Every assignment has an answer key / marking guide
- [ ] Every session has a knowledge check
- [ ] The capstone tests only skills taught in Sessions 1–11
- [ ] The assessment framework is transparent and measurable
- [ ] Differentiated support is planned for varied learning speeds
- [ ] Nigerian and African contexts are used throughout

- ☑ The workload fits within 105 minutes per session
- ☑ The course does not claim to produce experts, only competent beginners
- ☑ Safety precautions are included in every practical session

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Document Version:
1.0 **Date:** October 2026 **Author:** Curriculum Development Team, Cyber Elias Academy **Next Step:** Proceed to Part 2 — Complete Session-by-Session Teaching Materials, starting with Week 1 (Sessions 1, 2, and 3).

PART 2

Session Materials

SESSION 1 â€" COMPUTER COMPONENTS: WHAT IS INSIDE THE BOX

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 1 â€" COMPUTER COMPONENTS: WHAT IS INSIDE THE BOX

A. SESSION IDENTIFICATION

Item	Detail
Session number	1 of 12
Session title	Computer Components â€" What Is Inside the Box
Week / Module	Week 1 â€" Module: Components and Safety
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner (first contact with computer hardware)
Prerequisite knowledge	Basic computer skills: typing, saving files, opening programs. No hardware experience needed.
Connection to previous session	This is the first session of the Computer Repairs course. It assumes only the Typing & Computer Basics course.
Connection to the course	This session gives every learner the vocabulary of the trade â€" the names and jobs of every part inside a computer. Every later session (safety, disassembly, diagnostics, repair, the capstone) depends on knowing these parts by name and function.
Preparation before class	Physical samples of motherboard, CPU (+ cooler), RAM, HDD, SSD, PSU, GPU, SATA cable, power cable; at least one powered-off desktop computer; projector or large screen; printed Component Identification Chart (one per learner); anti-static wrist straps; component name cards for a labelling race.

Safety note for this session:
Learners will handle real components. Even though this is an "identification" session, every learner must wear an anti-static wrist strap or regularly touch the bare metal of the case before touching any board. The machine used for demonstration must be unplugged from the wall and the power button pressed once to discharge residual electricity. No component is opened, and no mains-powered work is done. All handling is done by the edges only. Where the academy has only one working demonstration machine, learners should be grouped and supervised.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Identify and name** all major internal components of a desktop computer â€" motherboard, CPU and its heatsink/fan, RAM, HDD, SSD, PSU, GPU, expansion cards, cooling fans, and the main ports and connectors.
2. **Explain the function** of each major component in their own words, using a familiar comparison.

4. **Distinguish clearly between RAM and storage**, and between HDD and SSD, including speed, durability and cost differences.
5. **Explain the relationship** between components â€" how they connect to the motherboard and why the system needs all of them to work.
6. **Compare a desktop and a laptop**, stating at least three differences in how components are arranged and serviced.
7. **Handle components safely** â€" holding boards by their edges, never touching gold contacts, and using anti-static protection.
8. **Identify components physically** on a real open machine or on labelled photographs and diagrams.

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- Physical samples of each component: motherboard, CPU with heatsink/fan, RAM stick(s), one HDD, one SSD (SATA type if available), one M.2/NVMe stick if available, PSU, a dedicated GPU card, a spare expansion card (sound/network card), a case cooling fan, SATA data cable, SATA power connector, 24-pin ATX power connector, and a 8-pin CPU power connector.
- At least one desktop computer, powered off and unplugged, with its side panel removed or removable, for live identification.
- A laptop (any working or non-working unit) for the desktop-vs-laptop comparison.
- Projector or large screen connected to the instructor's computer for showing photographs and diagrams.

Software:

- A simple slideshow or photo gallery of computer internals (instructor's own photos or freely available images). No special software required.
- Optional: a free system-information viewer already built into Windows (Task Manager â†' Performance tab) to show live CPU, RAM and disk activity. This is only a demonstration aid and is not essential.

Safety gear:

- Anti-static wrist straps (one per learner, or one per pair).
- Anti-static mat for the demonstration table (one is enough).
- Clean, dry, well-lit table surface. No food or drink on the table.

Sample / fault components:

- A dead/faulty RAM stick and a dead hard drive (clearly labelled "FAULT â€" DO NOT FIT") to show what failed parts look like. These are for looking only.
- A burnt-looking PSU (clearly labelled) to introduce the idea that components can fail. Do not open it.

Worksheets (printed):

- Worksheet 1.1 â€" "Component Identification Chart" (one per learner): a table with columns for component name, what it does, one key fact, and one common problem.
- Worksheet 1.2 â€" "Data Flow Diagram" (one per learner): a blank flow diagram the learner fills in during the guided exercise.
- Component name cards (small cards with one component name each) for the opening labelling race.

Tools:

- A Phillips PH2 screwdriver (only for the instructor to remove a panel; learners do not unscrew anything this session).
- A small torch/flashlight to point out parts inside the case.

Alternatives for a learner or academy with limited equipment:

- If no real components are available, use printed or projected high-quality labelled photographs and diagrams. The identification exercise can be done entirely from labelled photos.

- If a learner has low vision or shaky hands, pair them with a careful partner and let them identify and describe while the partner holds the part.

Suggested visuals:

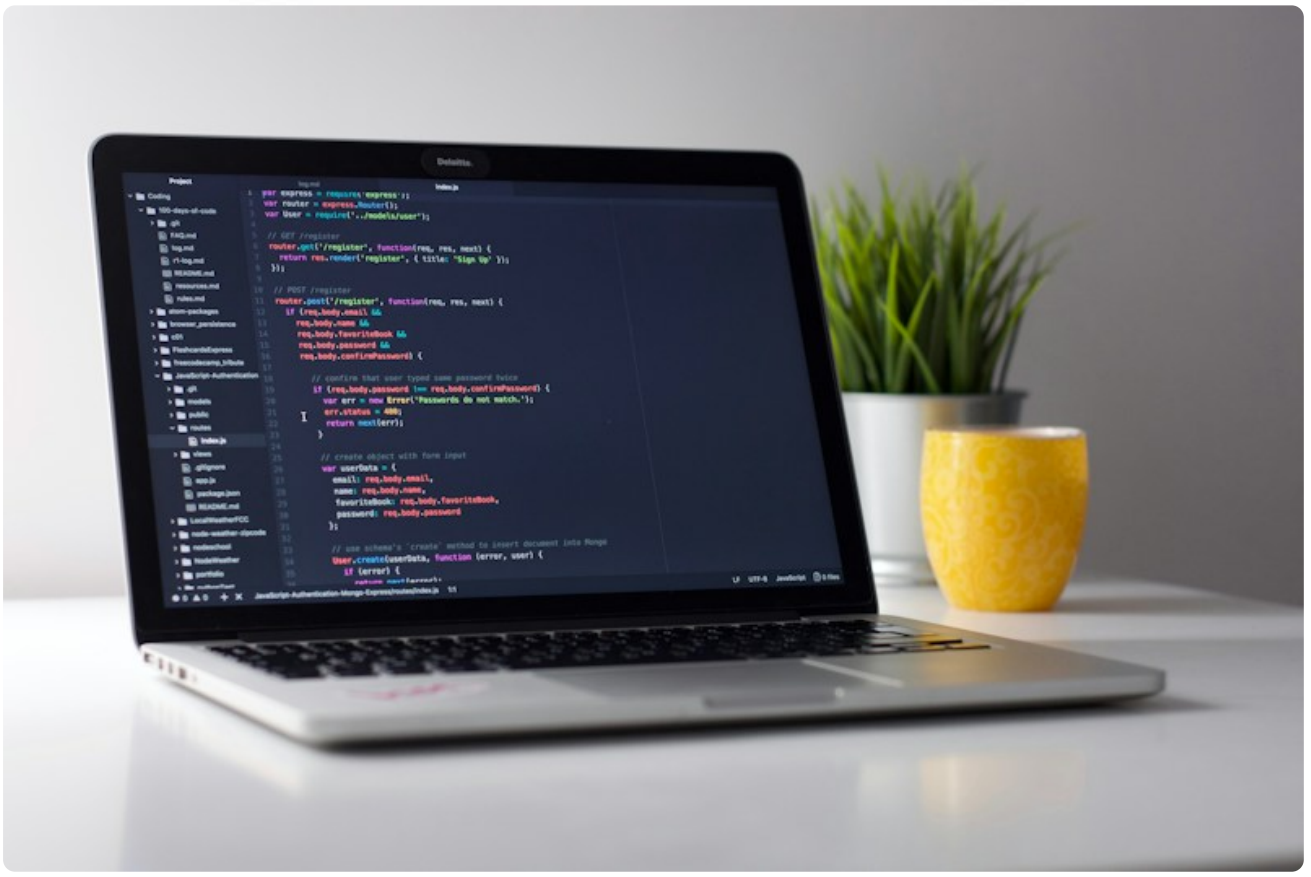
- 

â€” Identifying internal parts: CPU, RAM, storage, PSU

- 
â€” Open case used for the component tour



- â€" Screwdriver and torch for pointing out parts



- — Laptop components compared with desktop parts



— Component diagram and spec sheet shown on screen

D. OPENING ACTIVITY (6 minutes)

"What Is Inside the Box?"

Instructor setup:
Bring a sealed cardboard box to the front. On a side table, hidden under a cloth, place the loose components (motherboard, RAM, PSU, etc.). Do not reveal them yet. On the projector, keep a blank screen.

What the instructor says:

> *"Good morning, everyone. Welcome to the Computer Repairs course. Before we open a single computer, I want to play a game. I have a box here. Inside this box is a mystery item. I want you to guess what it is. I will give you one clue: it is something you all use, some of you every single day, and most of you have never seen the inside of it."*

Take three or four guesses from around the room.

> *"You are all close. The answer is: it is a computer â€" or at least, all the parts of one. Now here is the shocking part. Look at Chinedu's phone, look at the cyber cafÃ© machines down the road, look at the school computer lab â€" inside, they are all made of the same handful of parts, just arranged differently.*
>
> *Here is my question for you: if you took a computer apart, could you name every single thing inside it? Raise your hand if you could."*

Most hands stay down.

> *"By the end of today, every hand in this room will be up. You will know the name, the job, and the feel of every major component. And here is why that matters in Port Harcourt: when a customer walks into your shop and says 'my system is*

Now â€" let us meet the parts."

Then reveal the hook:
Pull the cloth off the side table and hold up the motherboard â€" the strangest-looking part. Ask: *"Is this a radio? A small road? A city map?"* Let them guess. Say:

> *"This is the*
>
> ***motherboard***
> *â€" the main board every other part plugs into. Everything we learn today connects back to this single board. By the end of the session, you will be able to point to every part on this board and tell me what it does."*

Expected learner responses:
Learners may guess "brain," "memory," "hard disk," or stay quiet. Accept all answers, because some will be partly right. Some will confuse the motherboard with a "chip" or the whole case. Note these misconceptions silently â€" you will address them in Section F.

Why this opener works:
It creates curiosity about a familiar object, connects the lesson to real Port Harcourt life (cyber cafÃ©, school lab), and turns a dry list of parts into a mystery to be solved. It also sets the promise of the session clearly and measurably.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a quick oral round-robin. Put the questions on the projector. Ask different learners each time; do not let one person answer everything.

Question 1:
"What is the difference between hardware and software?"

- Expected answer: Hardware is the physical parts you can touch (the keyboard, the screen, the parts inside). Software is the programs and instructions (Windows, a game, a browser) â€" you cannot touch it.
- Common gap: Some learners say software is "the internet." Gently correct: the internet is something software uses, not software itself.
- Bridge line: "Today is entirely about hardware â€" the parts you can touch."

Question 2:
"When you type a letter on the keyboard and it appears on the screen, is anything physical happening?"

- Expected answer: Yes â€" electricity moves through the parts inside the computer to turn the keypress into a letter on the screen.
- Common gap: "It is magic" or "the computer just knows." Explain that the computer is following instructions, and the parts we learn today are what carry those instructions.

Question 3:
"What happens to your typed document when the electricity goes off suddenly?"

- Expected answer: If you had not saved it, it is lost. Saved work stays. This introduces the idea of temporary vs permanent storage, which this session clarifies with RAM and HDD/SSD.
- Common gap: Some learners believe everything is always "saved." Use this as the hook for RAM vs storage.

Question 4:
"Name one part inside a computer. Even if you are not sure, guess."

- Expected answer: Any of: memory, hard disk, fan, "the thing that makes it fast." Accept everything.

Bridge to today (say this):

> *"Think of a computer like a restaurant kitchen. The kitchen has a head chef who does the cooking â€" that is the* ***CPU*** *. It has a prep table where ingredients sit while being used â€" that is the* ***RAM****. It has a store room that keeps ingredients even overnight â€" that is the* ***storage****. It has a gas cylinder that supplies the fire â€" that is the* ***PSU****. It has a manager who coordinates everyone â€" that is the* ***motherboard****. Today, we meet the whole kitchen staff."*

If more than a third of the class cannot answer Question 1 or 2, spend an extra three minutes re-explaining hardware vs software before continuing; the rest of the session depends on that distinction.

F. FULL LECTURE NOTES

F.1 What a computer actually is

The idea.
A computer is a machine that does three things over and over, extremely fast: it **takes in information** (input), it **works on that information** (processing), and it **produces a result** (output). It also **keeps information** for later (storage).

Plain-English definition.
A computer is an electronic machine made of separate physical parts that work together. Each part does one main job, and no single part can do everything alone. When people say "the CPU is the brain," that is a comparison, not the whole truth â€" a computer needs its memory, storage, power and board just as much as a person needs lungs, stomach and heart, not only a brain.

Why the parts are separate.
Have you ever noticed that you can buy a bigger hard disk, or add more memory, without buying a new computer? That is deliberate. By making the parts separate, a computer can be **repaired, upgraded and replaced part by part** instead of being thrown away whole. This modular design is the reason the computer repair trade exists â€" and the reason you have a job to learn.

Real-life comparison.
A computer is like a **Nigerian market stall**. The **stall itself** (the case) holds everything. The **stall owner** (CPU) makes the decisions. The **table in front** (RAM) holds only what is being sold right now. The **store behind** (storage) keeps the goods overnight. The **generator** (PSU) supplies power. The **signboard** (GPU/monitor) shows customers what is on offer. If the generator dies, the whole stall stops â€" even if the owner and goods are fine. That is why one failed part can stop a whole computer.

Important limitation / misconception.
Learners often think the "computer" is the screen. It is not. The screen (monitor) is an output device â€" a separate thing. The computer itself is the box (the system unit). Confusing the two causes wrong diagnoses later, for example blaming the monitor when the fault is inside the case.

F.2 The motherboard â€" the city where everything meets

Plain-English definition.
The **motherboard** is the large flat green (or sometimes black, blue or red) circuit board fixed inside the case. It is the main board that every other internal component plugs into or connects to. A circuit board is a flat board that has tiny metal pathways printed on it, along which electricity travels like traffic on roads.

Why it exists.

into the board, and the board carries signals between them.

What it does.
The motherboard:

- Holds the **CPU socket** â€" the special square socket where the processor sits.
- Provides **RAM slots** â€" long thin slots where memory sticks click in.
- Provides **expansion slots** (usually PCIe slots) where graphics cards and other add-on cards go.
- Provides **SATA ports** where hard drives and SATA SSDs connect by cable.
- Provides an **M.2 slot** (on modern boards) where NVMe SSDs slot in flat against the board.
- Carries the **main power connector** (the 24-pin ATX connector) where the PSU plugs in.
- Holds the **back I/O panel** â€" the row of ports at the back of the case for USB, network, audio and sometimes video cables.
- Often holds a small coin-shaped **CMOS battery** that keeps the clock and basic settings alive when the machine is off.

Real-life comparison.
The motherboard is like **Rumuola junction** in Port Harcourt â€" the central roundabout where many roads meet. Cars (signals) come from all directions and pass through the junction. If the junction is blocked, traffic stops everywhere at once, even though every car is fine.

Realistic repair-shop scenario.
A customer brings a desktop that powers on but never shows a picture. You open the case and notice the RAM stick is not clicked fully into its slot on the motherboard. You reseat it and the machine works. Nothing was "broken" â€" a connection on the main board simply was not complete. This is one of the most common quick fixes, and it is why knowing the board matters.

Common misconceptions.

- "The motherboard is the processor." No â€" the processor sits *on* the motherboard, but they are different parts.
- "All motherboards are the same." They are not. Different boards fit different CPU types and memory types, and they are not freely interchangeable. A beginner must never assume a random board fits a random CPU.

Limitations and safety.
The motherboard is delicate. Its surface has tiny tracks and small components. Never place a board on a carpet or a metal surface; never bend it; hold it by the edges; and always discharge static before touching it. Some repairs on the board itself (soldering, replacing capacitors) are advanced and are **not** part of this course â€" a beginner should refer those to a specialist.

F.3 The CPU â€" the processor (the brain)

Plain-English definition.
The **CPU (Central Processing Unit)** is the small square chip that carries out the instructions of every program. It does the arithmetic and the decision-making. It is often called the "brain" of the computer.

Why it exists.
Someone has to actually *do* the work â€" add numbers, compare values, move data. The CPU is the part that executes those instructions, billions of them every second.

What it does.
It reads instructions from programs, performs calculations, makes simple decisions (is this number bigger than that one?), and coordinates the rest of the parts by telling them what to do.

Key facts (plainly stated, no invented numbers):

- Two main makers of desktop CPUs are **Intel** and **AMD**.
- Speed is commonly described in **gigahertz (GHz)** â€" a measure of how many billions of cycles it can run per second. Generally a higher GHz means faster work, but it is not the only factor.
- A CPU runs **hot** when working. It therefore has a metal top (the **heat spreader**) and sits under a **heatsink and fan** (the cooler) that carries heat away.

The heatsink and fan â€” the CPU's cooling team.

A **heatsink** is a block of metal with many thin fins. It presses against the CPU to soak up heat. A **fan** blows air across the fins to carry that heat away. Between the CPU and the heatsink, a thin layer of **thermal paste** (a grey, slightly sticky compound) fills the tiny gaps so heat passes through properly. Too little, too old or dried-out paste is a common cause of overheating.

Real-life comparison.

The CPU is like the **head chef** in a busy kitchen. The chef cooks very fast and gets very hot. The **heatsink and fan** are like the kitchen's extractor fan and cooling system â€” without them, the chef collapses. The **thermal paste** is like oil in a frying pan, helping heat pass evenly.

Realistic repair-shop scenario.

A customer complains their laptop shuts itself off after about twenty minutes of use and the base gets very hot. You open it, find the fan clogged with dust and the thermal paste dried and crusty. You clean the fan and replace the paste, and the shutdown stops. The CPU was never faulty â€” it was simply being cooked. (Full cleaning is Session 8; today you only need to recognise the parts.)

Common misconceptions.

- "A faster CPU always makes the computer faster." Not always. If the computer has too little RAM or a slow hard drive, a faster CPU will not fix the slowness. The slowest part sets the pace, like a convoy that moves at the speed of its slowest vehicle.
- "The CPU is the whole computer." No â€” the CPU is one chip on the board.

Limitations and safety.

Never touch the pins under a CPU or the contacts in the CPU socket. Even a tiny bend can stop the machine working. Never power on a computer with the CPU cooler removed. CPU installation (Session 7 covers replacement) requires great care and should be practised only on a machine you are authorised to repair.

F.4 RAM â€” the short-term worktable

Plain-English definition.

RAM (Random Access Memory) is the computer's short-term, working memory. It holds the programs and data that the computer is using *right now*. While the power is on, RAM keeps this information; the moment the power goes off, everything in RAM is **erased**.

Why it exists.

The CPU works incredibly fast, but permanent storage is much slower to read from and write to. If the CPU had to fetch every tiny piece of data straight from the hard disk, it would spend most of its time waiting. RAM solves this: it is a fast, temporary holding area close to the CPU.

What it does.

When you open a program, the computer copies it from storage into RAM. As you work, the CPU reads from and writes to RAM. When you close the program or switch off, RAM is cleared.

Key facts:

- RAM comes in **sticks** (long, thin, flat modules) that click into the RAM slots on the motherboard.
- Capacity is measured in **gigabytes (GB)** â€” 8 GB, 16 GB and 32 GB are common in modern machines.
- More RAM lets more programs stay open at once without slowing down.
- RAM comes in **generations** called **DDR** (DDR3, DDR4, DDR5). Different generations are **not interchangeable** â€” a DDR4 stick will not fit a DDR5 slot, and forcing it will damage the board.

Real-life comparison.

RAM is the **prep table in a kitchen**. Ingredients being used right now sit on the table. When the kitchen closes (power off), the table is cleared â€” nothing stays on it overnight. The **store room** (storage) is where ingredients are kept permanently.

Realistic repair-shop scenario.

Port Harcourt repair shop, and it exists purely because people confuse "slow" with "broken."

Common misconceptions.

- "RAM and hard disk are the same thing." The single most common beginner mistake. RAM is temporary and small (GB) and empties on power-off; storage is permanent and large (hundreds of GB or TB).
- "More RAM makes the computer store more files." No â€" more RAM lets you *run more at once*. Storing files is the job of storage.
- "Adding RAM always speeds things up." Only if RAM was the bottleneck. If the slowness is from a failing hard drive or malware, more RAM will not help.

Limitations and safety.
RAM sticks are delicate. Hold them by their edges and never touch the **gold contacts** â€" the shiny strip along the bottom. Oils from your fingers on those contacts cause poor connections. Always handle RAM with anti-static protection.

(Note: "GB" for RAM and "GB" for storage use the same letters but mean different jobs â€" one is the size of the worktable, the other is the size of the store room.)

F.5 Storage â€" the store room (HDD, SSD, NVMe)

Plain-English definition.
Storage is where the computer keeps information **permanently** â€" the operating system, your programs, documents, photos and videos. Unlike RAM, storage keeps its contents when the power is off.

Why it exists.
You need your files to survive when the machine is switched off. Storage is the long-term memory of the computer.

The three main types a beginner must know:

1. HDD â€" Hard Disk Drive.
Inside an HDD are shiny magnetic platters that spin, with a read/write head that moves across them â€" like an old record player. Because parts move, an HDD is **slower**, can be **damaged by drops or knocks**, and makes soft whirring or clicking sounds while working. It is usually **cheaper per gigabyte** and comes in large sizes (for example 1 TB, 2 TB).

2. SSD â€" Solid State Drive.
An SSD has **no moving parts** â€" it stores data in electronic memory chips. It is **much faster** than an HDD, **more durable** (nothing to break when dropped), uses less power, and is silent. It is usually **more expensive per gigabyte** than an HDD.

3. NVMe â€" Non-Volatile Memory Express.
NVMe is the **fastest** type of SSD. It plugs flat into a special slot (the **M.2 slot**) directly on the motherboard, rather than connecting by cable like a SATA drive. It is commonly used for the operating system and the programs that must load fast.

SATA
â€" mention this because learners will hear it constantly. SATA is a standard for connecting storage drives and is the name of both the cable and the port. A SATA SSD uses the SATA cable; an NVMe SSD slots straight into the board.

Real-life comparison.
An HDD is like a **warehouse with a worker on a bicycle** fetching each item â€" it works, but takes time. An SSD is like a **warehouse where every item sits on a numbered shelf within arm's reach** â€" same goods, far faster to get. An NVMe is like a **shelf right beside your hand**.

Realistic repair-shop scenario.
A customer says "my computer takes forever to start up â€" I have to wait five minutes." You find the machine runs on an old HDD. You clone the system onto an SSD, and startup drops to seconds. The customer thinks you performed magic, but you simply moved the data from a spinning warehouse to an electronic shelf. Explaining this plainly is part of good customer service (Session 10).

Common misconceptions.

- "A 256 GB SSD holds less than a 256 GB HDD." Both hold about the same; they differ in speed and durability, not in how much they can hold for the same number.
- "I can drop an HDD because it is metal." Do not. The internal platters and head are extremely delicate; a fall can destroy it and your data.

Limitations and safety.
HDDs must be handled gently and never shaken while running. SSDs are more robust but can still fail, and when they fail they often give little warning. This is why backups matter. "Storage fails" is one of the most common real repair cases, so treat data with respect from the very first session.

F.6 The PSU â€" the power supply unit (the gas cylinder)

Plain-English definition.
The **PSU (Power Supply Unit)** is usually a metal box at the top or bottom-back of the case. It takes electricity from the wall socket (which in Nigeria is commonly around 220â€"240 volts alternating current) and converts it into the lower, steady **direct-current** voltages that the computer's parts need (commonly +3.3 V, +5 V and +12 V).

Why it exists.
The wall socket supplies one kind of electricity; the delicate components need a different, carefully controlled kind. Without the PSU, the computer cannot run. A computer has no internal battery to fall back on (a desktop, unlike a laptop, has no built-in battery).

What it does.
It converts and delivers power, through cables, to the motherboard (the big 24-pin connector), the CPU (the 4-pin or 8-pin connector), the drives and the graphics card.

Key facts:

- Capacity is measured in **watts (W)** â€" for example 400 W, 500 W, 650 W. The PSU must supply enough total power for all the components together.
- It provides several different voltages through different cables.
- A failing PSU can cause **random crashes, failure to start, sudden shutdowns, or damage to other components**.

Real-life comparison.
The PSU is like the **generator or gas cylinder** that supplies the fire for the kitchen. If it is weak, everything in the kitchen suffers. If it is faulty, nothing can cook at all, no matter how good the chef (CPU) is.

Realistic repair-shop scenario.
(This is the Session's case study, expanded here.) A customer's desktop will not power on at all â€" no lights, no fan spin. You check the wall socket and the cable first (always the simplest things first). Those are fine. You then test the PSU and find it has failed. You replace it. This is one of the most common reasons a computer shows "no sign of life."

Common misconceptions.

- "Any PSU will do; more watts is always better." A PSU must match the machine's needs and, importantly, must be of good quality. A cheap, badly built PSU can fail and damage everything else.
- "I can repair a PSU myself." **No.** Inside the PSU are components that can hold a dangerous charge even when unplugged. A beginner must **never open a PSU.** If it is faulty, replace it. This is a hard safety rule.

Limitations and safety.

- **Always unplug the computer from the wall before opening the case.**
- **Never open, drill, or tamper with the PSU itself** â€" the components inside can store a lethal charge.
- If you smell burning or see smoke, unplug immediately and do not touch it beyond that. Replace the PSU; never try to "fix" a burning one.
- This session does not test or replace the PSU. Those skills come later (Sessions 5 and 7), and even then, the PSU is replaced, not opened.

Plain-English definition.
The **GPU (Graphics Processing Unit)** handles images, video and everything shown on the screen. It turns the computer's data into the picture you see.

Two types:

1. **Integrated graphics** â€" built into the CPU or motherboard. Good enough for everyday work: browsing, typing, watching videos, light school work. It shares the main memory.
2. **Dedicated graphics card (GPU)** â€" a separate card that plugs into an expansion slot (PCIe) on the motherboard. It is far more powerful, has its own memory (called **VRAM**), usually has its own fan, and may draw extra power from the PSU. It is used for gaming, video editing, design and 3D work.

Why it exists.
Modern screens show millions of tiny dots (pixels) that must be calculated and redrawn many times per second. The CPU can do this, but a dedicated GPU does it far faster and frees the CPU for other work.

Key facts:

- The two main makers of dedicated GPUs are **NVIDIA** and **AMD**.
- A dedicated card has its own memory and its own cooling.
- Some powerful cards need a dedicated power cable from the PSU.

Real-life comparison.
If the CPU is the head chef, the GPU is the **artist who paints the signboard**. The chef cooks; the artist makes it look good to the customer. A busy restaurant needs both, and if the artist is weak, the signboard looks poor even if the food is perfect.

Realistic repair-shop scenario.
A customer says their monitor shows "no signal" even though the computer seems to be running (fans spin, lights on). You check the display cable â€" it is plugged into the motherboard's port, not the graphics card's port. Because a dedicated card is fitted, the motherboard port is often disabled. You move the cable to the card and the picture appears. The customer was sure the computer was "dead," but it was a one-minute fix â€" because you understood the GPU.

Common misconceptions.

- "A graphics card is only for games." Not true â€" many design and video jobs need one, and even everyday machines use graphics (built into the CPU).
- "No display always means the GPU is broken." No â€" it may be a loose cable, an unseated RAM stick, or the cable in the wrong port. This is why Session 4 teaches a proper workflow.

Limitations and safety.
A dedicated GPU can be large and heavy; support it when removing it so it does not bend in its slot. Some cards keep warm after use; let the machine cool before handling. Never touch the gold contact strip at the bottom of the card.

F.8 Expansion cards, cooling fans, ports and connectors

Expansion cards.
An **expansion card** is any add-on card that plugs into an expansion slot to add a feature the board lacks â€" for example a **sound card**, a **network (Ethernet) card**, or a TV tuner card. Many modern motherboards have sound, network and graphics built in, so dedicated cards are less common than they once were, but you will still meet them, especially in older or specialist machines.

Cooling fans.
A computer has several fans. The **CPU fan** sits on the CPU heatsink. The **case fans** move air in and out of the case. The **GPU fan** cools the graphics card. The **PSU fan** cools the power supply. **Purpose:** heat is the enemy of electronics; fans and heatsinks keep parts within safe temperatures. Dust is the enemy of fans â€" a dust-clogged fan cannot cool, and the machine overheats. (Cleaning is Session 8.)

Ports and connectors â€" where the outside world plugs in.

- **Network (Ethernet/RJ45) port** â€" for a network cable.
- **Audio jacks** â€" for speakers, headphones, microphone.
- **Display ports** â€" HDMI, DisplayPort or VGA, for monitors (on the motherboard if integrated graphics are used, or on the graphics card if one is fitted).
- **Power socket** â€" where the power cable plugs in (at the PSU).

Internal connectors to recognise:

- **24-pin ATX power connector** â€" the big one from the PSU to the motherboard.
- **4-pin/8-pin CPU power connector** â€" a smaller PSU cable near the CPU.
- **SATA data cable** â€" flat, with a small L-shaped connector, links drives to the board.
- **SATA power connector** â€" a wider, flat connector from the PSU to drives.
- **Fan connectors** â€" small 3-pin or 4-pin plugs for fans.
- **Front-panel connectors** â€" tiny plugs from the case's power button, reset button and front USB/audio to the board. These are fiddly and easy to misplace; we handle them carefully in Session 3.

Optical drive (legacy).
An **optical drive** (CD/DVD drive) is a bay device that reads and writes discs. It is now largely **legacy** â€" modern computers, especially laptops, often have none. You will still see them in older machines. Recognise it as a storage device with a disc tray, connected by SATA like a hard drive.

Real-life comparison.
Ports and connectors are the **doors and gates of a compound**. Each door is shaped for one purpose: the gate for cars (power), the pedestrian gate (USB), the network gate (Ethernet). You cannot drive a car through the pedestrian gate, and you cannot plug a USB cable into an Ethernet port â€" the shapes protect you from yourself.

Common misconceptions.

- "All USB ports are the same speed." They look alike but can differ in speed and purpose. Do not invent exact numbers; simply teach that they are not identical.
- "The VGA/HDMI port is on the wall." Some monitors and older machines use VGA; modern ones use HDMI/DisplayPort. Learn to look for the port on the machine and match it to the cable.

Limitations and safety.
Never force a connector â€" every plug has one correct orientation. USB plugs only fit one way (though USB-C fits either way). Forcing a connector bends pins and damages ports. Power off and unplug before plugging or unplugging internal cables.

F.9 How it all fits together â€" the relationship and data flow

The relationship in one paragraph.
The **motherboard** connects everything. The **CPU** does the thinking and sits under a **heatsink and fan** to stay cool. The **RAM** is the fast workspace the CPU uses right now. The **storage** keeps everything permanently. The **PSU** feeds electricity to all of them. The **GPU** paints what you see. **Fans** keep the heat down. **Ports and connectors** bring the outside world in and out. Remove any one and the whole system fails or runs badly â€" they are a team, not a list.

How data flows â€" follow one keypress from finger to screen.
This is the single most useful mental model a repair technician can carry.

1. **Input:** You press the letter **A** on the keyboard.
2. The signal travels through the **USB port** and along the motherboard to the CPU.
3. The **CPU** processes the instruction: "show an A."
4. The CPU uses **RAM** as its workspace while it decides what to do.
5. The CPU sends drawing instructions to the **GPU**.
6. The **GPU** turns that into the picture and sends it out through a **display port** (HDMI/DisplayPort/VGA).
7. The **monitor** shows the letter **A**.

9. Throughout all of this, the **PSU** supplies the electricity and the **fans** keep everything cool.

Why this matters in the shop.
When a customer says "my typing is delayed," you now have a map: is the keyboard/port at fault (step 2)? The CPU overloaded (step 3)? Too little RAM (step 4)? The GPU or cable (step 6)? The storage busy (step 8)? A good technician walks the flow, one step at a time. That is exactly the skill Session 4 builds into a formal workflow.

Real-life comparison.
Think of sending a **letter through the Nigerian postal system**. You write it (input), it goes to the local post office (motherboard), the sorting clerk reads the address (CPU), uses a desk to sort it (RAM), the delivery rider takes it out (GPU), and it arrives at the house (monitor). If the rider is missing, the letter never arrives, even though you wrote it perfectly.

Common misconceptions.

- "The CPU does everything." No â€" the CPU directs, but RAM, storage, GPU and PSU each do their own jobs.
- "Data flows only one way." It flows constantly in both directions: input comes in, output goes out, and the CPU reads and writes memory and storage continuously.

Limitations and safety.
Do not carry this table too far into advanced territory (buses, chipset, cache) â€" that is beyond a beginner course. The goal is a working mental map, not an engineering degree.

F.10 Desktop versus laptop â€" same parts, different arrangement

A

laptop
contains the same core parts as a desktop (CPU, RAM, storage, motherboard, cooling, power) but everything is **smaller, tightly packed and often harder to reach and replace**.

Aspect	Desktop	Laptop
Size and spacing	Large case, parts spaced out, easy to reach	Compact, tightly packed, few screws
Upgrades	Easy â€" RAM, storage, GPU, PSU often replaceable	Limited â€" RAM and storage sometimes replaceable; CPU and GPU usually fixed to the board
Power	PSU inside the case, fed from wall	Internal battery plus a charger; runs on battery when unplugged
Cooling	Larger fans and heatsinks	Small fans and heat pipes; clogs with dust more easily
Screen and input	Separate monitor, keyboard, mouse	Built in
Repair difficulty	Beginner-friendly	More delicate; needs care and sometimes special tools
Common faults	Failed PSU, dust, loose RAM, dead HDD	Overheating/dust, worn battery, broken screen/hinge, keyboard faults

Real-life comparison.
A desktop is like a **bungalow** â€" roomy, easy to walk around, easy to change furniture. A laptop is like a **well-packed suitcase** â€" everything fits, but to find one item you may have to disturb the others.

Safety note specific to laptops:
Laptops are more delicate and their batteries can be a hazard if punctured or swollen. This course's Week 1 disassembly practical (Session 3) uses **desktops**. Laptop disassembly should be attempted only with supervision and only once you have practised on desktops. Never open a laptop battery or pierce a swollen battery â€" report it to the instructor.

"Madam, this part is soldered to the board, so it cannot simply be swapped at" let me check what the real fault is before I promise anything." Honest assessment is part of the professional standard (Session 11).

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The unplugged desktop with side panel removed, on the anti-static mat, on a clean, well-lit table.
- Loose component samples in a row, each with a name card placed face-down or hidden so learners must recall.
- The wrist strap, drawn on and clipped to bare metal of the case.
- The projector showing a labelled motherboard diagram.
- The dead/fault parts (RAM, HDD, PSU) clearly labelled "FAULT â€" LOOK ONLY."
- A torch to point inside the case.

G.2 Demonstration 1 â€" live identification (8 minutes)

Say:
"I am going to open up a real computer and name every part inside it. Follow my finger with your eyes, then write the name on your chart."

Do this in order, pointing with the torch (never with a screwdriver, to avoid slips):

1. Point to the big board: "This is the **motherboard** â€" the main board everything plugs into."
2. Point to the square chip under the fan: "This is the **CPU**, under its **heatsink and fan**. The CPU is the brain; the fan keeps it cool."
3. Point to the long thin sticks: "These are **RAM** sticks. Short-term memory. Notice the shiny **gold contacts** along the bottom â€" we never touch those."
4. Point to the boxy drive: "This is a **hard disk drive** â€" permanent storage, with moving parts inside."
5. Point to an SSD or M.2 stick: "This is an **SSD** or **NVMe** â€" permanent storage with no moving parts, much faster."
6. Point to the metal box with many cables: "This is the **power supply** â€" it feeds electricity to everything. **We never open this box.**"
7. Point to the large card with its own fan: "This is the **graphics card** â€" it paints what you see on the screen."
8. Point to the spinning fan: "This is a **cooling fan** â€" it moves heat out of the case."
9. Point to the row of coloured plugs on the board: "These are **connectors** â€" power, SATA and front-panel cables plug in here."

Expected result:
Learners can repeat each name and a one-line job for each part.

Common errors to watch:
The instructor must not use a metal tool to point at a powered board (it is unplugged, but the habit matters). Do not let learners touch the CPU pins or the RAM gold contacts.

Questions to ask during the demo:

- "Which part stores your files permanently?" (Storage.)
- "Which part forgets everything when the power goes off?" (RAM.)
- "Which part would you change to make games run better?" (The GPU â€" or add RAM; accept both with reasoning.)
- "Why does the CPU need a fan?" (It gets hot when working.)

G.3 Demonstration 2 â€" correct, safe component handling (7 minutes)

Say:
"Now I will show you how to hold these parts so you do not damage them."

2. Pick up the RAM stick. "I hold it **by its edges**. I do not touch the gold contacts at the bottom."
3. Pick up the GPU. "I hold it by the metal bracket and the edges, and I support its weight so it does not bend."
4. Pick up the HDD. "I hold it firmly by the sides and I do **not** drop it â€" the inside is fragile."
5. Pick up the CPU (if a spare is available). "I hold it by the edges. I never touch the pins underneath or the contacts in the socket."
6. Point to the "FAULT" parts. "Remember, static damage can be invisible. A part can be dead and look perfect. That is why we protect every part, every time."

Expected result:
Learners can demonstrate the correct grip for RAM, GPU, HDD and CPU.

Reasoning to state aloud:
"I wear the strap because a static spark you cannot even feel can kill a chip. I hold boards by the edges because oils on my fingers corrode the gold contacts. I never open the PSU because it can hold a deadly charge even when unplugged."

Common errors to watch:
Learners grabbing the RAM contacts, touching the CPU pins, or resting boards on carpet or metal. Correct immediately and kindly.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "Component Identification Stations"

Objective:
With the instructor guiding step by step, learners identify each major component and write its name, function and one key fact on Worksheet 1.1.

Materials per learner/group:
Worksheet 1.1 (Component Identification Chart), Worksheet 1.2 (Data Flow Diagram), pencil, and access to the stations below.

Setup (instructor arranges four stations before class):

- Station A: the open desktop computer (unplugged, wrist strap provided).
- Station B: loose samples â€" CPU + cooler, RAM, expansion card.
- Station C: loose samples â€" HDD, SSD, NVMe.
- Station D: loose samples â€" PSU (sealed/closed), fans, SATA/power cables, ports photo sheet.

Procedure (instructor calls each step; learners move in small groups):

1. Put on your anti-static wrist strap and clip it to bare metal at the station. **If you have no strap, touch bare metal of the case before handling any part, and again before each new part.**
2. At **Station A**, point to and name: motherboard, CPU and cooler, RAM, storage, PSU, GPU, fans. Write each on your chart.
3. At **Station B**, pick up the RAM stick **by its edges** and describe what it does. Pick up the CPU and describe the role of the heatsink/fan and thermal paste.
4. At **Station C**, hold the HDD and the SSD. Write **two differences** between them (speed, moving parts, durability). Do not shake or drop the HDD.
5. At **Station D**, identify the PSU, the fan, the SATA data cable, the SATA power connector and the 24-pin power connector. **Never open the PSU.**
6. On Worksheet 1.2, fill in the **data-flow diagram**: keyboard â†' port â†' motherboard â†' CPU â†' RAM/GPU â†' monitor, with storage and PSU noted.
7. When finished, raise your hand for the instructor to check your chart.

Pause-and-predict points (instructor calls these aloud):

survives.

- After the group has seen HDD and SSD: *"Predict: which one would break if dropped while working?"* Expected: the HDD, because it has moving parts.
- After the data-flow diagram: *"Predict: if the GPU is removed, what happens to the picture?"* Expected: no picture on a dedicated-card setup, or lower performance if integrated graphics take over â€" accept reasoning.

Expected result:
Every learner completes a chart naming at least eight components with a correct function and one key fact each, and a filled data-flow diagram.

Troubleshooting while walking the room:

- Learner grabs RAM contacts â†' gently redirect to the edges; explain the oil problem.
- Learner says "RAM stores my files" â†' correct with the store-room/prep-table comparison.
- Learner calls the CPU "the whole computer" â†' point out it is one chip under the cooler.
- Learner cannot tell HDD from SSD â†' have them look for moving parts and the label; explain electronic vs mechanical.

Completion checklist (learner ticks each):

- ☐ Motherboard identified
- ☐ CPU + heatsink/fan identified
- ☐ RAM identified, held by the edges
- ☐ HDD identified and handled gently
- ☐ SSD (and NVMe if present) identified
- ☐ PSU identified (not opened)
- ☐ GPU identified
- ☐ At least one cooling fan and the main connectors identified
- ☐ Function and one key fact written for each
- ☐ Data-flow diagram completed
- ☐ Wrist strap worn or bare-metal touch done before handling

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I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
"The Repair Shop Intake"

Scenario:
You are the newest technician at **Elias Computer Clinic, Diobu, Port Harcourt**. A young man walks in carrying a desktop computer and complaining. He does not know the names of any parts. Your job is to translate his plain complaints into the correct technical component, and explain what each part does so he trusts your diagnosis.

Instructions:
For each complaint below, write (a) **which component** is most likely involved, (b) **what that component does**, and (c) **one question** you would ask to find out more. Work alone.

Complaints given to you:

1. "The machine makes a soft ticking sound and sometimes the files take long to open."
2. "It starts but the screen stays black, and I can hear the fan running."
3. "When I have Facebook, WhatsApp and music open at the same time, everything freezes."
4. "Nothing happens at all when I press the power button. No light, no sound."
5. "It gets very hot and switches itself off after some time."
6. "It works, but it is slow to start up in the morning and programs take long to open â€" it has never been upgraded."

- For each, give a one-sentence plain-English explanation a non-technical customer would understand.
- No copying the instructor's demonstration wording â€” use your own words.

Constraints:

- Assume this is a normal desktop computer from around a cyber cafÃ©.
- Keep each explanation to two or three sentences.
- Use correct component names.

Time allowance:
15 minutes.

Success criteria:

- All six complaints matched to a component with correct reasoning.
- Explanations use correct names and are understandable to a non-technical person.
- At least one question written for each complaint.

Hints if stuck:

- Hint 1: "No power at all, no light" points to the part that supplies electricity to everything.
- Hint 2: "Freezes when many programs are open" points to the fast temporary memory.
- Hint 3: "Slow to start up, never upgraded" points to permanent storage with moving parts.
- Hint 4: "Gets hot and switches off" points to the cooling parts and the CPU.

Instructor answer key (do not show learners):

1. **HDD (hard disk drive).** It stores files permanently using spinning platters and a moving head; a soft ticking and slow file access suggests a failing drive. Question: "When did the ticking start, and have you backed up your important files?" (Possible backup needed before any work â€” data safety first.)
2. **No display with fans running â†’** likely RAM not seated, GPU/cable issue, or display cable in the wrong port. Component focus: RAM / GPU / display connection. Question: "Did anything change before it happened â€” a move, a power cut, new parts?" (Accept a well-reasoned answer that names RAM, GPU or the cable and explains why.)
3. **RAM.** It is the fast temporary workspace; when too little, the machine slows badly. Question: "How much memory does it have, and does it happen only when many programs are open?"
4. **PSU (power supply).** It converts wall electricity to the voltages the parts need; if it fails, there is no sign of life. Question: "Have you checked the wall socket and the power cable, and does any light come on at all?" (Rule out the simplest things first.)
5. **Cooling system â€” heatsink/fan and thermal paste, plus the CPU.** Dust or dried paste causes overheating and shutdown. Question: "When did it last have its dust cleaned, and does the base or back feel very hot?"
6. **Storage â€” an old HDD.** Slow startup and program opening on a never-upgraded machine suggests an HDD; an SSD upgrade would help. Question: "How old is the machine, and how much free space is left on the drive?"

Marking guidance (out of 12 â€” 2 per complaint):

- Correct component named (1)
- Correct, plain explanation (1)
- Bonus credit within the same mark if a sensible question is also given.

J. REAL-WORLD CASE STUDY

Situation.
Chinedu runs a small business centre near the University of Port Harcourt. One Monday morning his main desktop will not come on at all. He presses the power button repeatedly â€” nothing: no fan, no light, not even a flicker. He has a queue of customers waiting to print, type and browse. He calls a technician friend, who says over the phone, "It is a motherboard

Relevant facts.

- The computer worked perfectly on Saturday evening.
- There was a heavy thunderstorm and a power "surge" (voltage spike) on Sunday night in the area.
- The machine is a normal desktop with an internal PSU; no battery.
- The customer has important files on the hard drive that he needs for his business.

Decision points.

1. Should you trust the phone diagnosis of "motherboard" and sell him a new machine, or investigate?
2. Where do you start â€" the most expensive part or the simplest checks?
3. What component is most consistent with "worked Saturday, power surge Sunday, completely dead Monday"?
4. What must you protect before doing anything else?

Possible approaches.

1. Accept the phone diagnosis and replace the whole computer (or motherboard). Fast, but expensive and possibly wrong.
2. Follow a basic order: check the wall socket and power cable, check the PSU, then look deeper. Cheap and methodical.
3. Try to repair the PSU yourself. **Unsafe and forbidden** â€" the PSU is replaced, never opened.

Recommended solution (for this stage).
Approach 2. Check the simplest things first (socket, cable, that the machine is actually receiving power), then test the PSU. Given the surge, a failed PSU is the most likely cause of a completely dead machine. **Replace** the PSU (never open it). Before returning the machine, confirm his hard drive is healthy so his business files are safe. Only if the machine is still dead after a known-good PSU is fitted would you suspect the motherboard â€" and even then, you confirm by testing, not guessing.

Trade-offs.

- Approach 1 is fast but risks replacing a healthy expensive board and charging the customer for the wrong repair â€" bad for reputation and for the customer.
- Approach 2 takes a little longer but is safe, cheap and usually correct, and it protects the customer's data and your reputation.
- Trying to repair the PSU is dangerous: the charge inside can injure or kill, and it is outside the scope of this course.

Professional connection.
This case is exactly why component knowledge matters. The technician who knows that the PSU converts and supplies *all* the power immediately suspects it after a surge, and checks it early and safely. The technician who only knows "it won't on, so buy a new one" costs the customer money and loses trust. In Port Harcourt, where power surges are common, this single insight saves customers thousands of naira â€" and wins you loyal customers who tell their friends.

Transfer exercise.
"A customer's computer runs but the display is blank and fans are spinning. Before blaming the GPU, what three simple things would you check?" (Display cable connected; cable in the correct port â€" motherboard vs graphics card; RAM seated fully.)

Safety reminder.
Everything above is done with the machine **unplugged**. The PSU is replaced, never opened. Working on mains-powered equipment or opening a PSU is never permitted in this course.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the live troubleshooting guide during the practical. When a learner struggles, ask them to check the relevant row rather than giving the answer immediately.

K.1 Touching the gold contacts on RAM or expansion cards

Symptoms:

part. **How to investigate:** Look at the contacts for smudges or dullness; check the part seats fully with clips closed. **Safe corrective steps:** (Instructor-supervised) remove the part with the machine unplugged, wipe the contacts gently with **isopropyl alcohol (90%+)** on a lint-free cloth if available, let it dry fully, and reseat holding only the edges. **How to confirm resolved:** The part is detected again and the machine behaves normally. **How to prevent recurrence:** Always hold boards by the edges; never touch the contacts.

K.2 Confusing RAM with storage

Symptoms:
The learner says "add a bigger RAM to store more videos" or "the RAM is full of files." **Likely cause:** The two concepts were never separated in the learner's mind. **How to investigate:** Ask the learner what happens to a document on the "RAM" when the power goes off. If they say "it stays," they are confusing RAM with storage. **Corrective steps:** Re-teach with the prep-table and store-room comparison. RAM empties on power-off; storage keeps data. **How to confirm resolved:** The learner can state the difference in one sentence using their own example. **How to prevent recurrence:** Use the comparison every time the topic returns.

K.3 Handling an HDD roughly or while it is running

Symptoms:
Clicking or grinding sounds after a knock; the drive stops being detected; files disappear. **Likely cause:** The internal platters or head were damaged by a shock; a mechanical failure. (Confirmed by the noise and by a drive test in a later session.) **How to investigate:** Listen for clicking, check whether the drive is detected, and (later, with tools) read its health. **Safe corrective steps:** Stop using the drive immediately; if there is important data, **do not keep power-cycling it** â€” refer for data recovery. Replace the drive (Session 7). **How to confirm resolved:** A replacement drive works normally. **How to prevent recurrence:** Never shake, drop or move a running HDD.

K.4 "No display always means the GPU is dead"

Symptoms:
Learner immediately blames the graphics card when the screen is blank. **Likely cause:** Skipping the simple checks (cable, port, RAM seating). **How to investigate:** Check the display cable is connected and in the correct port; check whether RAM is fully seated; test with a different monitor/cable if authorised. **Corrective steps:** Fix the simple cause first; only then consider the GPU. **How to confirm resolved:** The display returns. **How to prevent recurrence:** Always follow a simplest-first order (this becomes the Session 4 workflow).

K.5 Forgetting anti-static protection

Symptoms:
No immediate symptom â€” the danger is invisible. Later, a part mysteriously fails. **Likely cause:** Static electricity discharged into a component, causing latent (delayed) damage. **How to investigate:** You usually cannot see it; that is the point. Prevent it rather than detect it. **Corrective steps:** Always use the wrist strap or touch bare metal before handling parts. **How to confirm resolved:** You cannot always confirm â€” prevention is the only reliable answer. **How to prevent recurrence:** Make the strap part of the routine, like wearing gloves before handling a hot pot.

K.6 Forgetting to unplug before opening the case

Symptoms:
Risk of electric shock, short circuit, or damage; a fan may start unexpectedly. **Likely cause:** Rushing. **How to investigate:** N/A â€” this is a discipline issue. **Corrective steps:** Always unplug from the wall and press the power button once to discharge residual electricity before opening the case. **How to confirm resolved:** The machine is verified unplugged and discharged. **How to prevent recurrence:** Build the habit into every session's checklist.

Distinguishing likely vs confirmed:
In every row above, the "likely cause" becomes a "confirmed cause" only after you *check* it â€” see the contacts, hear the click, test the drive, reseat the part and see the fault disappear. Teach learners to confirm before they "fix."

Assignment title:
"My Component Reference Card"

Purpose:
To reinforce component knowledge by having the learner build a personal reference they will keep using all course. This becomes the foundation of the capstone service report.

Realistic scenario:
A customer walks into your new shop and asks, "Please, what exactly is inside this computer, and which part is likely causing my problem?" You must be able to answer confidently and simply. Your reference card is the tool you will use to explain it.

Detailed instructions:

1. Create a document in Microsoft Word or Notepad. Title it **"My Component Reference"**.
2. For **each** of these components, write:
    - The **name** of the component.
    - A short **description** of what it looks like (a sentence or a rough drawing; you may sketch on paper and photograph it).
    - Its **function** in your own words.
    - **One key fact** (for example, "RAM empties when the power is off").
    - **One common problem** linked to it (for example, "a dust-clogged fan causes overheating"). Components to include: motherboard, CPU (with heatsink/fan), RAM, HDD, SSD, PSU, GPU, cooling fan, ports/connectors.
3. Add a short paragraph (3â€“5 sentences) titled **"How data flows"** describing the path from a keypress to the screen.
4. Save the file as **"My Component Reference"** on the Desktop.
5. Show the saved file to the instructor before leaving, and be ready to answer one question about any component in it.

Required deliverables:

- One saved document with all nine components and the data-flow paragraph.

Submission format:
Saved on the Desktop, file named exactly `My Component Reference` . If the learner has no computer at home, they may complete it in the academy lab and show it before leaving.

Deadline:
Before Session 2 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
All components included	All nine present	Seven or eight present	Six or fewer
Function accuracy	All functions accurate	Mostly accurate	Several errors
Descriptions	Clear and detailed	Mostly clear	Vague or missing
Common problems	Relevant and accurate	Mostly correct	Incorrect or missing
Data-flow paragraph	Correct, ordered, clear	Mostly correct	Missing or wrong order
File saved correctly	Saved on Desktop, correct name	Saved, minor naming issue	Not saved

(Reduce the "All components" column to a 5/3/1 scale and keep total 25 by treating each row as equal weight.)

Common mistakes to avoid:
confusing RAM with storage; touching/claiming to touch contacts; calling the CPU "the whole computer"; leaving out the cooling system; saving the file in the wrong place.

Instructor reference solution (summary, do not show learners beforehand):

- RAM: fast temporary memory; empties on power-off; common problem = not fully seated, or too little for heavy use.
- HDD: permanent storage with moving parts; common problem = mechanical failure (clicking).
- SSD: permanent electronic storage, faster; common problem = can fail with little warning â€" hence backups.
- PSU: converts wall power to needed voltages; common problem = total failure (no sign of life) â€" and it is never opened.
- GPU: renders images; common problem = no display (often a cable/port issue, not the card).
- Cooling fan: moves heat out; common problem = dust clogging.
- Ports/connectors: connect outside devices; common problem = bent pins from forcing plugs.
- Data-flow paragraph: keyboard â†' USB port â†' motherboard â†' CPU â†' RAM works with CPU â†' GPU â†' monitor; storage and PSU noted.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below â€" do not read the key until you have attempted all questions.)

Q1. Short answer.
In one sentence each, say what the motherboard, the CPU and RAM do.

Q2. Multiple choice.
Which component stores your files **permanently**, even when the power is off? a) RAM b) CPU c) Storage (HDD/SSD) d) PSU

Q3. Explain in your own words.
Why is RAM described as "short-term" and storage as "permanent"? Give your own everyday example.

Q4. Practical identification.
A customer's computer makes a soft ticking sound from a spinning part inside and files open slowly. Which component is most likely involved, and why?

Q5. Error detection / truth-check.
A learner says: "I will add more RAM so I can store more videos." Find the mistake and correct it.

Q6. Scenario.
A desktop powers on â€" the fans spin and a light shows â€" but the screen stays black. List three things you would check, in order from simplest to most complex.

Q7. Short answer.
Why must the PSU never be opened, even when it is faulty?

INSTRUCTOR ANSWER KEY (Q1â€"Q7)

A1.
The **motherboard** is the main circuit board that connects all the other components; the **CPU** carries out the instructions of programs (the "brain"); **RAM** is the fast temporary workspace that holds data currently in use and empties when the power goes off.

A2.
(c) Storage (HDD/SSD).

A3.
RAM holds only what is being used right now and loses everything when the power goes off, like items on a kitchen prep table cleared at closing. Storage keeps files even with no power, like goods in a store room. Everyday example: an open, unsaved document lives in RAM; once you save it, it goes to storage and survives a power cut.

A4.
The **HDD (hard disk drive)**. It has moving parts â€" spinning platters and a read/write head â€" so a ticking sound and slow file access suggest a mechanical problem. (Advise protecting the customer's data before further use.)

empties on power-on. To store more videos you need a bigger **storage** drive (or an external drive), not more RAM. (More RAM helps when many programs are open at once.)

A6.
Accept any sensible, simplest-first order such as: (1) check the display cable is connected and in the correct port (motherboard vs graphics card); (2) check the monitor has power and is switched on; (3) reseat the RAM (a common cause of no display). GPU comes only after these.

A7.
The PSU contains components that can hold a dangerous electrical charge even when unplugged from the wall. Opening it risks a serious or fatal shock, and repairing it is outside the scope of a beginner course. If it is faulty, it is **replaced**, never opened.

Marking:
Award up to 2 marks per question (Q3 and Q6 up to 3 each). Total â‰ˆ 16â€"18; scale proportionally. Full credit requires the correct component name and correct reasoning.

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned what is inside a computer and what each part does. You met the **motherboard** (the board that connects everything), the **CPU** (the part that does the work, kept cool by a heatsink and fan), **RAM** (fast temporary memory that empties when the power goes off), **storage** (permanent memory â€" HDD and SSD), the **PSU** (the part that supplies electricity and must never be opened), the **GPU** (the part that paints the screen), the **cooling fans**, and the **ports and connectors** that bring the outside world in and out. You also learned how data flows from a keypress to the screen, how a desktop differs from a laptop, and how to handle parts safely by their edges with anti-static protection.

Five important takeaways:

1. A computer is a **team of parts**, each with one main job â€" none works alone.
2. **RAM is temporary; storage is permanent.** Never mix them up.
3. **HDD is mechanical and slower; SSD is electronic and faster; NVMe is the fastest SSD.**
4. The **PSU supplies all the power and is never opened** â€" replace it, never repair it.
5. Always **handle parts by the edges**, protect against static, and unplug before opening any case.

Questions to help you recognise what you can now do:

- Can I name at least eight components inside a computer without looking?
- Can I explain RAM vs storage to a customer in one sentence?
- Can I trace how a keypress becomes a letter on the screen?
- Can I state why the PSU must never be opened?

Connection to the next session:
Now that you know the parts, next session you will learn how to work around them **safely** â€" how to protect yourself from electricity and from ESD, how to choose and use the right tools, and how to set up a clean, safe workspace. Knowing the parts without knowing safety is dangerous; Session 2 makes you safe.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner names all major components without prompting and gives each a correct one-line function.
- The learner spontaneously holds RAM/boards by the edges and does not touch gold contacts.
- The learner correctly separates RAM from storage and HDD from SSD in their own words.
- The learner can trace the data flow from input to output and name the parts at each step.

- "The monitor is the computer." â†' The system unit is the computer; the monitor is output only.
- "RAM and hard disk are the same." â†' Temporary vs permanent; GB of workspace vs GB of store room.
- "The CPU is the whole computer / the case." â†' The CPU is one chip under the cooler.
- "A graphics card is only for games." â†' Used for design and video too; and integrated graphics exist.
- "The PSU can be fixed." â†' It is replaced; opening it is dangerous and forbidden.

Suggested remedial exercises for learners who struggle:

1. **Component flash cards:** write each component on one card and its function on another; match them, then shuffle and repeat.
2. **Draw-and-label:** sketch a simple desktop with three parts and label them; repeat from memory.
3. **Data-flow relay:** learners physically stand in a line holding component name cards (keyboard, port, CPU, RAM, GPU, monitor) and pass a paper "signal" along the line in the correct order.
4. **Analogy practice:** ask the learner to invent their own kitchen/market analogy for three components.

Differentiation:

- **Level 1 (supported):** provide a partially completed Component Chart (names filled, functions to finish) and work alongside the learner at the stations; pair with a careful partner.
- **Level 2 (standard):** complete the exercise as written, independently with the group.
- **Level 3 (challenge):** research and explain one additional component (for example a sound card, network card, optical drive, or the CMOS battery), and describe the data flow for saving a file, not just typing a letter.

Timing guidance:
The lecture (Section F) is long by design; if the room is slow, compress F.8 and F.10 to quick summaries and protect the hands-on time in Section H â€" the practical identification is the priority and is directly assessed. Keep Section J as a 3â€"4 minute story if time is short.

END OF SESSION 1

Next: Session 2 â€" Safety and Tools: Working on Computers Without Harming Yourself or the Machine.

SESSION 2 â€" SAFETY AND TOOLS: WORKING WITHOUT HARMING YOURSELF OR THE MACHINE

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 2 â€" SAFETY AND TOOLS: WORKING WITHOUT HARMING YOURSELF OR THE MACHINE

A. SESSION IDENTIFICATION

Item	Detail
Session number	2 of 12
Session title	Safety and Tools
Week / Module	Week 1 â€" Module: Components and Safety
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner
Prerequisite knowledge	Session 1 â€" Computer Components. The learner must already be able to name the major components.
Connection to previous session	Session 1 named the parts. This session teaches how to be safe around those parts: electrical safety, ESD (electrostatic discharge) protection, the correct tools, and a safe workspace.
Connection to the course	Every later session â€" disassembly (3), diagnostics (4â€"6), repair (7â€"8), the capstone (12) â€" requires these safety habits and tool skills. This is the session that makes the rest of the course possible without injury or damage.
Preparation before class	Tool kits per learner/group (Phillips PH1 and PH2, flathead, tweezers, torch, needle-nose pliers); anti-static wrist straps (one per learner); anti-static mat; safety glasses; a workspace table; a powered-off desktop to demonstrate on; printed Safety Procedures Checklist and Tool Identification Chart; a balloon for the opening ESD demonstration; optional multimeter (demonstration only).

Safety note for this session:
This session is *about* safety, so model it perfectly. Nothing is opened on mains power. The demonstration machine must be unplugged and discharged before any handling. The PSU is never opened. Learners must not test anything on a live socket. The multimeter is used only by the instructor on a low-voltage example (for example a battery) and never on mains wiring. If a learner has any medical condition that makes electrical work risky, speak to them privately and provide alternative observation roles.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain ESD (electrostatic discharge)** in plain words, describe how it damages components, and explain why damage can be invisible.

3. **Apply electrical safety rules** â€" unplugging from the wall, discharging residual electricity, never working inside a powered machine, and never opening the PSU.
4. **Identify and correctly name the common computer repair tools**, and state what each is used for.
5. **Use a screwdriver of the correct type and size** and explain why the wrong size damages screws.
6. **Set up a safe, organised workspace** with good lighting, screw organisation and safe handling.
7. **Practise safe lifting and handling** of computers and components.
8. **State clearly when a repair is beyond their skill level** and must be referred to a qualified technician or supervisor.

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- Tool kits, one per learner or pair: Phillips screwdrivers in sizes **PH1 and PH2**, a small flathead screwdriver, a pair of tweezers, a small torch/flashlight, and needle-nose pliers (optional pliers, one per table).
- Anti-static wrist straps â€" **one per learner**.
- An anti-static mat for each workspace table.
- A powered-off desktop computer to demonstrate safe setup on.
- A small container or screw tray for organising screws (an egg carton or ice-cube tray works well and is cheap).
- Safety glasses (one pair per learner where available; required when using compressed air).
- A balloon (for the ESD demonstration) and a woollen or synthetic jumper/cloth (optional).

Software:

- None required for this session. Safety and tools are physical skills.

Safety gear:

- Anti-static wrist straps and mat (as above).
- Safety glasses.
- A first-aid kit available in the room.
- A fire-safety reminder: know where the electrical point is and how to switch off the room's power in an emergency (point it out to learners).

Sample / fault components:

- A dead motherboard or dead expansion card labelled "FAULT â€" DEMO ONLY" to show a component that looks fine but is dead â€" useful for discussing invisible ESD damage.
- A stripped screw and a stripped screw hole to show what happens when the wrong driver is forced.
- A **closed, unopened** faulty PSU labelled "DO NOT OPEN" to reinforce the rule.

Worksheets (printed):

- Worksheet 2.1 â€" "Tool Identification Chart" (one per learner): a table with the tool name, what it is used for, and a "safe/unsafe for beginners" column.
- Worksheet 2.2 â€" "Safety Procedures Checklist" (one per learner): a step-by-step list the learner ticks during practice.
- Worksheet 2.3 â€" "Safe or Unsafe?" scenario sheet (10 short scenarios).

Tools (instructor demonstration kit):

- A complete tool kit, laid out neatly on the anti-static mat.
- A multimeter (instructor-only demonstration; used only on a low-voltage battery, never on mains).
- A can of compressed air (used outdoors or with good ventilation and safety glasses).

Alternatives for a learner or academy with limited equipment:

- If no anti-static straps are available, the fallback is the "touch bare metal before each part" method â€" but explain that a strap is far better and should be bought as soon as possible.
- If no real tools are available, use printed photographs of tools and have learners name and describe them; then practise with whatever basic screwdrivers exist.

Suggested visuals:

- 

â€" Tool kit identification on the bench



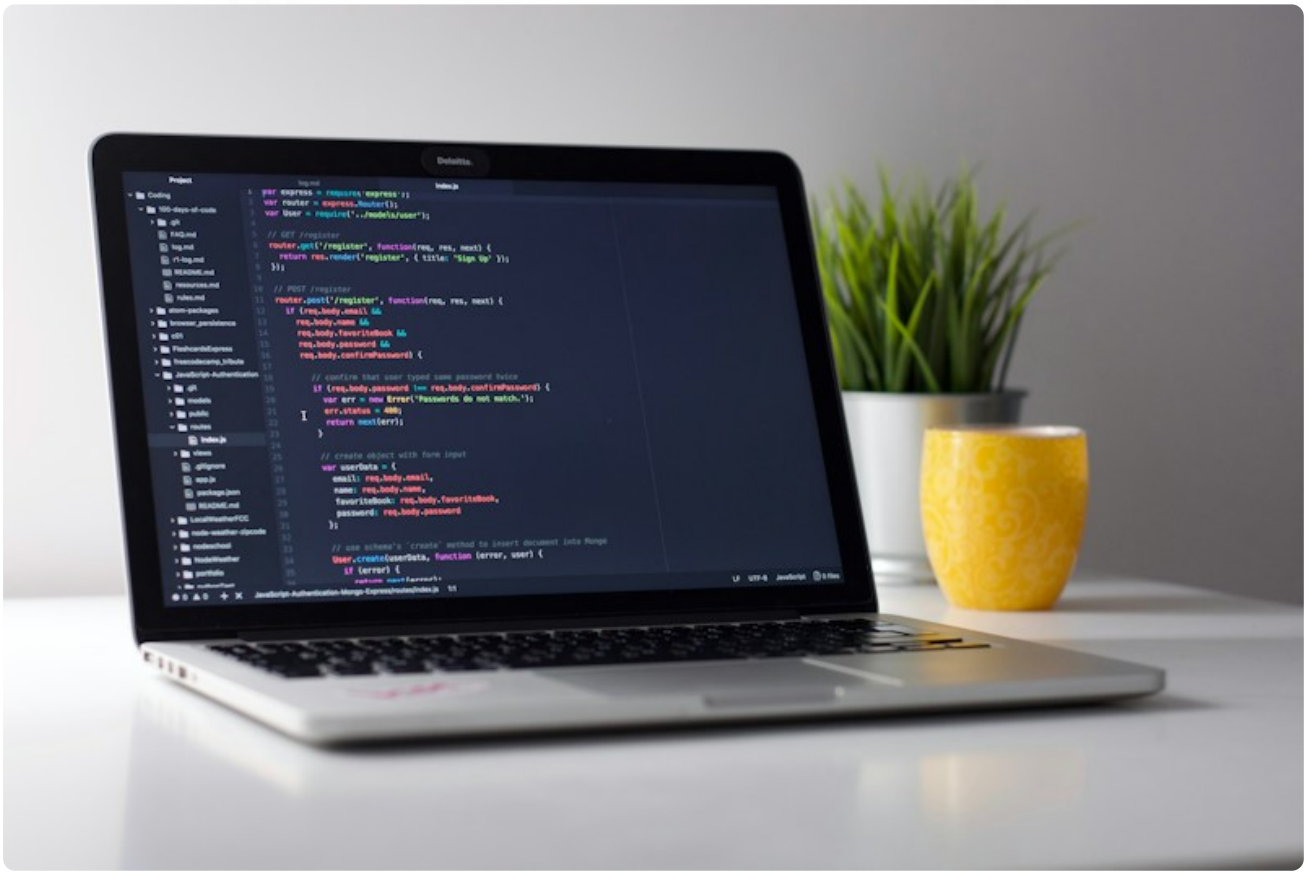
- â€" Safe workspace and ESD precautions



- â€" Fault parts used for the safety demonstration



— Safety and ESD checklist completed on the bench



— Laptop opened safely for the tools demonstration

D. OPENING ACTIVITY (6 minutes)

"The Invisible Danger"

Instructor setup:

show attraction.

What the instructor says:

> *"Watch this. [Rub the balloon.] I have just made electricity â€" but not from a socket. I made it by rubbing. You cannot see it on the balloon, but it is there. Now watch what it does to paper. [Bring the balloon near the paper; it jumps.] That invisible thing is called*
>
> ***static electricity***
>
> .
>
> *Now here is the serious part. Today, when we rub things together â€" our clothes, a plastic bag, our shoes on a carpet â€" we build up this same invisible charge on our bodies. It can be thousands of volts. When you touch something metal, that charge jumps across in a spark. Usually you feel a tiny shock, or nothing at all.*
>
> *But inside a computer, that same invisible spark â€" one you cannot even feel â€" can kill a chip. Dead. And you will not find out until weeks later, when the part fails for 'no reason.'*
>
> *We call this danger*
>
> ***ESD***
> *â€" electrostatic discharge. It is the number-one silent killer of computer parts in Nigeria and everywhere. And there is more than one danger today: plain electricity from the wall, tools used wrongly, and lifting machines carelessly.*
>
> *Today you learn how to touch a computer without harming it â€" or yourself. This is the most important session in the whole course, because if you are not safe, your skills do not matter."*

Then show the invisible damage:
Hold up the dead board labelled "FAULT â€" DEMO ONLY." Say:

> *"This board looks perfectly fine, does it not? Shiny. Clean. No burn marks. And yet it is dead. Static electricity leaves no mark. That is why we do not rely on our eyes â€" we rely on our*
>
> ***habits***
> ."

Expected learner responses:
Learners are often surprised that ESD damage is invisible and that they may have killed parts before without knowing. Some may say "but my phone works when I touch it" â€" accept this and explain that phones and finished products are more protected than exposed circuit boards during repair. Some may mention carpets at home. Praise that, because carpet is a real risk.

Why this opener works:
It makes an invisible, abstract danger physical and memorable using a simple balloon, and it immediately connects to the expensive consequences (dead parts, lost money). It also previews the whole session's theme: safety first.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a quick oral review. Ask different learners each time.

Question 1:
"Name at least six components inside a desktop computer and say what one of them does."

- Expected answer: Motherboard, CPU, RAM, HDD/SSD, PSU, GPU (and fans, connectors). Any correct function.
- Common gap: learners forget the cooling fan or connectors. Remind them that the fan is a component too.
- Bridge: "Today you will not just name these parts â€" you will learn to handle them without killing them."

Question 2:

- Common gap: learners think "unplugged means safe" for everything. Clarify: unplugging makes the machine safe to work in *except* inside the PSU, where a stored charge can remain; that is why we never open it.
- Bridge: "That rule starts today and stays with you for life."

Question 3:
"How should you hold a RAM stick, and why?"

- Expected answer: By the edges; do not touch the gold contacts, because finger oils damage the connection.
- Common gap: learners may not know *why*. Explain the oil problem.
- Bridge: "Correct handling is part of being safe â€" safe for the component. Today we also learn safe for you."

Question 4:
"What is the difference between RAM and storage?"

- Expected answer: RAM is temporary, empties on power-off; storage is permanent.
- Common gap: still mixing the two. Correct briefly; it comes up again in the practical.
- Bridge: "Keep this clear, because today's case study involves a mistake with physical parts, not memory."

Question 5:
"Have you ever received a small electric shock touching a car door or a metal tap? What caused it?"

- Expected answer: Static electricity discharging.
- Common gap: some say "faulty wiring." Clarify the difference between harmless static and dangerous mains electricity â€" both are real, but they behave differently, and today we handle both.

Bridge to today (say this):

> *"Think about a*
>
> ***hospital operating theatre***
> *. The surgeon has the best skills, but none of that matters if the equipment is not sterile and the theatre is not safe. A computer repair bench is our operating theatre. Today we set the rules and the tools so that every repair we do later is safe â€" for us and for the machine. Safety is not a lecture before the real work; safety* ***is*** *the real work."*

If more than a third cannot answer Question 2 or 3, spend an extra three minutes re-teaching the PSU rule and the handling rule before continuing, since both are safety-critical.

F. FULL LECTURE NOTES

F.1 ESD â€" the invisible killer

Plain-English definition.
ESD (electrostatic discharge) is the sudden jump of a small electric charge from one object to another when they touch. It is the same little "snap" you feel when you touch a car door or a metal gate after walking on a rug.

Why it happens.
When two different materials rub together (your shoes and carpet, your shirt and a chair, a plastic bag and your hand), tiny charges build up on your body. Your body becomes like a tiny battery. When you then touch something that conducts electricity â€" a metal object, or a circuit board â€" the charge jumps across in an instant.

How it damages components.
Computer chips and boards are designed for very small voltages. A static discharge that you cannot even feel can carry a voltage far higher than a chip can survive. Two things can happen:

1. **Immediate damage:** the chip dies instantly. The machine fails right away.

Why you cannot rely on your eyes or your senses.
A static shock that you can *feel* is only a small portion of the danger. You can damage a component with a discharge **you never feel and never see**. As the demonstration showed, a dead board can look perfect.

Important limitation â€" do not invent numbers.
It is tempting to say "10 volts kills a chip" or "your body holds 3,000 volts." Different sources quote different figures, and the exact numbers vary with the situation. The safe, honest teaching is this: **static electricity from ordinary daily activity can cause voltages far beyond what a chip can survive, even when you feel nothing.** Teach the *principle* and the *habit*, not a made-up exact figure.

Real-life comparison.
ESD is like **sand in an engine**. One grain is nothing; you cannot always see it going in; but over time, or at the wrong moment, it destroys the engine. You never allow sand near an engine, even though it looks harmless. In the same way, you never allow static near a board, even though you cannot see it.

Realistic repair-shop scenario.
A technician in a Port Harcourt shop reseats a customer's RAM on a hot, dry afternoon while standing on a tiled floor and wearing a synthetic shirt. Three weeks later the customer returns: the same machine now crashes randomly. The RAM had a latent ESD failure from that day. The technician blamed the customer. The truth is that a 30-second habit â€" wearing the strap â€" would have prevented it. This is why ESD is taught before any tool is picked up.

Common misconceptions.

- "ESD is not real; I have never destroyed anything." You may have â€" you just never connected the failure to your handling. Invisible damage is the whole problem.
- "Living in a humid climate, I am safe." Humidity reduces static buildup but does **not** remove the risk. Port Harcourt's humidity helps, but strap up anyway.
- "Only cheap parts are at risk." All exposed boards are at risk; expensive ones hurt the most when they die.

Safety implications.
ESD protection is a rule, not an option. No learner handles a board without the routine in F.2. A component that a learner is unsure about should be handled only under supervision.

F.2 How to prevent ESD â€" the routine

Principles.
The aim is to keep your body at the *same* electrical level as the machine, so that no charge jumps when you touch it. This is called **grounding** or **bonding** yourself to the equipment.

The ESD-safe routine (teach learners as a fixed sequence):

1. **Wear an anti-static wrist strap.** It is a band that goes on your wrist with a coiled wire that clips to bare metal of the computer case (or the anti-static mat). It slowly and safely drains any static from your body.
2. **Work on an anti-static mat** where possible. The mat is designed to carry charge away safely rather than let it build up.
3. **Touch bare metal of the case before handling any component** â€" every time, and again each time you come back to the machine after moving around. Unpainted metal, not the plastic or painted panels.
4. **Handle all boards by their edges.** Never touch gold contacts, chips or pins.
5. **Avoid building up static:** do not shuffle on carpets, do not rub balloons or plastic, do not work in socks on a synthetic rug, and avoid wearing loose woollen or synthetic clothing if possible.
6. **Keep components in anti-static bags** when they are not installed. The shiny grey or pink bags you receive parts in are antistatic; do not throw them away.
7. **Never place a board on a carpet, a foam mattress, a plastic tablecloth or a metal surface.** Use the mat, or the antistatic bag, or the original box.

Why the wrist strap must be clipped to the case, not to a wall socket.

tool, not a power wire.

Real-life comparison.
Wearing the strap is like **sharing a plate of food with a friend** â€” you are both at the same table, so nothing "jumps" between you. Without it, you are like a person holding a live electric fence and then touching your friend; the spark finds the path.

Common misconceptions.

- "I only need the strap when I am touching RAM." No â€” any board, and every time, including when you first sit down.
- "A cloth or aluminium foil works as well." No. Use a proper anti-static strap and mat.
- "If I keep the machine plugged in, it is grounded." Partly true for some systems, but the safe rule for beginners is: **unplug the machine first, then use the strap.** Never assume that being plugged in makes you safe â€” it can make you less safe.

Limitations and safety.
The strap protects equipment, not from mains electricity. It must be worn **in addition** to unplugging, never as a substitute. A damaged strap (torn wire, broken clip) must be replaced.

F.3 Electrical safety â€” the rules that keep you alive

Plain-English idea.
Inside the computer, the PSU and some parts use mains-derived electricity (from the wall, around 220â€“240 V in Nigeria). Touching certain parts while the machine is live can give a serious or fatal shock. The rules below are absolute.

The rules:

1. **Always unplug the power cable from the wall before opening the case.** Pull the plug out, do not just switch off at the socket â€” switch off first to be safer, then pull the plug.
2. **Press the power button once after unplugging.** This is called **discharging residual electricity**. Some parts (especially capacitors) can hold a small stored charge for a short time. Pressing the power button with the plug removed helps drain it so there is no surprise.
3. **Never work inside a powered-on computer.** Not even to "look." Not even to "just plug the fan." If the machine is on, stop and switch off.
4. **Never open the PSU.** It holds a dangerous charge even when unplugged, and mending it is beyond this course. If it is faulty, replace it â€” with supervision as you learn.
5. **Never use metal tools near live circuits.** No jewellery, no loose metal, no tools resting inside a live machine. Remove rings, watches and metal bangles before working.
6. **Keep liquids away.** Water and electricity do not mix. No drinks on the bench; if there is a spill, switch off at the wall before doing anything.
7. **If you smell burning, see smoke, or hear a loud pop â€” stop.** Switch off / unplug from a safe distance if possible, do not touch the machine, and call the instructor. Never move or open a burning or smoking machine that is still plugged in.
8. **Know the emergency switch.** Point out the room's main switch or fuse board to every learner. In an emergency, cut the power there â€” do not grab the machine.
9. **Watch the cables.** A trailing power cable or a laptop charger is a trip hazard. Keep the floor clear.
10. **Know your limits.** If a job involves mains wiring, the inside of the PSU, wall sockets or anything you are not confident about â€” stop and refer it to a qualified electrician or technician. No repair is worth your life.

Real-life comparison.
Electricity in a computer is like **a caged lion**. While the cage is closed (unplugged), the lion is still real and still dangerous if you poke into the wrong cage (the PSU), but the area around it is safe. Once the cage opens (live power), everything changes. Our rule is simple: we work only when the lion is asleep and the area is checked.

Realistic repair-shop scenario.
A technician in a hurry opens a case to swap a hard drive while the machine is still plugged in, intending to "just be quick." His metal wristwatch touches the exposed PSU connector. The result is a burn and a ruined machine. Ten seconds of unplugging would have prevented it. This is a true-to-life pattern â€” hurry causes more injuries than difficulty.

- "Low voltage = no danger." The motherboard and internal parts run on low voltage, but the **PSU input side and mains cable are high voltage**, and stored charges can still bite. Respect the whole machine.
- "I will just hold the tool carefully." Caution does not replace unplugging. The plug always comes out first.

Limitations and safety.
This course never asks you to work on mains wiring or inside a PSU. Those are outside scope and dangerous. When in doubt, stop and ask the instructor.

F.4 The correct tools â€" and what each one is for

Why tools matter.
The right tool makes a job safe and clean; the wrong tool strips screws, damages boards and causes injury. A professional technician's skills begin with good habits with basic tools.

1. Screwdrivers â€" the most important tool.

- **Phillips screwdriver (cross-shaped tip).** Used for most computer screws. Sizes **PH1** and **PH2** cover most jobs; **PH2** is the everyday size for case screws. Using the correct size matters: too small, and you slip and chew up the screw head; too large, and it will not seat. A magnetised tip helps hold screws in tight corners.
- **Flathead screwdriver (flat blade).** Used for prying open plastic clips and gentle levering, and for some slotted screws. **Use with great care** near boards â€" the flat blade can slip and scratch or cut circuit tracks. Never use a flathead as a chisel or to force a screw.
- **Precision (small) screwdrivers.** Needed for some laptops and small screws. If you do not have the correct size, do not force it â€" get the right driver.

Why screw size and type matter (teach this clearly).
Computer screws come in different lengths and thread types. A long screw forced into a short hole can damage the board. Always keep screws organised (see workspace) and always match screw to hole.

2. Tweezers.
Fine-point tweezers are used to retrieve dropped screws, hold small jumpers and connectors, and place tiny screws in awkward spots. Always use them with the machine unplugged, and never force a screw out with tweezers â€" you may bend the board.

3. Torch / flashlight.
A small torch is essential for seeing inside a dark case, checking for dust, and reading labels. It also helps spot a bent pin or a loose cable.

4. Needle-nose pliers.
Used for gripping, straightening and bending in tight spaces, e.g., carefully retrieving a part, or holding a nut while you turn a screw. **Never use pliers to force a screw head** â€" that is how screws get destroyed. Insulated handles are safer.

5. Anti-static wrist strap and mat.
Already covered â€" the ESD tools. Treat them as tools you use every single time.

6. Compressed air (canned air).
Used to blow dust out of heatsinks, fans and vents. **Important safety:** hold the can upright, never tilt it (liquid propellant can spray out and chill or damage parts), keep a safe distance, do not spin a fan with the air jet (which can over-speed and damage the bearing), and **wear safety glasses and use it in a well-ventilated area (ideally outdoors)** because dust can be harmful to breathe. Do not use a vacuum cleaner on electronics â€" it can create static.

7. Isopropyl alcohol (90% or higher).
Used to clean gold contacts, remove old thermal paste, and clean stubborn grime. It evaporates quickly and leaves no residue. **Do not use water**, and do not use petrol or kerosene. Use on a lint-free cloth or a cotton bud.

8. Microfibre / lint-free cloth.
Used for cleaning screens, contacts and surfaces without leaving fibres.

9. Thermal paste.

10. Cable ties / velcro straps.
Used to tidy cables so air can flow and the machine looks professional. Good cable management is part of good work.

11. Screw tray / container (or an egg carton).
Used to keep screws organised and separated by location, so you can put each one back in the right place.

12. Multimeter (instructor demonstration only).
A multimeter measures voltage, current and resistance. Professionals use it to test a PSU's output and check continuity. **In this course, the multimeter is demonstrated by the instructor only, on a safe low-voltage example (such as a battery), never on mains wiring.** Beginners do not test live circuits. Recognising the tool and knowing its purpose is the goal for now.

Real-life comparison.
Tools are like a **carpenter's plane and chisels**. The right tool for the right job produces clean work; using a chisel as a hammer ruins the wood and the tool. In computer repair, a wrong-size screwdriver is the most common self-inflicted damage.

Common misconceptions.

- "Any screwdriver will do." No â€” the wrong size strips screws and damages boards.
- "A bigger screwdriver gives more power." No â€” it gives more damage. Match the size.
- "A vacuum cleaner is fine for dust." No â€” it can generate static and zap parts. Use compressed air, carefully.
- "Water cleans contacts." No â€” isopropyl alcohol only.

Limitations and safety.
Do not use damaged tools (rounded tips, cracked handles, loose heads). Do not use a tool you do not understand. Ask before using anything unfamiliar. Keep tools dry and away from liquids.

F.5 Workspace setup and organisation

Why it matters.
A safe workspace prevents accidents and mistakes. On a cluttered, badly lit bench, screws vanish, cables get mixed, and slips happen.

A good computer-repair workspace has:

1. A **clean, flat, sturdy table** at a comfortable height (so you are not stooping).
2. An **anti-static mat** under the work.
3. **Good lighting** â€” a desk lamp plus overhead light, so you can see into the case.
4. A **screw tray or egg carton**, ideally with the compartments labelled by location (case, motherboard, drive, bracket).
5. **Tools laid out within reach**, not scattered where you might knock them into the case.
6. **Enough free space** to lay out removed components safely, ideally on the antistatic mat or on antistatic bags.
7. A **hard floor** or a non-static mat under your feet; avoid working directly on carpet if possible.
8. A **container for loose screws and small parts**, so nothing rolls onto the floor and into a fan.
9. **Good ventilation** if you will be using compressed air or cleaning fluids.
10. **No food or drink** on the bench.

Real-life comparison.
Setting up the bench is like a **surgeon preparing the operating theatre** before the patient arrives. Everything must be in its place, clean and ready, before the delicate work begins.

Common misconceptions.

- "Organisation slows me down." The opposite â€” it speeds you up, because you are not hunting for screws or re-opening a case.
- "Screws are all the same, one pile is fine." Different screws go in different places. One mixed pile means wrong screws in wrong holes and possible damage.

Computers and monitors can be heavy and awkward, and many benches are at the wrong height. Back injuries are a real workplace hazard.

Rules for safe lifting and handling:

1. **Assess the weight** before lifting. A desktop tower or a CRT monitor can be heavier than expected.
2. **Bend your knees, keep your back straight**, and lift with your legs, not your back.
3. **Hold the load close to your body** â€" do not stretch out your arms.
4. **Get help for heavy or awkward items** â€" two people for big monitors or full towers.
5. **Plan your path** â€" make sure the way is clear of cables and obstacles before you lift.
6. **Place the machine down gently** â€" never drop it on the bench.
7. **For components:** handle by the edges, support large cards and boards with two hands, and never carry a hard drive loosely where it can be knocked.
8. **Keep the bench tidy as you go**, so you do not trip over tools or cables.

Real-life comparison.
Lifting a computer is like lifting a **bag of cement**. If you bend at the waist and jerk it up, your back pays. If you bend your knees and lift smoothly, you stay safe.

Common misconceptions.

- "It is small, so it is light." Some small items (a heavy GPU, a full tower PSU) are deceptively heavy. Assess first.
- "I can carry three things at once to save a trip." That is how things get dropped. Take one at a time on the bench.

F.7 Working only on machines you are authorised to repair

Plain-English rule.
A technician must only work on equipment they have been **given permission** to open and repair. This protects the customer's property, the machine's warranty, and you.

Why it matters:

- A machine may still be under a **warranty**; opening it yourself can void the warranty and cause the customer to lose money.
- A machine may belong to a **school, church, company or government office**, where only approved staff may open it. Opening it without permission can cost you and the academy serious trouble.
- A customer's **data** is private. You must not browse their files, copy them, or share them. You only work on what is needed to fix the fault.
- If you damage an unauthorised machine, you may be personally liable.

The rule in practice:

1. Get the **owner's permission** before opening any machine.
2. Record what you received and its condition (with the intake process taught in Session 10).
3. Never open a machine labelled as belonging to someone else without the instructor's instruction.
4. Never plug a customer's drive into your own computer to "look" at their files without a good, authorised reason.
5. If you are unsure whether you are allowed, **ask before you touch.**

Real-life comparison.
It is like a **church keyboard or the school's projector**. You may use it, but you do not open it and take it apart unless the person in charge says you may. Respect for others' property is both professional and legal.

Common misconceptions.

- "It is a broken machine, nobody will mind." Someone owns it. Always get permission.
- "The customer said 'fix it,' so I can do anything." Fixing it does not mean rifling through private files or changing things they did not ask for.

G.1 What the instructor prepares

- A clean bench with the anti-static mat, the powered-off desktop, the full tool kit laid out, the screw tray, and safety glasses.
- The wrist strap ready to demonstrate.
- The balloon (from the opener) and the dead board.
- A bottle of isopropyl alcohol and a lint-free cloth (demonstration only).
- A can of compressed air (demonstrate outdoors or at a window with ventilation; learners wear safety glasses).

G.2 Demonstration 1 â€” the ESD-safe routine (6 minutes)

Say:
"I am going to handle a component the safe way. Copy my habits exactly, because these are the habits that will protect every computer you ever touch."

Do this slowly:

1. "First, I **unplug** the computer from the wall. I do not just switch it off â€” I pull the plug out."
2. "Then I **press the power button once**. This discharges residual electricity, like letting the last water drain from a pipe."
3. "Now I put on my **anti-static wrist strap** and clip it to **bare metal** on the case."
4. "I **touch the metal** of the case once more before I pick up any part."
5. "I pick up the RAM **by its edges**. My fingers never touch the gold contacts."
6. "If I walk away and come back, I **touch the metal again** before resuming."
7. "When a part is not installed, it lives in an **anti-static bag**, never on the carpet or a plastic sheet."

Show the wrong way, then correct it (hands-on contrast):

- Ruffle your hair, then pretend to grab a board by its contacts. Freeze. "Wrong." Now do it the right way. "Right."
- Ask: *"Why do we clip the strap to the case and not to anything else?"* Expected: to keep us and the machine at the same electrical level â€” and never to a socket, which would be deadly.

Expected result:
Learners can recite and copy the seven-step routine.

Common errors to watch:
clipping the strap to a wall socket (stop at once), touching the case's painted surface instead of bare metal, forgetting to re-touch metal after walking away.

G.3 Demonstration 2 â€” tools, one by one (6 minutes)

Say:
"Now, the tools of the trade. For each one, I will hold it up, name it, say what it is for, and say the main danger."

Hold up and explain, in this order:

1. **Phillips PH2** â€” "Most computer screws. The danger is using the wrong size and stripping the screw."
2. **Phillips PH1** â€” "Smaller screws. Match the size to the head."
3. **Flathead** â€” "For prying clips. The danger is slipping and scratching a board â€” use gently."
4. **Tweezers** â€” "For picking up dropped screws. Never force with them."
5. **Torch** â€” "For seeing inside the case. Always use light."
6. **Needle-nose pliers** â€” "For gripping and bending. Insulated handles are safer."
7. **Wrist strap** â€” "ESD protection â€” every single time."
8. **Compressed air** â€” "For dust. Upright, never tilt; wear glasses; ventilate; do not spin the fan with the jet."
9. **Isopropyl alcohol** â€” "For contacts and thermal paste. Never water."
10. **Multimeter** â€” "For measuring voltage. I demonstrate it **only** on a low-voltage battery. We never test mains wiring."

circuits. Later, with supervision, you will learn to test a PSU safely â€“ never inside it.

Expected result:
Learners can name each tool and its use, and remember at least one danger per tool.

Questions to ask during the demo:

- "Which screwdriver is the danger for scratching a board?" (The flathead.)
- "What is the golden safety rule with compressed air?" (Hold it upright, never tilt; ventilate; wear glasses.)
- "What do we never do with the PSU?" (Never open it.)

G.4 Demonstration 3 â€“ workspace setup (3 minutes)

Say:
"Watch how I set the bench. Notice everything has a place."

1. Wipe the table clean; place the anti-static mat.
2. Position the lamp so light falls into the case.
3. Place the tool kit to one side, screw tray in front-centre.
4. Unplug and discharge the machine; set it on the mat.
5. Lay out antistatic bags ready for removed parts.
6. Keep drinks off the bench and the floor clear.

Expected result:
Learners replicate the layout at their own bench.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "My Safe Bench and My Tool Kit"

Objective:
With the instructor guiding each step, learners set up a fully safe workspace, don the ESD protection, identify every tool, and complete the safety checklist and tool chart.

Materials per learner/group:
A bench, anti-static mat, anti-static wrist strap, a tool kit, a screw tray, a powered-off practice desktop, Worksheet 2.1 (Tool Identification Chart), Worksheet 2.2 (Safety Procedures Checklist), safety glasses.

Procedure (instructor calls each numbered step; learners follow; instructor walks the room):

1. **Clean and prepare** your bench: clear everything, wipe the surface, and place the anti-static mat.
2. **Position the light** so it shines into the case area. Place your tools to one side and the screw tray at the front.
3. **Unplug the practice computer** from the wall. Do not just switch it off â€“ remove the plug.
4. **Press the power button once** to discharge residual electricity.
5. **Put on the anti-static wrist strap** and clip it to bare metal of the case.
6. **Touch bare metal once more**, then place the machine on the mat.
7. **Lay out every tool.** On Worksheet 2.1, write each tool's name and what it is used for. Tick the "safe for beginners" column only where appropriate (note: the multimeter and PSU work are **not** beginner tasks).
8. **Practise the correct screwdriver choice:** find a loose screw (or a bolt on the practice stand) and pick the **correct size** Phillips driver. Test that it seats fully without wobbling.
9. **Practise safe handling:** pick up a loose RAM stick (if available) by its edges; return it to its anti-static bag.
10. **Complete Worksheet 2.2** (Safety Procedures Checklist), ticking each item as you do it: unplugged, discharged, strap on, mat in place, tools identified, bench tidy, drinks away.

12. When finished, raise your hand for the instructor to check your bench and both worksheets.

Pause-and-predict points:

- After step 4: "*Why do we press the power button after unplugging? Predict what could happen if we did not.*" Expected: to drain stored charge so we do not get a surprise or damage a part.
- After step 8: "*Predict what happens if I use a PH1 driver on a PH2 screw and force it.*" Expected: the screw head gets stripped and may become impossible to remove.
- After step 11: "*Predict: what is the risk if I skip touching the metal after walking on carpet?*" Expected: a static discharge into the part when I touch it.

Expected result:
Every learner has a safe bench, a completed tool chart, a completed safety checklist, and can demonstrate the ESD routine from memory.

Troubleshooting while walking the room:

- Learner still has the machine plugged in â†' stop and correct immediately; this is a critical safety error.
- Learner touches RAM contacts â†' redirect to the edges; explain the oil problem.
- Learner cannot name a tool â†' have them look at the chart, then cover the chart and try again.
- Learner uses the wrong driver size â†' show them the wobble, then the correct fit.

Completion checklist (learner ticks each):

- ☐ Bench cleaned and anti-static mat in place
- ☐ Light positioned; tools and screw tray laid out
- ☐ Computer unplugged (plug removed) and power button pressed
- ☐ Anti-static wrist strap worn and clipped to bare metal
- ☐ Bare metal touched before handling
- ☐ All tools named and their uses written on Worksheet 2.1
- ☐ Correct screwdriver size selected and tested
- ☐ RAM handled by the edges and returned to its bag
- ☐ Worksheet 2.2 completed
- ☐ Walk-away test passed (touched metal first on return)

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
"The Safety Audit"

Scenario:
You are now a junior technician at **Elias Computer Clinic, Diobu, Port Harcourt**. Your supervisor asks you to inspect the shop bench and a colleague's habits before you begin a repair, and to write a short audit that will be posted on the wall as a reminder for everyone.

Instructions:
Read the ten situations below. For each, decide whether it is **SAFE** or **UNSAFE**. If unsafe, write the correction. Then write the **single most important rule** you would put at the top of the wall poster.

Situations:

1. A technician unplugs the power cable but does not press the power button before opening the case.
2. A technician works barefoot on a thick woollen carpet while handling a motherboard.
3. A technician clips the anti-static strap to a screw on the case, but leaves the shop's power cable plugged into the wall "just for the light."

to look.

6. A technician picks up a RAM stick by its gold contacts to show it to a customer.
7. A technician holds the case fan still with a pencil while blowing it with compressed air, and wears safety glasses.
8. A technician using compressed air tilts the can sideways to reach under a heatsink.
9. A technician is unsure how to replace a power supply, so they ask their supervisor to supervise the job.
10. A technician opens a school's computer without asking anyone, saying "it was switched off anyway."

Requirements:

- Mark all ten SAFE/UNSAFE.
- Give the correction for every unsafe item.
- Write your single top rule.
- Write one sentence explaining *why* your top rule matters most.

Constraints:

- Use correct safety language: ESD, residual electricity, unplug, ground, anti-static.
- Keep each correction to one or two sentences.

Time allowance:
15 minutes.

Success criteria:

- All ten situations correctly judged.
- Corrections are accurate and specific.
- The top rule captures the spirit of the session.

Hints if stuck:

- Hint 1: Anything done while the machine is still plugged in is automatically unsafe.
- Hint 2: Static comes from carpets, woollen clothes and rubbing â€” anything that builds charge.
- Hint 3: The wrist strap grounds you to the *machine*, never to a socket.

Instructor answer key (do not show learners):

1. **UNSAFE.** After unplugging, press the power button once to discharge residual electricity.
2. **UNSAFE.** Do not work on carpets or barefoot on wool; work on a hard floor or anti-static mat and wear the strap.
3. **UNSAFE.** The machine should be unplugged while working. The strap is clipped to the case for grounding â€” but the mains plug must still be removed, and never rely on the machine being "plugged for the light." (If the "light" needs power, use a separate lamp.)
4. **UNSAFE.** Use the correct size driver (PH2 for that screw) and never force; the wrong size strips the screw head.
5. **UNSAFE.** With burning smell/smoke, unplug first (from a safe distance if needed) and do not open a live machine. Stop and get the instructor.
6. **UNSAFE.** Hold RAM by the edges; never touch the gold contacts (finger oils damage the connection).
7. **SAFE.** Holding the fan still while blowing and wearing safety glasses is correct â€” it prevents over-spinning the bearing and protects the eyes. (The machine should, of course, be unplugged.)
8. **UNSAFE.** Hold the compressed-air can upright; tilting can spray liquid propellant and damage parts.
9. **SAFE.** Knowing your limits and asking for supervision is exactly right.
10. **UNSAFE.** Never open equipment you are not authorised to repair, even if it is switched off. Get permission.

Top rule (accept any strong, defensible answer):
"Unplug the power and protect against static before you touch anything inside a computer." **Why it matters most:** because these two habits prevent the two biggest causes of injury and damage â€” electric shock and ESD.

Marking guidance (out of 12):

J. REAL-WORLD CASE STUDY

Situation.

Tunde is a talented young technician in a busy Port Harcourt shop. He has good hands and learns fast, but he has developed two dangerous shortcuts: he opens machines while they are still plugged in "to save time," and he rarely wears his anti-static strap because "Port Harcourt is humid and it never causes problems." One Tuesday, a customer brings in a laptop to have its RAM upgraded and its dust cleaned. Tunde opens the case with the charger still connected, drops a metal screwdriver near the PSU area, and then, without a strap, grabs the RAM by its contacts. The machine boots after the upgrade. The customer pays and leaves. Three weeks later, the customer returns furious: the laptop now shuts down randomly. Tunde insists the RAM "must be faulty from the shop."

Relevant facts.

- The charger was connected while Tunde worked (mains power present).
- No anti-static strap was worn; the RAM was touched by its gold contacts.
- A metal tool was used near a live PSU area.
- The failure appeared weeks later â€” the classic timing of a **latent** fault (ESD or a poor contact), not a part that was "faulty from the shop."

Decision points.

1. What are the *probable* causes of the weeks-later shutdown â€” and how would you confirm rather than guess?
2. Was the machine safe to open the way Tunde did it?
3. How should Tunde have handled the RAM?
4. Who is responsible for the failure, and what should the shop do for the customer?

Possible approaches.

1. Blame the RAM supplier and demand a replacement â€” but the evidence points elsewhere.
2. Re-open the machine safely, check the RAM seating and contacts, test the RAM properly, and inspect for damage. Honest and methodical.
3. Continue as before, since "it has always worked" â€” this is how a reputation dies.

Recommended solution.

Approach 2. A good technician would admit the machine was handled unsafely, re-open it **unplugged and with the strap**, reseat the RAM cleanly (checking for oily contacts), test the memory properly (a skill from Session 5), and confirm the fault. A latent ESD failure of the RAM stick is likely; the shop should replace the damaged part at the shop's cost and apologise honestly. Crucially, the shop should then **change its habits**: always unplug, always strap up, never bring metal tools near a live board.

Trade-offs.

- Approach 1 is fast and shifts blame, but it is dishonest and will not fix the machine if the cause is a poor contact or latent damage from handling.
- Approach 2 is slower but honest, and it protects the customer and the shop's reputation. It also teaches the lesson.
- Approach 3 is what actually causes shops to lose customers â€” word travels fast in Port Harcourt.

Professional connection.

In the repair trade, your **reputation is your business**. A technician who is careful earns trust and repeat customers; a technician who cuts corners saves seconds and loses customers. The two rules in this case â€” unplug, and strap up â€” cost nothing and prevent nearly all of the serious damage a beginner can cause. This is why CEA teaches safety before tools and tools before repair.

Transfer exercise.

"Name two habits from this case that, if Tunde had followed them, would have prevented the fault, and explain how each one helps." (Unplug â€” prevents shock and accidental short damage; strap and edge-handling â€” prevents ESD damage and contact contamination.)

open a machine that is on, never open the PSU, and never test live mains.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this as the live troubleshooting guide. Ask learners to find the relevant row rather than telling them the answer.

K.1 Working on a plugged-in machine

Symptoms:
Possible electric shock, a spark, a sudden fan spin, or a part that dies. In the worst case, serious injury. **Likely cause:** Rushing, or assuming "switched off" is the same as "unplugged." Caused by habit, not difficulty. **How to investigate:** There is no safe investigation â€" the rule is prevention. Confirm the plug is out before work begins. **Safe corrective steps:** Stop immediately. Unplug from the wall. Press the power button once to discharge. Only then continue. **How to confirm resolved:** The machine is verified unplugged and discharged. **How to prevent recurrence:** Make "unplug and discharge" the first item on every checklist.

K.2 Skipping the anti-static strap

Symptoms:
No immediate sign. Weeks later, random failures, a part that "died for no reason." **Likely cause:** ESD damage, often latent. Confirmed only indirectly â€" there is usually no visible mark. **How to investigate:** Look for the pattern (recent handling, dry conditions) and test the suspect part properly (later sessions). You cannot see ESD damage. **Safe corrective steps:** Adopt the routine: strap, mat, touch metal, hold by edges, antistatic bags. **How to confirm resolved:** Prevention, not repair â€" you cannot undo ESD damage. **How to prevent recurrence:** Treat the strap as non-negotiable, every time.

K.3 Using the wrong screwdriver size

Symptoms:
A screw that will not turn, a chewed-up screw head, or a screw stuck forever. **Likely cause:** Wrong driver size or type (PH1 on a PH2 screw, flathead on a Phillips). **How to investigate:** Try the correct size; if it does not seat snugly, the head may already be damaged. **Safe corrective steps:** Use the correct driver; if the head is rounded, use a proper screw extractor (instructor-level) â€" do not keep forcing. **How to confirm resolved:** The screw turns cleanly with the correct driver. **How to prevent recurrence:** Identify the screw type and match the driver before turning.

K.4 Touching gold contacts / mishandling boards

Symptoms:
A part not detected, an upgrade that "does not work," intermittent faults. **Likely cause:** Finger oils on gold contacts, or physical damage from bending. **How to investigate:** Inspect contacts; reseat holding only the edges. **Safe corrective steps:** Clean contacts gently with isopropyl alcohol (unplugged, supervised) and reseat. **How to confirm resolved:** The part is detected and works. **How to prevent recurrence:** Always hold by edges; never touch contacts.

K.5 Compressed air used wrongly

Symptoms:
Liquid spray on the board, a part that suddenly chills, a fan bearing that fails later, or dust in the eyes/lungs. **Likely cause:** Tilted can, too close, spinning the fan with the jet, or no eye protection/ventilation. **How to investigate:** Look for residue; check the fan spins freely by hand (with the machine off). **Safe corrective steps:** Hold the can upright; keep distance; hold the fan still; wear glasses; ventilate (ideally outdoors). **How to confirm resolved:** No residue; fan spins freely. **How to prevent recurrence:** Follow the compressed-air safety rules every time.

K.6 Opening a machine you are not authorised to repair

Symptoms:
A warranty voided, an angry owner, a data-privacy complaint, or damage you must pay for. **Likely cause:** No permission obtained before opening. **How to investigate:** Check with the owner or instructor before touching. **Safe corrective steps:**

permission.

Distinguishing likely vs confirmed:
Most safety problems are *prevented*, not diagnosed. Where a fault has already happened (a dead board, a stripped screw), the "likely cause" becomes "confirmed" only when you check it and the evidence matches. Teach learners that in safety, the goal is to never need the diagnosis.

L. SESSION ASSIGNMENT

Assignment title:
"My Safety and Tools Reference"

Purpose:
To build a personal, practical safety and tools guide that the learner will keep on the bench for the rest of the course and their career. This supports the capstone's emphasis on safe, professional work.

Realistic scenario:
You have just been hired as a junior technician. On your first day, the shop owner hands you a bench and says: "Before you touch any customer's computer, I want your personal safety rules and your tool list written down." This is that document.

Detailed instructions:

1. Create a document in Microsoft Word or Notepad titled **"My Safety and Tools Reference"**.
2. Write a section **"ESD Prevention"** with at least five numbered points (for example: wear a strap, work on a mat, touch bare metal, hold by edges, use antistatic bags).
3. Write a section **"Electrical Safety Rules"** with at least six numbered points (unplug, discharge, never work live, never open the PSU, insulated tools/no jewellery, know your limits).
4. Write a section **"Tool Chart"** as a table: for each of at least eight tools, give the **name**, what it is **used for**, and one **safety warning**.
5. Write a section **"Workspace Setup Checklist"** with at least six items (clean table, anti-static mat, lighting, screw tray, ventilation, no drinks).
6. Write a section **"When to Call a Professional"** with at least four clear situations (PSU internals, mains wiring, anything you are unsure of, a job requiring special tools).
7. Save the file as **"My Safety and Tools Reference"** on the Desktop.
8. Be ready to explain, in one sentence, why each of your top three rules matters.

Required deliverables:
One saved document containing all five sections.

Submission format:
Saved on the Desktop with the exact file name. If a learner has no home computer, complete it in the academy lab and show it before leaving.

Deadline:
Before Session 3 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
ESD prevention section	5+ accurate points with reasons	3â€“4 points	Fewer than 3 or inaccurate
Electrical safety section	6+ clear rules	4â€“5 rules	Fewer than 4
Tool chart	8+ tools, uses and warnings all correct	6â€–7 tools, mostly correct	Fewer than 6 or many errors

Workspace checklist	6+ complete items	4â€“5 items	Fewer than 4
Professional referral section	4+ clear situations	2â€“3 situations	Fewer than 2

Common mistakes to avoid:
confusing static electricity with mains electricity; saying "switch off" when you mean "unplug"; leaving out the PSU danger; forgetting the compressed-air warnings; copying the session notes word for word instead of using own words.

Instructor reference solution (summary, do not show learners beforehand):

- ESD: strap worn and clipped to bare metal; anti-static mat; touch metal before handling; hold by edges, never contacts; components kept in antistatic bags; avoid carpets/wool.
- Electrical: unplug from wall; press power button to discharge; never work on a live machine; never open the PSU; no metal jewellery/tools near live circuits; no liquids on the bench; know the emergency switch; know your limits.
- Tool chart (sample): Phillips PH2 â€” most computer screws â€” wrong size strips screws; flathead â€” prying clips â€” can slip and scratch; tweezers â€” retrieve screws â€” never force; torch â€” see inside â€” keep dry; wrist strap â€” prevent ESD â€” check for damage; compressed air â€” remove dust â€” upright, ventilated, glasses; isopropyl alcohol â€” clean contacts â€” never water; multimeter â€” measure voltage â€” instructor demonstration only, never mains.
- Workspace: clean flat table; anti-static mat; good lighting; screw tray; ventilation; tools within reach; no food/drink; clear floor.
- Professional referral: PSU internals; mains/wall wiring; unknown faults beyond experience; jobs needing special tools or training.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below.)

Q1. Short answer.
What does ESD stand for, and why is it dangerous to computer parts?

Q2. Multiple choice.
What is the correct first thing to do before opening a computer case? a) Put on safety glasses b) Unplug the power cable from the wall c) Remove the side panel d) Touch the RAM

Q3. Explain in your own words.
Why can ESD damage be worse than damage you can see?

Q4. Practical identification.
Name the tool and its purpose: (a) a cross-shaped screwdriver used for most computer screws; (b) a fine pair of pincers for picking up dropped screws; (c) a small lamp used to see inside a dark case.

Q5. Error detection.
A learner says: "After I unplug the computer, I do not need to press the power button, because no electricity is flowing." Find the mistake and correct it.

Q6. Scenario.
A friend asks you to open their church's computer to clean the dust, but tells you it is still under warranty. What should you do, and why?

Q7. Short answer.
Give two safety rules for using compressed air.

INSTRUCTOR ANSWER KEY (Q1â€“Q7)

A1.

the part may fail weeks later.

A2.
(b) Unplug the power cable from the wall. (Then press the power button to discharge.)

A3.
Visible damage (a burn, a crack) can be seen and dealt with immediately. ESD damage often leaves **no visible mark** and may be latent â€" the part works at first and fails later, so the technician blames the wrong cause and the customer loses trust. Invisible, delayed damage is the hard part.

A4.
(a) Phillips screwdriver (PH1/PH2) â€" used for most computer screws; (b) tweezers â€" for retrieving small screws and holding small parts; (c) torch/flashlight â€" for seeing inside the case.

A5.
The mistake is believing that unplugging removes all stored charge instantly. Some parts (capacitors) can hold a **residual charge** for a short time. The correction: after unplugging, **press the power button once** to help discharge residual electricity before handling anything.

A6.
Refuse to open it yourself and explain that opening a machine still under warranty may **void the warranty**, costing the owner money. Advise them to take it to the warranty provider (or get the owner's documented permission and check the terms). Getting permission and protecting the owner's property are the professional rules.

A7.
Any two of: hold the can **upright** (never tilt); use in a **well-ventilated** area (ideally outdoors); wear **safety glasses**; keep a safe distance; do **not spin the fan** with the jet; never use a vacuum cleaner on electronics.

Marking:
up to 2 marks each (Q3, Q6 up to 3). Total â‰ˆ 16; scale proportionally. Full credit requires correct names and reasoning.

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned how to work on computers without harming yourself or the machine. You learned that **ESD (electrostatic discharge)** is an invisible danger that can damage parts without any sign, and how to prevent it with a **wrist strap, an anti-static mat, touching bare metal, handling boards by the edges, and using antistatic bags**. You learned the **electrical safety rules**: always unplug and discharge, never work on a live machine, and **never open the PSU**. You learned the **tools of the trade** and what each is for, why the **correct screwdriver size** matters, and the safe use of compressed air and isopropyl alcohol. You set up a **safe, organised workspace**, learned **safe lifting and handling**, and learned to work only on machines you are **authorised** to repair.

Five important takeaways:

1. **ESD is invisible and can kill parts silently** â€" strap up every time.
2. **Unplug from the wall and press the power button** before opening any case.
3. **Never open the PSU** â€" replace it, never repair it.
4. **Use the correct tool and the correct screwdriver size** â€" the wrong one causes damage.
5. **Know your limits and get permission** â€" refer tricky or unauthorised jobs.

Questions to help you recognise what you can now do:

- Can I demonstrate the ESD-safe routine from memory?
- Can I explain in one sentence why ESD is dangerous?
- Can I name at least eight tools and one danger per tool?
- Can I state the rule about the PSU and the wall plug without hesitation?

reassemble a real desktop computer correctly â€" removing the panel, disconnecting cables safely, extracting and refitting components, recording screws and cables, and verifying the machine powers on. Everything you learned about safety today is used in every step of Session 3.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner unplugs and discharges **without being reminded**.
- The learner puts on the wrist strap and touches bare metal as a habit before each session of handling.
- The learner selects the correct screwdriver size by inspecting the screw first.
- The learner handles boards by the edges and never touches contacts.
- The learner can name all common tools and their purposes, with at least one danger each.
- The learner states the PSU rule and the authorisation rule confidently.

Common misconceptions to watch for and correct:

- "Static electricity is not dangerous / Port Harcourt is too humid." â†' Humidity reduces but does not remove risk; strap up anyway.
- "Switched off means safe." â†' Only **unplugged and discharged** is safe.
- "Any screwdriver will do." â†' The wrong size strips screws and damages boards.
- "I can repair the PSU." â†' Never open it; replace it.
- "If it is broken and off, I can open anyone's machine." â†' Get permission; respect warranty and data privacy.

Suggested remedial exercises for learners who struggle:

1. **Routine drill:** have the learner perform the unplug-discharge-strap-touch routine five times until it is automatic.
2. **Tool matching cards:** match tool names to pictures and to uses, then to dangers.
3. **Scenario sorting:** give ten cards (safe/unsafe) and have the learner sort them into two piles, explaining each.
4. **Screw-size drill:** present several screws and drivers and have the learner match them correctly without forcing.

Differentiation:

- **Level 1 (supported):** provide the worksheets partially filled and walk the learner through the bench setup step by step; pair with a steady-handed partner.
- **Level 2 (standard):** complete the exercise and the Safety Audit independently.
- **Level 3 (challenge):** research additional ESD/safety equipment (ESD gloves, ionisers, heel straps) and explain when each is used; also explain the difference between ESD safety and mains safety in detail.

Timing guidance:
Safety cannot be rushed. If the room is slow, compress the lecture sections F.5â€"F.7 into quick summaries, but **never** shorten the ESD and electrical rules (F.1â€"F.3) or the practical routine (H). The practical demonstration of the correct routine is the assessed priority.

END OF SESSION 2

Next: Session 3 â€" Disassembly and Reassembly: Opening a Computer Safely and Putting It Back Correctly.

SESSION 3 â€" DISASSEMBLY AND REASSEMBLY: OPEN IT AND PUT IT BACK CORRECTLY

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 3 â€" DISASSEMBLY AND REASSEMBLY: OPEN IT AND PUT IT BACK CORRECTLY

A. SESSION IDENTIFICATION

Item	Detail
Session number	3 of 12
Session title	Disassembly and Reassembly
Week / Module	Week 1 â€" Module: Components and Safety
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner, but hands-on and delicate; needs close supervision
Prerequisite knowledge	Sessions 1â€"2: component names and functions; ESD, electrical safety, tools.
Connection to previous session	The learner can name the parts and work safely. This session applies both skills by opening a real desktop, removing components in the correct order, and rebuilding it so it works.
Connection to the course	Disassembly and reassembly are the physical foundation of all repair. Session 7 (component replacement) and the Session 12 capstone both require exactly these skills. This session also completes the Week 1 mini-project, "My First Teardown."
Preparation before class	At least two practice desktop computers (ideally non-working or expendable units) openable by simple screws; tool kits; anti-static wrist straps per learner; anti-static mats; screw trays; a smartphone/camera for photographing connections; printed Disassembly and Reassembly checklists; tape and a marker for labelling cables; a working reference machine to test against.

Safety note for this session:
This is the first session where learners physically open a computer. Every rule from Session 2 applies and is enforced. The machine must be **unplugged from the wall and discharged** (power button pressed) before the case is opened. **No mains-powered work.** Never open the PSU. Never force a component. All work stops immediately if a learner is unsure. Learners work in pairs or small groups with an instructor circulating. Laptops are **not** dismantled in this session; desktop practice only. If a practice machine is a working one, test power-on only briefly and with the instructor present.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Prepare a machine safely** â€" unplug, discharge, set up the anti-static mat, wear the strap, and photograph the internals before starting.
2. **Safely disconnect all external cables** from the back of a desktop computer.

them.

5. **Remove components in the correct order** (expansion card, RAM, storage drives, then cooler) and set them aside safely.
6. **Reinstall components in the correct order**, applying the correct technique (for example, RAM clips snapping into place).
7. **Reconnect all cables correctly**, using photographs and labels to guide them.
8. **Verify the machine powers on** after reassembly, and know what to do if it does not.
9. **Record screw locations and cable positions** so reassembly is accurate and repeatable.

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- At least two practice desktop computers per class (non-working or expendable preferred; working units acceptable under close supervision).
- Tool kits: Phillips PH1 and PH2 screwdrivers, tweezers, torch, needle-nose pliers.
- Anti-static wrist straps (one per learner) and anti-static mats.
- Screw trays or labelled containers (egg cartons/trays) for keeping screws organised by location.
- A torch/desk lamp for each bench.
- A smartphone or camera (one per group) for photographing connections. If no camera is available, provide paper and pencil for sketching and labelling.

Software:

- None required. (Optional: a system-info tool is not needed for this session.)

Safety gear:

- Anti-static wrist straps and mats.
- Safety glasses if compressed air is used (optional this session).
- A first-aid kit in the room.

Sample / fault components:

- A spare dead hard drive and a spare RAM stick for practice handling, in case a group finishes early.
- A machine deliberately set up with a "loose RAM" fault for the troubleshooting discussion (optional, instructor-prepared).

Worksheets (printed):

- Worksheet 3.1 â€" "Disassembly Checklist" (one per learner).
- Worksheet 3.2 â€" "Reassembly Checklist" (one per learner).
- Worksheet 3.3 â€" "Screw and Cable Log" (a table: step, screw type/location, cable label, notes).
- Cable labels (small stickers or masking tape + marker).

Alternatives for a learner or academy with limited equipment:

- If only one machine is available, rotate small groups (3â€"4 learners) around it, while others watch the system-recorded demonstration or work with photo sheets and the ordering exercise.
- If no machine can be opened, run the session as a "dry" simulation: learners sequence cards showing each step, plus a photo-based identification of cable and screw types, and a walk-through of the demonstration machine.
- If a learner has shaky hands or low confidence, pair them with a careful partner: one removes, one records (writes the screw/cable log and takes photos), then swap.

Suggested visuals:



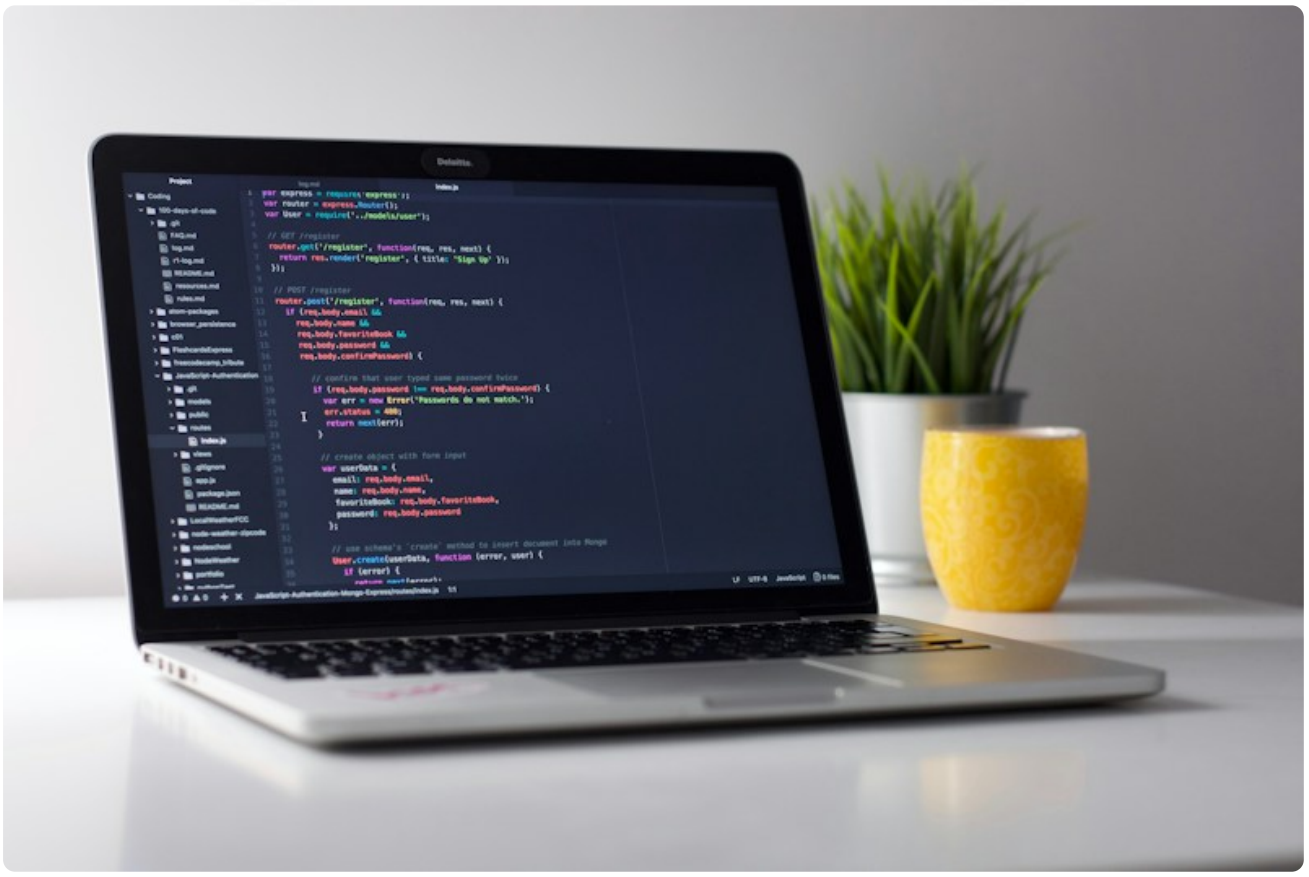
- â€" Opening the case step by step



- â€" Handling components during removal



- â€" Screw and cable log kept with the right tools



- — Laptop disassembly shown step by step



- — Removed parts and screws labelled on the bench

D. OPENING ACTIVITY (6 minutes)

"The Teardown Challenge"

Instructor setup:
Place a fully assembled desktop computer on the front bench, connected to a monitor. Power it on and let the class see it working. Then power it off properly and unplug it. Keep the tool kit nearby.

What the instructor says:

> *"Look at this computer. It is working. It has taken engineers years to design and factories hours to build. Now watch this."*

Unplug the machine and hold up a single screwdriver.

> *"With one small screwdriver and the right knowledge, I am going to take this whole thing apart â€" and then put it back together so that it still works. Anybody can break a computer by opening it. It takes a technician to open it and give it back working.*
>
> *Here is your challenge for today: by the end of this session, you will do exactly what I am about to do. You will open a real desktop, remove its parts, put them back, and press the power button â€" and it must come on. This is the skill that separates a technician from somebody who just 'carries' computers to the repair shop.*
>
> *But before we touch anything, I want you to remember Session 2. What is the first thing we do before opening any case?"*

Pause for answers.

> *"Correct â€" we unplug it. Then we press the power button to discharge. Then we strap up. Safety first, always. Today, safety and skill walk together."*

> *"Tell me â€" if you had to remove all these parts, which would you take out first? The RAM? The big card? The hard drive? Why does the order even matter? By the end of today, you will know the correct order and the reason for it."*

Expected learner responses:
Learners may guess "biggest first" or "whatever is easiest." Accept the answers. Some will remember the safety steps from Session 2 (unplug, discharge, strap) and should be praised. Note their guesses â€" you will correct the reasoning in the lecture.

Why this opener works:
It frames a mechanical task as a challenge with a clear, observable win (the machine works at the end), connects to the previous session's safety rules, and creates a genuine question (what order, and why) that the lecture answers.

E. PREREQUISITE REVIEW (7 minutes)

Run as a quick oral review; ask different learners.

Question 1:
"What are the first two things you must do before opening a computer case?"

- Expected answer: Unplug the power cable from the wall; press the power button to discharge residual electricity.
- Common gap: learners say "switch it off." Correct: **unplug** from the wall, then discharge. Off is not enough.
- Bridge: "Today, that is step one of everything."

Question 2:
"What is ESD, and name two ways you prevent it."

- Expected answer: Electrostatic discharge; prevent with an anti-static wrist strap clipped to bare metal, an anti-static mat, touching bare metal, holding by edges, antistatic bags.
- Common gap: learners remember the strap but forget to *touch bare metal* before handling. Emphasise.
- Bridge: "You will handle real boards today, so this matters every minute."

Question 3:
"Name three essential tools for opening a desktop."

- Expected answer: Phillips screwdriver (correct size), tweezers, torch (and anti-static wrist strap, pliers, screw tray).
- Common gap: not knowing the screwdriver size matters. Remind them: wrong size strips screws.
- Bridge: "Choosing the correct screwdriver is the first skill of disassembly."

Question 4:
"Where is the power supply, and what is the golden rule about it?"

- Expected answer: The metal box that supplies electricity to all parts; **never open it** â€" replace it if faulty.
- Common gap: thinking the PSU can be repaired. Correct firmly.
- Bridge: "You will meet the PSU today but you will never open it."

Question 5:
"Why do we keep screws organised and take photos?"

- Expected answer: So we know which screw goes where and how the cables were connected, making reassembly correct.
- Common gap: learners think it wastes time. Explain it saves time and prevents damage.
- Bridge: "Today, taking photos and logging screws is not optional â€" it is how professionals work."

Bridge to today (say this):

> *"Think of disassembling a computer like*

> *struggle to rebuild it. So the professional's secret is not strength or speed â€“ it is* ***recording and order****. Today you learn the order, and the discipline of recording."*

If more than a third cannot answer Question 1, spend an extra three minutes re-teaching the unplug-and-discharge routine, because it is safety-critical.

F. FULL LECTURE NOTES

F.1 Why order matters â€” the idea of reversibility

The idea.
Disassembly is the process of taking a computer apart; reassembly is putting it back together. The single most important principle is this: **reassembly is the reverse of disassembly**. If you removed A, then B, then C, you reinstall C, then B, then A.

Why it exists.
Components are often stacked or overlapping. A long graphics card may sit over the RAM clips; a cooler may sit over the CPU; a drive cage may block a cable. If you try to remove things in the wrong order, you will find you cannot â€” or you will force something and break it. The order is not arbitrary; it follows the physical layout.

Real-life comparison.
Think of **dressing a child for school**. You put on the vest first, then the shirt, then the jumper. To undress, you reverse it: jumper, shirt, vest. If you tried to remove the vest first, you would fail. A computer is no different.

Realistic repair-shop scenario.
A new technician at a Port Harcourt shop removes the CPU cooler first, before the RAM. He knocks the cooler into a RAM stick and bends a pin in a slot. Now, instead of a simple cleaning job, the board is damaged. Order is not a formality; it prevents damage.

Common misconceptions.

- "Order does not matter; I will just take out whatever I can reach." It matters â€” the layout dictates the order.
- "Faster is better." In repair, **correct** is faster, because mistakes cost far more time than care.

Limitations and safety.
Different machines have different layouts. The order below is the standard, but always **look first**, take photos, and adapt to your specific machine. If you are unsure, stop and ask the instructor.

F.2 Safety preparation before any disassembly

The non-negotiable pre-flight checklist (do this every time):

1. **Get permission.** Only open machines you are authorised to repair (Session 2, F.7).
2. **Record the machine.** Note the model, the condition, and any existing damage. Take a photo of the outside.
3. **Switch off** the operating system properly (do not just yank the power while Windows is running, unless it is already dead).
4. **Unplug** the power cable from the wall.
5. **Press the power button once** to discharge residual electricity.
6. **Set up the bench:** anti-static mat, good light, screw tray, tools within reach, no drinks.
7. **Wear the anti-static wrist strap** and clip it to bare metal of the case.
8. **Touch bare metal** before handling any internal part.
9. **Photograph the inside** before disconnecting anything. Take several photos from different angles, especially of the cable connections.
10. **Label cables** with tape and a marker if the machine is unfamiliar or complex.

Why each step matters:

- Proper shutdown protects the data and the operating system.
- Unplugging and discharging protect you from shock and the parts from surges.
- The mat, strap and bare-metal touch protect the parts from ESD.
- Photos and labels protect you from forgetting where things go.

Real-life comparison.
This is like the **pre-flight checklist a pilot runs before take-off**, no matter how many times they have flown. The checklist is not a sign of inexperience; it is what professionals rely on so that nothing is forgotten.

Common misconceptions.

- "I have done this before; I do not need the checklist." Experience is exactly why professionals *use* checklists â€" to stay consistent when they are confident and could become careless.
- "I can skip photos for a simple job." Even simple jobs go wrong; photos cost nothing and save everything.

Safety implications.
Never proceed without steps 4, 5, 7 and 8. These are the life-and-machine rules. If you are interrupted, re-touch bare metal before resuming, and re-check the plug is out.

F.3 Disassembly â€" the correct order, step by step

Below is the standard desktop disassembly order. Each step lists

what to do
and **why**.

Step 1 â€" External cables.
Disconnect everything from the back: monitor cable, keyboard, mouse, USB devices, network (Ethernet) cable, speakers, and the power cable (already unplugged). Label any that are unclear. *Why:* You must free the machine so you can move it to the bench safely. Cable labels help you reconnect exactly.

Step 2 â€" Move to the bench and open the panel.
Place the machine on its side on the anti-static mat (or stood up, per its design). Remove the screws holding the side panel (usually two at the back). Some cases use thumb screws (turn by hand) or slide-and-lift latches. *Why:* The side panel gives access. Keep the panel screws in the screw tray â€" do not put them on the floor.

Step 3 â€" Photograph and note the layout.
Before touching cables, photograph the whole inside, then close-ups of the connections: the 24-pin power connector, the CPU power connector, the SATA data cables, the SATA power cables, and the front-panel wiring. *Why:* This is your map for reassembly. The front-panel wires (power switch, reset, front USB/audio) are the most easily forgotten.

Step 4 â€" Disconnect internal cables.
Work carefully and grip the connectors, not the wires (pulling wires can break them):

- Unplug the **SATA data cables** from the drives (press the small latch if present).
- Unplug the **SATA power connectors** from the drives.
- Unplug the **power connectors** from the motherboard only if you will remove the board â€" otherwise leave them.
- If removing the board, note the **front-panel connectors** carefully. *Why:* Cables must be out of the way before components can be removed. Grip the connector to avoid damaging the wire and the socket.

Step 5 â€" Remove expansion cards (the graphics card first).
If a graphics card is fitted, it is usually the first component to remove because it often lies over other parts. Remove the screw holding its bracket to the case, release the clip at the back of the PCIe slot, and pull the card straight up by its edges, supporting its weight. *Why:* The card overlaps and blocks other parts; removing it early frees access. Supporting it prevents bending it or cracking the slot.

Step 6 â€" Remove RAM.

drives. Holding by the edges protects the contacts.

Step 7 â€” Remove storage drives.
Remove the screws holding each drive (or the drive cage screws), disconnect any remaining cables, and slide the drive out. Handle HDDs gently. *Why:* Drives sit in bays that may need the cables cleared first. Gentle handling protects the fragile HDD.

Step 8 â€” Remove the CPU cooler (only if the task requires).
Unplug the fan's power connector from the board. Then unscrew or release the cooler's clips. Gently twist the cooler slightly to break the seal of old thermal paste, then lift it off. Place it aside, fan-side up. *Why:* The cooler sits on top of the CPU; it comes off late because it is the most delicate and requires handling paste. Do this only when needed (for example cleaning or CPU work). *Warning:* With the cooler off, **never** power on the machine â€” the CPU can overheat in seconds.

Step 9 â€” Remove the motherboard (only if required).
Remove all screws holding the board to the case (note their positions), disconnect any remaining cables, and lift the board out carefully, holding by its edges. *Why:* This is the last step because the board is the foundation to which everything else attaches. Beginners usually leave the board in place unless the task demands otherwise.

A note on laptops.
Laptops follow the same *principle* (reverse order) but the parts are far smaller and more delicate, and many are held by hidden clips and tiny screws. **Do not dismantle laptops in this session.** Only attempt laptops under supervision and after you are confident on desktops.

Real-life comparison.
Disassembly is like **unpacking a well-packed suitcase**: you take out the top layers first, and you keep note of how things were folded. Reassembly is packing it the same way so it closes properly.

Common misconceptions.

- "The CPU cooler is always the first thing to remove." No â€” it is near the last, because it sits on top and handling is delicate.
- "I can pull cables by the wire." No â€” always grip the connector.
- "If it does not move, pull harder." Never. Check for a screw, clip or cable.

Limitations and safety.
Some machines use proprietary screws and clips (common in branded desktops). If you meet resistance you cannot explain, stop and ask. Never force.

F.4 Reassembly â€” the reverse order, with extra care

Reassembly is the reverse of disassembly, but with two extra emphases:

care with paste
and **verification before closing**.

Step 1 â€” Motherboard (if removed) first.
Place the board back, align it with the case's standoffs, and fit the I/O shield correctly at the back. Replace all screws evenly, **without over-tightening**.

Step 2 â€” CPU cooler.
If the cooler was removed, the old thermal paste must be cleaned off both the CPU and the cooler with isopropyl alcohol, and a **small pea-sized amount** of fresh paste applied to the centre of the CPU. Refit the cooler, tighten or clip it evenly (tighten screws in a cross pattern, a little at a time), and reconnect the fan cable. *Why:* Old, dried paste does not transfer heat; too much paste is also bad; even pressure seats the cooler properly.

Step 3 â€” Storage drives.
Slide each drive into its bay and secure the screws. Reconnect SATA data and SATA power cables, checking your photos.

Step 4 â€” RAM.
Open the clips, align the notch on the stick with the key in the slot (they only fit one way), and press down firmly and evenly at both ends until the clips **snap** into place. Listen and feel for the click. *Why:* A RAM stick not fully seated is one of the most

Remove the case's expansion slot cover if needed, align the card with the PCIe slot, press down firmly until the slot clip engages, and replace the bracket screw. Reconnect any power cable the card needs.

Step 6 â€” Internal cables.
Reconnect the SATA data and power cables, the motherboard power connectors if removed, and â€" most carefully â€" the **front-panel cables** (power switch, reset, USB, audio), using your photos and labels. The power switch wire is the one that lets the power button work; if it is misconnected, the machine will not start from the button.

Step 7 â€" Check before you close.
Before replacing the panel, **do a careful visual check**: Are all cables firmly seated? Is the RAM clipped? Are there any loose screws inside? Is the CPU cooler connected? Are the drive cables correct? A screw left loose inside can short-circuit the board. *Why:* This silent check catches most reassembly errors before power-on.

Step 8 â€" Close and reconnect.
Replace the side panel and its screws. Reconnect the external cables (monitor, keyboard, mouse, power). Plug in the power cable.

Step 9 â€" Power-on verification.
Press the power button. Check: Do the fans spin? Does a light show? Does the monitor display? Does it reach the operating system (or the BIOS/POST screen)? *Why:* This is the moment of truth. If it does not start, you diagnose (Section K) â€" you do not panic.

Real-life comparison.
Reassembly is like **reassembling a bicycle after fixing a puncture**. You fit the inner tube, the tyre, the wheel, then check the brakes before you ride. You always check before you trust it â€" you do not just close the case and hope.

Common misconceptions.

- "Any amount of thermal paste is fine." No â€" a pea-sized amount, not a thick layer.
- "The RAM will seat with a light push." Not always â€" it needs a firm, even push until it clicks. But never force it in the wrong orientation.
- "I can close the case and test later." Test as soon as possible, before fully working the machine, so faults are easier to trace.

Limitations and safety.
Do not over-tighten screws (they strip and can crack the board). Do not power on without the CPU cooler. If the machine does not start, **unplug again** before re-opening.

F.5 Recording screws, cables and the mirror-image rule

Recording screws.
Different screws have different lengths and threads. A long screw forced into a short hole can damage the board. Use a screw tray with compartments labelled by location (case, motherboard, drive, bracket) and place each screw where it belongs as you remove it. Keep a written **Screw and Cable Log** (Worksheet 3.3).

Recording cables.
Photograph before disconnecting, label awkward cables, and note the front-panel layout. The front-panel connectors are the most common source of "the power button does nothing" after reassembly.

The mirror-image rule.
Disassembly order and reassembly order are exact mirrors. Write them down as numbered lists so you can simply read one backwards to do the other.

Real-life comparison.
Recording is like a **tailor taking measurements before cutting cloth**. The few minutes spent measuring save the whole garment. In repair, a few photos and a screw log save the whole repair.

Common misconceptions.

- "I will remember where everything goes." You will not â€" especially on an unfamiliar machine, and especially after a break.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- One practice desktop on the bench with the anti-static mat, wrist strap, tool kit and screw tray.
- A phone/camera to demonstrate photographing.
- Tape and marker to demonstrate labelling.
- The projector showing the machine (or a second camera) so all learners can see.
- The Disassembly and Reassembly checklists on the wall or projected.

G.2 Demonstration â€" full teardown and rebuild, with narration (15 minutes)

Say:
"Watch every step, and listen to the reason behind it. I will disassemble, then reassemble, and the machine will work at the end."

Do and say (slowly, narrating reasons):

1. "I confirm this machine is mine to work on (permission). I note its condition."
2. "I shut it down properly, **unplug it from the wall**, and **press the power button** to discharge."
3. "I set up the mat, put on my **strap** and clip it to bare metal. I **touch bare metal** again."
4. "I remove the **side panel** screws and place them in the tray. Compare â€" the correct driver seats perfectly."
5. "I **photograph** the inside and the connections. This is my map."
6. "I disconnect the **SATA data** and **power** cables, gripping the connectors, not the wires."
7. "I remove the **graphics card** first â€" screw, slot clip, straight up by the edges, supporting it."
8. "I remove the **RAM** â€" clips down, straight out by the edges, into an anti-static bag."
9. "I remove the **storage drive** â€" screws out, slide out, handle gently."
10. "I set the components safely on the mat/antistatic bags."
11. "Now I reassemble in **reverse order**: drive first, then RAM (listen for the click), then the card."
12. "I reconnect the cables using my photo â€" especially the **front-panel** wires."
13. "Before closing, I **check**: all cables seated, RAM clipped, no loose screws inside, cooler connected."
14. "I close the panel, reconnect external cables, plug in, and **press power**. Fans spin, light on, display shows. It works."

Expected result:
Learners see the complete cycle and understand each reason. The machine powers on.

Reasoning to state aloud at key points:

- "I remove the graphics card first because it lies over everything else."
- "I grip connectors, not wires, because wires tear out of their plugs."
- "I press RAM until it clicks because a half-seated stick gives no display."
- "I check before closing because a loose screw can short the board."

Errors to watch and deliberately show (then correct):

- Not fully seating the RAM (simulate a loose stick) â†' "See how I check the clips. If they are not snapped, no picture."
- Forgetting the front-panel wire â†' "If I leave this off, the power button will do nothing."
- Cross-threading a screw â†' "I turn backwards gently first to find the thread, then tighten. Never force."

Questions to ask during the demo:

- "Why do we remove the card before the RAM?" (It blocks access.)
- "What should you do if a component will not come out easily?" (Check for a screw or clip; never force.)
- "How do you know the RAM is fully seated?" (The clips snap.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Exercise: "My First Teardown"

Objective:
With the instructor guiding each step, learners disassemble and reassemble a practice desktop without damage and verify it powers on. This is the Week 1 mini-project.

Materials per group:
Practice desktop, anti-static mat and wrist strap, tool kit, screw tray, torch, camera/phone, Worksheets 3.1, 3.2 and 3.3, cable labels, and a safe place for removed components.

Procedure (instructor calls each step; learners follow; instructor circulates continuously):

1. **Prepare:** Confirm permission. Note the machine's condition. Shut it down.
2. **Safety:** Unplug from the wall. Press the power button to discharge. Put on the wrist strap, clip it to bare metal, and touch bare metal.
3. **Bench:** Place the machine on the anti-static mat. Lay out tools and the labelled screw tray.
4. **Record:** Photograph the outside, then the inside, then each cable connection. Write the Screw and Cable Log (Worksheet 3.3).
5. **Open:** Remove the side panel screws (into the tray) and remove the panel. Set it aside safely.
6. **Disconnect:** Grip the connectors and disconnect SATA data, SATA power, and any power connectors you need to. Note the front-panel wiring.
7. **Remove the expansion card first** (if present), then the **RAM** (clips down, by the edges, into an antistatic bag), then the **storage drive** (gently).
8. **Set components safely** on the mat or antistatic bags. Do not stack boards.
9. **Reassemble in reverse order:** drive â†' RAM (push until the clips **snap**) â†' expansion card â†' reconnect all cables using your photos.
10. **Pre-close check:** All cables seated? RAM clipped? No loose screws inside? Cooler/fan connected?
11. **Close and reconnect:** Panel on, screws in, external cables connected, power cable plugged in.
12. **Power-on verification:** Press the power button. Confirm fans spin, lights show, and the display appears (or the BIOS/POST screen).
13. **Show the instructor** the working machine and your completed checklists.
14. **Reverse the teardown politely:** Power off properly, unplug, and leave the bench as you found it. If the machine is to remain assembled, leave it working.

Pause-and-predict points:

- After removing the graphics card: "*Predict: what happens if I reinstall the card but forget the bracket screw?*" Expected: it may not seat fully and could work loose or damage the slot.
- Before seating the RAM: "*Predict: what will happen if the clips do not snap?*" Expected: no display / the machine will not POST.
- Before closing the case: "*Predict: what is the danger of a loose screw inside?*" Expected: it can short-circuit the board or jam a fan.
- Before power-on: "*Predict: if the front-panel power wire is off, what happens when you press the button?*" Expected: nothing â€" the button will not start the machine.

Expected result:
The machine powers on after reassembly, and both checklists plus the screw/cable log are complete.

Troubleshooting while walking the room:

- Component will not move â†' check for a screw or clip; never force.
- RAM will not click â†' check the orientation (the notch/key); do not force the wrong way.

- A screw is left over â†' re-open and find its home before finishing; do not ignore loose screws.
- Learner rushes â†' slow them down; use the checklist aloud.

Completion checklist (learner ticks each):

- ☐ Permission confirmed and machine condition recorded
- ☐ Machine shut down properly
- ☐ Unplugged from the wall and discharged (power button pressed)
- ☐ Wrist strap worn and clipped; bare metal touched
- ☐ Machine on the anti-static mat; tools and screw tray ready
- ☐ Photos taken before disconnecting
- ☐ Screw and Cable Log completed
- ☐ Side panel removed without damage
- ☐ Internal cables disconnected by the connectors
- ☐ Expansion card removed first
- ☐ RAM removed by the edges, into an antistatic bag
- ☐ Storage drive removed gently
- ☐ Components reinstalled in reverse order
- ☐ RAM clips snapped into place
- ☐ All cables reconnected correctly (front panel included)
- ☐ Pre-close check done (no loose screws, cables seated)
- ☐ Side panel replaced and external cables reconnected
- ☐ Computer powers on and displays
- ☐ Bench left tidy

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
"Write the Shop Procedure"

Scenario:
You have been promoted to senior junior-technician at **Elias Computer Clinic, Diobu, Port Harcourt**. New apprentices are joining, and the owner wants a clear, numbered, step-by-step procedure pinned on the wall for opening and rebuilding a desktop computer. You must write it yourself, in your own words, including safety at every stage.

Instructions:
Write two numbered procedures:

1. **Disassembly procedure** (at least nine steps, from permission and safety to removing the components).
2. **Reassembly procedure** (at least seven steps, ending with power-on verification).

Then answer these questions in the same document:

- **Q-A:** Why is reassembly the reverse of disassembly?
- **Q-B:** Why do we remove the graphics card before the RAM and drives?
- **Q-C:** What are the three most important checks before closing the case?
- **Q-D:** What should an apprentice do if a component will not come out easily?

Requirements:

- Safety reminders at every stage where they apply (unplug, discharge, strap, bare metal, no forcing).
- Correct order throughout.
- Written in your own words (do not copy the instructor's demonstration sentences).

- Assume the machine is a standard desktop with a graphics card, two RAM sticks and one hard drive.
- Keep each step to one or two clear sentences.
- Include the photo/labelling step.

Time allowance:
15 minutes.

Success criteria:

- Both procedures are complete and in the correct order.
- Safety is built into the steps, not listed separately only.
- The four questions are answered correctly.
- The "common mistakes" list is realistic.

Hints if stuck:

- Hint 1: Start disassembly from the outside (cables, panel) and finish at the deepest part (cooler/motherboard).
- Hint 2: Reassembly simply reads your disassembly list backwards.
- Hint 3: The pre-close checks are: cables seated, RAM clipped, no loose screws (and cooler/fan connected).

Instructor answer key (do not show learners):

Disassembly procedure (model answer):

1. Confirm permission; record the machine's model and condition; shut it down properly.
2. Unplug from the wall; press the power button to discharge; wear the wrist strap and touch bare metal.
3. Set up the anti-static mat, tools, and a labelled screw tray.
4. Photograph the inside and each cable connection; label awkward cables.
5. Remove the side panel screws (into the tray) and remove the panel.
6. Disconnect internal cables by gripping the connectors (SATA data, SATA power, power connectors; note front-panel wiring).
7. Remove the graphics card first (screw, slot clip, straight up by edges, supporting it).
8. Remove the RAM (clips down, by the edges, into an antistatic bag).
9. Remove the storage drive (screws, slide out gently).
10. (If required) remove the CPU cooler â€” unplug the fan, release the clips, twist gently, lift off.
11. (If required) remove the motherboard â€” all screws noted, cables disconnected, lift by the edges.
12. Place all components safely on the mat/antistatic bags.

Reassembly procedure (model answer):

1. (If the board was removed) refit the motherboard, aligning standoffs and I/O shield; screws evenly, not over-tight.
2. Refit the CPU cooler with fresh thermal paste (pea-sized) if it was removed; reconnect the fan.
3. Refit the storage drive and reconnect SATA data and power using your photos.
4. Refit the RAM, aligning the notch, pressing until the clips **snap**.
5. Refit the graphics card into the PCIe slot until the clip engages; replace the bracket screw.
6. Reconnect all internal cables, especially the front-panel wires, following your photos.
7. Pre-close check: cables seated, RAM clipped, no loose screws, cooler/fan connected.
8. Replace the side panel and screws; reconnect external cables and the power cable.
9. Press the power button; verify fans spin, lights show, and the display appears.

Q-A:
Because the layout is stacked â€” the part removed last is the foundation and must be placed first; reversing guarantees nothing is trapped or blocked. **Q-B:** Because a graphics card often lies over the RAM clips and drives, so it blocks access; removing it first frees the area and prevents bending parts or slots. **Q-C:** All cables firmly seated (including front panel); RAM clips fully snapped;

Common mistakes (accept any four):
forcing components; not taking photos; mixing up screws; touching gold contacts; over-tightening screws; forgetting the I/O shield; not seating RAM fully; forgetting the front-panel wire; leaving a loose screw inside; powering on without the CPU cooler.

Marking guidance (out of 15):
Disassembly order and completeness (4); reassembly order and completeness (3); safety integrated (3); four questions (4); common-mistakes list (1).

J. REAL-WORLD CASE STUDY

Situation.
Adaeze is a careful learner who opens her home desktop to clean out the thick dust of Port Harcourt's dry season. She unplugs the machine, discharges it, straps up, and takes photos â€” good habits. She removes the side panel, the graphics card, the RAM and the drive, and cleans everything thoroughly with compressed air. Happy with her work, she rebuilds the machine: drive, RAM, graphics card, cables, panel. She plugs in and presses the power button. The fans spin and the lights come on, but the **screen stays black**. Nothing on the monitor. Her heart sinks â€” did she break it?

Relevant facts.

- The machine powers on (fans and lights) but there is no display.
- Adaeze removed and refitted the RAM and the graphics card.
- She used compressed air and cleaned thoroughly.
- The monitor cable was connected to the graphics card's port when she started.

Decision points.

1. Is the machine "dead," or is this a common, fixable reassembly issue?
2. What is the most likely cause given she removed and refitted RAM and the card?
3. What should she check, and in what order?
4. What could have damaged the machine â€” if anything?

Possible approaches.

1. Assume the worst and rush it to a shop for an expensive repair.
2. Panic and start removing everything again randomly.
3. Stay calm, unplug, and check the most likely, simplest causes first: RAM seating, card seating, display cable and port.

Recommended solution.
Approach 3. "Powers on but no display" after a teardown is very often a **RAM stick not fully seated** or the **graphics card not fully in its slot**, or the **display cable** in the wrong port or loose. The correct steps: unplug and discharge, re-open the case, press the RAM sticks down until the clips snap, press the graphics card firmly into its slot until the clip engages, check the bracket screw, and confirm the monitor cable is firmly in the **graphics card's** port (not the motherboard's). Close, reconnect, and test. In Adaeze's case, one RAM stick had not clicked home. With an even push, it snapped in, and the picture returned.

Trade-offs.

- Approach 1 wastes money and time on a fault she could fix in two minutes.
- Approach 2 risks causing real damage by random handling.
- Approach 3 is systematic, free, and teaches the exact lesson the course is built on: **check the simple, likely physical causes first.**

Professional connection.
"Powers on but no display" is one of the most common situations in any real repair shop. The technician who methodically reseats RAM, checks the card, and checks the cable solves it quickly and cheaply; the one who guesses replaces parts the customer did not need. This case is a direct bridge to **Session 4 (Diagnostic Workflow)** â€” the same logic, made formal.

Transfer exercise.

Safety reminder.
All of this checking is done with the machine **unplugged and discharged**, on the anti-static mat, with the strap worn. Never re-open or poke inside a machine that is still plugged in.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use as the live guide. Ask learners to find the relevant row.

K.1 Component will not come out

Symptoms:
A part feels stuck; pushing harder seems to work but you hesitate. **Likely cause:** A screw, clip, latch or cable you have not released. Cause is confirmed once you find and release it. **How to investigate:** Stop. Look all around the part (front, back, under) with a torch. Check for hidden screws and slot clips. **Safe corrective steps:** Release the missed fastener, then lift the part straight out. Never use force. **How to confirm resolved:** The part lifts out smoothly with no resistance. **How to prevent recurrence:** Study the machine's layout and take photos before removing; work slowly.

K.2 Powers on but no display after reassembly

Symptoms:
Fans spin, lights on, but the monitor shows nothing. **Likely cause:** RAM not fully seated, graphics card not fully seated, display cable loose or in the wrong port, or monitor off. **How to investigate:** Unplug and discharge; re-open; check RAM clips, the card's seating and clip, and the display cable/port. **Safe corrective steps:** Reseat RAM until it clicks; reseat the card; move the display cable to the graphics card's port; confirm the monitor is powered. **How to confirm resolved:** The display appears. **How to prevent recurrence:** Always check seating and cables before closing (Step 7 in reassembly).

K.3 Power button does nothing after reassembly

Symptoms:
Machine completely silent; no fans, no lights. **Likely cause:** The front-panel power switch connector is loose or misconnected, or the main power connectors are not seated, or the wall plug/socket is off. **How to investigate:** Check the wall socket and plug; check the 24-pin and CPU power connectors; check the front-panel power wire against your photo. **Safe corrective steps:** Reseat the front-panel power connector and the power connectors (machine unplugged). **How to confirm resolved:** The power button starts the machine. **How to prevent recurrence:** Photograph front-panel wiring before disconnecting; check it in the pre-close step.

K.4 Loose screw left inside the case

Symptoms:
A rattle when moving the machine; or a sudden short/failure on power-on. **Likely cause:** A screw fell inside during work. Confirmed by finding it. **How to investigate:** With the machine **unplugged and discharged**, tip and inspect carefully; use a torch. **Safe corrective steps:** Retrieve it with tweezers or by gently tilting; never shake a machine containing drives violently. **How to confirm resolved:** No rattle; the screw is found and put in its place. **How to prevent recurrence:** Always count screws; keep them in the tray; do the pre-close check.

K.5 Stripped or cross-threaded screws

Symptoms:
The screw turns but does not tighten, or bites the wrong way and jams. **Likely cause:** Wrong driver size, or forcing a screw into the wrong hole/angle. **How to investigate:** Try the correct size; inspect the head and the hole. **Safe corrective steps:** Turn the screw slowly backwards first to find the thread, then tighten gently. If it is badly stripped, get instructor help (a screw extractor may be needed). **How to confirm resolved:** The screw tightens cleanly without force. **How to prevent recurrence:** Match driver to screw; never over-tighten; start screws by hand.

K.6 Forgetting to reconnect a cable

A cable left disconnected or fully plugged. Confirmed by comparing with your photo. **How to investigate:** Compare the current connections with your pre-disassembly photos. **Safe corrective steps:** Unplug and discharge, re-open, reconnect the correct cable. **How to confirm resolved:** The function works again. **How to prevent recurrence:** Use photos and labels; do the pre-close check.

Distinguishing likely vs confirmed:
In every case, the "likely cause" becomes "confirmed" only when you actually check it (find the missed screw, see the RAM clip snap, trace the cable against the photo). Teach learners to check before "fixing."

L. SESSION ASSIGNMENT

Assignment title:
"My Disassembly and Reassembly Guide"

Purpose:
To produce a complete, safety-aware, step-by-step guide that the learner can follow on any standard desktop, and that reinforces the Week 1 skills needed for the capstone.

Realistic scenario:
A friend is starting a small repair business and has asked you for a written procedure he can pin above his bench. He is a beginner, so your guide must be clear and safe enough for him to follow without you present.

Detailed instructions:

1. Create a document titled **"My Disassembly and Reassembly Guide"**.
2. Write a **numbered disassembly procedure** (at least nine steps), including permission, recording the machine, and safety at every stage.
3. Write a **numbered reassembly procedure** (at least seven steps), ending with power-on verification and the pre-close check.
4. Include a **Safety Reminders** section (at least five points) that applies to the whole process.
5. Include a **Common Mistakes** section with at least five mistakes and how to avoid each.
6. Include a **Troubleshooting** section: what to do if (a) the machine will not power on, (b) it powers on but shows no display, (c) a component will not come out.
7. Save the file as **"My Disassembly and Reassembly Guide"** on the Desktop.
8. Be ready to explain one step in detail if asked.

Required deliverables:
One saved document with all sections.

Submission format:
Saved on the Desktop with the exact file name. If the learner has no home computer, complete it in the academy lab and show it before leaving.

Deadline:
Before Session 4 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Disassembly steps	Complete, correct order, 9+	Mostly correct, 6â€"8	Incomplete or wrong order
Reassembly steps	Complete, correct order, 7+	Mostly correct, 5â€"6	Incomplete or wrong order
Safety reminders	At every critical step	At most steps	Few or none
Common mistakes	5+ realistic, with solutions	3â€"4 mistakes	Fewer than 3

troubleshooting	3 realistic faults solved correctly	2 faults	1 or none

Common mistakes to avoid:
forgetting to unplug and discharge; jumping straight to reassembly without photos; forcing components; over-tightening screws; forgetting the front-panel wire; not seating RAM fully.

Instructor reference solution (summary, do not show learners beforehand):

- Disassembly: permission/record â†' shut down â†' unplug/discharge/strap â†' bench setup â†' photograph/label â†' remove panel â†' disconnect cables (by connector) â†' remove card first â†' remove RAM â†' remove drive â†' (cooler/board only if needed) â†' set parts aside safely.
- Reassembly: board first (if removed) â†' cooler with fresh paste â†' drive â†' RAM until clips snap â†' card until clip engages â†' reconnect cables (front panel per photo) â†' pre-close check â†' panel/external cables â†' power-on verification.
- Safety: unplug and discharge always; strap and bare-metal touch; hold by edges; no forcing; grip connectors not wires; never open the PSU; do not power on without the cooler; no liquids.
- Common mistakes: no photos; mixed screws; touching contacts; over-tightening; forget I/O shield; RAM not seated; missing front-panel wire; loose screw inside; forcing.
- Troubleshooting: (a) no power â†' check wall/plug, main power connectors, front-panel power wire; (b) no display â†' reseat RAM, reseat card, check display cable/port; (c) stuck part â†' find the missed screw/clip; never force.

M. KNOWLEDGE CHECK

(Learner version. Answers are in the Instructor Key below.)

Q1. Short answer.
What is the correct order for removing the main components during disassembly, and why?

Q2. Multiple choice.
Why do we take photographs before disconnecting internal cables? a) To show the customer b) To know where to reconnect them during reassembly c) To make the job look professional d) It is not necessary

Q3. Explain in your own words.
Why is reassembly described as "the reverse of disassembly"?

Q4. Practical identification.
How do you know a RAM stick is correctly installed?

Q5. Error detection.
A learner says: "If a component will not come out, I should pull harder." Find the mistake and correct it.

Q6. Scenario.
After reassembling a desktop, the fans spin and the lights come on, but the monitor shows nothing. List, in order, the first three things you would check.

Q7. Short answer.
Why should you never power on a computer with the CPU cooler removed?

INSTRUCTOR ANSWER KEY (Q1â€"Q7)

A1.
Remove the **graphics card first**, then the **RAM**, then the **storage drives**, and only later (if required) the **CPU cooler** and motherboard. The card often lies over other parts, so it must go first to free access; the RAM and drives come next; the cooler and board are last because they are the foundation and the most delicate.

A2.
(b) To know where to reconnect them during reassembly. (Photos are your map, especially for the front-panel wires.)

whatever was removed first (the top layer) is placed last. Reversing the order guarantees nothing is trapped or blocked, and that everything fits as before.

A4.
The retaining **clips at both ends snap into place**, and the stick sits level and firm in the slot. (You hear/feel the click.)

A5.
The mistake is using force. The correction: **stop and check** for a screw, clip, latch or cable you have missed; a part that will not move easily is being held by something. Never force it â€" forcing causes damage.

A6.
(1) Confirm the machine is unplugged/discharged, then check the **RAM is fully seated** (clips snapped); (2) check the **graphics card is fully seated** and its clip engaged; (3) check the **display cable is firmly connected to the correct port** (the graphics card, not the motherboard) and the monitor is powered on.

A7.
Because the CPU produces heat very quickly and relies on the cooler (heatsink/fan + thermal paste) to remove it. With the cooler off, the CPU can overheat within seconds and shut down or be permanently damaged.

Marking:
up to 2 marks each (Q1, Q6 up to 3). Total â‰ˆ 16; scale proportionally. Full credit requires correct order, correct component names, and correct reasoning.

N. REFLECTION AND RECAP

Plain-English summary.
Today you opened a real computer and put it back together. You learned the **safety preparation** (permission, recording, unplugging, discharging, strap, bare-metal touch), the **correct disassembly order** (external cables â†' panel â†' photograph â†' disconnect cables â†' card â†' RAM â†' storage â†' cooler only if needed), the **mirror-image rule** (reassembly is the reverse), the **correct reassembly technique** (RAM seated until it clicks, fresh thermal paste when the cooler is refitted, cables reconnected using photos), the **pre-close check**, and **power-on verification**. You also learned to record screws and cables so the job can be repeated correctly.

Five important takeaways:

1. **Safety first, every time:** unplug, discharge, strap, bare metal.
2. **Disassembly order matters**, and reassembly is its **mirror image**.
3. **Take photos and log screws** â€" they are your map and your memory.
4. **Never force a component**; check for the missed screw, clip or cable.
5. **RAM must click into place** and you must **check before you close** the case.

Questions to help you recognise what you can now do:

- Can I prepare a machine for work safely without reminders?
- Can I state the correct removal order and the reason for it?
- Can I reassemble a machine and get it to power on and display?
- Can I list the three checks to make before closing the case?

Connection to the next session:
You can now open, rebuild and safely test a machine. Next session you begin **Week 2: Diagnostics**. You will learn the **systematic diagnostic workflow** â€" how to gather symptoms, form a hypothesis, test one thing at a time, isolate the fault, verify the fix and document it. The "no display after reassembly" logic you used today is the seed of that whole method.

O. INSTRUCTOR EVALUATION NOTES

- The learner photographs and labels before disconnecting, and keeps a screw log.
- The learner follows the correct removal and refit order.
- The learner grips connectors (not wires) and never forces a part.
- The learner seats RAM until the clips snap and performs the pre-close check unprompted.
- The learner verifies power-on and knows the first checks if there is no display.

Common misconceptions to watch for and correct:

- "Order does not matter." â†' It does; the layout dictates it, and wrong order damages parts.
- "Force it if it is stuck." â†' Never; find the missed fastener.
- "Testing after reassembly is optional." â†' Always verify before declaring the job done.
- "Photos are for beginners." â†' Professionals photograph and label, especially on unknown machines.
- "Any amount of thermal paste works." â†' A pea-sized amount, applied correctly.

Suggested remedial exercises for learners who struggle:

1. **Order cards:** give the steps on shuffled cards and have the learner sequence them, then check against the checklist.
2. **One-component drill:** have the learner remove and refit just the RAM, then just the drive, until each is confident and clean.
3. **Photo-and-rebuild:** have the learner photograph a machine, then rebuild a second identical machine using only the photos.
4. **Fault-chasing drill:** the instructor creates a loose-RAM or disconnected-front-panel fault for the learner to find methodically.

Differentiation:

- **Level 1 (supported):** provide a partially disassembled machine for the learner to finish (for example, remove and refit only the RAM and drive), with a partner handling the trickier parts; use the checklists step by step.
- **Level 2 (standard):** complete the full disassembly and reassembly as written.
- **Level 3 (challenge):** disassemble and reassemble a **different model** of desktop (for example one with a tool-less case or a different drive bay), and/or remove and refit the CPU cooler with fresh thermal paste under supervision.

Timing guidance:
The practical (Section H) is the heart of this session and is directly assessed, so protect it. If the class is slow, compress the lecture sections F.4â€"F.5 into summaries and keep the demonstration and practical. If a group is fast, give them the healthy-machine-plus-simulated-fault challenge. Never shorten the safety steps.

END OF SESSION 3 â€" END OF WEEK 1 (COMPONENTS AND SAFETY)

Next: Session 4 â€" The Diagnostic Workflow: Finding the Fault Systematically.

SESSION 4 â€" THE DIAGNOSTIC WORKFLOW: FINDING THE FAULT SYSTEMATICALLY

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 4 â€" THE DIAGNOSTIC WORKFLOW: FINDING THE FAULT SYSTEMATICALLY

A. SESSION IDENTIFICATION

Item	Detail
Session number	4 of 12
Session title	The Diagnostic Workflow
Week / Module	Week 2 â€" Module: Diagnostics
Recommended duration	105 minutes (1 hour 45 minutes)
Difficulty level	Beginner (first contact with formal fault-finding)
Prerequisite knowledge	Sessions 1â€"3: component names and functions; ESD and electrical safety; tools; disassembly and reassembly.
Connection to previous session	In Session 3 the learner met the classic "powers on but no display" problem after reassembly. This session turns that informal troubleshooting into a **structured, repeatable method**.
Connection to the course	This workflow is the backbone of the whole course. Sessions 5 (hardware diagnostics), 6 (software diagnostics), 7 (replacement), and the Session 12 capstone all run through this method. The capstone is assessed *partly on whether the learner follows it*.
Preparation before class	Scenario cards (at least six faults); at least one practice computer with a **simulated fault** (for example RAM not fully seated) that can power on but show no display; a working reference machine; printed Symptom-to-Cause Reference and Diagnostic Questions Checklist; whiteboard for drawing the workflow; optional: a machine that beeps on power-on (to demonstrate POST beeps) or a diagram of BIOS beep codes.

Safety note for this session:
All physical diagnosis is done on **unplugged and discharged** machines, with the anti-static strap worn and boards handled by the edges. Never reseat or swap a component while the machine is powered. When testing power-on, the instructor supervises and only brief tests are made. Never open the PSU. Because this session involves deliberate faults, students must be told clearly that these are **practice machines** and that the same actions must never be attempted on a customer's machine without permission.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **State and follow the five-step diagnostic workflow** â€" gather symptoms, confirm the obvious, form a hypothesis, test one thing at a time, and verify and document.

4. **Explain POST (power-on self-test)** and what it means when a machine fails it.
5. **Interpret basic BIOS beep codes and no-display symptoms** as diagnostic clues (recognising that codes vary by maker and must not be invented).
6. **Use swap-testing** to isolate a faulty component safely.
7. **Apply a safe boot troubleshooting order** for common "no display" and "no power" situations.
8. **Document each step** of a diagnosis clearly, including what was tested, the result, and the conclusion.
9. **Explain why a structured method beats guessing**, using the two ideas of "likely cause" and "confirmed cause."

C. REQUIRED MATERIALS

Hardware (provided by CEA):

- At least one practice computer with a **simulated fault** that powers on but shows no display (for example, RAM deliberately left unseated, or the display cable in the wrong port).
- A working reference machine to compare against (very useful for swap-testing).
- One or two spare known-good components for swap-testing, e.g., a tested RAM stick, a tested display cable.
- A desktop for the instructor demonstration.
- Anti-static wrist straps and mats; tool kit; screw tray; torch.

Software:

- None essential. Optional: a free system-information viewer (Windows Task Manager â† ' Performance) to show live activity, but the session can run entirely without software. Note: no statistical or hardware-specific claims should be made.

Safety gear:

- Anti-static wrist straps and mats.
- First-aid kit available.

Sample / fault components:

- Known-good RAM stick for swap-testing.
- A known-good display cable and/or a second monitor if available.
- Clearly-labelled "FAULT â€" DEMO" parts where relevant.

Worksheets (printed):

- Worksheet 4.1 â€" "Symptom-to-Cause Reference" (one per learner): a table of symptoms and likely causes.
- Worksheet 4.2 â€" "Diagnostic Questions Checklist" (one per learner): the seven questions.
- Worksheet 4.3 â€" "Diagnostic Worksheet / Service Report Draft" (one per learner): a structured page with fields for symptoms, observations, questions, hypothesis, tests and results, resolution, and verification.
- Scenario cards (at least six) for the guided and independent exercises.

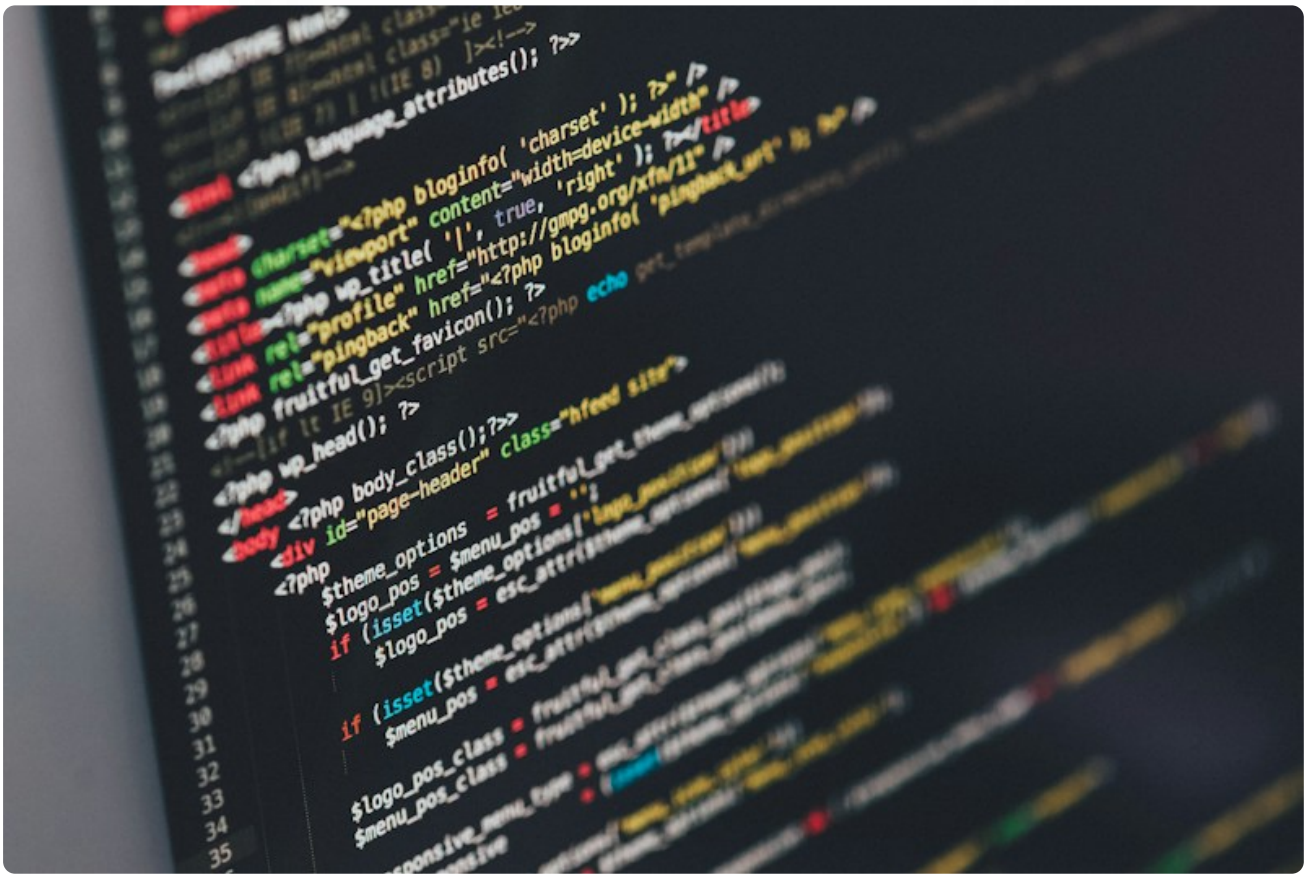
Alternatives for a learner or academy with limited equipment:

- The workflow can be taught almost entirely through **paper scenarios** if no faulty machine is available. Use the scenario cards and the diagnostic worksheet; role-play the customer and technician.
- If only one faulty machine exists, run it as a whole-class demonstration and let learners fill in the worksheet together, then give each learner different scenario cards to work through independently.
- If a learner is uncomfortable with hands-on fault work, let them be the "diagnostic recorder" and the "customer", which still practises the workflow and the questions.

Suggested visuals:



- â€" Symptom â†' hypothesis â†' test â†' verify in practice



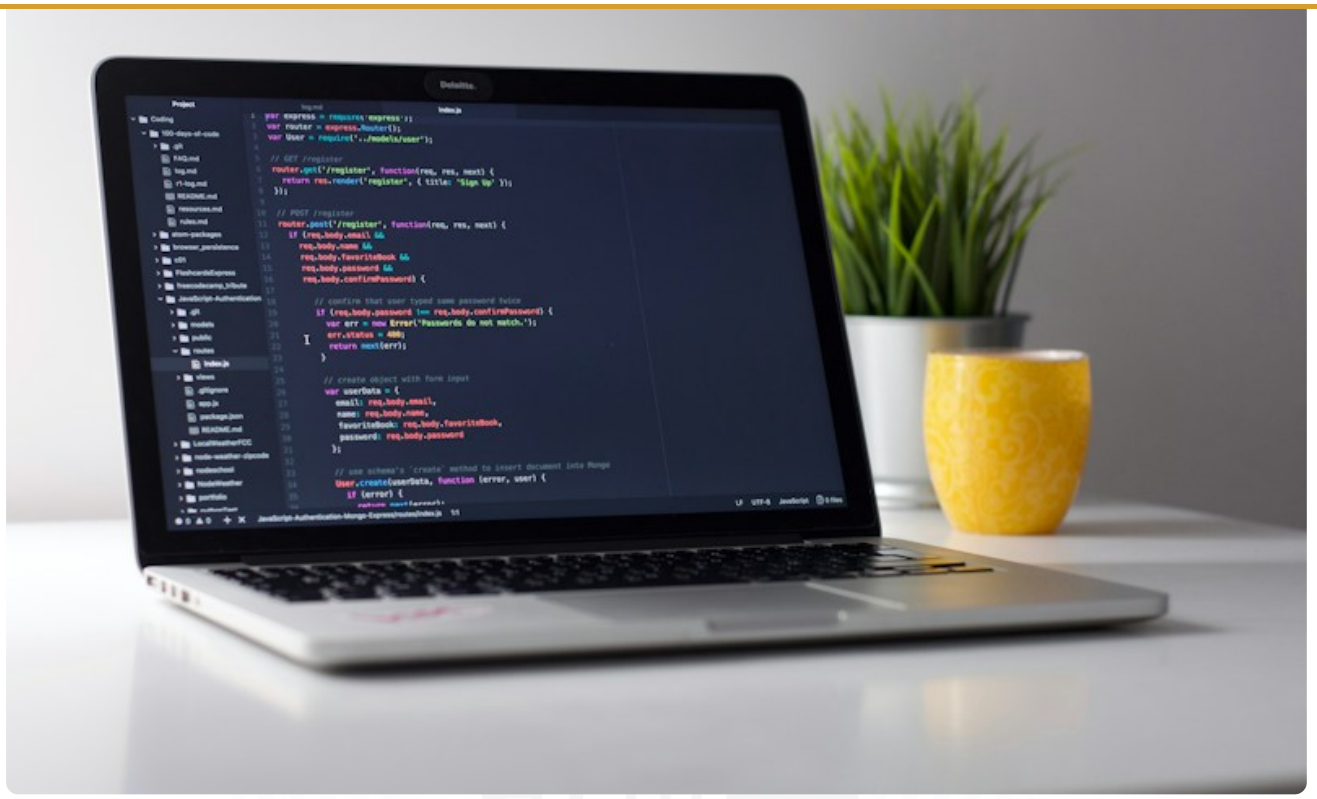
- â€" Recording observations on the diagnostic worksheet



- â€" Swap-testing known-good parts



- — Symptom-to-test flowchart drawn on paper



— Workflow applied to a laptop fault

D. OPENING ACTIVITY (6 minutes)

"The Detective Analogy"

Instructor setup:
On the projector, show a simple picture or drawing of a detective with a magnifying glass and a notebook, beside a computer with a question mark over it. Keep the faulty practice machine hidden or off to the side.

What the instructor says:

> *"Imagine a detective arrives at a crime scene. Does the detective walk in and immediately say, 'It was the gardener'? No. That is a guess. What does a good detective do? They look at the scene. They ask witnesses questions. They form a theory. They test the theory â€" fingerprints, alibi, evidence. And only when the evidence is solid do they say who did it.*
>
> *A computer technician is a detective. When a customer brings a faulty computer, the customer has given you a clue â€" a*
>
> ***symptom***
> *â€" like 'it will not come on' or 'the screen is black.' Your job is to find the **cause**. And here is the trap: the worst technicians are the ones who **guess**. They open the case, see a dirty fan, and say, 'Ah, it is the fan!' Maybe. Or maybe the real problem is somewhere else entirely.*
>
> *Today you learn the detective's method. You will learn a fixed set of steps â€" the same steps every single time â€" that stops you from guessing and leads you to the true cause.*
>
> *Let me ask you: which is better, a technician who guesses fast, or a technician who is right?"*

Pause for answers.

> *"Correct. Right beats fast, every time. A wrong guess costs the customer money and costs you your reputation. A method makes you reliable."*

> *"This computer is the crime scene. It powers on â€" you can see the fans spin â€" but there is no picture. Somewhere in here is the cause. By the end of today, using the method, you will find it â€" without guessing, and without replacing a single part."*

Expected learner responses:
Learners often suggest "the screen is broken" or "we need to check the graphics card." Accept all answers but point out that these are **hypotheses**, not yet confirmed. Some learners may guess correctly â€" praise them but insist on the method: "You may be right, but how will you *prove* it? That is the skill."

Why this opener works:
It uses a familiar role (detective) to make an abstract method concrete, draws a sharp contrast between guessing and method, and gives the class a real, solvable problem to look forward to.

E. PREREQUISITE REVIEW (7 minutes)

Run as a quick oral review; ask different learners.

Question 1:
"Name the major components and one thing each does."

- Expected answer: Motherboard (connects everything), CPU (does the work), RAM (temporary memory), storage (permanent), PSU (power), GPU (images), fans (cooling).
- Common gap: learners forget the PSU in the list, yet it is central to "no power" diagnosis. Remind them.
- Bridge: "All of these are suspects for today's detective."

Question 2:
"What is ESD, and why does it matter when we swap parts?"

- Expected answer: Electrostatic discharge; invisible damage; prevented by strap, mat, bare-metal touch, edge handling.
- Common gap: forgetting to re-touch bare metal between parts. Emphasise when swapping.
- Bridge: "When we swap components to test, we handle many parts â€" so the ESD habit matters even more."

Question 3:
"What are the first two things to check before opening a case?"

- Expected answer: Unplug from the wall; press the power button to discharge.
- Common gap: confusion between the power button and the wall plug. Clarify order.
- Bridge: "And note â€" the very *first* diagnostic clue is often outside the case entirely."

Question 4:
"From Session 3: if a freshly reassembled computer powers on but shows no display, what were the first things to check?"

- Expected answer: RAM seated (clips snapped); graphics card seated; display cable in the correct port; monitor powered.
- Common gap: jumping to "the GPU is broken."
- Bridge: "Today we turn that instinct into a proper order of operations."

Question 5:
"What is the difference between a 'likely' cause and a 'confirmed' cause?"

- Expected answer: A likely cause is a guess or probability; a confirmed cause is proven by a test or observation.
- Common gap: learners treat their first idea as confirmed. Clarify the difference â€" it is the heart of today.
- Bridge: "This distinction is the whole point of the method: confirm before you fix."

Bridge to today (say this):

> ***tracing a fault in the NEPA/PHED line***
> *after an outage. Would you climb the pole and start replacing cables immediately? No â€" you would first check whether the whole area is out, then whether your neighbour has light, then whether it is only your compound, then your meter, then your wiring. You follow an order from the biggest, simplest possibility to the smallest. Today's method is exactly that discipline, applied to a computer."*

If more than a third cannot answer Question 5, spend an extra three minutes on likely-vs-confirmed, as it underpins the whole session.

F. FULL LECTURE NOTES

F.1 Why a method beats guessing

The idea.
Diagnosis is the process of finding the cause of a fault. A **workflow** is a fixed sequence of steps you follow every time, so that you never skip something important and never jump to a conclusion.

Plain-English definition.
The diagnostic workflow is a step-by-step method: gather the symptoms, confirm the obvious, form a hypothesis, test one thing at a time, then fix and verify. It replaces "I think" with "I have proven."

Why it exists.
Computers have many parts that can cause similar symptoms. "No display," for example, can come from RAM, the graphics card, the cable, the monitor, the motherboard, or even the power supply. Guessing wastes time and money and often replaces healthy parts. A method narrows the possibilities in an efficient, safe order.

What it does.
It makes your work:

- **Repeatable** â€" the same fault is diagnosed the same way every time, by anyone.
- **Reliable** â€" you confirm the cause instead of assuming it.
- **Safe** â€" you test simple, safe things before complex or risky ones.
- **Accountable** â€" because you document each step, you can explain your reasoning to a customer or a colleague.

Real-life comparison.
The workflow is like a **doctor's consultation**. The doctor asks about symptoms, checks the basics (temperature, blood pressure), forms a suspicion, orders a test, confirms the diagnosis, treats it, and then checks you again. A doctor who prescribed medicine without checking would be dangerous. A technician who replaces parts without checking is the same.

Realistic repair-shop scenario.
Two technicians receive the same "no display" computer. The first guesses "graphics card" and fits a new one â€" the fault remains, and the customer is charged for nothing. The second follows the method: checks the cable and port, reseats the RAM (which was loose), and the display returns in two minutes for free. Same fault, opposite results â€" the difference is the method.

Common misconceptions.

- "Experienced technicians do not need a method; they just know." Experienced technicians *have internalised* the method; they still follow it. When a fault is unusual, even they return to the steps.
- "The most obvious cause is always the true one." Sometimes the obvious cause is a **red herring**. You only know by testing.

Limitations and safety.
The workflow guides you, but some faults (intermittent, or requiring special tools) need more time or specialist help. Recognising limits is part of the method â€" Step 5 explicitly includes referring onward when needed.

Collect every clue before you touch anything.

Observe (your own senses):

- **Look:** Is there visible damage? Swelling (swollen capacitors or batteries)? Burn marks? Dust? A loose cable? A fan not spinning? Any light at all?
- **Listen:** Do you hear beeps (POST codes)? Clicking or grinding (often a failing HDD)? Rattling (a loose fan or screw)? Silence (no power)?
- **Smell:** Is there a burning or "hot" smell (often an overheated part or a failing PSU)? **If you smell burning, stop and unplug safely â€" do not continue.**
- **Feel (carefully, only after unplugging):** Is a part unusually hot? Is a connector loose?

Ask the customer (the seven questions are detailed in F.4): What happened, when, what were they doing, what changed, does it happen always or sometimes, have they tried anything, is there any unusual sound/smell/damage.

Step 2 â€" Confirm the obvious (the simple things first).
Before assuming a deep fault, rule out the simple, common causes:

- Is the machine actually **plugged in** and is the **wall socket working** (test with a known-good device, or a lamp)?
- Is the **power switch** (on the PSU, on a laptop, on a power strip) in the correct position?
- Is the **monitor switched on** and set to the correct input (HDMI/VGA/DisplayPort)?
- Are the **external cables** connected (display, keyboard, power)?
- Is there **power in the building** (a blackout affects everything)? Many "faults" are none of these being right. Always check them first.

Step 3 â€" Form a hypothesis (an educated guess).
Based on your observations and the customer's answers, write down the **two or three most likely causes, in order of probability and ease of testing**. A hypothesis is not a conclusion; it is a smart guess to be tested. Example: "Given the machine powers on but shows no display, the most likely causes are (1) RAM not seated, (2) display cable/port, (3) graphics card not seated."

Step 4 â€" Test one thing at a time (and change one thing at a time).
Test your hypotheses, starting with the **simplest and most likely**. Crucially, **change only one thing per test**, then observe the result. If you change three things at once, you will not know which one fixed (or caused) the problem. Common safe tests:

- **Reseat** the suspected component (RAM, card, cable).
- **Swap-test:** replace a suspected component with a **known-good** one (for example, a tested RAM stick). See F.6.
- **Compare** with a working machine. After each test, note the result. If the fault is gone, you have likely found the cause; confirm it (Step 5).

Step 5 â€" Resolve, verify and document.
Fix the confirmed cause (or, at this stage, identify and record it). **Verify** the fix: does the machine now work fully? Test more than just the original symptom â€" for example, restart it, and check it behaves normally under use. Then **document** everything: the symptoms, the questions, your hypothesis, the tests and results, the cause, and the fix. Documentation protects the customer and you, and it is the basis of the professional service report (Sessions 11 and 12).

The golden rule of testing:
One change at a time. This single rule transforms confusing diagnosis into clear, logical progress.

Real-life comparison.
The five steps are like **finding which bulb has blown in a house**. You notice the lights are off (symptom). You check whether the whole street is off and whether the switch is on (confirm the obvious). You suspect a fuse or a bulb (hypothesis). You try one bulb at a time and one fuse at a time (test one thing). You confirm the light comes on (verify) and note what you changed (document). You would never replace every bulb in the house at once.

Common misconceptions.

- "Test many things at once to save time." This does the opposite â€" it creates confusion. One change at a time.

Limitations and safety.
Never test by working on a powered machine. All component reseating and swapping happens with the machine **unplugged and discharged**, wearing the strap. If a test requires mains-level work (for example, testing a PSU's output), that is a supervised, later-session activity and is not attempted here on live circuits.

F.3 POST â€” the computer's start-up self-check

Plain-English definition.
POST (Power-On Self-Test) is a basic self-check the computer runs automatically the moment it is switched on, before it tries to load the operating system. It is built into the motherboard's firmware (called the **BIOS** or, on newer machines, the **UEFI**). Firmware is small built-in software that runs before Windows does.

Why it exists.
Before handing control to the operating system, the computer checks that its essential parts are present and working: it detects the CPU, the RAM, the graphics, and the storage, and confirms basic settings. If something essential is missing or faulty, the POST fails, and the machine signals the problem instead of pretending everything is fine.

What the POST does.

- It checks that basic components are detected and respond.
- If the check **passes**, the machine usually continues to the BIOS/UEFI screen briefly and then boots the operating system (or shows a "no bootable device" message if there is no bootable system, which is a different issue from a dead machine).
- If the check **fails**, the machine can signal with a **beep code** (a pattern of beeps) if a small internal speaker is fitted, and/or a display message, and/or codes on the motherboard. Many modern machines have no internal speaker, so they cannot beep â€” instead they may show LED indicators or simply show no display.

How POST helps diagnosis.
When a computer **powers on but does not reach its operating system**, you can often tell *where* it stopped:

- **No display, no beep, fans spinning:** often a failure very early â€” RAM, graphics, or the POST itself.
- **Beep pattern:** a code that (on machines that do beep) points to a particular subsystem.
- **Reaches the BIOS screen but cannot find the OS:** the hardware is basically fine, but the boot drive or the operating system has a problem (a software or storage issue, not a dead board).

A crucial honesty note â€” beep codes vary.
BIOS beep codes are **not universal**. Different makers (AMI, Award, Phoenix, and different manufacturers) use different beep patterns, and modern machines often have no speaker at all. Therefore, **do not memorise invented "universal" beep codes.** The professional approach is:

1. Note the **make and model** of the machine or motherboard.
2. Look up that maker's **documented** beep codes (from the official manual or the manufacturer's website).
3. Interpret the beep against that specific documentation. Teach learners this discipline: *the code is only meaningful with the maker's own chart.* Do not invent or quote specific patterns as if they were universal.

Real-life comparison.
POST is like the **security check at the airport gate**. Before anyone boards, staff verify the plane's essential systems and passengers. If a check fails, the flight does not proceed until it is fixed. The beep code is like the specific alarm message: "gate 3 sensor fault" â€” but you must read *that airport's* manual to know what gate 3 means, because every airport numbers things differently.

Common misconceptions.

- "Beeps mean the same thing on every computer." No â€” they differ by maker. Always consult the maker's chart.
- "No beep means it is definitely broken." Not always â€” many machines have no speaker, so silence is normal on POST failure.
- "If it beeps, the POST passed." Not necessarily â€” some beeps indicate a fault.

must never rely on invented codes. If unsure, use the method: confirm simple things, then test components one at a time.

F.4 The seven diagnostic questions

The idea.
What the customer tells you is often the single biggest clue. Asking the right questions saves hours.

The seven questions:

1. **"When did the problem start?"** â€" Ties the fault to a time or event (a power cut, a move, a new update).
2. **"What were you doing when it happened?"** â€" Context reveals the trigger (gaming â†' overheating or GPU; a certain program â†' software or driver).
3. **"Has anything changed recently?"** â€" New hardware, a new program, an update, a new cable, a different room.
4. **"Does it happen all the time or only sometimes?"** â€" Intermittent faults are harder; consistent faults are usually easier to reproduce and test.
5. **"Have you tried anything to fix it?"** â€" You need to know what was already done or changed, so you do not undo it or chase your tail.
6. **"Is there any unusual sound, smell, or visible damage?"** â€" Physical clues (burning, clicking, swelling, a spill) are often decisive.
7. **"Does it happen with specific programs, or all the time?"** â€" Helps separate a software problem from a hardware one.

How to ask well.
Ask in plain, respectful language a customer will understand. Do not make the customer feel stupid. Repeat back what they say to confirm. Write down the answers â€" memory is unreliable, and notes become your report.

Real-life comparison.
These questions are like a **doctor asking about your symptoms and history**. The more accurate the information, the faster and more accurate the diagnosis.

Common misconceptions.

- "I will just open it and see." You will miss vital clues. Ask first.
- "The customer does not know anything." The customer may not know the *cause*, but they always know *what changed* and *when* â€" that is gold.

F.5 Safe boot troubleshooting order for "no display" and "no power"

When a machine shows

no picture
(or **no power at all**), follow a fixed, safe order from the outside inward and from simplest to most complex. This is an application of the workflow.

A. No display, but the machine powers on (fans spin, lights on).
Work in this order:

1. **Monitor and cable first:** Is the monitor powered on? Is the cable firmly connected at both ends? Is the monitor set to the correct input? (These are free and easy.)
2. **Cable port:** If a dedicated graphics card is fitted, is the display cable in the **graphics card's** port, not the motherboard's? (A very common mistake.)
3. **Signal check:** Does the monitor say "no signal" or is it completely dark? "No signal" points to a source/cable issue; complete darkness may point to the monitor itself.
4. **RAM reseating:** Unplug and discharge; reseat the RAM until the clips snap. (One of the most common no-display causes.)
5. **Graphics card reseating:** Reseat the card firmly into its slot until it clicks; check the bracket screw.
6. **Swap-test the cable and monitor** with known-good ones if available.

Work in this order:

1. **Wall power:** Is the socket working? Test with a known-good lamp or device. Is there power in the building?
2. **Power cable:** Is it firmly connected at both ends? Try a known-good cable.
3. **Power switch:** Is the PSU switch (if present) on? Is a power strip switched on?
4. **Front-panel power button:** Is the front-panel power connector seated on the board? (A common result of reassembly.)
5. **PSU test:** With supervision, and using a safe method, check whether the PSU is delivering power. **Never open the PSU.** At this stage, swap-testing with a known-good PSU is done **only under supervision**.
6. **Motherboard/board-level:** If a known-good PSU still gives nothing, suspect the board â€" but confirm carefully, do not guess.

The principle behind the order.
Always start with the **cheapest, simplest, safest, most likely** checks and move toward the **expensive, complex, riskier** ones. This saves money, avoids damage, and usually finds the fault sooner.

Real-life comparison.
This order is like **checking your phone will not charge**: first you check the socket (is there power?), then the cable (is it damaged?), then the charger, and only last the phone's charging port. You would not open the phone first.

Common misconceptions.

- "No display means the GPU is dead." The cable, port or RAM is more often the cause. Check those first.
- "No power means the PSU is dead." It may be the socket, the cable, the switch, or the front-panel button. Confirm first.

Limitations and safety.
Never test power-related faults while working inside a live machine. PSU testing is a supervised activity and the PSU is never opened. If you smell burning, stop and unplug safely.

F.6 Swap-testing â€" isolating the faulty component

Plain-English definition.
Swap-testing (also called substitution testing) means replacing a **suspected** component with a **known-good** one to see whether the fault moves or disappears. If the fault disappears after the swap, the suspected part was faulty. If the fault remains, the problem is elsewhere.

Why it works.
It turns a doubt into a fact. Instead of guessing whether the RAM is faulty, you prove it by fitting a stick you know works.

How to swap-test safely.

1. Identify the component you suspect, based on the workflow.
2. **Power off, unplug and discharge** the machine.
3. Wear the strap and touch bare metal. Handle parts by the edges.
4. Remove the suspected part and fit a **known-good** part (from a working machine or a tested spare).
5. Power on and observe. **One component per test.**
6. If the fault is gone, you have strong evidence the original part was faulty. If it remains, put the original part back and move to the next hypothesis.

Critical cautions.

- Only swap-test components you are **authorised** to handle, and only on **practice machines** in this course.
- Never swap a part from a customer's working machine into another machine without permission â€" you could introduce a fault or lose data.
- When swapping storage drives, be extremely careful: a drive may contain important data. Do not swap drives around casually.
- After swap-testing, restore the machine to its correct configuration â€" do not leave a "spare" part installed and forgotten.

was faulty. If it still does not charge, the problem is the phone, not the charger.

Common misconceptions.

- "If the new part fixes it, the old part is definitely dead." Usually true, but confirm â€” sometimes the act of reseating (not the part itself) fixes a poor connection.
- "Swap many parts at once to be quick." Never â€” you will not know which change did what. One at a time.

Limitations and safety.
Swap-testing needs known-good spares. Without them, rely on the other steps. Never swap a PSU or open it without supervision. Keep track of which part went where.

F.7 Documenting the diagnosis (the beginning of the service report)

Why document.
A diagnosis you cannot explain is of little professional value. Documentation:

- Records what you did, so you or a colleague can pick up the job later.
- Protects you if the customer claims you did something you did not.
- Becomes the **service report** the customer receives (Sessions 11â€“12).
- Helps you learn â€” patterns you record today become experience tomorrow.

What to record (a simple structure):

- **Customer and machine details:** name, contact, model, any visible condition.
- **Reported symptoms:** in the customer's words.
- **Your observations:** look/listen/smell/feel.
- **Questions and answers:** the seven questions.
- **Hypotheses:** the likely causes you listed.
- **Tests and results:** what you tested, one at a time, and what happened.
- **Confirmed cause:** the root cause once proven.
- **Fix/plan:** what was done or recommended.
- **Verification:** how you confirmed the machine is now working.

Real-life comparison.
Documentation is like a **doctor's case notes**. If you see a different doctor next time, the notes tell them your whole history. Without notes, every visit starts from zero.

Common misconceptions.

- "Documentation is paperwork that wastes time." It saves time, prevents repeated mistakes, and is what separates a professional from a tinkerer.
- "I will remember." You will not â€” especially after several jobs in a day.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The practice desktop with a **simulated fault** (recommended: RAM left unseated so the machine powers on but shows no display; alternative: display cable plugged into the motherboard port while a dedicated card is fitted).
- A working reference machine nearby, plus a known-good RAM stick and a spare display cable for swap-testing.
- The whiteboard with the five steps drawn.
- The Diagnostic Worksheet (4.3) on the projector to show how to fill it in.
- The wrist strap, mat and tools.

"Watch me follow the five steps exactly. Notice that I do not guess, and I test one thing at a time."

Do and say, step by step:

Step 1 â€” Gather symptoms.

- "I power on the machine. The fans spin; a light comes on; the monitor says 'no signal.'"
- "I listen â€” no beeps. I look â€” no burn marks, no loose drive. I smell â€” nothing burning."
- "I ask the 'customer' (I will play both roles): 'When did it start? What were you doing? Has anything changed?' The customer says: 'It worked yesterday; today I opened it to clean dust and now there is no picture.'"

Step 2 â€” Confirm the obvious.

- "Is the monitor on? Yes. Is the cable connected? I check both ends. Is it on the correct input? Yes. Is the cable in the right port? I check â€” the machine has a graphics card, and I see the cable is in the graphics card port, which is correct."
- "So the simple things are fine. Now I move inward."

Step 3 â€” Form a hypothesis.

- "Given it worked until the case was opened for cleaning, and now there is no display, my likely causes are: (1) RAM not fully seated, (2) graphics card disturbed, (3) a cable knocked loose. I write these down in order."

Step 4 â€” Test one thing at a time.

- "I **power off, unplug and discharge**. I wear the strap and touch bare metal."
- "Test 1: I reseat the **graphics card** first? No â€” I start with the most likely and simplest: the **RAM**. I press each stick until the clips **snap**." (If the simulated fault is loose RAM, one stick was not clipped.)
- "I check the graphics card is seated â€” it is fine."
- "I make sure I changed only the RAM seating. Nothing else."

Step 5 â€” Resolve, verify, document.

- "I close, reconnect, and power on. The display appears. The fault is gone."
- "I verify beyond the symptom: I restart it; the display returns and the machine behaves normally."
- "I document: symptoms, the customer's answer, my hypothesis, the test (reseat RAM), the result (display returned), the confirmed cause (RAM not fully seated), and the verification."

Expected result:
The machine shows a display; the class sees the method find the fault without any part being replaced.

Reasoning to state aloud at key points:

- "I start with the cable and monitor because they are outside the machine, free, and easy."
- "I reseat RAM before touching the graphics card because in a case that was just cleaned, RAM is very commonly nudged loose, and reseating it is simple and safe."
- "I change one thing at a time so I know exactly what fixed it."
- "I verify by restarting, not just by seeing one picture."

Errors to watch and mention:

- Do not test by opening the machine while it is plugged in â€” narrate the unplug/discharge clearly.
- Do not change two things at once â€” demonstrate the discipline.
- Do not declare success after one picture â€— verify.

Questions to ask during the demo:

- "Why did I check the monitor and cable first?" (Simplest, outside the machine, free.)
- "After reseating the RAM, how do I know the fix is real?" (The display returns and it restarts normally â€— verified.)
- "What would I do if reseating the RAM had not worked?" (Put it back, move to the next hypothesis â€— reseat/test the card, then swap-test.)

Exercise: "Diagnostic Scenarios"

Objective:
With the instructor guiding, learners apply the five-step workflow to written scenarios and (if a faulty machine is available) to a real simulated fault, filling in a Diagnostic Worksheet for each.

Materials per learner/group:
Worksheets 4.1 (Symptom-to-Cause), 4.2 (Questions Checklist) and 4.3 (Diagnostic Worksheet); scenario cards; access to the faulty practice machine (rotating); a working reference machine for comparison.

Procedure (instructor guides each part; learners work in pairs; instructor circulates):

1. **Review the worksheet structure.** The instructor shows Worksheet 4.3 on the projector: fields for symptoms, observations, questions, hypothesis, tests/results, cause, and verification.
2. **Real machine (whole class, guided):** The class diagnoses the instructor's simulated fault together, using the worksheet. The instructor calls the steps; learners fill in each field and predict what the instructor will do next.
3. **Written scenarios (pairs):** Each pair takes **two scenario cards** and completes a full Diagnostic Worksheet for each, using all five steps.
4. **Swap and check:** Pairs swap worksheets with another pair and check each other's logic (did they confirm the obvious? did they test one thing at a time? did they verify?).
5. **Report back:** Two or three pairs read out their best diagnosis. The class gives feedback.
6. **Instructor checks** each pair's worksheets before they finish.

Scenario cards (the six core scenarios):

- **Card 1:** Computer powers on (fans spin) but shows no display.
- **Card 2:** Computer shows a blue error screen (BSOD) occasionally, then restarts.
- **Card 3:** Computer is very slow to start and programs take a long time to open.
- **Card 4:** Computer makes a clicking noise from the drive area.
- **Card 5:** Computer shuts down randomly after about thirty minutes of use.
- **Card 6:** Nothing happens at all when the power button is pressed (no fans, no lights).

Pause-and-predict points:

- After reading Card 1: *"Before touching anything inside, what three obvious things will you confirm?"* (Monitor on; cable connected and in the correct port; monitor input correct.)
- After forming a hypothesis on Card 5: *"If a machine shuts down after about thirty minutes, what is your leading hypothesis and why?"* (Overheating/cooling â€" because it happens after time under load; but confirm, do not assume.)
- After a test on Card 6: *"If the socket and cable are fine and there is still no power, what is your next safety-aware step, and what must you never do?"* (Check the front-panel power wire and, under supervision, consider PSU testing; never open the PSU.)

Expected result:
Each pair completes two full, sensible Diagnostic Worksheets, and the class completes the real-machine worksheet, each showing all five steps.

Troubleshooting while walking the room:

- Learner jumps to a cause without observing â†' point them back to Step 1 and ask for the clues.
- Learner tests several things at once â†' remind them of "one change at a time."
- Learner forgets safety when handling the machine â†' stop and reinforce unplug/discharge/strap.
- Learner declares success without verifying â†' ask "how do you know the fault is truly gone?"
- Pair disagrees â†' encourage them to state the evidence for each idea.

Completion checklist (learner ticks each):

- [ ] Two scenario cards completed with full worksheets
- [ ] For each scenario: symptoms gathered and obvious things confirmed
- [ ] For each scenario: at least two hypotheses listed in order
- [ ] For each scenario: tests described one at a time
- [ ] For each scenario: cause stated as likely or confirmed, with evidence
- [ ] For each scenario: verification described
- [ ] Peer-check completed and feedback given
- [ ] Safety followed while handling the real machine

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
"The Cyber CafÃ© Fault"

Scenario:
You are the technician at **Elias Computer Clinic, Diobu, Port Harcourt**. A cyber cafÃ© owner, *Mama Ngozi*, brings in one of her ten computers. She says: "*This one just stops working after some time. The customer will be browsing, and then the screen freezes and it restarts by itself. It happens more when many people are using the cafÃ© at once, and in the afternoon when the place is hot. In the morning it works fine for a long time.*" She wants to know the cause before she pays for any parts.

Your task:
Apply the **five-step diagnostic workflow** to Mama Ngozi's computer and produce a written diagnosis.

Requirements:

1. **Symptoms:** List the reported symptoms and the observations you would make yourself (look/listen/smell/feel).
2. **Obvious checks:** State which simple things you would confirm first.
3. **Questions:** Write at least **five** diagnostic questions you would ask Mama Ngozi, and the likely answer you are hoping to hear.
4. **Hypotheses:** List at least **three** likely causes ranked from most to least likely, with a reason for each.
5. **Test plan:** Describe, one test at a time, how you would confirm or rule out your top hypothesis. Include one swap-test.
6. **Safety:** State every safety step you would take before handling components.
7. **Documentation:** Complete a one-page diagnostic note (you may use Worksheet 4.3).
8. **Conclusion:** State your **leading** cause, and whether it is *likely* or *confirmed*, and how you would confirm it.

Constraints:

- This is a computer that must keep serving customers, so your tests must be safe and quick.
- Do not replace any part without confirming the cause first.
- Never open a live machine or the PSU.
- Use the workflow steps as headings in your answer.

Time allowance:
15 minutes.

Success criteria:

- All five workflow steps are present and correctly applied.
- Symptoms are distinguished from confirmed observations.
- At least three ranked hypotheses with reasons.
- A test plan with one change at a time and a swap-test.
- A correct safety section.
- A clear leading cause marked likely or confirmed, with a confirmation method.

- Hint 2: A very common cause of a computer shutting down under load is **overheating from dust, a failing fan, or dried thermal paste.**
- Hint 3: A safe, quick test for overheating is to observe whether it fails sooner with the case side off in a cool spot with a fan directed at it â€" but always under supervision, and the machine must be handled safely.
- Hint 4: A swap-test here could be testing with a known-good case fan or checking the CPU fan spins; do not swap the CPU casually.

Instructor answer key (do not show learners):

1. **Symptoms:** Freezes and restarts by itself; worse in the hot afternoon and under heavy use; fine in the morning and light use. Self-observations: check for dust-clogged heatsink/fan; listen for a weak or noisy fan; feel (after shutdown) for unusual heat at the vents; look for error messages on restart.
2. **Obvious checks:** Confirm the power supply from the wall is steady (a cyber cafÃ© may have many machines and a weak extension); confirm the machine is in a well-ventilated spot; confirm nothing is blocking the vents.
3. **Questions (any five):** When did it start? What were you doing? Has anything changed (new machines, new programs, an update)? Does it happen all the time or only when busy/hot? Does the machine feel very hot at the back/top? Did you see any message when it restarted? Have you cleaned it recently? (Accept well-chosen questions.)
4. **Hypotheses (ranked):** (1) **Overheating** â€" dust in the heatsink/fan or dried thermal paste, worsened by heat and load; (2) a **failing power supply** struggling under load (common where many machines share a weak supply); (3) a **failing RAM stick** or (3) **drivers/malware** causing restarts under certain tasks. (Any sensible ranked list with reasons is acceptable, but overheating should be prominent given the pattern.)
5. **Test plan:** First, with the machine off, inspect and clean the dust from the heatsink and fan (Session 8 skill; in this session, describe it). Confirm the CPU fan spins when powered. Monitor the behaviour in a cooler environment. Swap-test: if a known-good PSU is available, test the machine with it under supervision. Test one thing at a time and observe.
6. **Safety:** Unplug from the wall and discharge before opening; wear the anti-static wrist strap and touch bare metal; handle boards by the edges; never open the PSU; never work inside a live machine; ensure ventilation when using compressed air.
7. **Documentation:** A filled Worksheet 4.3 with the above.
8. **Conclusion:** Leading cause: **overheating** (likely). Confirm by cleaning and observing that the shutdown stops, and/or by showing temperature-related behaviour; if it persists after cleaning, test the PSU and RAM by swap-testing. Because a single observation can mislead, the cause remains *likely* until the fix removes the fault reproducibly.

Marking guidance (out of 15):
five steps applied (5); symptoms vs observations (2); ranked hypotheses with reasons (3); test plan with one-at-a-time and a swap-test (3); safety completeness (2).

J. REAL-WORLD CASE STUDY

Situation.
A customer brings a desktop to the shop with a complaint that sounds familiar: "It works fine for a few minutes, then the picture freezes or the machine restarts." A young technician at the shop tests the RAM and finds nothing wrong. He tests the hard drive and finds nothing wrong. He tests the power supply and finds nothing wrong. Frustrated, he tells the customer: "Your machine is fine â€" I cannot find any problem." The customer leaves annoyed, and the fault continues. A day later, the same customer returns, even more upset.

Relevant facts.

- The fault is **intermittent** â€" it does not appear during casual testing.
- The machine works for a while, then fails.
- Standard component tests (RAM, drive, PSU) show "nothing wrong."
- The young technician tested for the fault without understanding **when** and **why** it happens.

Decision points.

1. Why did "testing everything" fail to find the fault?

4. How should he confirm the cause once he has a hypothesis?

Possible approaches.

1. Keep testing components randomly, replacing parts until something works.
2. Give up and tell the customer "it is fine."
3. Return to Step 1: gather better symptoms, especially *what task* triggers the fault, then form a targeted hypothesis.

Recommended solution.
Approach 3. Intermittent faults are diagnosed by **reproducing** them and by asking sharper questions â€" not by testing parts at random. The technician should ask: *"Does it happen when you are doing a particular thing, like watching videos or playing games, or when the room is hot?"* Suppose the customer says: *"Yes â€" it happens when I watch videos for a while, especially in the afternoon."* That points strongly to **overheating under load**. The technician then monitors the machine's behaviour under load in a controlled way and finds the CPU temperature climbing and the fan struggling/blocked. **Confirmed cause:** overheating from a dust-clogged fan and dried thermal paste. **Fix:** clean the cooling system and replace the thermal paste (Session 8 skill), then verify by running the machine under load for a longer period and confirming the fault no longer occurs.

Trade-offs.

- Approach 1 (random replacement) wastes the customer's money and may introduce new faults. It is guesswork disguised as testing.
- Approach 2 (giving up) destroys trust; the fault is real, even if it is intermittent.
- Approach 3 (better symptoms â†' targeted hypothesis â†' reproduce â†' confirm) is slower at the start but finds the true cause and builds a lasting reputation.

Professional connection.
Intermittent faults are common and are where a real technician earns respect. The skill is patience and structure: ask **when** and **under what conditions**, reproduce the fault, test one variable at a time, and confirm before charging the customer for anything. This is also where the "likely vs confirmed" distinction matters most â€" you must *reproduce* the fault and see it disappear after the fix.

Transfer exercise.
"Why are intermittent problems harder to diagnose than problems that happen every time, and what two techniques help?" (They do not appear during casual testing; techniques: ask sharper questions to find the trigger and reproduce the fault deliberately; and test one variable at a time, often over a longer period or under load.)

Safety reminder.
Any handling of the machine during this diagnosis is done **unplugged and discharged**, with the strap worn, and without opening the PSU. If testing under load, it is supervised, and the machine is never left unattended while running a stress test on a practice unit.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use as the live guide. Ask learners to find the relevant row.

K.1 Jumping to conclusions (guessing)

Symptoms:
Replacing a part that was probably fine; the fault returns; the customer is charged for the wrong work. **Likely cause:** Skipping Steps 1â€"3 and going straight to a favourite guess. Confirmed when the fault persists after the "fix." **How to investigate:** Ask: what evidence did I actually have? Did I confirm the simple things first? **Safe corrective steps:** Return to Step 1. Gather symptoms, confirm the obvious, list hypotheses, and test one at a time. **How to confirm resolved:** The fault disappears **and** you can explain the evidence. **How to prevent recurrence:** Always write the hypothesis list before touching a part.

K.2 Not asking enough questions

Symptoms:

seven questions, write the answers. **Safe corrective steps:** Re-interview the customer and revise the hypothesis. **How to confirm resolved:** The new information explains the pattern. **How to prevent recurrence:** Make the seven questions a fixed script.

K.3 Testing the most complex cause first

Symptoms:
Wasted time and money on deep checks while a simple cause is ignored. **Likely cause:** Excitement, or a bias toward a favourite part. Confirmed when a simple check (like a loose cable) fixes it. **How to investigate:** Always start from the outside in, simplest first. **Safe corrective steps:** Follow the safe boot order (F.5). **How to confirm resolved:** The simple cause is found and fixed. **How to prevent recurrence:** Memorise the "cheapest, simplest, safest, most likely first" rule.

K.4 Changing more than one thing at a time

Symptoms:
The fault disappears but you do not know why, or it returns unpredictably. **Likely cause:** Multiple changes in one test. Confirmed by the confusion when it fails again. **How to investigate:** Recall exactly what you changed; re-test one variable at a time. **Safe corrective steps:** Redo the tests, one change each time. **How to confirm resolved:** You can point to the single change that fixed the fault. **How to prevent recurrence:** Adopt the golden rule: one change per test.

K.5 Not verifying the fix

Symptoms:
The machine seems fine, then the fault returns; the customer loses trust. **Likely cause:** Declaring success after one observation or a single picture. Confirmed by the return of the fault. **How to investigate:** Test the machine properly â€” restart it, use it, run it under the conditions that caused the fault. **Safe corrective steps:** Verify under real conditions before returning the machine. **How to confirm resolved:** The fault does not return after proper verification. **How to prevent recurrence:** Make verification the final, mandatory step.

K.6 Not documenting

Symptoms:
You cannot explain your work later; a colleague must start from scratch; the customer disputes what was done. **Likely cause:** Believing documentation is a waste of time. Confirmed when memory fails. **How to investigate:** N/A â€” it is a habit. **Safe corrective steps:** Fill in the Diagnostic Worksheet during the job, not after. **How to confirm resolved:** Your notes completely describe the job. **How to prevent recurrence:** Make documentation part of the workflow, not an extra.

K.7 Misreading beep codes / inventing universal codes

Symptoms:
A wrong diagnosis based on a beep pattern from the wrong maker. **Likely cause:** Assuming all BIOS beeps are universal. Confirmed when the maker's actual chart contradicts the assumption. **How to investigate:** Identify the make/model and consult that maker's documented beep codes. **Safe corrective steps:** Interpret the beep using the maker's own chart; if unavailable, fall back to the workflow. **How to confirm resolved:** The interpretation matches the maker's documentation and the tested fault. **How to prevent recurrence:** Never memorise invented "universal" beep codes.

Distinguishing likely vs confirmed:
Every "likely cause" above becomes "confirmed" only when checked and verified. In diagnosis, an unverified conclusion is just another hypothesis.

L. SESSION ASSIGNMENT

Assignment title:
"My Diagnostic Reference and Worked Case"

Purpose:

Realistic scenario:

You have just started work at a repair shop. Your supervisor hands you a reference book to maintain: a symptom-to-cause table, the workflow, the questions, and one full worked example that new apprentices can study.

Detailed instructions:

1. Create a document titled **"My Diagnostic Reference and Worked Case"**.
2. **Symptom-to-Cause Table:** list at least **eight** symptoms, each with its likely causes (for example: no power; no display; blue screen; slow performance; overheating/shutdown; strange drive noise; random restarts; no internet).
3. **The Five-Step Workflow:** write the five steps in your own words, with one sentence each on why the step matters.
4. **The Seven Diagnostic Questions:** list all seven, and for each, write what kind of clue it gives you.
5. **Safe Boot Order:** write the safe order for (a) "no display but powers on" and (b) "no power at all."
6. **Worked Case:** choose **one** scenario (you may invent a realistic one), and write a full diagnosis using the five steps as headings, including symptoms, observations, questions, ranked hypotheses, a test plan (one change at a time, with a swap-test), the likely-or-confirmed cause, and the verification method.
7. Save the file as **"My Diagnostic Reference and Worked Case"** on the Desktop.
8. Be ready to explain your worked case aloud in the next session.

Required deliverables:

One saved document containing all six parts.

Submission format:

Saved on the Desktop with the exact file name. (If no home computer, complete in the lab and show before leaving.)

Deadline:

Before Session 5 begins.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Symptom-to-cause table	8+ symptoms, accurate	5â€“7 symptoms	Fewer than 5 or inaccurate
Five-step workflow	Complete and clear, with reasons	Mostly complete	Incomplete or vague
Seven questions	All seven with useful clues	5â€–6 with clues	Fewer than 5
Safe boot order	Both cases correct and in order	One correct	Both unclear or wrong
Worked case	All five steps applied, verified, coherent	Mostly applied	Missing steps or illogical

Common mistakes to avoid:

confusing symptoms with causes; listing hypotheses as if they were confirmed; testing several things at once; forgetting the verification step; omitting safety; inventing universal beep codes.

Instructor reference solution (summary, do not show learners beforehand):

- Symptom-to-cause examples: no power â†' wall/cable, front-panel switch, PSU, board; no display â†' cable/port, RAM seating, card seating, monitor; blue screen â†' RAM, driver, OS, overheating; slow â†' malware, full disk, insufficient RAM, failing drive, overheating; shutdown under load â†' overheating/cooling, PSU; drive noise â†' failing HDD; random restart â†' RAM, PSU, overheating, driver; no internet â†' cable/router/driver.
- Workflow: gather symptoms; confirm the obvious; form a hypothesis; test one thing at a time; resolve, verify and document.
- Seven questions with their clues (as in F.4).
- Safe boot order (as in F.5).
- Worked case: any coherent five-step diagnosis with a clear likely/confirmed cause and a verification method.

Q1. Short answer.
List the five steps of the diagnostic workflow in order.

Q2. Multiple choice.
Why should you confirm the obvious things (power, cables, switches) first? a) Because they are the most interesting b) Because they are the simplest, cheapest, safest and often the cause c) Because the customer likes it d) Because it is a legal requirement

Q3. Explain in your own words.
What is POST, and what does it mean when a machine "fails POST"?

Q4. Practical identification.
A computer powers on (fans spin, lights on) but shows no display. Give the first three things you would check, in order.

Q5. Error detection.
A learner says: "I will replace the RAM, the graphics card and the cable all at once to save time." Find the mistake and correct it.

Q6. Scenario.
A customer says their computer restarts by itself, only when the room is hot. Which part of the diagnostic workflow did most technicians forget, and what would you do next?

Q7. Short answer.
Why must you not treat BIOS beep codes as universal?

INSTRUCTOR ANSWER KEY (Q1â€“Q7)

A1.
(1) Gather the symptoms (observe and ask); (2) confirm the obvious (power, cables, switches); (3) form a hypothesis; (4) test one thing at a time; (5) resolve, verify and document.

A2.
(b) Because they are the simplest, cheapest, safest and often the actual cause â€” and checking them first prevents wasting time and money on deeper faults.

A3.
POST (Power-On Self-Test) is the basic self-check the computer runs when switched on, before loading the operating system, to confirm its essential parts are present and working. If a machine "fails POST," it means this early check did not pass â€” often shown by a beep code (on machines with a speaker), LED indicators, or simply no display. It tells you the problem is very early, before the system loads.

A4.
(1) Monitor and cable (is the monitor on; is the cable connected at both ends and in the correct port â€” graphics card vs motherboard; is the input correct); (2) reseat the RAM until the clips snap; (3) reseat the graphics card. (Order of the first two may vary; the point is simplest/outside-first.)

A5.
The mistake is changing more than one thing at a time. The correction: **change one thing per test**, observe the result, then move on. Otherwise, if the fault changes, you will not know which change caused it. The golden rule is one change at a time.

A6.
The technician forgot **Step 1 â€” gather symptoms** properly (especially the context: "only when hot"). Next: ask more targeted questions and **reproduce** the fault under the trigger conditions, form the hypothesis of **overheating**, and confirm it by checking the cooling system and observing whether the fix removes the fault. (This is a context-rich, intermittent fault.)

A7.
Because BIOS/UEFI beep codes differ between makers (AMI, Award, Phoenix, and manufacturers), and many modern machines have no speaker at all. A beep pattern is only meaningful when read against the **specific maker's documented chart**. Do not memorise invented "universal" codes.

and correct reasoning.

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned to diagnose faults like a professional â€" not by guessing, but by a method. The **five-step workflow** is: **gather the symptoms** (observe and ask), **confirm the obvious** (power, cables, switches), **form a hypothesis**, **test one thing at a time**, and **resolve, verify and document**. You learned about **POST (Power-On Self-Test)** â€" the computer's automatic start-up self-check â€" and how to interpret basic clues (remembering that **beep codes differ by maker** and must not be treated as universal). You learned a **safe boot order** for "no display" and "no power," the technique of **swap-testing** to isolate a faulty component, and the discipline of **documentation**. Above all, you learned the difference between a **likely** cause and a **confirmed** cause.

Five important takeaways:

1. **Follow the five-step workflow every time** â€" method beats guessing.
2. **Confirm the obvious first** â€" power, cables and switches are often the cause.
3. **Test one thing at a time** (and swap-test with known-good parts).
4. **Beep codes are not universal** â€" always check the maker's own chart.
5. **Verify and document** â€" an unverified fix is not a fix, and undocumented work is not professional.

Questions to help you recognise what you can now do:

- Can I list and explain the five steps of the workflow?
- Can I gather symptoms using the seven questions?
- Can I give the safe order of checks for "no display" and "no power"?
- Can I explain the difference between a likely and a confirmed cause?

Connection to the next session:
You now have the method. Next session you put it to work with **hardware diagnostics** â€" using tools and software to test RAM (for example MemTest86), check a hard drive's health (for example CrystalDiskInfo), monitor temperatures (for example HWMonitor), and check a power supply. The workflow you learned today decides *when* and *why* each of those tools is used.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner states and follows the five steps without prompting.
- The learner gathers symptoms before opening anything, and asks good questions.
- The learner confirms the obvious (power/cables/switches) first.
- The learner tests one thing at a time and uses swap-testing correctly.
- The learner distinguishes "likely" from "confirmed" causes.
- The learner documents each step clearly.

Common misconceptions to watch for and correct:

- "The obvious cause is always right / guessing is fine." â†' Form a hypothesis and test it; the obvious cause can be a red herring.
- "Experienced technicians skip the method." â†' They follow it faster; it is what makes them reliable.
- "Test many things at once." â†' One change at a time.
- "Once it works, stop." â†' Always verify; and always document.
- "Beep codes are universal." â†' They are maker-specific; consult the maker's chart.

sequence is automatic.

2. **Question drill:** role-play the customer while the learner practices the seven questions and takes notes.
3. **Likely vs confirmed sorting:** give ten statements and have the learner mark each as "likely" or "confirmed" and justify it.
4. **Safe-order drill:** shuffle the checks for "no display" and "no power" and have the learner re-sequence them.

Differentiation:

- **Level 1 (supported):** provide a partially completed Diagnostic Worksheet with the symptoms pre-filled and the steps labelled, and work through one scenario together before letting the learner try independently.
- **Level 2 (standard):** complete the guided and independent exercises as written.
- **Level 3 (challenge):** diagnose a **real intermittent** fault on a practice machine (for example a partially blocked fan), timing how long it takes to reproduce the fault, and produce a full written report; or design a new scenario card and model answer for the class.

Timing guidance:

The value of this session is the method, so protect the guided practical (Section H) and the independent exercise (Section I). If the room is slow, compress the lecture sections on swap-testing (F.6) and documentation (F.7) to summaries, but do **not** shorten the five steps (F.2), safe boot order (F.5), or the POST honesty note (F.3). Keep the case study as a short story if time is tight.

END OF SESSION 4

Next: Session 5 â€" Hardware Diagnostics: Testing RAM, Storage, Temperatures and the Power Supply.

SESSION 5 â€" HARDWARE DIAGNOSTICS

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 5 â€" HARDWARE DIAGNOSTICS

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	5 of 12
Session title	Hardware Diagnostics
Week / Module	Week 2 â€" Module: Diagnostics
Recommended duration	105 minutes
Difficulty level	Beginner (first structured hardware fault-finding)
Prerequisite knowledge	Sessions 1â€"4: the six major components, safe handling and tools, disassembly/reassembly, and the five-step diagnostic workflow (Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve). If a learner cannot name the six major components or list the workflow, run the 10-minute refresher in Appendix 5A before this session.
Connection to previous work	Session 4 gave you the *workflow*. This session gives you the *tools and tests* you apply at the "Test" step when you suspect the fault is hardware.
Preparation before class	Practice desktops on the benches; diagnostic software installed (CrystalDiskInfo, HWMonitor); a bootable MemTest86 USB prepared by the instructor; one known-good and one known-faulty RAM stick; one known-good and one failing hard drive; one known-good PSU; a working monitor and one spare VGA/HDMI cable; anti-static wrist straps; a PSU tester if the academy has one (optional); printed Diagnostic Report Forms (one per learner); Component Identification Charts on the wall.

A note on wording used in this course.
In CEA materials the word "diagnostics" means *the act of finding out what is wrong*, not only software. A **diagnostic tool** is any device or program that helps you find a fault. This session is about hardware: physical parts that can fail. Session 6 is about software.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain**, in plain language, what a hardware fault is and how it differs from a software fault.
2. **Describe** the common symptoms of a faulty RAM stick â€" failure to boot, beep codes, random restarts, blue screens â€" and run a memory test using Windows Memory Diagnostic or MemTest86.
3. **Read** a storage drive's S.M.A.R.T. health in CrystalDiskInfo and decide whether the drive is Good, Caution or Bad; state when to back up and when to replace.
4. **List** the basic checks for a power supply fault (no power, random shutdowns) and explain why the PSU must never be opened.

7. **Apply** the swap-testing and isolation method to prove which component is faulty.
8. **Record** findings on a Diagnostic Report Form using clear, plain language a customer can understand.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least two practice desktop computers set up for diagnostics (they may be opened during the practical).
- One **known-good** RAM stick and one **known-faulty** RAM stick (instructor keeps these labelled; learners are not told which is which).
- One **known-good** hard drive and one drive that shows wear (the instructor can supply a drive with a rising reallocated-sector count, or simulate a "Caution" reading using a training drive).
- One known-good power supply and, if available, one PSU that has failed.
- A working monitor plus one spare display cable (VGA, HDMI or DisplayPort, matching the practice machines).
- Anti-static wrist straps (one per learner) and a clean, hard work surface.
- Screwdrivers, tweezers, flashlight, and small parts trays.
- Bootable MemTest86 USB drive (prepared by the instructor). One USB can be shared around the class.
- Optional: a PSU tester (a small device that plugs onto the PSU's main connector and lights up the voltage lines). This is a luxury item; the course works without it.

Software (provided by CEA)

- CrystalDiskInfo (free) â€" reads the health of a storage drive.
- HWMonitor (free) â€" shows temperatures, fan speeds and voltages.
- Windows Memory Diagnostic â€" already built into Windows (no install needed).
- MemTest86 (free, on the USB drive) â€" a deeper memory test run from outside Windows.

Printed materials

- Diagnostic Report Form (one per learner).
- Symptom-to-Cause quick reference card (one per learner).

Safety equipment

- Anti-static wrist straps, safety glasses, first aid kit.

Internet

- Not required. Optionally, the instructor may show a search for a beep-code meaning at the very end, but do not depend on it.

Alternative for learners who lack a facility

- Any working desktop can be used for the temperature and storage checks. Learners without a personal machine share a bench machine in pairs: one person runs the tool, the other records the findings, then they swap.

Suggested visuals:



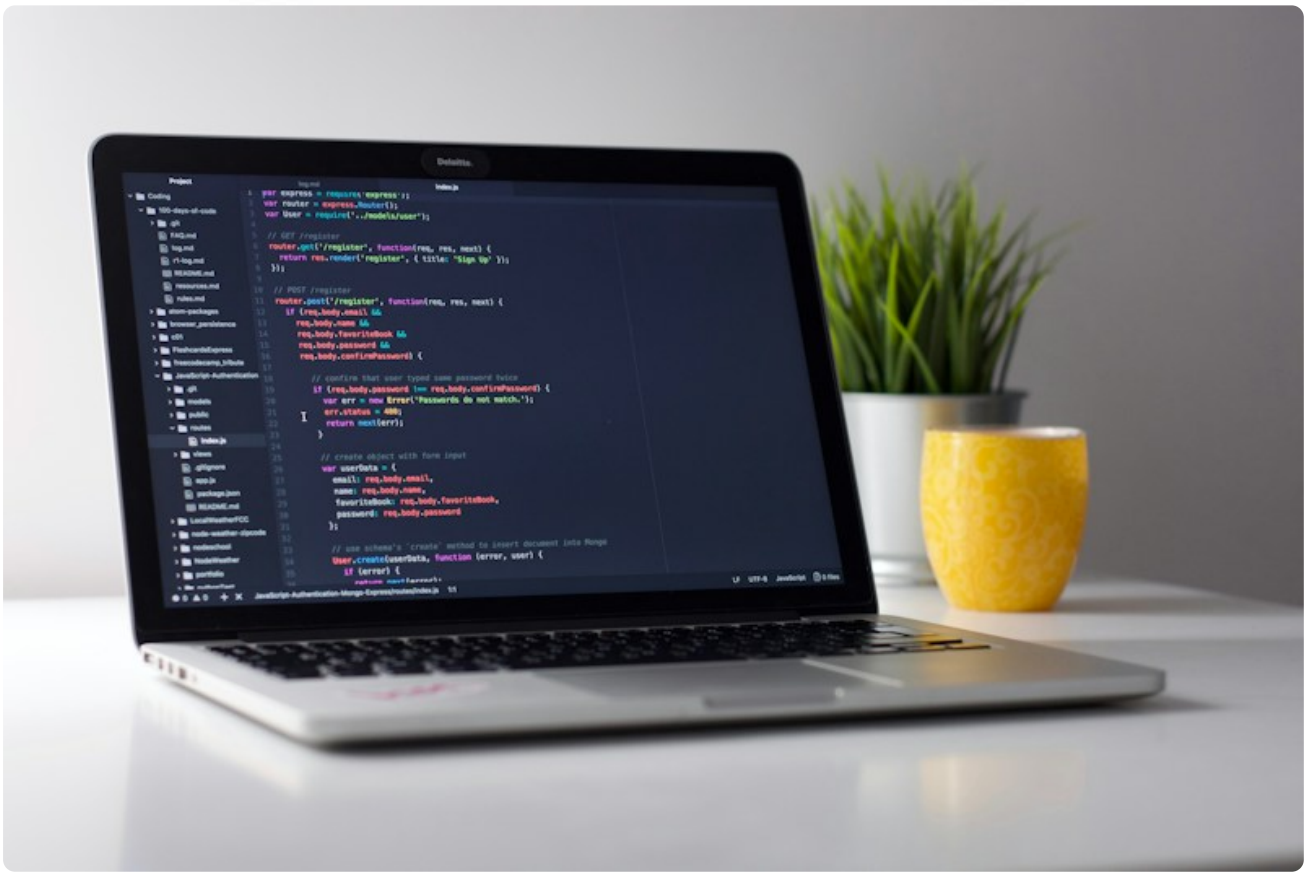
- â€" Checking RAM, storage and fans



- â€" Hardware diagnostic session at the bench



- â€" Tools and straps ready for safe testing



- — Laptop hardware under test



— Diagnostic software results on screen

D. OPENING ACTIVITY (6 minutes)

Instructor setup:
On the projector, show a photo or drawing of a car dashboard with the **check-engine light** on. Do not explain yet.

What the instructor says:

> *"Good morning. When your car's check-engine light comes on, you do not open the bonnet and start changing parts at random. You take it to a mechanic. The mechanic plugs in a small computer and it gives codes â€" it tells him exactly which sensor is complaining. He does not guess; the tool does the guessing for him.*
>
> *A computer is the same. When a computer misbehaves, we do not start replacing parts blindly. We use tools to ask the machine a question and read the answer. Sometimes the machine answers with a*
>
> ***beep***
> *instead of words. Sometimes it answers with a **temperature reading**. Sometimes it answers with a **health status: Good, Caution or Bad**.*
>
> *Today you become the mechanic. You will learn the tools that make a computer tell you what is wrong with its hardware."*

Expected learner response:
Some may mention "blue screen", "it beeps", "loud fans" or "the screen just shows nothing". Accept all answers and write them on the board. Say: "*These are all symptoms. Today we turn symptoms into a diagnosis.*"

Then demonstrate the hook:
On the projector, open CrystalDiskInfo. Point at the **blue "Good"** status and say:

> *"See this word â€" Good. That is the drive telling us it is healthy. If I open a drive that is failing, this turns yellow, or even red. One colour is the difference between a fine computer and five years of somebody's photos lost forever. By the end of*

Why this opener works:
It connects a familiar everyday tool (the mechanic's code reader) to computer work, it sets the expectation that we *test rather than guess*, and it ends with a concrete, exciting promise (reading drive health) that the learners will actually do in the practical.

E. PREREQUISITE REVIEW (7 minutes)

Run this as an oral round-robin. Put the questions on the projector. Keep it fast â€" this is revision, not new teaching.

Question 1:
"Name the five steps of the diagnostic workflow."

- Expected answer: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).
- Common gap: learners remember "test" but forget "ask the customer" or "document". Remind them that a diagnosis you cannot explain is not finished.

Question 2:
"What are the six major components inside a computer?"

- Expected answer: **Motherboard, CPU, RAM, storage (hard drive/SSD), PSU, GPU.**
- Common gap: omitting the PSU. Probe: *"What turns the wall electricity into the voltages the parts need?"*

Question 3:
"Why do we always start by checking the simplest possible cause first?"

- Expected answer: Because the simplest cause is quick to check and is very often the real one. We save time and money by ruling out easy things (a loose cable, a dead socket) before opening the case.
- Common gap: learners jump straight to "replace the hard drive". Correct this â€" a loose monitor cable looks exactly like a dead GPU, and costs â‚¦0 to fix.

Question 4:
"A computer fails to start and you hear repeated beeps. What is the machine trying to tell you?"

- Expected answer: The motherboard lights up and runs a self-test before loading Windows; if something basic fails, it beeps a pattern instead of showing a picture. Beeps usually point to RAM, the GPU, or another key part.
- Common gap: learners think the beeps are "just a noise". Introduce the term **POST** now (Power-On Self-Test) â€" we define it properly in Section F.

Question 5:
"Why must we document every finding, even when we are not yet sure?"

- Expected answer: So the work can be verified later, so a colleague can continue where you stopped, and so the customer gets an honest, clear report. Also, comparing "before" and "after" readings proves the repair worked.

Bridge to today (say this):

> *"In Session 4 you learned to think like a technician. Today you learn to test like one. Think of a doctor. The workflow is the doctor asking questions and forming a theory. The diagnostic tools today are the stethoscope and the blood test â€" the things that turn a theory into a fact."*

Gap handling:
If more than a third of the class cannot answer Question 1 or 2, pause and run the Appendix 5A refresher now (10 minutes), then continue. Do not push on with learners who cannot name the parts â€" today depends on it.

F. FULL LECTURE NOTES

Everything that goes wrong with a computer is either **hardware** or **software**. Getting this right early saves hours.

Plain-English definition.

- A **hardware fault** is a problem with a physical part: a stick of RAM that is damaged, a hard drive that is wearing out, a fan that has stopped spinning.
- A **software fault** is a problem with the instructions or data: a virus, a corrupted Windows file, a wrong driver.

Why the distinction exists.
The word "fault" describes *what is wrong*, but hardware and software are fixed in completely different ways. Hardware faults are usually fixed by **replacing** or **repairing a part**. Software faults are usually fixed by **cleaning, correcting or reinstalling** the software. If you fix the wrong type, you waste time and money.

Analogy.
A car is hardware; the driver's bad habit of always pressing the brake is "software". If the brake pedal itself is broken, that is a hardware fault â€” you replace the pedal. If the pedal is fine but the driver keeps stalling, that is a "software" fault â€” you have to change the behaviour. Replacing a perfectly good pedal because of a driving habit would be a waste.

Realistic Port Harcourt scenario.
A customer in Rumuokoro says his laptop "is very slow and sometimes beeps and restarts". The "slow" part sounds like software (too many startup programs, or a virus). But the "beeps and restarts" part sounds like **hardware** (RAM or overheating). This session's job is the second half â€” the hardware. Session 6 handles the first half. A good technician separating the two is what makes a diagnosis fast.

Important misconception.
"If Windows is slow, it must be a virus." Not always. A failing hard drive and too little RAM both make a computer slow, and both are hardware. Always *test* before you decide.

Key limitation.
Some faults look like one type but are the other. Example: a failing hard drive can corrupt Windows files, so you see software symptoms caused by a hardware problem. This is why we test the hardware *first* â€” if the foundation (RAM, drive, power, cooling) is bad, no amount of software fixing will hold.

F.2 POST and beep codes â€” the very first test

The idea.
When you press the power button, before Windows ever appears, the computer runs its own quick self-check.

Plain-English definition.
POST stands for **Power-On Self-Test**. It is a tiny set of instructions stored on the motherboard that checks that the basic parts â€” CPU, RAM, graphics, keyboard â€” are present and responding before handing over to the operating system.

Why it exists.
Imagine a bus company that will not move the bus until it has checked that the engine, brakes and doors work. POST is the computer doing that check on itself. If a critical part fails the check, the computer cannot show a picture, so it uses **beeps** (or a light on the motherboard, depending on the model) to tell you which basic part failed.

What it does.
It runs in the first one or two seconds after power-on. If everything passes, you hear a single short beep (on some models) and the screen shows the maker's logo. If something basic fails, you hear a pattern of beeps, or nothing at all.

Analogy.
POST is like the fuel-pump priming sound you hear for a moment when you switch on many cars before the engine starts â€” the car checking itself quietly before it lets you drive.

Realistic Port Harcourt scenario.
A cyber cafÃ© PC will not show anything on screen and beeps three times. The cafÃ© owner assumes the monitor is dead and is about to buy a new monitor. A technician recognises the beep pattern as a **memory (RAM)** warning, reseats the RAM stick,

The beep pattern depends on the **make of the motherboard's BIOS/UEFI** (for example, the number and length of beeps). Because this differs between makers, the correct professional method is: **count the beeps, then look up that maker's beep code in the manual or on the maker's support page.** Do not memorise one pattern and assume it fits every computer. In this course we teach you to *count and look up*, not to guess.

Important misconception.
"No beep means the computer is completely dead." Not always. Many modern motherboards have no built-in speaker, so they cannot beep at all. On those, a small **debug light** or **LED code** on the board does the same job. Also, some working computers never beep. Absence of a beep is a clue, not a verdict.

Limitation.
POST only checks very basic things. It does not test your storage deeply, it does not test the display cable, and it cannot tell you a drive is slowly failing. For those, we use the tools below.

F.3 Diagnosing RAM faults

The idea.
RAM (Random Access Memory) is the fast scratch-pad where the computer holds whatever it is working on right now. When RAM goes bad, the computer becomes unstable in a very particular way.

Plain-English definition.
RAM faults are failures in the memory sticks â€” the long, flat strips plugged into the motherboard. A stick may be entirely dead, or it may work most of the time and fail only occasionally (called an **intermittent fault**, one of the hardest to catch).

Why RAM matters.
Windows and every program keep their current work in RAM. If the memory returns a wrong value even once, calculations go wrong, programs crash, or Windows shows a blue screen.

Common symptoms of faulty RAM:

- Computer will not start, or starts then restarts again and again ("boot loop").
- Repeated **beep patterns** during POST.
- **Blue screen** errors (in Windows, the "Blue Screen of Death" or **BSOD**) â€” especially stop codes that mention memory.
- Programs closing suddenly with no warning.
- Files becoming corrupted for no clear reason.
- Random freezes under load.

Analogy.
RAM is like the work-table of a tailor in the market. If one corner of the table is warped, most sewing is fine, but every time you lay cloth on that corner the measurement comes out wrong. The whole table is not useless â€” only that spot â€” but it spoils everything that touches it.

How we test RAM â€” Tool 1: Windows Memory Diagnostic.

1. Save all work and close all programs.
2. Press the **Windows key**, type **Windows Memory Diagnostic**, and open it.
3. Choose **Restart now and check for problems**.
4. The computer restarts and runs the test outside Windows. This can take 10â€“30 minutes.
5. When Windows restarts, the results appear (also visible later in Event Viewer).

What it does: it writes known patterns into RAM and reads them back, looking for any mismatch.

How we test RAM â€” Tool 2: MemTest86.
MemTest86 is the deeper, more trusted test. It runs from a USB drive before Windows starts.

1. Insert the MemTest86 USB prepared by the instructor.

3. The test starts automatically and shows a screen of moving numbers.
4. **Let it complete at least two full passes** (this can take 30+ minutes, so in class we usually start it and observe the first pass only).
5. **Any red line or "errors" count above zero means the RAM is faulty.**

Isolating the bad stick (swap testing).
If the machine has more than one RAM stick:

1. Remove one stick, leaving only stick A, and run the test.
2. If errors appear, stick A is bad. If not, remove stick A and test stick B alone.
3. Test one at a time until you find the faulty one.
4. You can also **swap** a known-good stick in: if the faults disappear, the original stick was the problem.

Important misconception.
"The computer counts 8 GB of RAM, so the RAM is fine." Capacity is not health. A stick can be fully detected and still return wrong values occasionally. Only a test proves health.

Limitation.
Memory testers cannot find every intermittent fault, because a bad spot may only fail when hot or under a certain load. If a memory test passes but the symptoms strongly suggest RAM, an experienced technician may still swap the stick as a test. Also, a blue screen can be caused by a bad **driver** rather than bad RAM â€” Session 6 covers that.

F.4 Diagnosing storage faults (hard drives and SSDs)

The idea.
Storage is where your files live permanently. When a spinning hard disk is wearing out, it usually warns you before it dies â€” if you know how to read the warning.

Plain-English definition.

- A **hard disk drive (HDD)** has spinning metal platters and a moving read/write head. Being mechanical, it can wear out and make noise.
- A **solid-state drive (SSD)** has no moving parts; it stores data in memory chips. It is faster and quieter, but it can still fail, often with less warning.
- **S.M.A.R.T.** stands for **Self-Monitoring, Analysis and Reporting Technology** â€” a built-in system where the drive continuously measures its own health and reports it.

Why it exists.
Drive makers realised that most drives die slowly. S.M.A.R.T. was created so that a technician's tool could read these self-measurements and warn the owner *before* total failure â€” giving time to back up.

What it reports (the important parts).

- An overall **Health Status**: **Good (blue)**, **Caution (yellow)** or **Bad (red)**.
- **Temperature** â€” a drive running too hot may need better airflow or cleaning (see F.6).
- **Reallocated sector count** â€” sectors the drive has quietly set aside because they became unreadable. **Any number above zero is a warning sign**; a rising number means the drive is dying.
- **Power-on hours** â€” how long the drive has been running in its life, useful context.

Analogy.
S.M.A.R.T. is like the oil-warning light in a car. It does not fix the engine, but it tells you "the engine is not happy" early enough for you to act. Ignore it, and you get a broken-down car and a big bill.

Tool: CrystalDiskInfo.

1. Open CrystalDiskInfo.
2. Read the **Health Status** word and colour at the top.

5. If it says **Good** â€" continue normal checks. If **Caution** â€" **back up the customer's data immediately** and recommend replacement. If **Bad** â€" stop writing to the drive, back up if it still reads, and replace it.

Symptoms a storage drive is failing:

- The computer became very slow, especially at start-up or when opening files.
- Clicking, ticking or grinding noises from the drive (HDD only).
- The drive is **not detected** by the BIOS or by Windows.
- "No boot device" or "operating system not found" errors.
- Files and folders disappearing or becoming corrupted.

Clicking noise â€" what to do.
A repetitive **clicking** sound from an HDD almost always means the head is failing to read the platter. **Do not keep powering it on and off trying to make it work** â€" each attempt can scrape more data away. If the data is important, power it down and take it to a **data recovery** specialist. Explain to the customer, honestly, that data recovery is not something a general repair technician should attempt casually, and that it can be expensive.

Important misconception.
"SSDs do not fail." They do fail. They are more shock-resistant than HDDs, but they can still stop working, often suddenly. Backups matter for both.

Limitation of CrystalDiskInfo.
A drive can show "Good" and still fail suddenly (especially some SSDs). S.M.A.R.T. is a warning tool, not a guarantee. The only true protection is a **backup**. Say this to every customer: "*Health tools reduce risk; only a backup removes it.*"

F.5 Diagnosing power supply (PSU) faults

The idea.
The **PSU (Power Supply Unit)** converts the wall electricity (240 V in Nigeria) into the low voltages (12 V, 5 V, 3.3 V) the components use. If it is weak or dead, the whole computer is affected.

Plain-English definition.
The **PSU** is the metal box, usually at the top or bottom-back of the case, that the power cable plugs into. A **PSU fault** means it is not delivering correct, stable power.

Why it exists.
Components cannot use 240 V directly â€" they would be destroyed instantly. The PSU safely steps the voltage down and steadies it.

Common symptoms of a PSU fault:

- The computer does not power on at all (no lights, no fans).
- The computer powers on for a second then dies.
- **Random shutdowns** or restarts, especially when the computer is working hard.
- Burning smell, or scorch marks near the PSU.
- The fans twitch but the machine will not start.

Analogy â€" and this analogy is a warning.
The PSU is like the transformer that feeds a whole street. If your mains voltage is unstable, every appliance in the house behaves strangely â€" flickering lights, a fridge that hums then dies. The PSU is the same for the computer. But unlike a street transformer, you must **never** open the PSU to check inside.

> ***SAFETY â€" READ ALOUD AND POST ON THE WALL.***

general technician â€“ it is a ***replace-only*** *item.*

Basic PSU checks a beginner may perform (all with the computer unplugged unless stated):

1. **Visual check (unplugged):** look for bulging capacitor tops, burn marks, melted plastic or a strong burnt smell. A bulging or burnt PSU means replace it, full stop.
2. **Smell test:** a sharp burning smell around the PSU = failed, replace it.
3. **Connection check (unplugged):** make sure the big 24-pin motherboard cable and the CPU power cable are fully seated. A loose connector causes no-power symptoms.
4. **Socket check:** confirm the wall socket actually has power â€” test it with another appliance or a known-good lamp. This sounds obvious, but it is the most common "no power" cause.
5. **Swap test (the safest and most reliable):** replace the suspect PSU with a **known-good PSU** of suitable wattage. If the computer then works, the original PSU was the fault.

The paperclip test â€” advanced and optional.
There is a way to force a PSU on by bridging two pins on its main connector with a bent paperclip, to see if the fan spins. This test:

- Should only ever be done by the **instructor**, on a bench, as a demonstration; it is **not** a learner task.
- Requires care because you are working with a live PSU once it starts.
- Proves only that the PSU powers up; it does **not** prove the voltages are within spec.

Because of the danger and its limited usefulness,

this course uses the swap test as the standard method.
If the academy has a **PSU tester** (a small device that plugs onto the main connector and shows the voltage readings), the instructor may demonstrate it â€” it is safer and more informative than the paperclip method.

Important misconception.
"A bigger wattage PSU always fixes random shutdowns." Wattage is a *capacity*, not a *health measure*. A failing 700 W PSU is still failing. Also, choosing the correct wattage matters (Session 7 covers this).

Limitation.
A PSU can fail in subtle ways that basic checks miss â€” delivering slightly wrong voltages that cause random faults. When symptoms strongly suggest PSU and the swap test confirms it, replace it; do not try to "tune" it.

F.6 Diagnosing overheating

The idea.
Every chip produces heat when it works. If that heat cannot escape, performance drops and, eventually, parts are damaged.

Plain-English definition.
Overheating means a component is running hotter than it should. The usual cause is dust, a failed fan, dried-out thermal paste, or blocked airflow.

Why it matters.
Modern CPUs and GPUs can slow themselves down on purpose when hot (called **thermal throttling**) â€” that is why a dusty computer *feels slow*. If it gets hotter still, the computer shuts itself off to protect the part (a **thermal shutdown**).

Analogy.
Imagine wearing a thick coat (northern-style) in the Port Harcourt heat while doing hard labour. You cannot work well, and if you keep going you collapse. The computer "collapses" by shutting down.

Tool: HWMonitor.
This free tool displays **temperatures**, **fan speeds** and **voltages** for the sensors it can find on the machine.

- **Normal ranges (rough guide):**

  - **Hard drive:** roughly 25â€“45 Â°C.
  - **Motherboard:** roughly 30â€“50 Â°C.
- These are guides, not exact rules; different models differ. What matters is: a reading well above the guide, or a big gap between "before" and "after" cleaning, tells you overheating is happening. If a value keeps climbing toward 90 Â°C or above, treat it as a fault.

Common symptoms of overheating:

- Random shutdowns after some minutes of use.
- Fans running loudly all the time.
- The case or the base of a laptop feeling very hot.
- Slowness that gets worse the longer the computer runs.

Basic checks (with the computer off and unplugged):

1. **Dust:** open the case and look at heatsinks and fan blades. Thick dust "felt" between the fins blocks airflow (the "dust blanket").
2. **Fans:** check that every fan spins freely when the machine is running and is not grinding. A stopped CPU fan is serious â€” the computer may run for a minute then shut down.
3. **Thermal paste:** the grey paste between the CPU and its heatsink can dry out over years and lose its ability to transfer heat. Re-applying it (Session 8) restores cooling.
4. **Airflow:** confirm intake and exhaust vents are not blocked by a wall, a curtain, or a laptop sitting on a soft bed.

Important misconception.
"If the fan is loud, the fan is broken." Often the fan is loud because it is working hard to fight overheating â€” the *cause* is dust, not the fan. Clean first, then judge the fan.

Limitation.
Not every machine exposes every temperature sensor, and some older machines report only a few values. Absence of a reading is not proof that temperature is fine. Where sensors are missing, rely on symptoms (shutdowns, hot case) and physical inspection.

F.7 Diagnosing display (no signal) faults

The idea.
"No signal" or a black screen does not mean the computer is dead. It means the picture is not reaching the screen. There are several possible breaks between the computer and your eyes.

Plain-English definition.
A **display fault** is anything that stops a correct picture appearing: the cable, the port, the GPU, the monitor, or the computer not finishing POST.

Analogy.
Think of a DSTV or GOtv setup. If the TV shows "no signal", you do not immediately buy a new TV. You check the cable, the decoder, the dish. The same order of thinking applies to a computer screen.

Order of checks (cheapest and simplest first):

1. **Is the monitor on, and on the right input?** Many monitors have several inputs (HDMI1, HDMI2, VGA). The input may have been switched by mistake. Check the monitor's menu.
2. **Cable:** unplug and firmly reseat the display cable at **both** ends. Try a **known-good spare cable**. A damaged or loose cable is a very common "no signal" cause.
3. **Try a different port:** if the computer has both a motherboard video output and a GPU output, try the other. On a desktop with a dedicated GPU fitted, the monitor should be connected to the **GPU**, not the motherboard â€” connecting to the wrong one can give no signal.

5. **Swap the monitor:** try a **known-good monitor**. If the picture appears, the original monitor was faulty.
6. **Swap the GPU:** on a desktop, fit a **known-good GPU** (or use the motherboard output) to test. If the picture appears, the original GPU was faulty.

Common symptoms and what they suggest:

- "No signal" but the PC fans spin and lights come on â†' cable, monitor, GPU or POST failure.
- Picture appears only after wiggling the cable â†' damaged cable or port.
- Screen shows strange colours or lines â†' often the monitor or cable (sometimes the GPU).
- Laptop screen dark but an external monitor works â†' the laptop's screen or its ribbon cable, not the GPU.

Important misconception.
"Black screen = dead computer." Very often the computer is running fine and only the display path is broken. Also, on a laptop, a very dim screen may just be backlight or brightness â€" shine a torch at an angle to see if a faint image exists; that points to backlight rather than a dead machine.

Limitation.
Diagnosing laptop screens in detail (screen replacement, ribbon cables) is beyond this session; here we focus on isolating *whether* it is the screen and passing that finding on.

F.8 Motherboard and laptop battery issues

The idea.
The motherboard is the main circuit board everything plugs into. The laptop battery stores power so the laptop can run without the charger.

Motherboard â€" plain-English definition and symptoms.
The **motherboard** is the large board that carries the CPU socket, RAM slots, expansion slots and connectors. Motherboard faults are usually the hardest to diagnose because the board controls so much.

- Symptoms: no POST at all, no beeps, no display, no power lights; sometimes power comes on for one second and dies; sometimes some ports dead while others work.
- Common causes: a failed capacitor, physical damage, a power surge, or liquid spillage.
- **Beginner rule:** because the motherboard interacts with everything else, you must first rule out the *other* parts (RAM, PSU, GPU, display) using swap tests. Only when every other part is known-good do you conclude "the motherboard is faulty". Motherboard-level repair (soldering, chip replacement) is **advanced work** and outside the scope of this certificate â€" the honest action is to replace the board or advise the customer.

Laptop battery â€" plain-English definition and symptoms.
The **battery** is the rechargeable pack that powers a laptop when it is not plugged in. A worn battery is not a "fault" in the same way, but it causes problems:

- Symptoms: laptop only works when plugged in; battery percentage jumps; the laptop shuts down suddenly; the battery swells (bulges) and may push the trackpad out.
- **Safety warning:** a **swollen battery** is a fire risk. Stop using the laptop immediately, do not press on it, do not puncture it, and advise the customer to have it replaced by a technician. Do not attempt to force a swollen battery out of a laptop without proper care and supervision.
- Diagnostic note: a "not charging" laptop might be the battery, the charger cable, or the charging port. Try a **known-good charger** first (cheapest test), then consider the battery, then the port.

Important misconception.
"A swollen battery just looks a bit fat, it is still fine." It is not fine. Swelling means the cells are failing and can catch fire. Treat it as a priority safety issue.

Limitation.
Battery health detail (cycle count, capacity percentage) can be read with some built-in laptop tools, but tool names and menus differ by make. In this course we rely on the clear physical signs (only works on charger, swollen pack) rather than model-

F.9 Swap testing and isolation â€” the professional method

The idea.
The fastest, safest way to prove which part is faulty is to **change one thing at a time**.

Plain-English definition.
Swap testing means replacing a suspect part with a **known-good** part to see if the fault disappears. **Isolation** means removing or disconnecting parts to narrow down where the fault is.

Why it exists.
Guessing wastes money. If a customer is told "buy a new GPU" and the real fault was a â‚¬500 cable, you lose the customer's trust. Swap testing turns a guess into a **confirmed diagnosis**.

Rules of swap testing:

1. **Change only one thing at a time.** If you change two parts and it works, you do not know which one fixed it.
2. **Use a known-good part** â€” one you are sure works. If your "test" part is also faulty, you will reach a wrong conclusion.
3. **Power off and unplug** before swapping internal parts, and wear your anti-static wrist strap.
4. **Record each step** on the Diagnostic Report Form: what you changed, and what happened.
5. **Return the machine to its original state** if the swap does not fix it, before trying the next test.

Analogy.
A doctor testing for malaria does not treat for five diseases at once. She tests one thing, reads the result, then decides the next test. Swap testing is the same discipline.

Important misconception.
"I do not need to use a known-good part; any spare will do." No. Your conclusion is only as good as your test part. Always use a part you trust.

Limitation.
Some parts are hard to swap quickly (a soldered laptop battery, an integrated screen). For those, isolation (disconnecting, checking connectors, using external equivalents) replaces swapping.

F.10 Summary of the session

- Every fault is **hardware** or **software**; today we prove hardware faults.
- **POST** runs first; **beep codes** (count, then look up) point to basic failures.
- **Faulty RAM** â†’ boot loops, beeps, blue screens, crashes. Test with **Windows Memory Diagnostic** or **MemTest86**; isolate one stick at a time.
- **Failing storage** â†’ slowness, clicks, not detected. Read **S.M.A.R.T.** in **CrystalDiskInfo** (Good / Caution / Bad). Back up on "Caution".
- **PSU faults** â†’ no power, random shutdowns, burning smell. **Never open a PSU.** Use the **swap test**.
- **Overheating** â†’ shutdowns, loud fans, slowness. Check dust, fans, paste, airflow and temperatures in **HWMonitor**.
- **Display faults** â†’ check monitor, cable, port, GPU, POST, in that order.
- **Motherboard / laptop battery** â€” rule out everything else first; swollen batteries are a fire risk.
- **Swap testing + isolation** turns guesses into confirmed diagnoses.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- One practice desktop, opened on the bench, with an anti-static wrist strap.
- One known-good RAM stick and one stick designated "suspect" (the class is not told which).

- A known-good monitor and a spare display cable.
- The Diagnostic Report Form on the projector.

G.2 Demonstration 1 â€” Reading drive health with CrystalDiskInfo (4 minutes)

Say:
"Watch me ask this drive a question and read its answer."

Steps:

1. Open **CrystalDiskInfo**. Point to the top line: the **Health Status** word and colour.
2. Read the **Temperature**. Say: "38 Â°C â€” that is cool and normal."
3. Scroll to **Reallocated Sectors Count**. Say: "It is zero. That is what we want â€” zero means the drive has not yet set aside any bad spots."
4. Now open CrystalDiskInfo on the **training drive** that shows wear. Point at **Caution (yellow)** and the non-zero reallocated count.
5. **Say:** "Caution means this drive is telling us it is getting sick. Our action is simple and urgent: **back up the data today**, and tell the customer to plan for a replacement."

Expected results:
"Good" (blue) on the healthy drive, "Caution" (yellow) with a non-zero reallocated sector count on the training drive.

Reasoning to say aloud:
"I am not guessing. The drive measured itself and I am reading the measurement. That is the whole philosophy of diagnostics."

Common errors to show and correct:

- Reading only the word and ignoring the attribute list. Show that Caution can appear even if the summary word is slow to update.
- Confusing a warm drive with a failing drive â€” temperature and health are different columns.

Questions to ask during the demo:

- "If the status is Caution, what do we do *first* â€” back up or replace?" (Back up first, so we do not lose data.)
- "Is a drive with zero reallocated sectors guaranteed to live forever?" (No â€” that is why we back up.)

G.3 Demonstration 2 â€” Temperature check with HWMonitor (3 minutes)

Say:
"Now the health of the cooling system."

Steps:

1. Open **HWMonitor**. Point to the **CPU** temperature and the **fan speed** rows.
2. Say: "CPU at 45 Â°C idle â€” normal for these machines."
3. Point to fan speeds. Say: "These are the CPU and case fan speeds. If a fan shows zero while the machine is on, that fan is not spinning â€” investigate."
4. Optional: gently run a heavy task (or just say what *would* happen) to show the CPU number rising under load.

Expected results:
a CPU reading within the normal idle guide, and fan speeds above zero.

Questions to ask:
"If the CPU sits at 85 Â°C while doing nothing, what are the first two things to check?" (Dust and fan rotation.)

G.4 Demonstration 3 â€” RAM test and swap isolation (5 minutes)

Say:
"The hardest fault to catch is the one that only happens sometimes. Memory is famous for this."

2. Put in one **suspect** stick. Say: "Let us say the customer complains of random restarts."
3. Start **Windows Memory Diagnostic** (or insert the MemTest86 USB and begin booting it). Explain that it runs outside Windows so it can test all of memory.
4. While it runs (or while describing the result), reveal the known-faulty stick result.
5. Perform the **isolation**: power off, unplug, strap on, remove the suspect stick, fit the known-good stick, power on.
6. **Say:** "The fault is gone. We have not guessed â€" we have **proven** the original stick was bad by swapping in a known-good one."

Expected results:
the memory test reports errors on the faulty stick; the machine behaves normally with the known-good stick.

Reasoning to say aloud:
"One change at a time. Known-good part. Record every step. That is how we avoid blaming the wrong component."

Common errors to show and correct:

- Forcing the stick in the wrong way (show the **notch** alignment).
- Forgetting to unplug and to wear the wrist strap.
- Concluding "RAM is bad" from a single blue screen without any test.

Questions to ask:

- "Why do we run the test outside Windows?" (So Windows is not using the memory we are testing.)
- "If there were two sticks and only one were bad, how would we find it?" (Test/isolation one at a time.)

G.5 Demonstration 4 â€" No-signal display isolation (3 minutes)

Say:
"Black screen â€" do not panic, and do not buy a monitor yet."

Steps:

1. With the machine running and the monitor showing "No Signal", first check the monitor's **input menu**.
2. **Reseat** the display cable at both ends. Still no signal.
3. Swap in the **known-good spare cable**. Picture appears.
4. **Say:** "We found it: a bad cable. Cost to the customer: the price of a cable, not a monitor or a GPU. This is why we test from cheapest to most expensive."

Expected results:
picture returns after the cable swap.

Questions to ask:
"Why do we try the cheap cable before swapping the expensive GPU?" (Cost, and the cheapest cause is very often the real one.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective:
With the instructor guiding, each learner (or pair) runs the hardware diagnostic tools on a practice machine and records findings on a Diagnostic Report Form.

Safety first â€" read aloud before starting:

- **Switch off and unplug** the computer before opening the case or touching any internal part.
- Wear your **anti-static wrist strap** whenever you handle internal components.
- **Never open the PSU.**
- Keep drinks away from the bench.

Procedure (instructor calls out each step; learners follow; instructor walks the room):

1. On the **running** machine, open **CrystalDiskInfo**. Record: Health Status word and colour; drive temperature; and the **Reallocated Sectors Count** value.
2. Open **HWMonitor**. Record the **CPU temperature**, the **GPU temperature** (if listed) and the **fan speeds**. Note any fan showing zero.
3. **Pause and predict:** "Before we start a memory test, what would you expect to see if this stick is healthy?" (Zero errors, machine passes.) Write your prediction before running.
4. Start **Windows Memory Diagnostic** (or insert the MemTest86 USB and begin the boot). If time is short, observe the first pass and note the running errors count. Do **not** force the machine to keep testing if it is needed for the next group â€” coordinate with the instructor.
5. **Display isolation drill:** with the machine running, deliberately loosen the display cable at one end (or start from a "no signal" setup) and follow the order: check input â†’ reseat cable â†’ known-good cable â†’ known-good monitor. Record which step fixed it.
6. **Swap-test drill (instructor supplies parts):** power off, unplug, strap on. Remove the suspect RAM stick and fit the known-good stick. Note the difference in behaviour (or in the memory-test result).
7. **Overheating inspection:** with the machine off and unplugged, open the case and inspect the CPU heatsink, fans and vents for dust. Record what you see (light dust / heavy dust / clear).
8. Complete the **Diagnostic Report Form**, writing each finding in plain language a customer could understand.

Expected results:

- CrystalDiskInfo shows the drive health word and a temperature.
- HWMonitor shows at least one CPU temperature and fan speeds.
- The memory test reports a pass (or, on the faulty stick, an error count).
- The display isolation drill identifies the specific fault (input/cable/monitor/GPU).
- The swap test changes the machine's behaviour in the expected direction.
- A completed report form.

Points to pause on and predict:

- After step 1: "If the status is Caution but the count is zero, what could still be wrong?" (Temperature, or a different flagged attribute.)
- After step 5: "We changed two things â€” which one fixed it?" (Correct answer: we must change one at a time to *know*.)
- After step 6: "If the machine still fails with the known-good stick, what does that tell us?" (The RAM was probably not the fault â€” look elsewhere.)

Troubleshooting while walking the room:

- CrystalDiskInfo shows no drives â†’ the tool may need to be run as administrator; also confirm the drive cable is connected.
- HWMonitor shows very few values â†’ this machine has few sensors; record what is shown and say so.
- Windows Memory Diagnostic will not restart â†’ save work and make sure no other user is logged in.
- MemTest86 USB does not boot â†’ confirm the boot key and that "USB boot" is enabled; try another USB port.
- Learner forces RAM â†’ stop immediately, show the notch, re-seat gently.

Completion checklist (learner ticks each):

- ☐ CrystalDiskInfo opened; health status, temperature and reallocated count recorded
- ☐ HWMonitor opened; temperatures and fan speeds recorded
- ☐ A memory test started and its result noted
- ☐ Display isolation drill completed and the cause identified
- ☐ Swap test completed (one change at a time, known-good part)
- ☐ Overheating inspection done (dust, fans, vents noted)
- ☐ Diagnostic Report Form completed in plain language

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
Diagnose three **simulated hardware faults** without copying the instructor. For each one, use the correct tool, decide whether the component is healthy or faulty, and write your conclusion in plain language.

Scenario cards (each learner or pair receives these; the instructor can set up each as a bench station).

Card A â€" "The Beeping Cyber CafÃ© PC".
A desktop will not display a picture. When switched on, it beeps three times, then pauses, then beeps three more times. All fans spin. The drive and PSU are known-good. There are two RAM sticks fitted.

Card B â€" "The Slow Office Computer".
A desktop started fine but now boots to Windows very slowly and sometimes says "operating system not found". CrystalDiskInfo is available. The machine's drive makes a soft ticking sound occasionally.

Card C â€" "The Machine That Dies".
A desktop runs for about 20 minutes, then suddenly switches off with no warning. It restarts when you press the power button, then dies again after some minutes. Fans spin during use. CrystalDiskInfo and HWMonitor are available.

Requirements:

1. For each card, state the **most likely faulty component**.
2. State **which tool or test** you used (or would use) to confirm it.
3. State the **next safe action** for a technician (e.g., back up, clean, replace).
4. Write each answer as a short paragraph in plain English.

Time allowance:
15 minutes.

Constraints:

- Do not open the PSU.
- Use a **known-good part** for any swap test.
- Record the finding on the Diagnostic Report Form.

Success criteria:

- Each card is matched to the correct component.
- The confirmation test chosen makes sense.
- The recommended action is safe and honest.

Hints available if stuck:

- Hint 1 (Card A): repeated beeps before any picture usually point to one basic component that the machine needs before it can show anything.
- Hint 2 (Card B): ticking plus "operating system not found" points to a mechanical component that stores data.
- Hint 3 (Card C): a machine that dies after a set time, then dies again, is the classic sign of a heat or power problem.

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

Card A â€" Most likely faulty component: RAM.
Three repeating beeps during POST usually indicate a memory fault; all fans spinning means the PSU is probably fine and the machine is trying to start.

- Confirmation test: **MemTest86** (or Windows Memory Diagnostic), and **isolation** by removing one stick at a time, testing each alone.

Card B â€" Most likely faulty component: the storage drive (HDD) is failing.
Slow boot, "operating system not found" and ticking all point to a dying mechanical drive.

- Confirmation test: **CrystalDiskInfo** â€" look for **Caution** and a **non-zero reallocated sector count**; note the ticking (mechanical failure).
- Next safe action: **back up the customer's data immediately** if the drive still reads; then replace the drive (an SSD upgrade is ideal â€" see Session 7) and reinstall/the-clone the system. Warn the customer that a ticking drive may fail fully at any moment.

Card C â€" Most likely faulty components: overheating (dust/dried paste) or a weakening PSU.
Dying after a fixed time is the signature of heat buildup; random shutdowns can also be a PSU problem.

- Confirmation test: **HWMonitor** for temperatures under load; inspect **dust, fan rotation and thermal paste**; then a **PSU swap test** with a known-good PSU.
- Next safe action: clean the dust and check fans/paste first (free and safe); if it still dies, perform the PSU swap test and replace the PSU if confirmed. Never open the PSU.

Marking guidance (out of 12):

- Correct component identified for each card (3 Ã— 2 = 6)
- Sensible confirmation test chosen for each (3 Ã— 1 = 3)
- Safe, honest next action for each (3 Ã— 1 = 3)
- Plain-language writing and completed report form (bonus up to +1, cap at 12)

Accept any answer that shows correct testing logic
, even if the wording differs. Penalise unsafe actions (such as opening the PSU or "replace everything") heavily.

J. REAL-WORLD CASE STUDY

Situation.
Emeka owns a small business centre along Ikwerre Road, Port Harcourt. His main office computer holds his customer invoices, his accounting records and the software he uses for printing. For two weeks it has been slow and sometimes restarts on its own. He ignores it â€" "it is still working." One Friday evening, the machine refuses to boot: "Boot device not found." He calls a technician.

Relevant facts.

- The computer is seven years old and still has its original mechanical hard drive.
- It had begun to make a soft ticking noise some weeks earlier.
- Nobody had ever run a health check on the drive.
- No backup exists.

The decision point.
The technician opens CrystalDiskInfo. The drive shows **Bad (red)**, with a high reallocated sector count and repeated read errors. The shop's invoice and accounting data were only on this one drive.

Possible approaches.

1. Keep trying to boot the drive repeatedly to copy files.
2. Power the failing drive down, back up what can still be read with care, then replace the drive.
3. Send the drive for data recovery immediately.

Recommended solution (for this stage).
Approach 2, with honest limits. The technician explains that a drive in this state may copy some files but is at high risk of dying completely at any moment, and each attempt shortens its life. He backs up the most critical data first (invoices and accounts), then attempts the rest. He advises Emeka that if any important data cannot be copied, only a professional **data recovery** service should open the drive â€" not the shop technician.

- Approach 2 preserves the best chance of saving data and is affordable; but some files may already be lost.
- Approach 3 is expensive and unnecessary if the important files are still readable â€” it should be the last resort.

Professional connection.
Real repair shops earn trust by doing exactly this: reading S.M.A.R.T. health, being honest that a failing drive must be replaced, and telling the customer to **back up**. The course certificate says a graduate can "diagnose common hardware faults" â€” this is precisely that skill, plus the professional honesty that goes with it.

Transfer exercise (ask the class):
"If a customer's drive shows Caution, and the customer says *'I will bring it back next month,'* what should the technician do or say?"

Suggested answer: explain clearly, in writing, that Caution means the drive is already showing wear, that it can fail before next month, and that the data should be backed up **now**. Record the advice on the service sheet so the customer cannot later say they were not warned.

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the session's live troubleshooting guide. When a learner hits a problem, ask them to check the relevant row first.

K.1 Black screen / "No Signal" but the machine seems to run

- **Symptoms:** fans spin, keyboard lights come on, monitor shows "No Signal".
- **Likely causes:** monitor on the wrong input; loose or damaged display cable; monitor connected to the motherboard instead of the GPU; a POST fault (RAM/GPU); or a dead monitor.
- **How to investigate:** check the monitor input menu; reseat the cable at both ends; try a known-good cable; try the other video output; check for beeps; try a known-good monitor.
- **Safe corrective steps:** work **from cheapest to most expensive** â€” input, cable, port, then monitor/GPU. Power off before changing cables.
- **How to confirm resolved:** a picture appears and stays stable.
- **Prevention:** keep a spare known-good cable and monitor on the bench.

K.2 The computer does not power on at all

- **Symptoms:** no lights, no fans, nothing.
- **Likely causes:** wall socket dead; power cable loose or faulty; PSU switch (on the back) off; PSU failed.
- **How to investigate:** test the socket with another appliance; check the power cable and the PSU switch; then try the swap test with a known-good PSU.
- **Safe corrective steps:** replace the power cable, use a working socket, swap the PSU. **Never open the PSU.**
- **How to confirm resolved:** the machine powers on and completes POST.
- **Prevention:** use a surge protector (Session 8); label good cables.

K.3 Random shutdowns and restarts

- **Symptoms:** works, then dies; sometimes restarts by itself.
- **Likely causes:** overheating (dust, dead fan, dried paste); weak PSU; loose power connectors; faulty RAM.
- **How to investigate:** check temperatures in HWMonitor; inspect dust and fans; reseat the 24-pin and CPU power connectors; run a memory test; do a PSU swap test.
- **Safe corrective steps:** clean cooling first (free); re-seat connectors; swap PSU. Power off and unplug for all internal checks.
- **How to confirm resolved:** the machine runs a heavy task (or a long memory test) without dying.
- **Prevention:** regular cleaning (Session 8), surge protection, correct PSU wattage.

- **Likely causes:** a faulty RAM stick; a dirty or faulty RAM slot; RAM not fully seated; incompatible/mismatched RAM.
- **How to investigate:** reseat the stick; clean the slot with compressed air; test one stick at a time; test each stick in each slot.
- **Safe corrective steps:** remove the faulty stick; if the slot itself is bad, use a different slot; replace the faulty stick.
- **How to confirm resolved:** re-run the test and get zero errors.
- **Prevention:** hold by the edges, never force a stick, power off first.

K.5 CrystalDiskInfo shows Caution or Bad

- **Symptoms:** yellow or red health status; non-zero reallocated sectors; read errors.
- **Likely causes:** the drive is wearing out or already failing.
- **How to investigate:** read the full attribute list; listen for clicks; note the temperature.
- **Safe corrective steps: back up the customer's data immediately**; stop writing new data; replace the drive. For clicks or unreadable important data, hand over to a data recovery specialist.
- **How to confirm resolved:** the new drive shows Good and the system runs normally.
- **Prevention:** regular health checks (Session 8) and a backup habit.

K.6 Distinguishing likely from confirmed

In every row above, a *likely* cause becomes a *confirmed* cause only after a test proves it: a swap test that changes the behaviour, a S.M.A.R.T. reading, a temperature reading, or a memory-test error count. **Teach learners never to write "confirmed" on a report based on a guess.** Saying "likely cause" honestly is professional; overclaiming is not.

L. SESSION ASSIGNMENT

Assignment title:
"Hardware Diagnostic Report on a Real Machine"

Purpose:
To practise the full hardware diagnostic process â€” inspecting, testing with tools, isolating with swap tests, and writing findings in plain language â€” on a real computer.

Realistic scenario:
A customer brings you a desktop computer and says: *"It is slow, sometimes it restarts on its own, and last week the screen went black once."* You must produce a professional diagnostic report before any repair is done.

Detailed instructions:

1. Choose a real computer you have access to (a practice machine at CEA, a family computer, or a machine at your work). **Get permission before opening any machine that is not yours.**
2. **Observe and record:** note the make/model if visible, the symptoms, and anything the "customer" told you.
3. **Run the software checks** (safe, no case opening needed): CrystalDiskInfo (health, temperature, reallocated sectors) and HWMonitor (CPU/GPU temperatures, fan speeds).
4. **Run a memory test** (Windows Memory Diagnostic or MemTest86) and record the result.
5. **Physical inspection** (only if you may open the machine): with it **off and unplugged**, note dust levels, fan condition, and whether the case has good airflow. **Do not open the PSU.**
6. **Write your conclusion:** state the most likely faulty component (or that all tested parts look healthy), and your recommended next action.
7. Complete the **Diagnostic Report Form** and hand it in.

Required deliverables:

- One completed Diagnostic Report Form (the CEA template), including all readings.
- A one-paragraph plain-language summary for the "customer".

Deadline:
Before Session 6.

Marking rubric (20 marks):

Criterion	Marks	Description of a strong answer
Observation & symptom record	3	Clear, specific symptoms and customer description
CrystalDiskInfo findings	3	Health status, temperature and reallocated count all recorded
HWMonitor findings	3	CPU (and GPU if available) temperatures and fan speeds recorded
Memory test result	3	Test named, run, and result clearly stated
Physical inspection (if permitted)	2	Dust, fans and airflow noted honestly
Conclusion & recommendation	4	Logical, safe, honest conclusion with a sensible next action
Plain-language summary	2	A customer could understand it; no unexplained jargon

Common mistakes to avoid:
writing "confirmed" without a test; confusing temperature with health; recommending "replace everything"; opening the PSU; forgetting to record the actual numbers.

Instructor reference solution (abbreviated example):

> ***Machine:***
> *CEA practice desktop (7th-gen CPU, 4 GB RAM, 500 GB HDD).* ***Symptoms:*** *slow boot; one random restart last week; one black screen.* ***CrystalDiskInfo:*** *Health* ***Caution (yellow)****; temperature 41 Â°C; Reallocated Sectors Count* ***18*** *(and rising).* ***HWMonitor:*** *CPU 52 Â°C idle; CPU fan 1,250 RPM; one case fan reads 0 RPM.* ***Memory test:*** *Windows Memory Diagnostic â€" **0 errors** (passes).* ***Physical inspection:*** *heavy dust between CPU heatsink fins; one case fan not turning.* ***Conclusion â€" likely cause:*** *the* ***hard drive is wearing out*** *(Caution, non-zero reallocated sectors) and the* ***cooling is compromised*** *(dust + one dead case fan).* ***Recommendation:*** *back up the customer's data* ***today****; replace the hard drive (an SSD will also make it much faster); clean the dust and replace/repair the stopped case fan; re-test temperatures after cleaning.* ***Plain-language summary:*** *"Your computer's storage is showing early signs of failure, and dust is making it run hot. Please let us back up your files now, then we will replace the storage and clean the cooling."*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key below.)

Q1. Short answer.
What does **POST** stand for, and why does it use beeps instead of the screen? **Q2. Multiple choice.** A drive in CrystalDiskInfo shows **Caution (yellow)** with a non-zero reallocated sector count. What is the correct *first* action? a) Replace the drive immediately without copying anything. b) Back up the customer's data immediately. c) Ignore it â€" Caution is normal. d) Open the drive to look inside. **Q3. True or false, with a reason.** "A PSU can be safely opened to check the capacitors as long as it is unplugged." **Q4. Practical identification.** A desktop has two RAM sticks, shows random restarts and produces repeated beeps at start-up. Describe, step by step, how you would find the faulty stick. **Q5. Scenario.** A laptop shuts down after about 30 minutes every time. Give **two** likely causes and the test you would use for each. **Q6. Explain in your own words.** Why do we change **only one thing at a time** during swap testing?

INSTRUCTOR ANSWER KEY (Q1â€"Q6)

screen. If a basic part fails, a beep pattern (or a debug light) tells the technician where to look. (The beep pattern depends on the motherboard maker, so we count and look it up.)

A2.
(b) Back up the customer's data immediately. Caution means the drive is already showing wear and could fail soon. Data first, then replace the drive.

A3.
False. Even unplugged, the large capacitors inside a PSU can hold a **lethal charge** for a long time. A PSU is replace-only for a general technician.

A4.
Step-by-step: power off and unplug; wear the anti-static strap. Remove one stick, leaving only stick A; run MemTest86 (or Windows Memory Diagnostic); note errors. If errors appear, stick A is bad. If not, power off and test stick B alone. Alternatively, swap in a **known-good** stick; if the faults disappear, the original stick was bad. Test one stick (one change) at a time and record each result. (Accept any answer that follows isolation + known-good swapping.)

A5.
Likely causes: **overheating** (dust, failed fan, dried thermal paste) and a **weakening PSU**. Test for overheating with **HWMonitor** while the machine runs (watch the temperature climb before shutdown) and by inspecting dust/fans/paste. Test for the PSU with a **swap test** using a known-good PSU of suitable wattage. (Also accept "faulty RAM" if supported by a memory test.) Never open the PSU.

A6.
Because if you change two things at once and the fault disappears, you cannot tell **which** change fixed it. One change at a time gives a **confirmed** answer, saves money, and produces an honest report.

Marking:
award for correct concept even if wording differs. Deduct for unsafe answers (opening the PSU, "replace everything").

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned to prove hardware faults instead of guessing them. You saw how a computer checks itself at start-up (**POST**) and tells you via **beeps** when a basic part fails. You learned the symptoms and tests for faulty **RAM** (boot loops, blue screens â€” tested with Windows Memory Diagnostic or MemTest86), failing **storage** (slowness, clicks, "not found" â€” read with **CrystalDiskInfo** using **S.M.A.R.T.**), **PSU** faults (no power, random shutdowns â€” **swap test only**, never open it), **overheating** (shutdowns, loud fans â€” **HWMonitor**, dust, fans, paste), **display** faults (no signal â€ check input, cable, port, monitor), and **motherboard / laptop battery** issues (rule out everything else; swollen batteries are a fire risk). Above all, you learned **swap testing and isolation**: change one thing at a time, use a known-good part, and record every step.

Five important takeaways:

1. Test, do not guess â€ a diagnosis you cannot prove is not finished.
2. **Never open the PSU**; capacitors can hold a lethal charge.
3. **Caution** in CrystalDiskInfo means back up **now**, then replace.
4. Change **one thing at a time** with a **known-good** part.
5. Always **back up** before replacing storage, and always **record** your findings.

Questions to help you recognise what you can now do:

- Can I read a drive's health and decide the next action?
- Can I find a faulty RAM stick using isolation?
- Can I explain why we never open a PSU?
- Can I check temperatures and spot overheating?
- Can I find a "no signal" fault from cheapest to most expensive cause?

- ☐ I wore my anti-static wrist strap when handling parts.
- ☐ I never opened a PSU.
- ☐ I backed up before touching storage data.
- ☐ I stopped and called the instructor when unsure.

Connection to the next session.
Today you proved faults in the **hardware**. But many "broken" computers have perfectly healthy parts â€” the problem is the **software**. Next session, Session 6 â€” Software Diagnostics, you will diagnose slow performance, viruses, boot problems and driver faults using Windows' own tools, and learn how to tell a software fault from a hardware one.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner checks **cheapest/simplest causes first** (socket, cable, monitor input) before opening the case.
- The learner reads the **actual number** in CrystalDiskInfo (reallocated count, temperature) rather than only the colour.
- The learner uses a **known-good** part for a swap test and changes **one thing at a time**.
- The learner states findings in plain language and distinguishes "likely" from "confirmed".
- The learner automatically powers off, unplugs, and wears the wrist strap.
- The learner refuses unsafe actions (opening the PSU, poking a swollen battery).

Common misconceptions to watch for and correct:

- "Black screen means the computer is dead." â†’ Often just the display path or POST.
- "Caution is fine." â†’ Caution means back up now.
- "A loud fan is a broken fan." â†’ Usually it is fighting overheating; fix the dust/heat first.
- "A bigger PSU fixes shutdowns." â†’ Wattage is capacity, not health.
- "I can open the PSU if it is unplugged." â†’ Never. Lethal stored charge.

Suggested remedial exercises for learners who struggle:

1. **Tool-drill:** with a working machine, repeatedly open CrystalDiskInfo and HWMonitor and read aloud the health, temperature and fan values until fluent.
2. **Swap-test tracing:** give a written scenario where two parts were changed at once and ask "what can we conclude?" (Answer: nothing certain â€” repeat one at a time.)
3. **Symptom sorting:** hand out ten symptom cards and ask learners to sort them into "RAM / storage / PSU / heat / display" buckets.

Differentiation:

- **Level 1 (supported):** provide a step-by-step tool card and a partially completed report form; pair the learner with a stronger partner for the swap test.
- **Level 2 (standard):** the exercise as written.
- **Level 3 (challenge):** introduce a **mixed** fault (e.g., a failing drive *and* overheating at once) and ask the learner to diagnose both and prioritise which to fix first, justifying the order.

Timing guidance:
This is a practical-heavy session. If the room is slow, protect the guided practical (Section H) and trim the case study to a two-minute mention. Memory tests can be time-consuming â€” start the test early, let it run while other checks happen, and do not let one test consume the whole session. If a real memory test cannot finish in class, record the running result and note honestly that a full pass is required.

APPENDIX 5A â€” 10-MINUTE REFRESHER (only if needed)

RAM, Storage, PSU, GPU.

2. **The five-step workflow** â€" write on the board and make the class chant: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).
3. **Safety** â€" rehearsal: "Before I touch anything inside, I do two things: I _____ (power off and unplug) and I _____ (wear my anti-static wrist strap). I never open the _____ (PSU)." Confirm each learner can answer before continuing.

END OF SESSION 5

Next: Session 6 â€" Software Diagnostics.

SESSION 6 â€" SOFTWARE DIAGNOSTICS

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 6 â€" SOFTWARE DIAGNOSTICS

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	6 of 12
Session title	Software Diagnostics
Week / Module	Week 2 â€" Module: Diagnostics
Recommended duration	105 minutes
Difficulty level	Beginner
Prerequisite knowledge	Sessions 1â€"5: components, safety and tools, disassembly/reassembly, the five-step diagnostic workflow, and hardware diagnostics (RAM, storage, PSU, heat, display). If a learner cannot explain the difference between a hardware and a software fault, do the 5-minute recap in Appendix 6A before this session.
Connection to previous work	Session 5 proved faults in the **hardware** using physical tools. This session diagnoses the **software** â€" slow computers, viruses, boot problems and driver faults â€" using Windows' own built-in tools. Together the two sessions let you decide whether a fault is hardware or software.
Preparation before class	Practice computers with Windows 10/11 ready; at least one machine pre-loaded with a **safe simulated software fault** (e.g., several unnecessary startup programs enabled, a full Recycle Bin, a large unnecessary file, or a deliberately misconfigured/disabled device driver â€" do **not** use real malware); Windows Defender active and updated; Command Prompt access tested; Event Viewer accessible; a restore point created before the simulation so it can be undone. Keep a **second, clean** machine untouched for comparison.

Important safety and ethics note.
In this session we *simulate* software faults safely for teaching. We do **not** intentionally download viruses, and we never disable a customer's antivirus. Where the course talks about malware, it teaches **recognition and safe removal**, not creation. All work is on academy machines with permission.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain** the difference between a software fault and a hardware fault, and give an example of each.
2. **List** the common software faults â€" slow performance, malware infection, boot failure, driver issues, full disk â€" and match each to its usual symptoms.
3. **Use Task Manager** to identify programs using too much CPU, memory or disk, and safely end a misbehaving task.
4. **Use Event Viewer** to find error events and note their codes.

7. **Describe** the safe approach to malware removal, including when to use a full scan and when a second-opinion scanner is appropriate.
8. **Use System Restore / recovery options** to return a machine to a previous working state when appropriate.
9. **Decide** whether a fault is more likely software or hardware, and explain the reasoning.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least one practice desktop or laptop running Windows 10 or Windows 11 for each pair of learners.
- One machine pre-loaded with the **simulated** software faults described in Section A.
- One clean machine for comparison.
- A powered network is **not** required, but if available it helps update Defender's definitions.

Software (all built into Windows unless noted)

- **Task Manager** (Ctrl+Shift+Esc).
- **Event Viewer**.
- **Command Prompt** (run as Administrator) for `sfc /scannow` and `chkdsk`.
- **Disk Cleanup**.
- **System Restore / Recovery** options.
- **Device Manager** (for driver issues).
- **Windows Defender** (built-in antivirus) or the academy's approved antivirus.
- Optional second-opinion scanner (e.g., Malwarebytes) â€" instructor only, pre-installed and updated, used for demonstration.
- `msconfig` / **Startup Apps** settings.

Printed materials

- Software Diagnostic Report Form (one per learner).
- Windows Tools quick-reference card (one per learner).

Safety equipment

- None special for software, but keep drinks away from machines and never disable antivirus.

Internet

- Optional. Needed only if the instructor chooses to show a driver download from a manufacturer's site at the end. Do not depend on it.

Alternative for learners who lack a facility

- Any Windows computer works for Task Manager, Disk Cleanup and Event Viewer; `sfc` needs Administrator rights. Learners without a personal machine share a bench machine and swap roles.

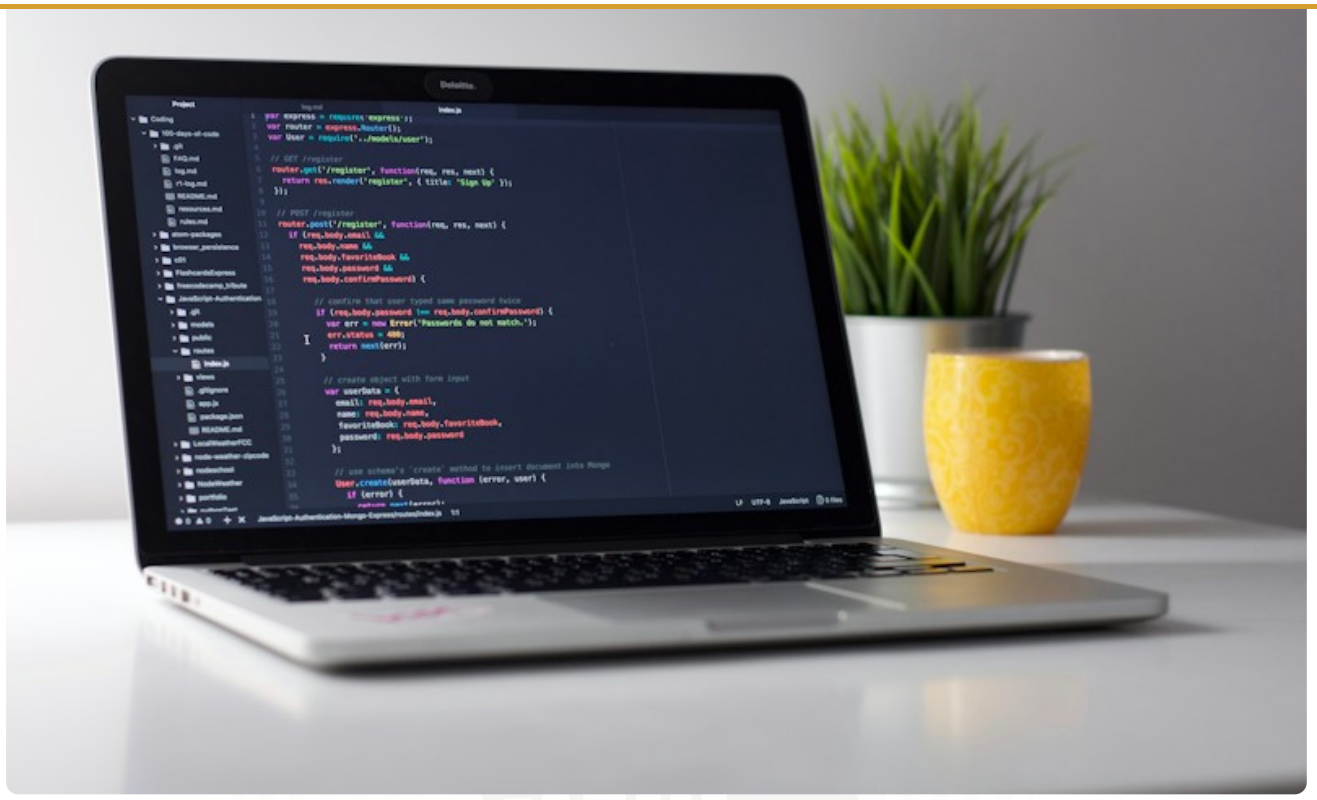
Suggested visuals:



- â€" Running Task Manager and Event Viewer



- â€" Guided software diagnostic walkthrough



- â€" Recording results on the software report form



- — Performance graphs read from Task Manager



— Results logged in the software diagnostic form

D. OPENING ACTIVITY (6 minutes)

Instructor setup:
On the projector, open the pre-prepared **simulated faulty** machine. Have several browser-like pop-up windows already sitting on the desktop (harmless images or text files the instructor arranged), and open Task Manager to show a high CPU figure. Keep it controlled and safe.

What the instructor says:

> *"Good morning. Look at this computer. The screen is crowded with pop-ups. It is crawling â€" everything takes forever. Programs close on their own. The owner is panicking.*
>
> *Last week we learned to test hardware with tools. But tell me something: if I open this case right now, will I find a broken part? No. The RAM is fine, the drive is fine, the power supply is fine. And yet the computer is very sick.*
>
> *This computer is sick in its*
>
> ***software***
> *. Software problems are just as common as hardware problems â€" and do you know what makes them harder? You cannot see them. There is no clicking noise, no beep, no hot part you can feel. Today you learn to see software problems using Windows' own tools â€" the tools that are already inside every computer, free."*

Expected learner response:
Learners may shout "virus!", "it needs formatting", or "delete everything". Accept all answers. Write them on the board. Say:

> *"Maybe. But a good technician does not format first and ask questions later â€" that destroys the customer's data. We investigate first. That is what today is about."*

Why this opener works:

safer, investigative approach.

E. PREREQUISITE REVIEW (7 minutes)

Run as a fast oral round-robin. Put the questions on the projector.

Question 1:
"Name the five steps of the diagnostic workflow."

- Expected answer: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).
- Common gap: forgetting to **document**. Remind learners: not documented = not done.

Question 2:
"What hardware fault causes a computer to run slowly and make a soft clicking sound?"

- Expected answer: a **failing hard drive**. Note that this is a **hardware** cause of a symptom people usually blame on software.

Question 3:
"What does it mean if CrystalDiskInfo shows 'Caution'?"

- Expected answer: the drive is showing wear; **back up now** and plan to replace it.

Question 4:
"Why do we never open the PSU?"

- Expected answer: **lethal stored charge** in the capacitors, even when unplugged.

Question 5:
"A computer beeps repeatedly and shows no picture. Is that more likely hardware or software â€" and why?"

- Expected answer: **hardware** â€" software has not even loaded yet when POST runs. This bridges neatly into today.

Bridge to today (say this):

> *"Session 5 was about a sick*
>
> ***body***
> *. Today is about a sick **mind**. A computer can have perfect parts and still behave badly â€" because the instructions it is following are wrong, infected or broken. Your job today is to use the tools that let you look inside the mind."*

Gap handling:
If learners cannot clearly separate a hardware from a software example (Question 5), spend up to three extra minutes with the Appendix 6A recap, because this distinction is the backbone of today's session.

F. FULL LECTURE NOTES

F.1 What a software fault is â€" and why it is different

The idea.
A computer is hardware **plus** the instructions that run on it. Those instructions and data are the **software**. When the instructions or data go wrong, the machine misbehaves even though every physical part is fine.

Plain-English definition.
A **software fault** is anything wrong with the programs, the operating system, the drivers, or the files â€" as opposed to the physical parts. Examples: a virus, a corrupted Windows file, an out-of-date or wrong driver, a full disk, or too many programs starting automatically.

Why the distinction exists.

decide which type you are facing.

Analogy.
Think of a football team. The **players** are the hardware. The **coach's instructions** are the software. If a player breaks a leg (hardware), no new instructions will help â€" you need a new player or rest. If all the players are strong but the coach gave the wrong formation (software), replacing players will not help â€" you must fix the instructions. Same team, same "hardware"; the problem is elsewhere.

Realistic Port Harcourt scenario.
A student in Elelenwo buys a "free" cracked version of a design program from a friend's USB flash drive. Two weeks later her laptop is slow, opens strange pages in her browser, and shows adverts on the desktop. The parts are all fine; the machine is infected. A technician who understands software faults knows to scan, remove, and teach the student about trusted sources â€" not to replace her hard drive.

Important misconception.
"Software problems are the easy ones â€" just restart." Sometimes a restart clears a temporary glitch, but many software faults (viruses, corruption, failing drivers) persist and get worse. Do not rely on restarts as a fix.

Key limitation.
Software faults often *look like* hardware faults. A corrupted system file can cause a blue screen; a failing drive can corrupt system files. So a good technician still runs the **hardware** checks (Session 5) when symptoms are severe, to make sure a hidden hardware problem is not the root cause.

F.2 Slow performance â€" the most common complaint

The idea.
"My computer is slow" is the number one thing customers say. It has several possible causes, and the skill is to find *which* one.

Plain-English definition.
Slow performance means the computer takes much longer than expected to start, open programs, or respond.

The main causes of a slow computer (each with its own symptom pattern):

Cause	Type	Typical sign
Too many **startup programs**	Software	Slow only at start-up; settles after a few minutes
Full or nearly full disk	Software/storage	Slow saves; "low disk space" warnings; slow everywhere
Malware	Software	Slow *plus* pop-ups, redirects, strange behaviour
Failing drive	Hardware	Slow *plus* clicking/errors; check S.M.A.R.T. (Session 5)
Too little RAM	Hardware	Slow with many programs open; disk light constantly busy
Overheating	Hardware	Slow *plus* loud fans, hot case, random shutdowns
Fragmented HDD / old hardware	Both	Generally slow but healthy; an SSD upgrade helps

How we investigate:

- **Task Manager** (Ctrl+Shift+Esc) â†' **Processes** tab shows what is using CPU, memory and disk *right now*.
- **Startup Apps** tab (or `msconfig`) shows what runs automatically at start-up.
- **Disk space** check (This PC) shows how full the drive is.
- **CrystalDiskInfo** (Session 5) rules out a failing drive.
- **HWMonitor** (Session 5) rules out overheating.

Analogy.

which it is.

Important misconception.
"Slow means I need a new computer." Often a cleanup, more RAM, or moving Windows onto an SSD makes an old machine feel new again â€” far cheaper than replacing it (Session 7).

Limitation.
Task Manager shows the *present moment*. A program that spikes the CPU only sometimes may look innocent when you glance. Watch it for a while, and use Event Viewer for the history.

F.3 Malware â€” viruses, spyware and ransomware

The idea.
Malware is any software written to harm you or to take your money or data. A "virus" is only one type.

Plain-English definitions (keep these simple for beginners):

- **Virus:** malicious code that spreads by attaching itself to files and programs.
- **Spyware:** software that secretly watches what you do (passwords, banking, browsing).
- **Ransomware:** software that locks your files and demands money (a "ransom") to unlock them.
- **Adware / unwanted pop-ups:** programs that flood you with adverts, often bundled with "free" software.
- **Trojan:** a harmful program disguised as something useful (e.g., a "free game" that is really spyware).

Why it exists and how computers get infected:

- Downloaded from untrusted websites, especially "cracked" or "free" versions of paid software.
- From an **infected USB flash drive** shared between computers.
- From email attachments or links that look real (called **phishing**).
- From clicking fake pop-ups that pretend to be "your antivirus found 5 viruses â€” click here".

Common symptoms of infection:

- Pop-ups and adverts appearing even when the browser is closed.
- The browser sending you to strange websites (redirects).
- The computer suddenly slow, with high CPU/disk even at idle.
- Strange programs opening, or unfamiliar icons/programs appearing.
- Files, folders or settings changing on their own.
- Antivirus switching itself off and refusing to turn back on.
- For ransomware: files suddenly renamed or encrypted and a "pay us" note.

The safe removal approach (step by step):

1. **Do not panic and do not format yet.** First preserve the customer's data.
2. **Disconnect from the internet** (unplug the network cable / turn off Wi-Fi) to stop the malware communicating or spreading.
3. **Run a full scan** with the updated antivirus (Windows Defender or the academy's approved tool). A **Full Scan** checks the whole system; a Quick Scan checks only common locations.
4. **Review the results**, then **quarantine** or **remove** detected threats. Quarantine moves the threat to a safe place rather than deleting it immediately.
5. **Restart** if prompted, and **scan again** to confirm nothing remains.
6. **Second-opinion scan (optional):** if symptoms continue but the first scan found nothing, run a trusted second scanner (e.g., a reputable free malware-removal tool). Different tools catch different things.
7. **Change important passwords** afterwards if spyware was suspected (banking, email), because they may have been captured.

Analogy.
Malware is like rats in a store-room. Poisoning (scanning) is not enough if you leave the door (the unsafe download habit) open. You must remove the rats *and* close the door.

Important misconception.
"One antivirus is enough, forever." Antivirus **definitions** (the list of known threats) must be kept **updated**. An out-of-date antivirus is like a watchman with an old list of faces â€" new intruders walk straight past.

Limitation.
No antivirus catches every threat, especially brand-new ones. Also, some "vaccines" people download are themselves malware ("fake antivirus"). Only install antivirus from the built-in Windows Defender or a well-known, trusted brand. Never pay a random pop-up that claims to clean your PC.

F.4 Boot problems â€" when Windows will not start

The idea.
Sometimes the computer powers on (so hardware and POST are fine) but Windows does not finish loading. That is a **boot problem**.

Plain-English definition.
A **boot problem** is a fault in starting Windows. The machine passes POST, but Windows does not reach the desktop â€" either it hangs, shows an error, restarts, or offers a repair screen.

Common symptoms:

- A black screen with a spinning dots animation that never finishes.
- An error such as "Boot device not found" (often **hardware** â€" check S.M.A.R.T. first).
- "Your PC ran into a problem and needs to restart" (a **blue screen**).
- An automatic "Automatic Repair" / "Recovery" screen.
- A **boot loop**: Windows starts to load, then restarts, over and over.

Why it happens:
a corrupted system file, a bad update, a wrong or corrupted driver, a power cut during an update, or an underlying failing drive.

Recovery options (what each one does):

- **Start-up Repair:** Windows tries to repair common start-up problems automatically.
- **Safe Mode:** Windows starts with only the basic drivers and services. It is how you get in to fix a problem when normal mode fails. *(Safe Mode is defined properly in F.5.)*
- **System Restore:** returns system files and settings to an **earlier point** without touching the customer's personal files (see F.7).
- **Reset / reinstall:** the last resort. **Always back up first.**

Analogy.
A boot problem is like a car that starts (engine turns) but the gear will not engage. The engine is fine; the path from engine to wheels is broken. You do not replace the engine â€" you find and fix the broken link.

Important misconception.
"If Windows will not start, the computer is finished â€" format it." Not true, and formatting too quickly loses data. "Boot device not found" is often a hardware problem (or a loose cable), and a blue screen is often a driver or file problem. Investigate before you wipe.

Limitation.
Some boot faults are actually hardware â€" a failing drive or bad RAM. That is why every boot problem starts with the Session 5 checks.

When normal Windows will not start, **Safe Mode** gives you a stripped-down Windows so you can still get in and fix the problem.

Plain-English definition.
Safe Mode starts Windows with only the **essential** drivers and services. Extra programs and fancy drivers are left out, so the system is more likely to start even when something is wrong.

Why it exists.
If a bad driver or a program that starts automatically is crashing Windows, Safe Mode does not load it, so the system comes up and you can remove or disable the culprit.

Analogy.
Safe Mode is like a hospital's **emergency room**. Everything non-essential is switched off so the doctors can focus on the problem. Normal mode is the whole hospital running at once â€" noisier, busier, harder to work in when something is broken.

When to use it:

- To remove stubborn malware that will not delete in normal mode.
- To uninstall a program or roll back a driver that broke Windows.
- To create a System Restore point or run recovery tools.

How to reach it (Windows 10/11):
If Windows will not start, it often offers a **Recovery environment** after failing to boot two or three times. From the recovery screen choose **Troubleshoot â†' Advanced options â†' Startup Settings â†' Restart**, then select **Enable Safe Mode**. (Exact wording is stable but the paths depend on the Windows version; the instructor demonstrates on the practice machine.)

Important misconception.
"I can work normally in Safe Mode forever." No â€" Safe Mode is temporary and limited. It is a tool for getting in and fixing the cause, not a place to stay.

Limitation.
Some problems (a failing drive, bad RAM) are not fixed by Safe Mode. Safe Mode proves the fault is *software* â€" a useful clue in itself.

F.6 Driver issues â€" the translator problem

The idea.
A **driver** is a small piece of software that tells Windows how to talk to a specific piece of hardware.

Plain-English definition.
A **driver** is the translator between Windows and a hardware device (printer, graphics card, Wi-Fi adapter, sound chip). Each device needs the right driver to work properly.

Why drivers exist.
Windows cannot contain instructions for every device ever made. Instead, each device's maker writes a driver, and Windows loads it. New hardware needs a driver; that is why after a fresh Windows install "some things may not work until drivers are installed."

Common symptoms of a driver problem:

- A device does not work at all (no sound, no Wi-Fi, printer not found).
- A yellow triangle with an exclamation mark next to a device in **Device Manager**.
- Poor performance (especially graphics: the GPU works but slowly, or with wrong resolution).
- Crashes or blue screens after installing or updating a driver.

Tool: Device Manager.
Open it by pressing the Windows key and typing **Device Manager**. It lists every device. A problem device shows a **yellow warning triangle**.

- **To roll back a driver:** if a problem started after an update, right-click the device â†' **Properties** â†' **Driver** tab â†' **Roll Back Driver**. This returns to the previous version.
- **To identify an unknown device:** right-click â†' Properties â†' Details â†' note the hardware ID and look it up (or search it online) to find the correct driver.

Analogy.
A driver is like a person who speaks both languages. Windows speaks "Windows"; the printer speaks "printer". Without a translator, neither understands the other, and nothing prints. Updating the driver is getting a better translator.

Important misconception.
"Always get drivers from any site that offers a free 'driver updater'." Avoid random driver-updater websites and tools; they often bundle unwanted software. Get drivers from **Windows Update** or the **device manufacturer's official site**.

Limitation.
Not every old device has a driver that works on the latest Windows; sometimes the honest answer is "this device is too old for this version of Windows." Also, the *newest* driver is not always the *best* â€" if the machine was stable before an update and broke after, rolling back is the right move.

F.7 The Windows diagnostic tools

These tools are already inside Windows. Learn what each does before you use it.

1. Task Manager (Ctrl+Shift+Esc).

- Shows **Processes**: every running program and how much **CPU**, **memory**, **disk** and **network** it is using.
- The **Startup Apps** tab shows what launches automatically at start-up (see F.8).
- Use it to spot the culprit: a program using 80â€"100% CPU or disk while you do nothing is suspicious.
- To stop a frozen or suspect program: select it â†' **End task**. *Safety:* end tasks only for programs you recognise as misbehaving; ending a Windows system process can unbalance the machine.

2. Event Viewer.

- A **log** of everything Windows has recorded: information, **warnings** and **errors**.
- Look under **Windows Logs â†' System** and **Windows Logs â†' Application**.
- **Errors** show a red circle with an "X". Note the **error code** and the **source** and search the exact code online for its meaning.
- Use it to find *what happened* when a crash occurred, even after the machine restarted.

3. System File Checker (`sfc /scannow`).

- A built-in tool that scans Windows' own system files and repairs any that are corrupted, using a cached good copy.
- Run it in a **Command Prompt opened as Administrator**: press Windows key â†' type **cmd** â†' right-click **Command Prompt** â†' **Run as administrator** â†' type `sfc /scannow` â†' Enter.
- It takes roughly 10â€"20 minutes. When finished it reports one of: no problems found; problems found and repaired; or problems found that could not be repaired (then you may need to check `DISM`, an advanced repair â€" instructor demonstration only).
- *Safety:* never run it on a failing drive without backing up first â€" repairs write to the disk.

4. Chkdsk (`chkdsk`).

- Checks a **disk** for file-system errors and bad sectors. `chkdsk C: /f` fixes errors on drive C.
- Advanced, writes to the disk; back up first and use under supervision.

5. Disk Cleanup.

- Press Windows key â†' type **Disk Cleanup** â†' choose the drive â†' tick categories (Temporary files, Recycle Bin, Windows Update leftovers, etc.) â†' **OK** to delete.
- Frees storage space safely. Larger cleanup ("Clean up system files") can also remove old Windows update files.

- Choose a **restore point** from *before* the problem began â†' follow the prompts.
- Restores system files, drivers and settings to that earlier time. It **does not delete the customer's personal documents** â€" but warn the customer that programs installed after the restore point may need reinstalling.

7. Startup Apps / `msconfig`.

- **Startup Apps** (in Task Manager) lists programs that start automatically. Disabling unnecessary ones speeds up boot time.
- `msconfig` (System Configuration) is an older, more powerful tool; use it under supervision. Disabling critical services can break Windows.

Important misconception.
"`sfc /scannow` fixes everything." It repairs *system files* only. It does not remove viruses, it does not fix a failing drive, and it does not replace missing drivers. It is one tool among many.

Limitation.
These tools work only if Windows can start at least in Safe Mode. If Windows cannot start at all, you use the recovery environment (F.4) first.

F.8 Startup programs and disk space

Startup programs.
Many programs (chat apps, updaters, "helper" tools) add themselves to the start-up list. Each one slows boot and uses memory. Review the **Startup Apps** tab in Task Manager and **disable** anything the user does not need at start-up. To disable: select the app â†' **Disable**. *Safety:* do not disable items you do not recognise if they appear related to security, audio, or trackpad â€" ask first.

Disk space.
Windows needs **free space** to work. When a drive is nearly full, everything slows. Check free space in **This PC**. Free it with **Disk Cleanup**, emptying the **Recycle Bin**, uninstalling unused programs, and moving personal files to an external drive or cloud. A common real-world cause of a "slow" office PC is a drive that is 98% full.

Analogy.
Startup programs are like guests who arrive at your shop every morning and occupy chairs before you open. A few are helpful; dozens block the shop. Disk space is the shop floor: if it is packed to the ceiling with boxes, you cannot work.

Important misconception.
"Deleting files always frees space permanently." Files in the Recycle Bin still occupy space until the bin is emptied.

F.9 Software fault or hardware fault? â€" deciding

The idea.
The most valuable skill in Week 2 is choosing the correct category before you act.

Quick decision guide:

Symptom	Lean towards	First check
Beeps, no picture, no power	Hardware	POST, PSU, RAM (Session 5)
Slow + clicking + "not found"	Hardware	CrystalDiskInfo S.M.A.R.T.
Slow + hot + loud fans	Hardware	HWMonitor; dust/fans
Pop-ups, redirects, strange behaviour	Software	Antivirus full scan
Blue screen after installing a program/driver	Software	Safe Mode; roll back driver
Device not working (sound/Wi-Fi/printer)	Software	Device Manager

Low disk space, slow saves	Software	Disk Cleanup
Random restarts even in Safe Mode	Hardware	RAM, PSU, heat

Why this matters.
If you format a machine whose real problem is a failing drive, the fault returns and you have destroyed data. If you replace a drive whose real problem is a virus, you have wasted the customer's money. Deciding correctly saves both.

Analogy.
A doctor first decides whether the problem is a broken bone or an infection, because the treatments are completely different. Decide "hardware or software" first, every time.

Important misconception.
"If it started after I installed something, it must be software." Usually true â€” but installing a heavy program can also reveal a *hardware* weakness (e.g., more RAM pressure causing crashes). Keep testing.

Limitation.
Mixed faults exist. A failing drive can cause software symptoms; a virus can cause hardware-like crashes. When in doubt, test the hardware foundation first.

F.10 Summary of the session

- A **software fault** is a problem with instructions or data, fixed by cleaning/correcting/restoring â€” not by replacing parts.
- **Slow performance** has many causes: startup programs, full disk, malware, or failing hardware. Investigate with **Task Manager**.
- **Malware** symptoms: pop-ups, redirects, slowness, strange programs. Safe approach: disconnect, **full scan**, quarantine, restart, re-scan, update passwords, educate.
- **Boot problems**: use recovery options, **Safe Mode**, and **System Restore**. "Boot device not found" is often hardware.
- **Drivers**: use **Device Manager** to update (from the maker) or **roll back** a bad update.
- Windows tools: **Task Manager**, **Event Viewer**, `sfc /scannow`, **chkdsk**, **Disk Cleanup**, **System Restore**, **Startup Apps**.
- Always decide **software vs hardware first**, and always **back up** before destructive steps.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The pre-prepared **simulated faulty** machine (extra startup programs enabled, a full Recycle Bin, a large unnecessary file, and one device driver deliberately disabled or mis-set â€” no real malware).
- The clean machine for comparison.
- A restore point created earlier so the simulation can be undone.
- Command Prompt (Administrator) tested; Event Viewer accessible.
- The Software Diagnostic Report Form on the projector.

G.2 Demonstration 1 â€” Task Manager and startup programs (5 minutes)

Say:
"Watch me find out *what* is making this computer slow â€” without guessing."

Steps:

1. Press **Ctrl+Shift+Esc** to open **Task Manager**.
2. On the **Processes** tab, click the **CPU** column header to sort by CPU use. Point at the top process.
3. **Say:** "This is the culprit *right now* â€” it is using a huge share of the CPU. On a healthy idle machine, the top numbers should be small."

Watch: I disable the ones marked 'High' impact, but I leave security and audio alone."

6. Disable two or three clearly unnecessary startup apps.
7. **Say:** "Notice I did not delete anything. Disabling start-up does not uninstall the program; it just stops it launching by itself. That is a safe fix."

Expected results:
the startup list shortens; the class can see the "Impact" column (High/Medium/Low) that Windows provides.

Reasoning to say aloud:
"We are finding the *cause*, not just making the number look smaller."

Common errors to show and correct:

- Ending a critical system process and destabilising the machine. Show the warning that a program might lose data.
- Disabling a start-up item that the user actually needs.

Questions to ask:
"If the CPU is high but settles down after five minutes, is that a fault?" (Not necessarily â€" start-up work is normal; it becomes a fault if it never settles or stays extreme.)

G.3 Demonstration 2 â€" Event Viewer and `sfc /scannow` (5 minutes)

Say:
"Now let us find errors that already happened."

Steps:

1. Open **Event Viewer**. Expand **Windows Logs â†' System**.
2. **Say:** "Red = Error, Yellow = Warning. I look at the errors around the time of the crash."
3. Open one error, point to the **Source** and the **Event ID** (error code). **Say:** "I copy this code and search it exactly, so I learn what it means."
4. Open **Command Prompt as Administrator** (show right-click â†' Run as administrator). Type `sfc /scannow` and press Enter.
5. **Say:** "This scans every Windows system file and repairs any that are corrupted. It takes about 10 to 20 minutes. While it runs, let us do another task."
6. (Optional) Open **Disk Cleanup** and show the categories and the "Total amount of disk space you gain" figure.

Expected results:
Event Viewer shows red error entries with Event IDs; `sfc /scannow` begins scanning and reports a result; Disk Cleanup shows reclaimable space.

Reasoning to say aloud:
"Event Viewer is the *history*; Task Manager is the *present*. sfc repairs *system files*. Each tool answers one question."

Common errors to show and correct:

- Running `sfc` in a normal (non-Admin) Command Prompt â†' it may fail or do nothing. Show the difference.
- Panicking at *any* red error â€" some errors are harmless and old. Read the time and source.

Questions to ask:
"Does `sfc /scannow` remove viruses?" (No â€" that is the antivirus's job.)

G.4 Demonstration 3 â€" Simulated malware check and Second opinion (3 minutes)

Say:
"Here is how we handle a suspected infection safely."

Steps:

1. On the simulated machine, open **Windows Defender** (or the academy antivirus) â†' **Virus & threat protection**.

If symptoms continue but the scan is clean, we use a **second-opinion** scanner, because no single tool catches everything."

4. Point out the importance of **keeping definitions updated** (there is usually an "up to date" indicator).

Expected results:
Defender runs a scan; the class sees the quarantine/review screens.

Common errors to show and correct:

- Clicking fake "you have 5 viruses" web pop-ups. **Say:** "Real antivirus does not pop up in your browser demanding a click. That is a trick."
- Disabling the antivirus "just for a minute" to download something. **Say:** "Never â€” that is exactly when you get infected."

Questions to ask:
"Why disconnect from the internet first?" (To stop the malware communicating or spreading.)

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective:
With the instructor guiding, learners use Windows' built-in tools on a practice machine and record findings on the Software Diagnostic Report Form.

Safety and ethics first â€” read aloud before starting:

- Only work on the **practice machines**; do not experiment on a stranger's or school's live computer.
- **Never disable the antivirus.**
- **Back up** or confirm a backup before running any tool that writes to disk (chkdsk, restore).
- **Do not click** unknown pop-ups or links.
- If in doubt about a step, ask the instructor before doing it.

Procedure (instructor calls each step; learners follow; instructor walks the room):

1. **Task Manager.** Open **Task Manager** (Ctrl+Shift+Esc). On the **Processes** tab, sort by **CPU**, then by **Memory**, then by **Disk**. Record the top process in each column and its percentage.
2. **Startup Apps.** Go to the **Startup Apps** tab. Record how many apps are enabled and which are marked **High** impact. Disable **one** clearly unnecessary app. Record what you disabled.
3. **Disk space.** Open **This PC** and record the free space on the C: drive and whether Windows warns of low space.
4. **Pause and predict.** "If I disable five startup programs, will the computer get more RAM or just boot faster?" Write your prediction. (Answer: it frees up some memory and CPU and speeds boot; it does not add physical RAM.)
5. **Disk Cleanup.** Open **Disk Cleanup** for C:. Record the "Total amount of disk space you gain". Do **not** click OK until the instructor confirms; then clean the **temporary files and Recycle Bin only**.
6. **Event Viewer.** Open **Event Viewer â†' Windows Logs â†' System**. Find **three** red Error entries. Record the **Event ID** and **Source** of each.
7. **System File Checker.** Open **Command Prompt as Administrator**. Run `sfc /scannow`. Record the final message (no problems / problems repaired / could not repair). This may take some minutes â€” start it and continue other checks.
8. **Driver check.** Open **Device Manager**. Look for any device with a **yellow warning triangle**. Record the device name if found.
9. **Antivirus check.** Open the antivirus. Confirm definitions are **up to date**, then start a **Quick Scan** (a Full Scan is too slow for class time; the instructor may run a Full Scan on the demo machine). Record the result.
10. Complete the **Software Diagnostic Report Form**.

Expected results:

- Task Manager shows running processes and resource percentages.
- At least one startup app disabled.
- Free disk space recorded; Disk Cleanup shows reclaimable space.

- Device Manager checked; any yellow-triangle device noted.
- Antivirus scan completed with a result.
- A completed report form.

Points to pause on and predict:

- After step 1: "If the Disk column is pinned at 100% while nothing is happening, what should you suspect?" (A failing drive â€" check S.M.A.R.T. â€" or a heavy background task.)
- After step 4: "Will disabling start-up items change the installed programs?" (No â€" the programs remain installed; they just do not auto-launch.)
- After step 8: "A device has a yellow triangle. Is that hardware or software?" (Software â€" it needs a driver or has a conflict.)

Troubleshooting while walking the room:

- Command Prompt will not run `sfc` â†' confirm it says "Administrator" in the title; right-click and choose Run as administrator.
- Event Viewer is empty â†' expand the correct log; some machines log few errors. Use another log (Application) if System is quiet.
- Disk Cleanup shows 0 bytes â†' this machine may already be clean; note that honestly.
- Antivirus scan hangs â†' it may be waiting for updates; skip and record it as "not completed".
- Learner wants to "just format it" â†' stop them; remind them of the data-first rule.

Completion checklist (learner ticks each):

- ☐ Task Manager opened; top CPU, memory and disk processes recorded
- ☐ Startup Apps reviewed; one unnecessary app disabled and noted
- ☐ Disk free space recorded
- ☐ Disk Cleanup run (temp files and Recycle Bin only)
- ☐ Three Event Viewer errors recorded with Event IDs
- ☐ `sfc /scannow` run and its result recorded
- ☐ Device Manager checked for yellow-triangle devices
- ☐ Antivirus scan run and result recorded
- ☐ Software Diagnostic Report Form completed
- ☐ All safety and ethics steps followed (no disabling antivirus, no unknown pop-ups)

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
Diagnose a **slow/faulty software scenario** on your own, using the correct Windows tools, and write a short diagnosis. Do not copy the instructor's demonstration.

Scenario.
Mr. Bassey, who runs a small printing business in Mile 3, Port Harcourt, brings you a Windows desktop. He reports:

- The computer takes **over five minutes** to reach the desktop in the morning.
- Once running, it is **slow to open files**.
- He sees **adverts popping up** even when his browser is closed.
- Yesterday a message said **"Low disk space on C:"**.
- Last week, **his sound stopped working** after a Windows update.

Requirements:

1. List the **three most likely software causes** for the slowness, and say which Windows tool you would use to confirm each.
2. Explain what the **pop-up adverts** most likely indicate, and describe the **safe removal approach** in numbered steps.

5. Decide: are these problems **all software**, or should any of them prompt a **hardware** check? Justify your answer.

Time allowance:
15 minutes.

Constraints:

- Use only the Windows tools from today's session.
- Do not disable the antivirus.
- Write in plain language a customer could follow.

Success criteria:

- The three slowness causes are plausible and each is matched to a suitable tool.
- The malware approach is safe (disconnect â†' full scan â†' quarantine â†' restart â†' re-scan â†' passwords â†' educate).
- The disk-space explanation is correct and names a real tool.
- The sound issue is correctly identified as a driver problem.
- At least one hardware check is justified (the disk-space + slow pattern should prompt a S.M.A.R.T. check).

Hints available if stuck:

- Hint 1: The slow *morning* start-up points at one particular tab in Task Manager.
- Hint 2: "Low disk space" + slow file opening is a **storage** issue â€" but is a *full* drive the same as a *failing* drive? Think about it.
- Hint 3: Sound stopping right after an update screams "the translator between Windows and the sound chip".

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

1. Likely causes of slowness + tools:

- Too many **startup programs** â†' confirm in **Task Manager â†' Startup Apps** (High-impact entries).
- **Malware** (supported by the pop-ups) â†' confirm with a **full antivirus scan**.
- **Low disk space / full drive** â†' confirm by checking **free space in This PC** and using **Disk Cleanup**. *(Also acceptable: a failing drive, since "slow + full + adverts" can hide hardware â€" confirmed by CrystalDiskInfo S.M.A.R.T.)*

2. Pop-up adverts:
most likely **malware/adware**. Safe removal approach:

1. Disconnect from the internet.
2. Run a **full** antivirus scan (definitions updated).
3. Review results; **quarantine/remove** detected threats.
4. Restart if prompted.
5. **Scan again** to confirm.
6. If symptoms persist, run a **trusted second-opinion** scanner.
7. Warn the customer to change important passwords (spyware may have captured them).
8. Educate the customer: only download from trusted sources.

3. Low disk space:
Windows needs free space for temporary files, updates and paging (using disk as extra memory). When the drive is nearly full, these operations slow down, so the whole computer feels slow. **Disk Cleanup** (and emptying the Recycle Bin / uninstalling unused programs) helps.

4. Sound after an update:
It is a **driver** problem. Open **Device Manager**, find the sound device (look for a yellow warning triangle), then either **Update driver** (from the manufacturer's official site is best) or **Roll Back Driver** if the update caused it.

should also run a **hardware** check ac- **CrystalDiskInfo S.M.A.R.T.** ac- to rule out a failing drive (the same "slow" symptom can be a dying HDD). *This is the key insight of the exercise:* never assume "software" without a quick hardware sanity check.

Marking guidance (out of 15):

- Three plausible slowness causes + matching tools (3)
- Safe malware approach, in sensible order (4)
- Correct disk-space explanation + real tool (2)
- Correct driver identification + update/rollback (3)
- Correctly notes a hardware check is warranted and why (3)

Accept any answer showing correct reasoning
, even if wording differs. Penalise unsafe steps (formatting first, disabling antivirus, clicking pop-ups).

J. REAL-WORLD CASE STUDY

Situation.
Chidinma is a final-year student at a school in Port Harcourt. She needs a "free" copy of a design program for her project. A friend gives her a downloaded installer on a USB flash drive. She installs it. Within days, her laptop slows down badly, her browser opens strange pages by itself, and pop-ups appear on the desktop. She is afraid she has "spoiled the laptop" and is about to pay a shop to "reset everything" â€” which would wipe her project files.

Relevant facts.

- The slowness and pop-ups began **right after** she installed the "free" program.
- Her project files are stored on the same drive, with **no backup**.
- The machine still boots and works, just slowly.

The decision point.
Should the shop format the laptop and lose her project, or is there a safer path?

Possible approaches.

1. Format and reinstall Windows immediately (fast, but her project files would be at risk if not backed up).
2. Disconnect from the internet, back up her project files, run a full antivirus scan, quarantine the threats, restart, re-scan, and remove the "free" program politely.
3. Ignore the pop-ups and keep using the laptop.

Recommended solution (for this stage).
Approach 2. First **back up her project files** (this is not optional). Then disconnect the internet and run a **full scan**, quarantine anything found, restart, and scan again to confirm. Remove the "free" program. Explain how she got infected and why "free cracked software" from a USB is one of the most common infection routes in Nigeria.

Trade-offs.

- Approach 1 is the "nuclear option": it certainly removes the malware but risks her data and wastes time. It should be the **last** resort, after backup â€” not the first reaction.
- Approach 2 is safer, cheaper, faster, and preserves her work; but it depends on removing the *cause* (the habit) or the infection will return.
- Approach 3 lets the infection and the risk grow â€” spyware could steal her passwords.

Professional connection.
A technician earns a customer's trust by refusing to destroy data for convenience. The CEA graduate is trained to preserve the customer's files, remove the threat safely, and educate the customer â€” exactly the "honest assessment and recommendations" the certificate stands for.

Transfer exercise (ask the class):
"Chidinma asks, 'How do I know which downloads are safe?' What three rules would you give her?"

updated and never click pop-ups that claim to "clean your PC".

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the session's live troubleshooting guide. When a learner has a problem, direct them to the relevant row.

K.1 The computer is slow but Task Manager looks normal

- **Symptoms:** user complains of slowness; CPU/memory/disk look modest.
- **Likely causes:** failing drive (slow reads); intermittent background task; too little RAM despite low average use; overheating causing throttling.
- **How to investigate:** look at the **Disk** column over time; run **CrystalDiskInfo** (S.M.A.R.T.); check temperatures with **HWMonitor**; review Event Viewer for disk warnings.
- **Safe corrective steps:** back up; replace/upgrade the drive if failing; clean cooling if hot; add RAM if memory pressure is confirmed.
- **How to confirm resolved:** normal speed returns and the S.M.A.R.T./temperature readings are healthy.
- **Prevention:** regular health checks and cleaning (Session 8).

K.2 Pop-ups and browser redirects

- **Symptoms:** adverts appear; browser opens strange sites; new toolbars.
- **Likely cause: malware/adware**, usually from an untrusted download or an infected USB.
- **How to investigate:** check recently installed programs and browser extensions; run a full scan; check **Startup Apps** for unknown entries.
- **Safe corrective steps:** disconnect internet â†’ full scan â†’ quarantine â†’ restart â†’ re-scan â†’ remove the unwanted program/extension â†’ change passwords â†’ educate.
- **How to confirm resolved:** pop-ups and redirects stop after a restart and a clean re-scan.
- **Prevention:** trusted downloads, scan USB drives, keep antivirus updated.

K.3 "Low disk space" and slow saving

- **Symptoms:** warning messages; slow saves; small free space on C:.
- **Likely cause:** the drive is nearly full (temp files, downloads, Recycle Bin, unused programs).
- **How to investigate:** check free space in **This PC**; run Disk Cleanup to see reclaimable space.
- **Safe corrective steps:** empty Recycle Bin; Disk Cleanup; uninstall unused programs; move personal files to external storage/cloud. **Do not delete system files or the Windows folder.**
- **How to confirm resolved:** free space rises and saving is fast again.
- **Prevention:** monthly Disk Cleanup; keep at least ~15% of the drive free.

K.4 Blue screen after a driver or program install

- **Symptoms:** blue screen ("Your PC ran into a problem"); recent install/update.
- **Likely cause:** a faulty driver or an incompatible program (sometimes faulty RAM â€” always rule it out).
- **How to investigate:** note the stop code; start **Safe Mode**; check **Device Manager**; check **Event Viewer** around the crash time.
- **Safe corrective steps:** in Safe Mode, **roll back** the recent driver or uninstall the recent program; if unstable, use **System Restore** to an earlier point; run a **memory test** to rule out RAM.
- **How to confirm resolved:** the machine runs normally for a full session without a blue screen.
- **Prevention:** get drivers from the maker; avoid untrusted software; keep System Restore points.

K.5 A device stopped working after a Windows update

- **Symptoms:** no sound / no Wi-Fi / printer not found, right after an update.

- **Safe corrective steps: Roll Back Driver**; or **Update driver** from the manufacturer's official site; then restart.
- **How to confirm resolved:** the device works again.
- **Prevention:** install manufacturer drivers deliberately; create restore points before big updates.

K.6 Distinguishing likely from confirmed

As in Session 5, a

likely
cause becomes **confirmed** only after evidence: an Event Viewer error with a code, a scan result, a `sfc` message, a driver roll-back that fixes the device, or a hardware reading. Teach learners to write "likely" until a test proves it. Say clearly: **"I have not confirmed this yet."**

L. SESSION ASSIGNMENT

Assignment title:
"Software Diagnostic Report on a Real Machine"

Purpose:
To practise using Windows' built-in diagnostic tools, to diagnose common software faults safely, and to write findings plainly for a customer.

Realistic scenario:
A customer brings you a Windows computer and says: *"It is very slow to start, sometimes shows adverts, and the sound stopped working last week."* Produce a professional software diagnostic report before any destructive action.

Detailed instructions:

1. Use a real Windows computer you have permission to work on (a CEA practice machine, a family machine, or a machine at your workplace). **Do not work on a computer without permission.**
2. **Observe and record** the reported symptoms in your own words.
3. **Task Manager:** record the top CPU, memory and disk processes and the number of startup apps enabled.
4. **Startup Apps:** disable **one** unnecessary app and record what you disabled. (Re-enable it afterwards if the machine is not yours.)
5. **Disk space & Disk Cleanup:** record free space; run Disk Cleanup on temporary files and the Recycle Bin only; record space reclaimed.
6. **Event Viewer:** record three error Events (Event ID + Source).
7. `sfc /scannow`**:** run it as Administrator and record the final message.
8. **Device Manager:** list any device with a yellow warning triangle.
9. **Antivirus:** confirm definitions are up to date; run a **Quick Scan** and record the result. *(Do not run a Full Scan on a machine you do not own without permission â€" it is slow.)*
10. **Conclusion:** state the most likely software cause(s) and your recommended action, in plain language for the customer.

Required deliverables:

- One completed **Software Diagnostic Report Form** with all readings and results.
- A one-paragraph plain-language summary for the "customer".

Submission format:
Written report on the CEA template (paper or class folder).

Deadline:
Before Session 7.

Marking rubric (20 marks):

Observation & symptom record	2	Clear, specific symptoms recorded
Task Manager & Startup findings	3	Top processes and startup app count recorded; one safely disabled
Disk space & Disk Cleanup	3	Free space and reclaimed space recorded; safe cleanup only
Event Viewer findings	3	Three errors with Event ID and Source
`sfc /scannow` result	3	Run as Administrator; final message recorded
Device Manager & antivirus	3	Yellow-triangle device noted (if any); scan result recorded
Conclusion & recommendation	2	Logical, safe, honest conclusion in plain language
Report completeness & safety	1	No unsafe actions; form fully filled

Common mistakes to avoid:
running `sfc` without Administrator rights; deleting system files; disabling the antivirus; formatting before backup; writing "confirmed" without a test; ignoring a possible hardware cause.

Instructor reference solution (abbreviated example):

> ***Symptoms:***
> *slow start-up; occasional adverts; sound stopped after last week's update.* ***Task Manager:*** *top process "Update Helper" at 45% CPU and 70% disk;* ***9*** *startup apps enabled, 3 marked High impact.* ***Startup Apps:*** *disabled "Photo Startup Tool" (unnecessary).* ***Disk:*** *only* ***4 GB free*** *of a 500 GB drive â€" very low. Disk Cleanup reclaimed* ***6.1 GB*** *(temp files + Recycle Bin).* ***Event Viewer:*** *Error â€" Event ID* ***7000*** *(Service Control) source Service Control Manager; Error â€" Event ID* ***7009****; Error â€" Event ID *1001 (Windows Error Reporting).* `sfc /scannow` ***:*** *"Windows Resource Protection did not find any integrity violations."* ***Device Manager:*** *yellow triangle on* ***Realtek High Definition Audio****.* ***Antivirus:*** *definitions up to date; Quick Scan â€" no threats.* ***Conclusion â€" likely causes:*** *(1) too many startup programs + a very full disk â†' slow start-up and slow file opening; (2) the sound device has a* ***driver*** *problem after the recent update.* ***Recommendation:*** *keep three unnecessary startup apps disabled; empty the drive further and set up monthly Disk Cleanup;* ***roll back*** *the audio driver or install the correct one from the laptop maker's site; restart and re-test the sound. Also run a S.M.A.R.T. check (Session 5) because the drive is nearly full and slow reads can mask a failing drive.* ***Plain-language summary:*** *"Your computer is starting too many programs at once and the storage disk is almost full, which is why it is slow. The sound stopped because of a driver problem after an update. We can fix all three without losing your files."*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key below.)

Q1. Short answer.
In one sentence each, what is a **software fault** and what is a **hardware fault**? **Q2. Multiple choice.** Which Windows tool lets you see which programs are using the most CPU right now? a) Disk Cleanup b) Event Viewer c) Task Manager d) System Restore **Q3. Explain in your own words.** Why is it dangerous to "just format" a computer first when the owner complains of slowness? **Q4. Practical instruction.** What exact command checks and repairs corrupted Windows system files, and how must the window be opened to run it properly? **Q5. Matching.** Match each symptom to the most likely cause:

1. Pop-ups and browser redirects
2. Sound stopped right after a Windows update
3. "Low disk space on C:"
4. Device shows a yellow triangle in Device Manager A. Driver problem B. Malware C. Full drive D. Missing/incorrect hardware driver **Q6. Scenario.** A computer is slow and makes a soft clicking sound. Is this more likely software or hardware, and what should you check first?

A **software fault** is a problem with programs, drivers, the operating system or data; a **hardware fault** is a problem with a physical part. Software faults are fixed by cleaning/correcting/restoring the software; hardware faults are fixed by repairing or replacing a part.

A2.
(c) Task Manager.

A3.
Because formatting erases everything, including the customer's personal files, and it may not even fix the problem â€" the real cause could be a failing **hard drive**, in which case the fault returns *and* the data is gone. Always back up and diagnose first; format only as a last resort.

A4.
`sfc /scannow`. It must be run in a **Command Prompt opened as Administrator** (right-click Command Prompt â†' Run as administrator).

A5.
1â€"B (malware), 2â€"A (driver problem), 3â€"C (full drive), 4â€"D (missing/incorrect driver). *(Note both 2 and 4 are driver issues; 2 is a driver problem triggered by an update, 4 is a general missing/wrong driver.)*

A6.
More likely **hardware** â€" a soft clicking sound plus slowness is the classic sign of a **failing hard drive**. Check **CrystalDiskInfo S.M.A.R.T.** first, and **back up the data immediately** before anything else. (A memory/CPU fault is possible but the click strongly points to the drive.)

Marking:
award for correct concept. For Q5, accept 2â€"D and 4â€"A if the learner explains the driver reasoning clearly.

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned that a computer can have perfect parts and still be very sick â€" because the problem is in its **software**. You learned the common software faults (slow performance, malware, boot problems, driver issues, full disk) and the Windows tools that find them: **Task Manager** (what is using CPU, memory and disk right now), **Startup Apps** (what launches automatically), **Event Viewer** (the history of errors), `sfc /scannow` (repairing corrupted system files), **chkdsk**, **Disk Cleanup**, **System Restore**, **Safe Mode**, and **Device Manager** (drivers). You also learned the safe approach to malware â€" disconnect, full scan, quarantine, restart, re-scan, change passwords, educate â€" and, most importantly, how to decide whether a fault is **software or hardware** before you act.

Five important takeaways:

1. A software fault is fixed by cleaning/correcting/restoring â€" never by replacing parts blindly.
2. **Back up first.** Never format before you have a backup and a diagnosis.
3. Task Manager shows the present; Event Viewer shows the past; `sfc` repairs system files.
4. Safe malware removal: disconnect â†' **full scan** â†' quarantine â†' restart â†' re-scan â†' passwords â†' educate.
5. Decide **software or hardware first**; a slow machine with clicking may be a failing drive.

Questions to help you recognise what you can now do:

- Can I open Task Manager and name the biggest culprit?
- Can I run `sfc /scannow` correctly as Administrator?
- Can I explain the safe malware-removal steps in order?
- Can I use Device Manager to fix a device that stopped working?
- Can I tell whether a fault is software or hardware and say why?

Safety self-check (tick honestly):

- ☐ I did not click unknown pop-ups or links.
- ☐ I worked only on machines I had permission to use.
- ☐ I asked the instructor when unsure.

Connection to the next session.
You can now find faults â€" hardware and software. But finding a fault is only half the job. Next session, Session 7 â€" Component Replacement, you will learn to **choose compatible parts and physically replace them** â€" RAM, storage (including upgrading an old hard drive to a fast SSD), power supplies and GPUs â€" and verify the machine afterwards. Diagnosis told you *what* is wrong; replacement makes it *right*.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner opens Task Manager **first** when investigating slowness, and reads the actual percentages.
- The learner checks for a **hardware** cause (S.M.A.R.T., temperature) even when the problem "looks like software".
- The learner runs `sfc /scannow` in an **Administrator** window without being reminded.
- The learner follows the safe malware sequence in order and refuses to disable antivirus.
- The learner backs up before any destructive action and says so out loud.
- The learner explains the difference between "likely" and "confirmed".

Common misconceptions to watch for and correct:

- "Slow always means virus." â†' Could be startup programs, a full/failing disk, too little RAM, or heat.
- "Formatting is the fastest fix." â†' It destroys data and may not fix a hardware cause.
- "`sfc /scannow` removes viruses." â†' No; it repairs system files only.
- "Any free driver-updater website is fine." â†' Use Windows Update or the manufacturer's official site.
- "Safe Mode is a permanent way to use the PC." â†' It is a temporary rescue tool.

Suggested remedial exercises for learners who struggle:

1. **Tool-naming drill:** show five symptoms and have the learner name the right tool for each, repeatedly, until fluent.
2. **Order-the-steps drill:** shuffle the malware-removal steps on cards and have the learner arrange them correctly.
3. **Present-vs-past drill:** give scenarios and ask "which tool â€" Task Manager or Event Viewer â€" tells me this?"

Differentiation:

- **Level 1 (supported):** provide a step-by-step tool card with the exact clicks, and a partially completed report form; pair with a stronger learner.
- **Level 2 (standard):** the exercise as written.
- **Level 3 (challenge):** present a **mixed** fault (e.g., a full disk *and* a suspicious process *and* a yellow-triangle device) and require the learner to prioritise the fixes and justify the order, including a hardware sanity check.

Timing guidance:
`sfc /scannow` and antivirus full scans are slow. Start `sfc` early in the practical and let it run while learners do other checks; do **not** let a scan consume the whole session. Protect the guided practical (Section H). If short on time, convert the case study to a two-minute discussion and keep the independent exercise, which is where the real diagnosis skill is practised.

APPENDIX 6A â€" 5-MINUTE HARDWARE-vs-SOFTWARE RECAP (only if needed)

If learners cannot distinguish hardware from software faults, do this before Section D.

1. Write on the board: **Hardware = physical part. Software = instructions and data.**

3. Quick drill â€" call out and have the class answer "H or S":

   - "Hard drive making a clicking sound." (H)
   - "Pop-ups on the desktop." (S)
   - "No power at all." (H)
   - "Sound stopped after a Windows update." (S)
   - "CPU at 95 Â°C." (H)
   - "USB flash drive infected from a friend." (S) Confirm each learner can answer before continuing.

END OF SESSION 6 â€" End of Week 2 (Diagnostics)

Next: Session 7 â€" Component Replacement.

SESSION 7 â€" COMPONENT REPLACEMENT

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 7 â€" COMPONENT REPLACEMENT

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	7 of 12
Session title	Component Replacement
Week / Module	Week 3 â€" Module: Repair and Maintenance
Recommended duration	105 minutes
Difficulty level	Beginner (hands-on, safety-critical)
Prerequisite knowledge	Sessions 1â€"6: components, safety and tools, disassembly/reassembly, the diagnostic workflow, hardware diagnostics, and software diagnostics. If a learner cannot demonstrate safe case-opening and ESD precautions, give the 10-minute safety refresher in Appendix 7A before this session.
Connection to previous work	Sessions 5 and 6 taught you to **find** the fault. This session teaches you to **fix** it by choosing compatible parts and physically replacing them, then verifying the machine works.
Preparation before class	One or more practice desktops that may be opened; matching **known-good** replacement RAM (same generation as the practice machines) and a spare known-good stick; at least one **2.5-inch SSD** and a matching **SATA cable + power cable**, or a spare HDD; a known-good PSU of suitable wattage for the practice machine; thermal paste; cleaning cloths; anti-static wrist straps (one per learner); screwdrivers with the correct heads; a **known-good** USB or SATA cable; a bootable Windows USB (for the OS-reinstall branch) and, if available, a **cloning tool** (a program that copies one drive to another) pre-installed on a bootable USB or on the running machine. Printed **Compatibility Check Sheet** and **Replacement Service Log** (one per learner).

Important wording used in this session.

- A **component** is a physical part of the computer (RAM, drive, PSU, GPU, CPU cooler).
- To **verify** a replacement means to power on the machine and *confirm* that the new part is detected and working correctly â€" never assume it just because it fitted.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain** why a replacement part must be **compatible** with the machine, and check compatibility for RAM (generation, speed, size), storage (interface) and PSU (wattage and connectors).
2. **State** the difference between a **hard disk drive (HDD)** and a **solid-state drive (SSD)**, and between **SATA** and **NVMe**, and choose the correct one for a given machine.

the old drive or **installing Windows** on the new drive.

5. **Explain "clone" and "fresh install"**, including when each is appropriate and why backing up comes first.
6. **Replace a power supply safely** using a known-good unit of suitable wattage, reconnecting every cable and double-checking before power-on.
7. **Re-apply thermal paste** correctly (clean, apply a small amount, refit the cooler) and confirm improved cooling.
8. **Verify** a machine after any replacement using the tools from Sessions 5 and 6 (POST, Windows, Task Manager, HWMonitor).
9. **Follow safety and ESD procedures** at every step, and know when a task should be handled by a qualified technician.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least one practice desktop per two learners, safe to open and reassemble.
- **Replacement RAM** that matches the practice machines' generation (e.g., DDR4 if the boards use DDR4). Never mix mismatched generations.
- A **known-good** spare RAM stick kept aside as the "test" stick.
- At least one **2.5-inch SATA SSD** with a **SATA data cable** and a **power cable**; or a spare known-good HDD if SSDs are not available.
- One known-good **PSU** of suitable wattage for the practice machine, plus its cables.
- A tube of **thermal paste** (a little goes a long way).
- Cleaning cloths and **isopropyl alcohol** (for cleaning old paste).
- **Anti-static wrist straps** (one per learner), screwdrivers with correct heads, tweezers, a small parts tray, a flashlight.
- A **bootable Windows USB** (for the fresh-install branch).
- Optional: a bootable **cloning tool** (a program that copies one drive to another) on a USB or installed, and a USB-to-SATA adapter cable if available (useful for cloning from an old drive that is still readable).

Software

- Windows 10/11 installation media.
- A cloning tool (optional), used under supervision.
- CrystalDiskInfo and HWMonitor (from Sessions 5â€“6) for verification.

Printed materials

- Compatibility Check Sheet (one per learner).
- Replacement Service Log (one per learner).

Safety equipment

- Anti-static wrist straps, safety glasses, first aid kit.
- Fire safety: know the location of the fire extinguisher and the first-aid kit.

Internet

- Optional, for looking up a motherboard's supported RAM or a PSU wattage calculator. Do not depend on it; the instructor prepares compatibility notes in advance.

Alternative for learners who lack a facility

- Desktop replacement is best learnt on a desktop. Learners without access to a personal desktop share a CEA practice machine. Learners who own a laptop may observe laptop RAM/SSD replacement when the instructor demonstrates it (laptop parts are smaller and differ by model).

Suggested visuals:



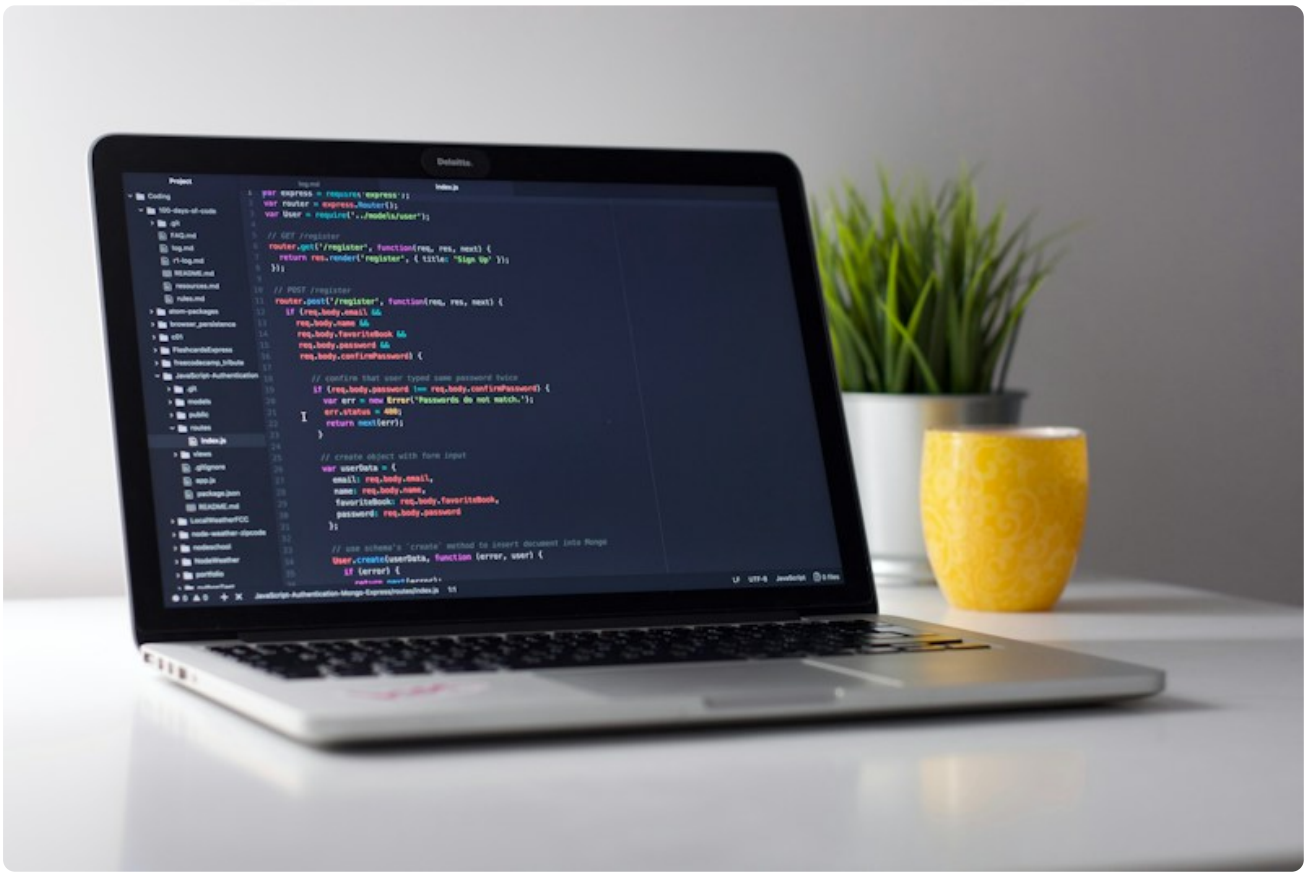
- â€" Fitting a compatible upgrade part



- â€" Replacement carried out with ESD protection



- â€" Right screwdriver for the job



- — Laptop component replaced with a compatible part



— Compatibility and spec check before fitting the part

D. OPENING ACTIVITY (6 minutes)

Instructor setup:

On the bench, place two objects side by side: an old mechanical **hard drive (HDD)** and a **solid-state drive (SSD)** of the same size. Also write on the board the price of a typical small SSD versus the cost of a new budget computer (use realistic Port Harcourt market figures the academy has prepared â€" do not invent exact prices).

What the instructor says:

> *"Good morning. This week you stop watching and start doing. Last week you learned what is wrong. Today you make it right.*
>
> *Look at these two things. One is an old spinning hard drive. The other is a solid-state drive â€" an SSD. Same size, similar price. But this one [SSD] can wake an old, tired computer up so well that the owner will swear you replaced the whole machine. Customers love this job more than almost any other, because it is cheap and the change is dramatic.*
>
> *Here is the question for today: How do you know the right part to buy? Buy the wrong RAM and it will not fit. Buy the wrong SSD and it will not connect. Replace a power supply wrongly and you can damage the whole computer â€" or hurt yourself.*
>
> *Today you learn to*
>
> ***choose compatible parts**, to **replace them safely**, and to **verify** that the machine is truly fixed. This is the day you become a technician, not just a diagnoser."*

Expected learner response:

Learners will be interested in the SSD speed claim; some may ask "how much does an SSD cost?" or "can my laptop take one?". Accept the questions and say they are exactly what a technician must learn to answer for customers.

part, and it sets the safety tone for a hands-on session where mistakes have real consequences.

E. PREREQUISITE REVIEW (7 minutes)

Run as a fast oral round-robin. Put the questions on the projector.

Question 1:
"What are the two ESD safety steps before you touch any internal component?"

- Expected answer: **Power off and unplug** the machine; **wear an anti-static wrist strap** (or touch a bare metal part of the case first).
- Common gap: learners remember the strap but forget to unplug. Say: both, every time.

Question 2:
"In what order did we remove components during teardown?"

- Expected answer: a sensible order â€" e.g., remove power cables first, then the GPU, then RAM, then storage, and the CPU cooler only when needed (or the reverse for reassembly).
- Common gap: forgetting to disconnect power first. Emphasise: **power first, always**.

Question 3:
"What is the five-step diagnostic workflow?"

- Expected answer: **Observe â†' Ask â†' Hypothesise â†' Test â†' Resolve** (then document).

Question 4:
"A machine is slow and CrystalDiskInfo shows a healthy drive, RAM passes, and temperatures are normal. What might make it fast again without replacing it?"

- Expected answer: cleaning up software (startup programs, full disk, malware) from Session 6 â€" and, if the drive is an old HDD, replacing it with an **SSD** (today's topic) even though it is not "faulty".

Question 5:
"Why must we never open a PSU, and how do we test one?"

- Expected answer: **lethal stored charge**; test by **swap** with a known-good PSU, never by opening.

Bridge to today (say this):

> *"Diagnosis is detective work. Replacement is surgery. Before a surgeon operates, they check that the new part matches the patient â€" blood type, size, and so on. Today, 'matching' means RAM generation and speed, storage interface, and PSU wattage. Get the match wrong and the operation fails."*

Gap handling:
If more than a third of the class cannot state the ESD steps in Question 1, stop and run the Appendix 7A safety refresher now. This session must not proceed with learners who cannot work safely.

F. FULL LECTURE NOTES

F.1 Why compatibility matters â€" the foundation of replacement

The idea.
Not every part fits every computer. Before buying or fitting any part, you must confirm it is **compatible** with the machine.

Plain-English definition.
Compatible means the part is of the correct type and size to work with that specific motherboard, case or system. An incompatible part will not fit, will not connect, or will fit but not work (or even damage something).

matched power supply) can damage the computer. Checking compatibility *first* is what separates a professional from a guesser.

Analogy.
Buying a replacement part is like buying shoes: the right brand is not enough â€” you need the right **size**. A beautiful shoe that is the wrong size is useless. Compatibility is the "size" of a computer part.

What we always check, by part:

- **RAM:** the **generation** (DDR3, DDR4 or DDR5), the size (GB per stick), and ideally the **speed** (MHz) and whether it is a **desktop (DIMM)** or **laptop (SO-DIMM)** stick.
- **Storage:** the **interface** (SATA or NVMe) and the physical size/format (2.5-inch, 3.5-inch, M.2).
- **PSU:** the **wattage** (must be enough and not excessive) and the **connectors** the machine needs.
- **GPU:** the **slot type (PCIe)**, the physical length that will fit the case, and whether the PSU can supply enough power.
- **Laptop parts:** model-specific; always match the exact laptop model.

Important misconception.
"Any RAM will fit any computer." False. **DDR3, DDR4 and DDR5 are different shapes and are not interchangeable.** A DDR4 stick will not go into a DDR3 slot, and forcing it will damage the slot.

Limitation.
Even compatible parts can differ in quality; a cheap part of the right type may fail early. Compatibility is necessary, not sufficient â€” quality matters too.

F.2 Choosing RAM â€” generation, speed and size

Plain-English definition.
RAM (Random Access Memory) is the fast working memory that holds whatever the computer is doing right now. It comes in **sticks**. More RAM lets you run more programs at once before the machine slows down.

The three things to match:

1. **Generation: DDR3**, **DDR4** or **DDR5**. These are different physical shapes and **cannot** be mixed. Check the motherboard's manual or look up the model to know which generation the machine uses.
2. **Size:** how many gigabytes (GB) per stick, and the total the machine supports. A 32-bit version of Windows can only use about 4 GB, so extra RAM is wasted there; 64-bit Windows uses more.
3. **Speed:** measured in **MHz** (e.g., 2666 MHz, 3200 MHz). Ideally use the same speed as the existing RAM. If you mix speeds, the machine usually runs all sticks at the **slowest** speed â€” it should still work, but matching is best.

Desktop vs laptop.
Desktops use **DIMM** (longer) sticks; laptops use **SO-DIMM** (shorter) sticks. They are not interchangeable.

Running in pairs (dual-channel).
Many motherboards run faster when two matched sticks are installed in the correct paired slots (often the same colour slots). This is optional for beginners but good practice.

Realistic Port Harcourt scenario.
A shop owner's desktop with 4 GB RAM slows to a crawl when he opens his accounting program and a browser with many tabs. The technician checks the motherboard is DDR4, buys one matching 8 GB DDR4 stick of the same speed, installs it, and the machine runs smoothly. The cost is a fraction of a new computer.

Analogy.
RAM is the size of a trader's work-table. A small table cluttering up with goods slows every task; a bigger table lets you spread out and work faster. But the table must fit your shop â€” you cannot buy a table bigger than the room.

Important misconception.
"More RAM always makes it faster." It helps when the machine is *short* of memory (many programs open, disk light always busy). If the real problem is a slow HDD or a virus, adding RAM will not fix it.

GB) beyond which extra RAM is not recognised. Always check first.

F.3 Choosing storage â€” HDD vs SSD, SATA vs NVMe

Plain-English definitions.

- **HDD (Hard Disk Drive):** stores data on spinning metal platters with a moving head. Cheap per gigabyte, but slower and mechanical (it can click and wear out).
- **SSD (Solid-State Drive):** stores data in memory chips with **no moving parts**. Much faster, quieter and more shock-resistant. More expensive per gigabyte, but the speed difference is huge.
- **SATA:** a common connection type used by both 2.5-inch/3.5-inch HDDs and 2.5-inch SSDs. It uses a flat **SATA data cable** and a **SATA power cable**. Typical speed: around 6 Gb/s (the interface limit), though a mechanical HDD is far slower than that in real use.
- **NVMe:** a newer, much faster connection used by **M.2** SSDs that slot directly onto the motherboard, with no data or power cable. NVMe drives can be several times faster than SATA SSDs.

Why this matters for choosing a part.
You must match the **interface** the machine supports:

- If the machine has SATA ports and drive bays â†’ a **2.5-inch SATA SSD** is the safe, universal upgrade.
- If the motherboard has an **M.2 slot** that supports NVMe â†’ an **NVMe SSD** is faster (but check the slot's key type and supported length).
- A 3.5-inch bay takes a **3.5-inch HDD**; a 2.5-inch bay takes a **2.5-inch SSD/HDD** (a 2.5-inch drive can usually be mounted in a 3.5-inch bay with an adapter bracket).

Why an SSD upgrade speeds up an old PC.
An HDD must physically spin up and move a head to find data, which is slow. Windows, on start-up and when opening files, reads thousands of small pieces of data. On an HDD each read waits for the mechanical parts; on an SSD the data is found instantly. That is why replacing an HDD with an SSD makes an old computer feel new â€” the mechanical delay disappears. **The CPU and RAM are unchanged; the speed came from removing the mechanical bottleneck.**

Analogy.
An HDD is like fetching a book from a huge library by walking down long shelves. An SSD is like having the book already on your desk. The book is the same; the *time to get it* is completely different.

Important misconception.
"SSD means I can never lose data." SSDs fail too, often suddenly and with less warning than HDDs. **Backups remain essential** for both.

Limitations of SSD choice.

- An old motherboard with only SATA ports cannot use NVMe â€” you must use a SATA SSD.
- Some very old machines do not support a drive large enough to hold the customer's data comfortably; check capacity needs.
- Cloning requires the new drive to have **enough capacity** to hold everything on the old drive.

F.4 Choosing a PSU â€” wattage and connectors

Plain-English definition.
The **PSU (Power Supply Unit)** converts wall electricity into the low voltages the components need. **Wattage (W)** is its power capacity â€” how much total power it can supply.

The two things to match:

1. **Enough wattage:** the PSU must supply at least as much as the machine needs. A rough guide: a basic office desktop with no dedicated GPU needs far less than a gaming machine with a big graphics card. If the PSU is too weak, the machine

2. **The correct connectors:** the PSU must have the connectors the machine needs â€" the big **24-pin** motherboard connector, the **CPU power** connector (often 4-pin or 8-pin), **SATA power** connectors for drives, and **PCIe power** connectors for a graphics card if fitted.

Safety â€" the golden rule.
Never open a PSU. Even unplugged, its capacitors hold a **lethal charge**. A PSU is a **replace-only** item.

Realistic Port Harcourt scenario.
A customer's desktop shuts down during printing (when the printer and CPU work together). Diagnosis by swap test shows the PSU is failing. The technician checks the new PSU has enough wattage and the right connectors, fits it, and the shutdowns stop.

Analogy.
The PSU is like the generating set for a building. If the generator is too small for all the appliances, everything dims or the generator cuts out when many appliances run at once. It must be big enough *and* have the right sockets to plug everything in.

Important misconception.
"A bigger-wattage PSU always makes the machine more reliable." Wattage is capacity, not health. A failing 700 W PSU is still failing. Match the wattage sensibly and choose a quality unit.

Limitation.
Cheap "no-name" PSUs often overstate their wattage and can damage components. Buy from a reputable brand. If unsure about wattage for a big build, a technician uses a wattage calculator or the GPU maker's recommendation â€" not guesswork.

F.5 Upgrading RAM â€" the procedure

Safety first:
power off, unplug, press the power button for a second to discharge leftover power, wear your anti-static wrist strap, and work on a hard flat surface.

Steps:

1. **Confirm compatibility** using the Compatibility Check Sheet: correct generation (DDR3/4/5), desktop or laptop type, size and speed.
2. **Open the case** and locate the RAM slots on the motherboard.
3. **Remove the old stick (if replacing):** push the retaining **clips** at both ends outward until the stick pops up, then lift it out by its edges. **Do not touch the gold contacts.**
4. **Align the new stick:** note the **notch** (the off-centre gap in the row of contacts). It only fits one way. Line the notch up with the slot's key.
5. **Seat the stick:** press down **firmly and evenly** at both ends until the clips snap up into place. You should hear/feel a click. If it resists, **stop** â€" the notch is probably the wrong way round; do not force it.
6. **Run dual-channel (optional):** if fitting two sticks, use the paired slots indicated in the motherboard manual (often the same colour).
7. **Close the case**, reconnect power, power on.
8. **Verify:** the machine should POST and start. Check the new total in Windows: **Task Manager â†' Performance â†' Memory**, or the **About** page, or the BIOS. Confirm the full amount is recognised (some is reserved by the system/hardware, so a small shortfall is normal).

Analogy.
Seating RAM is like fitting a key into a lock â€" the key only goes in one way (the notch), and it needs a firm, straight push, not a twist or a shove.

Common mistake.
Forcing the stick. If it will not seat easily, it is almost always the wrong orientation or wrong generation. Back off and re-check.

F.6 Replacing storage â€" monitoring, cloning vs fresh install

cloning can fail, and a fresh install erases the old system.

Definition: cloning.
Cloning means copying the entire contents of the old drive (Windows, programs, settings and files) to the new drive, so the machine starts up exactly as before but on the new hardware. It saves the customer from reinstalling everything.

Definition: fresh install.
A **fresh install** means putting a brand-new copy of Windows on the new drive (Session 9), then reinstalling the customer's programs and restoring their files from backup. It is cleaner and often faster, but it is more work and the customer must have their data backed up.

When to clone vs fresh install:

Situation	Better choice
Old drive still readable, customer wants everything unchanged	**Clone**
Old drive is failing and barely reads (may fail mid-clone)	Back up files; **fresh install** is safer
Customer's Windows is full of junk/malware	**Fresh install** (cleaner)
Customer has no backup and the drive is healthy	**Clone** (but still back up first if possible)

Steps to replace the drive:

1. **Back up** the data (clone tool copies everything, but backup protects against clone failure).
2. Power off, unplug, discharge, strap on, open the case.
3. **Locate the drive bay** and the current drive.
4. **Disconnect** the **SATA data cable** and the **SATA power cable** from the drive.
5. **Remove the screws** holding the drive (note where they go; use a tray).
6. **Slide the old drive out** and set it aside carefully (handle it flat; do not drop an HDD).
7. **Fit the new drive** into the bay (use an adapter bracket if a 2.5-inch drive goes into a 3.5-inch bay). Secure the screws.
8. **Reconnect** the SATA data cable (to a motherboard SATA port) and the SATA power cable.
9. **Close the case**, reconnect power, power on.
10. **Now either clone or install Windows:**
    - **Clone path:** boot from the cloning tool (or run it from Windows, with the old drive attached via a USB adapter) and copy old â†' new. Then set the new drive as the boot drive and remove the old one. Verify it boots and everything is present.
    - **Fresh-install path:** boot from the Windows USB and install Windows on the new drive (Session 9), then reinstall programs and restore files.
11. **Verify:** the machine boots from the new drive, Windows runs, and the drive shows up as healthy (CrystalDiskInfo) and with the expected capacity.

Analogy.
Cloning is like moving your whole house to a new building â€" every piece of furniture already in place. A fresh install is like moving into an empty new building and furnishing it yourself. Cloning is quicker to settle in; a fresh install gives you a clean start.

Important misconception.
"Cloning always works." It can fail â€" especially with a dying source drive, a drive that is larger than the destination, or some Windows/partition setups. That is why **backing up first** is not optional.

Limitation.
Cloning needs the new drive to be **at least as large** as the used space on the old drive. If the old drive is full, a same-size new drive may not fit the clone.

Power off, **unplug the power cable from the wall and from the PSU**, press the power button to discharge, wear your anti-static wrist strap. **Never open the PSU case. Never work with it plugged in.**

Steps:

1. **Confirm compatibility:** the new PSU has **enough wattage** and the **correct connectors** (24-pin, CPU power, SATA power, PCIe power if needed).
2. Open the case. **Photograph or note every cable** before removing it â€" a phone photo is a technician's best friend.
3. **Disconnect ALL PSU cables** from the motherboard and every component: the 24-pin, the CPU power, SATA power to each drive, and PCIe power to the GPU. Also disconnect any cables to extra fans or lights.
4. **Remove the four screws** at the back of the case holding the PSU (usually at the top-back or bottom-back).
5. **Slide the old PSU out** and set it aside. Remember: it is not repairable â€" it is going to be replaced.
6. **Slide the new PSU into place** and fit the four screws.
7. **Reconnect every cable**, following your photo/notes: 24-pin first, then CPU power, then SATA power to drives, then PCIe to the GPU. Make sure each connector is **fully seated** and not forced into the wrong socket.
8. **Double-check every connection.** Count them against the photo. A missed cable (especially the CPU power) is the number-one cause of "it will not start".
9. Keep cables tidy so they do not touch fan blades.
10. **Close the case, reconnect the power cable, and power on.**
11. **Verify:** the machine POSTs, fans spin, and it reaches Windows/shows a picture. Check that all drives are detected.

Analogy.
Replacing a PSU is like replacing the main breaker of a house: you shut off the supply, label every wire, fit the new box, and reconnect each wire to exactly the right place. One wrong wire and things do not work (or worse).

Important misconception.
"The PSU plugs in one way, so I cannot make a mistake." You can plug the **wrong connector into the wrong socket** with force, especially PCIe power into CPU power or vice versa. Check shapes â€" do not force.

Limitation.
Modular PSUs (where cables detach from the PSU) mean you must use **only the cables supplied with that PSU model**. Mixing cables from different brands/models can damage components because the pin wiring differs. Never reuse one PSU's cables with a different PSU.

F.8 Re-applying thermal paste correctly

Plain-English definition.
Thermal paste (also called thermal compound) is a grey paste applied between the CPU and its **heatsink/fan** (the cooler). It fills the tiny invisible gaps between the two metal surfaces so heat can pass from the CPU into the cooler more efficiently.

Why it exists.
Metal surfaces look flat but are microscopically rough. Air is a poor heat conductor. The paste fills those gaps, giving a much better heat path. It dries out over years, so replacing it every so often restores cooling.

When to re-paste:
when the CPU and cooler are separated (e.g., during a cleaning or a cooler replacement), or when temperatures have crept up over the years despite clean fans.

Steps:

1. Power off, unplug, discharge, wrist strap on.
2. Remove the CPU cooler (unscrew or unclip it â€" types differ; note the order).
3. **Clean** the old paste off both the CPU (top surface) and the cooler base using isopropyl alcohol and a lint-free cloth or cotton bud, until both are clean and dry. **Do not bend the CPU pins** (on older socket types the pins are on the CPU; on newer ones they are in the socket â€" be careful either way).

a small line; the point is a *small* amount that will spread under pressure.

6. **Refit the cooler**, pressing evenly so the paste spreads. Tighten screws gently and evenly (a little on each screw in turn, not one fully tight).
7. **Reconnect the cooler's fan cable** to the CPU fan header on the motherboard.
8. **Power on and verify** with **HWMonitor**: temperatures should be normal or lower than before.

Analogy.
Thermal paste is like adding a thin layer of cooking oil between a frying pan and the burner â€" it helps heat travel evenly where the surfaces do not perfectly touch. Too little and heat does not transfer; too much and it makes a mess and can squeeze out.

Important misconception.
"More paste is better." No â€" **too much** can squeeze out onto the motherboard and can actually reduce contact. A small pea-sized amount is correct.

Safety limitation.
Removing a cooler on some machines also removes the CPU (on a few older designs). If the CPU comes out stuck to the cooler, **do not pull it apart** â€" ask the instructor. Twisting or pulling can bend pins and destroy the CPU.

F.9 Verifying after replacement â€" the professional finish

The idea.
A replacement is not finished when the part is fitted; it is finished when the machine is **verified working**. Never hand back a machine you have not tested.

A verification checklist for any replacement:

1. **POST and boot:** power on; the machine completes POST and reaches Windows (or the installer). No beeps, no errors.
2. **Detected correctly:** the OS sees the new part.
   - RAM: Task Manager â†' Performance â†' Memory shows the new total.
   - Storage: This PC / Disk Management shows the new drive with the right capacity; CrystalDiskInfo shows it as **Good**.
   - PSU: all drives and the GPU power up; no random shutdowns during a quick load test.
   - Thermal paste: HWMonitor shows normal (or improved) CPU temperature.
3. **Run for a while:** use the machine for a few minutes; open programs; watch for crashes, restarts or overheating.
4. **Document:** record the old part, the new part, the reason, and the verification results on the Replacement Service Log.
5. **Return the machine to the customer** with a plain-language explanation and any advice (e.g., back up regularly).

Important misconception.
"If it powers on, it is fixed." Powering on is step one. A drive might not be detected, a RAM stick might be faulty, or the machine might overheat ten minutes later. Verify *and* test under use.

F.10 Summary of the session

- **Compatible** means the part matches the machine (RAM generation/size/speed; storage interface; PSU wattage/connectors; laptop = model-specific).
- **RAM:** match DDR generation, DIMM vs SO-DIMM, size and speed; seat by the **notch**, click the clips, verify the total.
- **Storage: HDD** is mechanical and slow; **SSD** is fast; **SATA** uses cables; **NVMe** uses an M.2 slot. An SSD upgrade speeds up an old PC by removing the mechanical bottleneck.
- **Cloning** copies everything to the new drive; a **fresh install** is cleaner but more work. **Back up first either way.**
- **PSU:** match wattage and connectors; **never open it**; reconnect **every** cable and double-check.
- **Thermal paste:** clean off the old, apply a **pea-sized** amount, refit the cooler, check temperatures.
- **Verify** every replacement before returning the machine.

G.1 What the instructor prepares

- A practice desktop on the bench, opened, with wrist straps available.
- A matching replacement RAM stick and a known-good test stick.
- A 2.5-inch SSD with SATA data and power cables (drive replacement demonstration).
- A known-good PSU (for the connector explanation; the instructor may demonstrate or narrate the replacement depending on time and safety).
- Thermal paste, cleaning materials.
- A bootable Windows USB and, if available, a cloning tool.
- The Compatibility Check Sheet and Replacement Service Log on the projector.

G.2 Demonstration 1 â€” Compatibility check (3 minutes)

Say:
"Before I touch a single screw, I check that the part fits."

Steps:

1. Show the **Compatibility Check Sheet**. Fill in the practice machine's details: RAM generation (e.g., DDR4), memory type (DIMM), storage interface (SATA), PSU wattage needed.
2. Hold up the replacement RAM. **Say:** "DDR4, DIMM, correct speed â€” this will fit. If it were DDR3, I would stop here, because DDR3 and DDR4 are different shapes."
3. Hold up the SSD. **Say:** "2.5-inch SATA SSD. The board has SATA ports, so this fits. If the board had an M.2 NVMe slot, I could go faster."

Expected results:
the sheet completed; learners able to state one reason a part is compatible.

Reasoning to say aloud:
"Checking first protects the customer's money and my reputation."

Questions:
"What are the two things we match for a PSU?" (Wattage and connectors.)

G.3 Demonstration 2 â€” RAM upgrade (4 minutes)

Say:
"Watch the notch â€” it is the safety feature."

Steps:

1. Power off, unplug, press power to discharge, put on the wrist strap.
2. Push the RAM clips outward; remove the old stick by its edges.
3. Hold up the new stick and point to the **notch**. **Say:** "The notch is off-centre. It only goes in one way. If I try the wrong way, it will not seat â€” and I will not force it."
4. Align and press firmly at both ends until the clips snap up. **Say:** "Both clips up = fully seated."
5. Close the case, reconnect power, power on.
6. Show the total in **Task Manager â†’ Performance â†’ Memory**. **Say:** "Verified. The machine sees the new amount."

Expected results:
the machine boots and shows the increased memory.

Common errors to show and correct:

- Trying to force the stick the wrong way (demonstrate stopping).
- Touching the gold contacts.
- Forgetting the wrist strap.

G.4 Demonstration 3 â€” Drive replacement and clone/install (5 minutes)

Say:
"This is the upgrade customers love."

Steps:

1. Show the old HDD and the new SSD side by side. **Say:** "Same size, but one has moving parts and one does not. That is the source of the speed."
2. Power off, unplug, discharge, strap on.
3. Disconnect the **SATA data** and **SATA power** cables from the old drive. Point them out clearly.
4. Remove the screws, slide out the old drive, fit the new SSD, secure it, reconnect the SATA data and power cables.
5. **Say:** "Two choices now: **clone** â€” copy everything across â€” or **fresh install**. If I clone, the customer's Windows, programs and files come along. If I install fresh, it is cleaner but I must reinstall everything."
6. If a cloning tool is available, show the first screen (or describe the process): source drive â†' destination drive, start.
7. Otherwise, begin booting the **Windows USB** and show the install screen (full install is Session 9).
8. Close up, power on, and show the new drive detected and CrystalDiskInfo reporting **Good**.

Expected results:
the new drive is detected and healthy; the clone/install path is understood.

Common errors to show and correct:

- Not backing up before starting. **Say:** "If the clone fails and we wiped nothing yet, the old drive still has the data. But if we had deleted it first, the customer's files would be gone. Backup first."
- Connecting the data cable to the wrong port or leaving the power cable off.
- Handling the HDD roughly.

Questions:
"When would a fresh install be better than cloning?" (When the old Windows is full of junk/malware, or the source drive is barely readable.)

G.5 Demonstration 4 â€” Thermal paste and PSU connector safety (3 minutes)

Steps (thermal paste):

1. Show the dry old paste on a spare CPU/cooler. Clean it with isopropyl alcohol and a cloth.
2. Apply a **pea-sized** dot in the centre. **Say:** "Small. Too much squeezes out and can make things worse."
3. Show how the cooler presses down and spreads it. Reconnect the fan cable.

Steps (PSU connectors â€” narration/short demonstration):

1. Show the **24-pin**, **CPU power**, **SATA power** and **PCIe** connectors. **Say:** "Each shape is different so you cannot easily connect the wrong one â€” but with force you can. Never force."
2. **Say loudly:** "We NEVER open a PSU. Even unplugged it can kill you. We replace it, we do not open it."
3. If demonstrating removal, unplug, discharge, note/photo every cable, disconnect all, remove four screws, slide out, slide in the new one, reconnect all, double-check, power on.

Expected results:
learners see correct paste amount and understand PSU connector matching and the danger.

Questions:
"Why must we use only the cables that come with the PSU?" (Different models wire the pins differently; mixing can damage parts.)

Objective:

With the instructor guiding, each learner (or pair) replaces a part on a practice machine â€" a **RAM upgrade** and/or a **storage drive replacement** â€" and verifies the machine afterwards.

Safety first â€" read aloud before starting:

- **Power off, unplug**, press the power button to discharge, **wear your anti-static wrist strap.**
- Handle RAM and drives **by the edges**; never touch gold contacts.
- **Never open the PSU.** If a PSU task is set, it is a swap only.
- **Back up data before replacing any storage drive.**
- Work on a hard, flat surface; keep screws in a tray.
- Stop and call the instructor if a part resists or if you are unsure.

Procedure (instructor calls each step; learners follow; instructor walks the room):

1. **Compatibility check.** Complete the **Compatibility Check Sheet** for the practice machine: RAM generation and type, storage interface, PSU wattage needed. Confirm with the instructor before proceeding.
2. Power off, unplug, discharge, strap on. Open the case.
3. **RAM upgrade.** Locate the RAM slots. Push the clips and remove the old stick by its edges. Note the **notch**. Align the new stick and press firmly at both ends until the clips snap. Record what you did.
4. **Pause and predict.** "After this, what number will Task Manager show for Memory?" Write your prediction, then power on (after closing up) and check.
5. **Verify RAM.** Close the case, reconnect power, boot. Open **Task Manager â†' Performance â†' Memory** and record the total. Did it match your prediction?
6. **Storage drive replacement.** Power off, unplug, discharge. Disconnect the **SATA data** and **SATA power** cables from the old drive. Remove the screws and slide the old drive out. Fit the new SSD, secure it, and reconnect the **data** and **power** cables.
7. **Pause and predict.** "Will the machine boot straight into Windows, or will it ask for an operating system?" Write your prediction. (If the new drive is blank, there is no Windows on it yet.)
8. **Verify storage.** Power on. Check that the new drive is detected (BIOS and/or Windows Disk Management) and, if Windows is running, that **CrystalDiskInfo** shows the new drive as **Good** with the correct capacity.
9. **Thermal paste (if the instructor sets this task):** remove the CPU cooler, clean the old paste with isopropyl alcohol, apply a **pea-sized** dot, refit the cooler, reconnect the fan cable. Record the CPU temperature before and after using **HWMonitor**.
10. **PSU swap (instructor-supervised only, if time allows):** note every cable; disconnect all; remove four screws; slide out; fit the known-good PSU; reconnect every cable; **double-check against your notes**; power on. Verify POST. **Never open the old PSU.**
11. Complete the **Replacement Service Log**.

Expected results:

- Compatibility sheet completed and confirmed.
- RAM replaced, clips seated, new total visible in Windows.
- Drive replaced, cables connected, new drive detected and healthy (Good in CrystalDiskInfo); the machine either boots (if the drive was cloned/OS installed) or shows an OS prompt (if blank).
- Thermal paste replaced and temperatures checked (if this task was set).
- Machine POSTs and runs; service log completed.

Points to pause on and predict:

- After step 3: "What tells you the stick is fully seated?" (Both clips snap up.)
- After step 6: "Which cable is data and which is power?" (The thinner flat cable is SATA data; the wider multi-coloured connector is SATA power.)
- After step 7: "Why might the machine not boot?" (The new drive is blank until Windows is cloned or installed.)

Troubleshooting while walking the room:

- No picture / beeps after RAM change â†’ the stick may not be fully seated or is the wrong type; reseat it; check the notch.
- New drive not detected â†’ check both SATA data and power cables; try a different SATA port; confirm cables are fully pushed in.
- Machine boots to "no operating system" â†’ expected if the drive is blank; the OS must be cloned or installed (Session 9).
- Temperatures high after re-paste â†’ the cooler may not be seated evenly, or too little paste; re-check the cooler fit.
- Learner forces RAM â†’ stop immediately; show the notch; reseat gently.

Completion checklist (learner ticks each):

- ☐ Compatibility Check Sheet completed and confirmed
- ☐ Safety followed: unplugged, discharged, wrist strap worn, no PSU opened
- ☐ RAM removed and new stick fully seated (both clips up)
- ☐ New RAM total verified in Windows
- ☐ Old drive removed, new drive fitted and cables (data + power) connected
- ☐ New drive detected; CrystalDiskInfo shows Good (if Windows available)
- ☐ Thermal paste replaced and temperature verified (if set)
- ☐ PSU swap completed and verified (if set; instructor-supervised)
- ☐ Replacement Service Log completed
- ☐ All tools and screws accounted for; case closed properly

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
Work alone. Read a customer scenario, choose the **compatible** parts, plan the replacement, and write the plan and the verification steps. You do **not** perform a new physical task here â€" you produce the professional **plan** a technician would use. (If time allows, the instructor may let you execute part of your plan on a practice machine.)

Scenario.
Mrs. Nwosu, who runs a small fashion business from home in Woji, brings in a desktop with the following facts:

- It is an older tower PC. The motherboard label says it supports **DDR4** memory and has **SATA** ports.
- It currently has **4 GB RAM** and a **500 GB mechanical hard drive (HDD)**.
- Windows and the customer's files are on the HDD, which is **healthy** (CrystalDiskInfo: Good).
- Complaint: "It is very slow to start and to open files, and when I open my design program and a browser it freezes."
- She wants to keep **all her files and programs exactly as they are**, and she has a **1 TB external drive she has never used** for backups.
- Her budget is limited.

Requirements:

1. **Choose the RAM:** state the generation, type (DIMM/SO-DIMM), a sensible size, and why. (Remember 32-bit vs 64-bit Windows for capacity.)
2. **Choose the storage:** decide between an **HDD and an SSD**, choose the **interface** (SATA or NVMe) based on her machine, and state the capacity you would recommend.
3. **Decide clone or fresh install**, and justify it using her wish to "keep everything exactly as they are".
4. **List the key steps** of the replacement, in order, including the two safety must-dos at the start.
5. **State your verification steps** â€" how you will prove the machine is fixed after the replacement.
6. **State one backup action** you will take before touching the drive, and why.

Time allowance:
15 minutes.

Constraints:

- Write in a way you could explain to Mrs. Nwosu in plain language.

Success criteria:

- RAM choice is DDR4, DIMM (desktop), sensible size, correct reasoning.
- Storage choice is a **SATA SSD** (the machine has no NVMe slot), sensible capacity.
- **Clone** is chosen and justified (she wants everything unchanged), with a backup first.
- Steps are in the correct order and include safety.
- Verification is specific (Task Manager memory, CrystalDiskInfo Good, faster boot, no freezing).

Hints available if stuck:

- Hint 1: "Keep everything exactly as they are" points strongly to one of clone/fresh install.
- Hint 2: The machine does not have an NVMe slot, so which SSD type must you choose?
- Hint 3: Before copying or installing anything, what should she do with that unused 1 TB external drive?

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

1. RAM choice.
DDR4, **DIMM** (desktop form factor), a sensible **8 GB or 16 GB** stick (or a matching pair). Because the machine has only 4 GB and freezes when several programs are open, more memory will help. Note: if the Windows is **32-bit**, it can only use about 4 GB, so check the Windows type first â€" if 32-bit, explain that upgrading RAM alone will not help much and a 64-bit Windows would be needed. **(Key teaching point: check 32-bit vs 64-bit.)**

2. Storage choice.
A **2.5-inch SATA SSD** (the board has SATA ports, not NVMe). Recommend a capacity at least as large as her used space (e.g., 500 GB or 1 TB). Explain that the SSD removes the mechanical bottleneck that makes the HDD slow.

3. Clone.
Choose **cloning**, because she wants to keep all her files and programs exactly as they are. (A fresh install would require reinstalling everything and restoring her files.) **But back up first** â€" cloning can fail.

4. Key steps (in order):

1. **Back up** her data to the 1 TB external drive (and/or clone).
2. **Power off, unplug, discharge, wear the anti-static wrist strap.**
3. Open the case; disconnect the SATA data and power cables from the old HDD.
4. Remove the screws, slide out the HDD.
5. Fit the SSD (adapter bracket if needed), secure it, reconnect **SATA data and SATA power**.
6. Close up, power on.
7. **Clone** old HDD â†' new SSD using the cloning tool (source â†' destination).
8. Set the SSD as the boot drive; remove the old HDD (or keep it as a spare after wiping, with the customer's permission).
9. Boot from the SSD and check everything is present.

5. Verification steps.
Boot successfully from the new SSD; confirm **Task Manager â†' Performance â†' Memory** shows the new RAM total; confirm **CrystalDiskInfo** shows the SSD as **Good**; measure that start-up and file-opening are clearly faster; run the design program + browser together without freezing; open a few customer files to confirm the clone worked.

6. Backup action.
Back up her files to the 1 TB external drive before replacing the drive, because if the clone fails, the files are safe, and because it is simply professional practice: never touch a customer's storage without a backup.

Marking guidance (out of 15):

- Correct DDR4/DIMM RAM choice + sensible size + checks 32/64-bit (3)

- Steps in correct order with safety must-dos (3)
- Specific verification steps (2)
- Backup action stated and justified (1)

Accept any answer that shows correct compatibility reasoning and safety.
Penalise choosing NVMe for a SATA-only board, choosing a fresh install while ignoring "keep everything the same", or omitting the backup.

J. REAL-WORLD CASE STUDY

Situation.
Mr. Okafor runs a small cyber cafÃ© and business centre near a school in Port Harcourt. He has five desktop PCs used for typing, printing and browsing. They are about six years old: each has 4 GB RAM and a 500 GB mechanical hard drive. Customers complain that the machines take "forever" to start and freeze when two people use them. Mr. Okafor is considering buying five new computers, which he cannot really afford.

Relevant facts.

- The hard drives are **healthy** (CrystalDiskInfo: Good) â€" they are not faulty, just slow (old mechanical technology).
- The machines are **short of RAM** for modern browsing and office work.
- The CPUs and other parts are still serviceable.
- A few machines were recently diagnosed with malware, now cleaned.

The decision point.
Buy five new computers, or upgrade the five old ones?

Possible approaches.

1. Buy five brand-new desktop computers (expensive).
2. Upgrade each machine: add matching RAM and replace the HDD with a **SATA SSD** (clone the existing system so nothing is relearned).
3. Do nothing and tell customers to be patient (bad for business).

Recommended solution (for this stage).
Approach **2**. For each machine: check compatibility (DDR generation, SATA ports), add a matching RAM upgrade, and clone each HDD onto a SATA SSD. The machines will start in seconds instead of minutes, and the RAM upgrade stops the freezing. The cost is a small fraction of five new computers, and each upgrade is a job Mr. Okafor can even pay a CEA graduate to do.

Trade-offs.

- Approach 1 solves everything but costs many times more; the old CPUs would still be fine for cafÃ© work.
- Approach 2 is affordable, dramatic in results, and preserves the existing setups; it requires care (compatibility, backup, cloning) and takes time per machine.
- Approach 3 loses customers.

Professional connection.
This is exactly how a real repair business grows: offering a **cost-effective upgrade** that gives the customer most of the benefit of a new machine at a fraction of the price. The certificate holder who can correctly choose compatible parts and clone a drive is the person a cyber cafÃ© owner trusts with five machines.

Transfer exercise (ask the class):
"Mr. Okafor asks, 'Can I just put a bigger HDD instead of an SSD?' What do you tell him, and why?"

Suggested answer: "Yes, a bigger HDD will give you more space, but it will **not** fix the slowness â€" an HDD is still mechanical. For speed, the drive type matters more than the size. An SSD of the same or smaller capacity will make the machines far faster. If you need both speed and space, use an SSD for Windows and programs and keep a large HDD for storing files."

K.1 Machine will not power on after a replacement

- **Symptoms:** no lights, no fans after closing up and pressing power.
- **Likely causes:** the 24-pin or CPU power cable is loose or not reconnected; the PSU switch is off; the wall socket/cable is faulty; a cable was left disconnected during the swap.
- **How to investigate:** re-open and check the 24-pin and CPU power connectors are fully seated; check the PSU switch and power cable; compare against your photo/notes.
- **Safe corrective steps:** reseat connectors; use a working socket/cable. **Never open the PSU.**
- **How to confirm resolved:** the machine POSTs and boots.
- **Prevention:** photograph every cable before removal; double-check all connections before power-on.

K.2 RAM not detected / wrong amount shown

- **Symptoms:** the boot fails, beeps, or Windows shows less memory than fitted.
- **Likely causes:** the stick is not fully seated; wrong orientation (notch); wrong generation; a faulty stick; or the OS is 32-bit and cannot use the full amount.
- **How to investigate:** reseat and press until both clips click; confirm generation; check the OS type (32/64-bit) in Windows.
- **Safe corrective steps:** reseat correctly; replace a faulty stick; explain 32-bit limits to the customer.
- **How to confirm resolved:** Task Manager shows the expected total (minus a small normally-reserved amount).
- **Prevention:** match generation and type on the Compatibility Check Sheet first.

K.3 New drive not detected

- **Symptoms:** BIOS/Windows does not see the new drive.
- **Likely causes:** SATA data or power cable not connected; loose cable; a bad SATA port; a dead drive (rare, if known-good).
- **How to investigate:** check both cables; try a different SATA port; try a known-good cable.
- **Safe corrective steps:** reconnect both cables fully; change port/cable.
- **How to confirm resolved:** the drive appears in BIOS and in Windows Disk Management.
- **Prevention:** label the data vs power cable; push each connector fully home.

K.4 Machine boots to "no operating system" after fitting a blank drive

- **Symptoms:** POST passes, but there is no Windows.
- **Likely cause:** the new drive is blank (expected if not cloned/no OS installed).
- **How to investigate:** confirm the drive is the new one and is empty of an OS.
- **Safe corrective steps: clone** the old drive to the new one, or install Windows (Session 9) and restore data.
- **How to confirm resolved:** Windows boots from the new drive.
- **Prevention:** plan clone vs install *before* fitting; back up first.

K.5 Random shutdowns return after fitting a new PSU

- **Symptoms:** machine still shuts down under load after a PSU swap.
- **Likely causes:** the new PSU is underpowered for the load; a connector is loose; the real cause was overheating or RAM, not the PSU.
- **How to investigate:** check the PSU wattage vs the machine's needs; reseat all connectors; re-check temperatures (HWMonitor) and RAM (memory test).
- **Safe corrective steps:** fit a correctly-rated PSU; fix the actual cause.
- **How to confirm resolved:** the machine runs a heavy load without shutting down.
- **Prevention:** size the PSU correctly; diagnose thoroughly before replacing (Session 5).

K.6 Temperatures worse after re-pasting

- **How to investigate:** check the cooler is flat and tight; check the fan cable is on the CPU fan header; check the fan spins.
- **Safe corrective steps:** reseat the cooler evenly; add a small correct amount of paste; reconnect the fan cable.
- **How to confirm resolved:** HWMonitor shows normal/lower temperatures.
- **Prevention:** even tightening; pea-sized amount; always reconnect the fan cable.

K.7 Distinguishing likely from confirmed

As always: a replacement is only

confirmed good
after **verification** (POST, OS detection, temperature/health reading, and a short period of real use). Never hand a machine back based on "it powered on once".

L. SESSION ASSIGNMENT

Assignment title:
"Compatibility and Replacement Plan for a Real Upgrade"

Purpose:
To practise choosing compatible parts, planning a safe replacement, and writing a professional plan a customer could approve.

Realistic scenario:
A customer brings you a working but slow desktop (or laptop) and wants it faster **without buying a new computer**. You must produce a written **upgrade plan**: which parts to buy, why they are compatible, how you will fit them, how you will keep their data, and how you will verify the result.

Detailed instructions:

1. Choose a real machine you have access to and permission to inspect (a CEA practice machine or your own). If it is a laptop, note that parts are model-specific.
2. **Compatibility Check Sheet:** record (where known) RAM generation and type, storage interface, available drive bays/M.2 slot, and the PSU wattage needed or fitted.
3. **Choose the parts:** propose a RAM upgrade and/or a storage upgrade (SSD). For each, state **why it is compatible** and what you are matching (generation, interface, connectors).
4. **Decide clone vs fresh install** (for a storage change) and justify it, including the **backup step**.
5. **Write the steps** in order, including the two safety must-dos at the start and the PSU golden rule.
6. **Write the verification plan** â€" exactly how you will prove the machine is fixed.
7. **Write a plain-language paragraph** for the customer explaining what you will do and any risks.

Required deliverables:

- One completed **Compatibility Check Sheet**.
- A written **upgrade and replacement plan** (steps + verification + plain-language summary).
- One completed **Replacement Service Log** (the plan may be a "dry run" if you do not physically perform the upgrade).

Submission format:
Written plan and completed sheets (paper or class folder).

Deadline:
Before Session 8.

Marking rubric (20 marks):

Criterion	Marks	Description of a strong answer
Compatibility Check Sheet	3	Correct generation/type/interface/wattage recorded

RAM choice & reasoning	3	Correct generation, DIMM/SO-DIMM, sensible size, notes 32/64-bit
Storage choice & reasoning	3	Correct interface (SATA/NVMe) for the machine; sensible capacity
Clone vs fresh install + backup	3	Correct choice, justified; backup step included
Replacement steps in order	3	Correct order; safety must-dos; PSU never opened
Verification plan	3	Specific: POST, OS detection, CrystalDiskInfo, memory total, speed/no crashes
Plain-language summary	2	A customer could understand and approve it

Common mistakes to avoid:
choosing DDR3 RAM for a DDR4 board; choosing NVMe for a SATA-only board; forgetting the backup; omitting the safety steps; forgetting to verify; forcing a part.

Instructor reference solution (abbreviated example):

> ***Machine:***
> *CEA practice desktop, DDR4, SATA ports, no M.2 slot. Currently 4 GB RAM, 500 GB HDD (healthy), 64-bit Windows.* ***Compatibility sheet:*** *RAM = DDR4, DIMM; storage = SATA; PSU needed â‰ˆ (per practice-machine note) modest, no dedicated GPU.* ***RAM choice:*** *add one 8 GB DDR4 DIMM of matching speed (total 12 GB). Compatible: correct generation and desktop form factor; Windows is 64-bit so it will use the memory.* ***Storage choice: 2.5-inch SATA SSD****, 500 GBâ€"1 TB. Compatible: the board has SATA ports; the drive fits a 2.5-inch bay (adapter if needed). Reason for speed: the SSD removes the mechanical HDD bottleneck.* ***Clone vs install: Clone*** *â€" the customer wants everything unchanged; SSD capacity chosen â‰¥ used space on the HDD.* ***Backup*** *to external drive first.* ***Steps:*** *1) Back up data. 2) Power off, unplug, discharge, wear wrist strap. 3) Open case; remove RAM, fit matching stick (notch, clips click). 4) Disconnect SATA data + power from HDD; remove screws; slide out. 5) Fit SSD; reconnect SATA data + power; secure. 6) Close up; power on. 7) Clone HDD â†' SSD; set SSD as boot drive. 8) Boot and check. (Never open the PSU.)* ***Verification:*** *POST passes; Task Manager shows 12 GB (minus reserved); CrystalDiskInfo shows Good; boot and file-opening clearly faster; run design program + browser together without freezing; open sample files to confirm the clone.* ***Plain-language summary:*** *"We will add more memory so the computer can run several programs at once, and replace the old spinning hard drive with a modern solid-state drive so it starts in seconds. We will copy everything across exactly as it is first, and back up your files before we begin, so nothing is lost."*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key below.)

Q1. Short answer.
What does it mean for a replacement part to be **compatible**? **Q2. Multiple choice.** A motherboard supports **DDR4** memory. Which RAM can you fit? a) DDR3 b) DDR4 c) DDR5 d) Any of them, RAM always fits **Q3. Explain in your own words.** Why does replacing an old mechanical hard drive with a solid-state drive make a computer faster, even though the CPU and RAM are unchanged? **Q4. Practical instruction.** What are the two safety must-dos before you open a computer to replace a part? And what is the one thing you must **never** open? **Q5. Short answer.** Give one reason to **clone** a drive and one reason to do a **fresh install** instead. **Q6. Scenario.** After replacing RAM, the machine beeps and shows nothing on screen. What is the most likely cause, and what do you do? (Remember the golden rule about forcing.)

INSTRUCTOR ANSWER KEY (Q1â€"Q6)

A1.
It means the part is of the correct **type and size** to work with that specific machine â€" matching, for example, RAM generation/form factor, storage interface, or PSU wattage and connectors â€" so it will fit, connect and work (and not damage anything).

A3.
An HDD must physically spin and move a reading head to find data, so every read â€" and Windows makes thousands of small reads when starting up and opening files â€" waits for mechanical parts. An SSD has no moving parts and finds data almost instantly. Removing that mechanical wait is what makes the machine feel new; the CPU and RAM did not change.

A4.
The two must-dos: **power off and unplug** (and press power to discharge), and **wear an anti-static wrist strap** (ESD protection). The one thing you must **never** open is the **PSU** (power supply) â€" its capacitors hold a lethal charge even when unplugged.

A5.
Clone: when the old drive is healthy and the customer wants Windows, programs and files kept exactly as they are (saves reinstallation). **Fresh install:** when the old Windows is full of junk/malware, or the old drive is failing and barely readable, so a clean start is safer.

A6.
Most likely the RAM stick is **not fully seated**, or it is the **wrong type/orientation**. Do **not** force it. Power off and unplug, reseat the stick correctly (check the **notch** and press until both clips click), confirm it is the correct generation, then power on and re-check.

Marking:
award for correct concept; deduct heavily for unsafe answers (opening the PSU, forcing parts).

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned to **replace components safely** â€" the skill that turns a diagnosis into a repair. You learned that every part must be **compatible**: RAM must match the motherboard's **generation** (DDR3/4/5), type (DIMM for desktop, SO-DIMM for laptop), size and speed; storage must match the **interface** (**SATA** uses cables, **NVMe** uses an M.2 slot); and a PSU must have enough **wattage** and the right **connectors**. You learned to upgrade **RAM** by seating it on the **notch** until the clips click; to replace a **drive** and choose between **cloning** (copy everything) and a **fresh install** (clean start) â€" always **backing up first**; to replace a **PSU** by reconnecting **every** cable and double-checking, while **never opening** it; and to re-apply **thermal paste** with a **pea-sized** amount. Finally, you learned that a replacement is only done when the machine is **verified** working.

Five important takeaways:

1. **Compatibility first** â€" matching saves money and prevents damage.
2. **An SSD removes the mechanical bottleneck** and makes old PCs feel new.
3. **Back up before replacing storage**; choose clone vs fresh install on purpose.
4. **Never open a PSU**; replace it and double-check every cable.
5. **Verify** every replacement before returning the machine.

Questions to help you recognise what you can now do:

- Can I check whether a RAM stick is compatible before buying it?
- Can I fit RAM and confirm the new amount in Windows?
- Can I choose between clone and fresh install and explain why?
- Can I replace a PSU safely and reconnect every cable?
- Can I re-paste a CPU and confirm the temperature improved?

Safety self-check (tick honestly):

- ☐ I powered off, unplugged and discharged before opening the machine.
- ☐ I wore my anti-static wrist strap.
- ☐ I never opened a PSU.
- ☐ I backed up before touching a storage drive.

You can now replace parts. Next session, Session 8 â€" Cleaning and Maintenance, you will learn to **keep** computers healthy: safe dust removal, ESD care, cleaning fans, heatsinks, keyboards, ports and screens, when to re-apply thermal paste, and how to build **preventive maintenance schedules** for homes, small offices and cyber cafÃ©s, including protection from dust, humidity and power surges (surge protectors and UPS). Prevention is the cheapest repair of all.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner checks **compatibility** (generation, interface, wattage) *before* fitting anything.
- The learner seats RAM by the **notch** and confirms both clips are up, without forcing.
- The learner correctly identifies SATA **data** vs **power** cables and reconnects both.
- The learner chooses clone or fresh install with a clear reason and **backs up first**.
- The learner reconnects **every** PSU cable and double-checks before power-on.
- The learner **verifies** (POST, OS detection, CrystalDiskInfo, temperature) before declaring success.
- The learner automatically follows ESD and power-off safety and refuses to open a PSU.

Common misconceptions to watch for and correct:

- "Any RAM fits any PC." â†' DDR3/4/5 are different shapes; DIMM â‰ SO-DIMM.
- "More RAM always makes it faster." â†' Only if the machine is short of memory.
- "An SSD cannot fail." â†' It can; back up regardless.
- "Cloning always works." â†' It can fail; back up first.
- "A bigger PSU is always more reliable." â†' Wattage is capacity, not health.
- "More thermal paste is better." â†' A pea-sized amount is correct.

Suggested remedial exercises for learners who struggle:

1. **Compatibility sorting:** give ten part cards and ten machine cards; have the learner match them and explain each match.
2. **Cable tracing:** on a real machine, have the learner point to and name SATA data, SATA power, 24-pin, CPU power and PCIe power connectors.
3. **Order-the-steps drill:** shuffle the RAM and PSU replacement steps and have the learner reorder them, including the safety steps.

Differentiation:

- **Level 1 (supported):** provide a partially completed Compatibility Check Sheet and step card; the learner completes the RAM upgrade with hands-on help.
- **Level 2 (standard):** the exercise as written (RAM + drive).
- **Level 3 (challenge):** perform the full **clone** to an SSD *and* the RAM upgrade, then verify both; or replace a PSU and GPU as well, documenting every step on the service log.

Timing guidance:
This is a hands-on, safety-critical session. Protect the guided practical above all. If time is short, do the **RAM upgrade** as the core practical and demonstrate the drive replacement while learners watch; move the PSU swap to an instructor-supervised demonstration only. Never rush a safety step. Cloning can be slow â€" start it early or demonstrate the process and let learners complete it in Session 9 if needed.

APPENDIX 7A â€" 10-MINUTE SAFETY REFRESHER (only if needed)

If learners cannot demonstrate safe case-opening and ESD precautions, run this before Section D.

1. **The two must-dos** â€" the class chants them: **"Power off and unplug. Wear the anti-static wrist strap."**
2. **Discharge** â€" press the power button after unplugging to drain leftover electricity.

5. **Screws and trays** â€” keep screws in a tray; know where each goes. Demonstrate once, then have each learner demonstrate the two must-dos before they touch a machine.

END OF SESSION 7

Next: Session 8 â€” Cleaning and Maintenance.

SESSION 8 â€" CLEANING AND MAINTENANCE

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 8 â€" CLEANING AND MAINTENANCE

A. SESSION IDENTIFICATION

Item	Detail
Institution	Cyber Elias Academy (CEA), 24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State
Session number	8 of 12
Session title	Cleaning and Maintenance
Week / Module	Week 3 â€" Module: Repair and Maintenance
Recommended duration	105 minutes
Difficulty level	Beginner (hands-on, safety-critical)
Prerequisite knowledge	Sessions 1â€"7: components, safety and tools, disassembly/reassembly, the diagnostic workflow, hardware and software diagnostics, and component replacement (including thermal paste and safe handling). If a learner cannot demonstrate the two safety must-dos, run the 10-minute refresher in Appendix 8A before this session.
Connection to previous work	Session 7 taught you to **replace** parts. This session teaches you to **prevent** faults â€" keeping a machine cool, clean and protected so it lasts longer and costs the customer less. Cleaning is the cheapest repair of all.
Preparation before class	A deliberately **dusty practice desktop** (or one that can be made dusty for demonstration); canned **compressed air**; a **soft anti-static brush** (or a clean, soft paintbrush); **isopropyl alcohol** (90%+); lint-free/microfiber cloths; **cotton buds**; thermal paste; a small **screwdriver and anti-static wrist straps** (one per learner); HWMonitor installed; a **surge protector** and, if available, a **UPS** to show; an old keyboard and an old monitor/phone screen for safe cleaning practice; printed **Cleaning Checklist** and **Maintenance Plan Template** (one per learner). Optional: a copy of a PC cleaning video is not required â€" keep it live.

A note on what cleaning means here.
Cleaning a computer is not just making it *look* nice. It is a **maintenance action** that removes the main causes of overheating (dust, blocked airflow, dried thermal paste) and protects the machine from environmental risks (humidity, power surges). Done wrong, cleaning can damage a computer with **static electricity** or **liquid**; done right, it extends a machine's life by years.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain** how dust, heat and blocked airflow damage a computer, and how these lead to slowness, shutdowns and part failure.
2. **List** the correct cleaning tools and **state** which tools are unsafe (e.g., ordinary vacuum cleaners).
3. **Perform** a safe internal clean â€" powering off, unplugging, using ESD care, and using compressed air, brushes and isopropyl alcohol correctly.

heatsink).

6. **Describe** the environmental threats to a computer â€" dust, humidity, heat and power surges â€" and how a **surge protector** and a **UPS** help.
7. **Create** a practical **preventive maintenance schedule** for a home, a small office and a cyber cafÃ©, including who does what and how it is documented.
8. **Produce** a written **maintenance recommendation** for a real machine after cleaning and a cooling check.

C. REQUIRED MATERIALS

Hardware (provided by CEA)

- At least one **dusty practice desktop** safe to open; ideally a second, cleaner machine for comparison.
- **Compressed air** cans (one or more). Keep them upright; do not let learners shake or tilt them.
- A **soft anti-static brush** (or clean, soft paintbrush) for loosening dust from fan blades and heatsink fins.
- **Isopropyl alcohol (90% or higher)** â€" for cleaning old thermal paste and stubborn grime.
- **Microfiber / lint-free cloths** and **cotton buds**.
- **Thermal paste** (a little goes a long way).
- **Anti-static wrist straps** (one per learner), a small screwdriver set, a parts tray.
- An **old keyboard**, an **old monitor or phone screen**, and (optionally) an old keyboard with removable keys â€" for practising safe external cleaning.
- A **surge protector** and, if available, a **UPS** to demonstrate.
- **Safety glasses** and a **first-aid kit**.

Software

- **HWMonitor** (from Sessions 5â€"6) to record temperatures before and after cleaning.

Printed materials

- Cleaning Checklist (one per learner).
- Maintenance Plan Template (one per learner).

Safety equipment and environment

- Anti-static wrist straps, safety glasses, first-aid kit.
- Well-ventilated area for using compressed air (dust flies); avoid blowing dust into the room.
- Know the location of the **fire extinguisher** and the **first-aid kit**.

Internet

- Not required. Optionally, the instructor may look up a specific motherboard's clean-airflow guidance, but do not depend on it.

Alternative for learners who lack a facility

- Any working computer can be cleaned. A desktop gives the fullest practice (internal dust). Learners who own only a laptop may practise **external** cleaning (vents, keyboard, screen, ports) safely; internal laptop cleaning is more delicate and should be instructor-led beside them.

Suggested visuals:



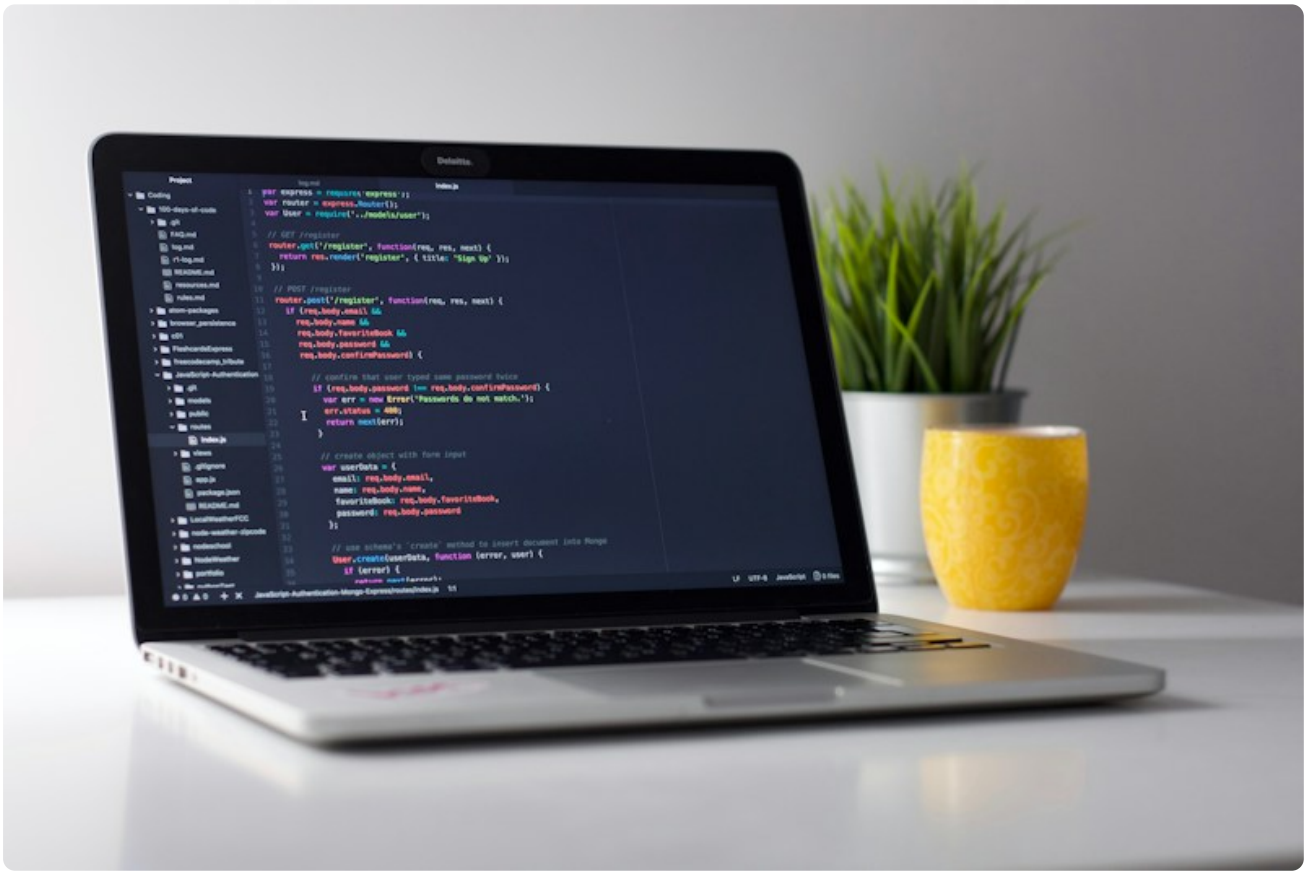
- â€" Internal dust removal with compressed air



- â€" Cleaning kit and maintenance supplies



- â€" Checking fans and filters after cleaning



- — Laptop vents and fans cleaned



— Bench tidied and tools put away after maintenance

D. OPENING ACTIVITY (6 minutes)

Instructor setup:
On the bench, place the **dusty practice desktop** with its side panel removed so the class can see the inside. Beside it, place a clean, dust-free machine (or a clean fan/heatsink) for contrast. Have HWMonitor ready on the dusty machine (or a note of its earlier temperature).

What the instructor says:

> *"Good morning. Look inside this computer. What do you see? Dust. A grey blanket of dust sitting on the heatsink, in the fan blades, in every corner.*
>
> *Now here is the important part. Dust does not just look bad. That dust blanket is a*
>
> ***heat trap***
> *. Heat cannot escape. The CPU gets hot, so the computer slows itself down on purpose â€" that is why a dusty machine feels slow. If it gets hotter still, it shuts itself off to save its own life. And year after year, that heat quietly shortens the life of every part inside.*
>
> *In Nigeria, in Port Harcourt, with our heat and harmattan dust, this is one of the most common reasons a computer dies young. But here is the good news: cleaning is the*
>
> ***cheapest repair***
> *there is. No parts to buy. Just proper tools, proper care, and a habit of doing it regularly.*
>
> *Today you will learn to clean a computer*
>
> ***safely***
> *â€" because done wrongly, cleaning can damage a computer with static or liquid â€" and you will learn to build a* ***maintenance schedule*** *so the computer never gets this bad again."*

Expected learner response:

> *"By the end of today you will know exactly why a vacuum cleaner is dangerous, why blowing with your mouth is a bad idea (moisture), and what to use instead."*

Then demonstrate the hook:
on the compressor or with a canned air, give one short, controlled puff across the dusty heatsink over a bin or outside airflow. The cloud of dust that rises is the moment the lesson lands.

Why this opener works:
It shows the real, physical enemy (dust) instead of describing it, it names a very common Nigerian reality (heat + dust), and it challenges several common bad habits (vacuuming, blowing) with a promise to teach the correct tool.

E. PREREQUISITE REVIEW (7 minutes)

Run as a fast oral round-robin. Put the questions on the projector.

Question 1:
"Name the six major components."

- Expected answer: **Motherboard, CPU, RAM, storage, PSU, GPU.**
- Common gap: forgetting the PSU. Probe and move on.

Question 2:
"What are the two safety must-dos before opening a computer?"

- Expected answer: **Power off and unplug** (press power to discharge); **wear an anti-static wrist strap.**
- Common gap: forgetting the discharge step. Correct it â€" cleaning involves moving air and cloths, so static and loose power are real risks.

Question 3:
"What is ESD, and why does it matter when cleaning?"

- Expected answer: **Electrostatic discharge** â€" a tiny static shock from your body or a cloth that can silently damage components. Cloths and brushes rub and build static, so we must ground ourselves and use anti-static tools.
- Common gap: learners think ESD is only a concern when replacing parts. Make clear it is *especially* relevant to cleaning, because rubbing a cloth on a case builds static.

Question 4:
"When do we re-apply thermal paste, and what amount do we use?"

- Expected answer: when the CPU and cooler are separated, or when temperatures rise despite clean fans; use a **pea-sized** amount. (From Session 7.)

Question 5:
"A computer shuts down after 30 minutes and the case is hot. What are two likely causes?"

- Expected answer: **overheating** (dust, a failed fan, dried thermal paste) and possibly a **failing PSU**. This bridges directly into today.

Bridge to today (say this):

> *"Session 5 taught you to find overheating with tools. Session 7 taught you to replace parts. Today, you learn to prevent the overheating in the first place. A surgeon can operate; a good doctor keeps you from needing the operation."*

Gap handling:

F. FULL LECTURE NOTES

F.1 Why dust and heat destroy computers

The idea.
A computer produces heat when it works and must get rid of that heat. Dust blocks the path heat takes to escape. Block the path, and the heat builds up â€" and heat is the slow killer of electronics.

Plain-English definitions.

- **Dust** is a mixture of tiny particles â€" skin, cloth fibres, and in Nigeria especially **harmattan dust** and road dust. It settles on every surface inside a computer.
- **Heat** is a by-product of electricity moving through components. It must be carried away by the cooling system (fans, heatsinks, airflow) and out of the case.
- A **heatsink** is a block of metal fins attached to a hot chip (like the CPU). The metal absorbs heat from the chip, and the fins give a large surface area for a **fan** to blow air across and carry the heat away.

Why dust causes damage â€" the chain of events:

1. Dust settles on fan blades and between heatsink fins.
2. The dust "felt" blocks airflow and insulates the metal, so heat cannot escape.
3. The CPU/GPU gets hotter than it should.
4. The computer **slows itself down on purpose** to make less heat (this is called **thermal throttling**) â€" so a dusty computer *feels slow*.
5. If it gets hotter still, the computer **shuts itself off** suddenly to protect the chip (a **thermal shutdown**) â€" which looks like random restarts or shutdowns.
6. Over months and years, the constant high heat **shortens the life** of the chip, the drive and other parts.

Analogy.
Imagine doing hard labour in Port Harcourt midday heat while wrapped in a thick blanket. You cannot work well, you feel weak, and if you keep going you collapse. The dust blanket does exactly this to a computer.

Realistic Port Harcourt scenario.
A cyber cafÃ© near Mile 1 has machines that keep shutting down during busy evenings. The cafÃ© owner thinks the machines are "old and spoiled". A technician opens one and finds the CPU heatsink completely packed with dust â€" the fins look like a carpet. After cleaning, the machines run cool and no longer shut down. The owner was about to buy new computers; he only needed a cleaning.

Important misconception.
"Cleaning is just to make it look nice." No â€" the purpose is **cooling and reliability**. A machine can look clean on the outside and be clogged inside.

Limitation.
Cleaning cannot reverse damage already done. If a fan bearing is worn out or thermal paste has dried and temperatures are already very high, cleaning helps a lot but may need to be combined with a **fan replacement** or **re-pasting** (Session 7). And cleaning will not fix a computer whose real problem is software (Session 6) or a failing part (Session 5).

F.2 Safe cleaning tools â€" and the dangerous ones

The idea.
Not every cleaning method is safe. Some tools can destroy a computer. Know which is which.

SAFE tools:

- **Antistatic brush / soft paintbrush** â€“ for gently loosening dust from fan blades and heatsink fins, and for brushing the motherboard surface.
- **Isopropyl alcohol (90%+)** on a **lint-free cloth or cotton bud** â€“ for stubborn grime and for cleaning old thermal paste. It evaporates quickly and does not conduct electricity when dry.
- **Microfiber / lint-free cloths** â€“ for wiping the case interior, screens and exterior.
- **Cotton buds** â€“ for small crevices and ports.
- **Anti-static wrist strap** â€“ always, to protect against ESD.

DANGEROUS / UNSAFE tools:

- **Ordinary vacuum cleaner** â€“ **never** use inside a computer. The plastic nozzle and the moving air **build static electricity**, which can silently destroy components. A vacuum can also physically suck small parts and jumpers off the board. (Special anti-static vacuums exist for professional use; the ordinary household one does not.)
- **Wet cloths / water** â€“ water conducts electricity and causes corrosion. Never use water inside a computer.
- **Your own mouth / blowing with breath** â€“ your breath carries **moisture** (which can cause corrosion) and saliva droplets; use compressed air instead.
- **Sharp metal tools** â€“ can scratch circuit traces and short connections.
- **Household cleaning sprays** (e.g., strong surface cleaners, bleach solutions) â€“ can leave residues that damage components and screens. Use only approved screen cleaners or a barely-damp microfiber cloth.

Why compressed air can is safer than a vacuum.
A can of compressed air is a **sealed, non-sparking** source; it does not generate static the way a vacuum's plastic nozzle does. Keep the can upright so it does not spray its liquid propellant onto the parts.

Important misconception.
"A vacuum cleaner is fine if I do not touch the parts." The **static** builds on the nozzle through the moving air and can discharge onto the board even without contact. The safe rule is: **compressed air inside, vacuum only outside and never near open electronics.**

Limitation.
Compressed air can push dust *deeper* into crevices if used carelessly. Blow dust **outward, toward the exit**, and work in a well-ventilated place, or ideally take the machine outside or near a window so the dust leaves the room.

F.3 The safe internal cleaning procedure

Safety first â€“ read aloud and follow every time:

- **Power off, unplug** the power cable, then press the power button for a second to discharge leftover electricity.
- **Wear an anti-static wrist strap.**
- Use only **safe tools** (F.2).
- Work in a **well-ventilated area**; dust will fly.
- **Never open the PSU.**
- Handle components by the edges; do not spin fans with the air jet.
- If anything smells burnt or you see a swollen battery, **stop and call the instructor**.

Step-by-step procedure:

1. **Prepare the workspace.** Unplug everything. Move the machine to a hard, flat surface in a ventilated area (outside if possible). Put on the wrist strap.
2. **Open the case.** Remove the side panel (method depends on the case; the instructor demonstrates on the practice machine).
3. **Loosen, then blow.** Use the **brush** first to gently loosen the dust "carpet" on the heatsink fins and fan blades. Then use **compressed air** (short bursts, can upright) to blow the loosened dust **out of the case**.

5. **Clean the case interior.** Wipe surfaces with a **microfiber cloth**; use a brush for corners. Clean the **air filters** if the case has them (many can be removed and rinsed and dried â€" only refit when fully dry).
6. **Clean the connectors and slots lightly.** Use a brush or a short gentle burst of air on RAM slots and expansion slots. Do not push dust into them.
7. **Inspect while you are in there.** Look for: a fan that does not spin when the machine runs; bulging capacitors; a swollen laptop battery; burnt smells; loose cables.
8. **Reassemble.** Replace the side panel, reconnect only the power cable and needed cables, and power on.
9. **Verify and record.** Open **HWMonitor** and record the CPU/GPU temperatures. Compare with the "before" reading. A drop of several degrees is a clear success.

Analogy.
Cleaning the inside is like clearing a blocked drainage channel before the rainy season. You do not wait for the flood; you clear the channel so the water (heat) can flow away freely.

Important misconception.
"I should spin the fan with the air jet to clean it faster." No â€" forced spinning damages the fan bearing. Hold the blade steady.

Limitation.
Some heatsinks are so clogged that air alone cannot clear them; the cooler may need to be **removed** to clean properly (and then thermal paste re-applied on refitting â€" Session 7). Some laptops require deeper disassembly to reach the fan; that is more delicate work and should be supervised.

F.4 Cleaning keyboards, ports and screens

The idea.
Not everything can be cleaned the same way. Blowing compressed air into a keyboard's electronics area, or spraying liquid on a screen, can cause damage. Match the method to the surface.

Keyboard (external cleaning, safe for learners):

- Power off the computer.
- Tip the keyboard upside down and gently shake out crumbs.
- Use **short bursts of compressed air** along the rows of keys, at a slight angle, to lift dust and crumbs out.
- Wipe the key tops and the frame with a **barely-damp microfiber cloth** (dampened with a little isopropyl alcohol or plain water â€" **not dripping**). Never spray liquid directly onto the keyboard.
- Between keys, use a **cotton bud**. Do not force it under keys and do not bend key mechanisms.
- **Removable-key keyboards:** the instructor may show removing a keycap to clean beneath; **do not** remove keys on a laptop keyboard (they break easily).

Ports (USB, HDMI, charging ports):

- Power off and unplug.
- Use **short bursts of compressed air** to blow dust out of the port. Do **not** insert sharp metal objects, which can bend the internal contacts.
- For stubborn dust, a **soft dry brush** or a **wooden toothpick** used *very* gently along the edge â€" never force anything into the port.
- **Never** put liquid into a port.

Screens (monitors, laptop screens, phone screens):

- Power off the screen (this makes dust easier to see and is safer).
- Use a **microfiber cloth**, dry for light dust, or **very slightly** dampened with **plain water or an approved screen cleaner** for marks.
- **Never spray liquid directly on the screen** â€" spray onto the cloth, not the screen, and never let liquid run into the edges.
- Use **gentle, circular** motion with light pressure. Do not press hard â€" screens are fragile.

Analogy.
Different surfaces need different care, like cleaning a pair of glasses versus cleaning a car engine â€" same idea (remove dirt), completely different method.

Important misconception.
"More liquid cleans better." For screens, less is more. Excess liquid is the main cause of screen and keyboard damage during cleaning.

Limitation.
Some marks (scratches, screen-print wear) cannot be cleaned away. On laptops, deep cleaning behind the keyboard requires disassembly and is not a beginner task.

F.5 When to re-apply thermal paste

Plain-English definition.
Thermal paste (thermal compound) is the grey paste between the CPU and its heatsink (cooler). It fills the microscopic gaps between the two metal surfaces so heat passes efficiently from the chip into the cooler.

When to re-apply:

- Whenever the **CPU and cooler have been separated** (for example, during a deep clean where the cooler is removed, or a cooler replacement).
- When temperatures have **crept up over the years** despite clean fans and good airflow.
- On a machine that has run hot for a long time and shows signs of dried, cracked paste.

How often (a guide, not a strict rule):
many technicians recommend re-pasting roughly **every 1â€"2 years**, or whenever the cooler is removed. For a well-cooled home PC it may last longer; for a hot, dusty cyber-cafÃ© machine that is opened regularly, more often. Judge by **temperature readings**, not only by a calendar.

Signs the paste needs replacing:

- CPU temperatures are high even after cleaning dust and confirming fans spin.
- The old paste, when the cooler is lifted, looks dry, cracked or powdery (instead of soft and grey).
- Fan noise is high and the machine still runs hot.

The steps (recap from Session 7):

1. Power off, unplug, discharge, wrist strap on.
2. Remove the cooler carefully (do not twist if the CPU is on an older socket design where the CPU may come away stuck to the cooler â€" ask the instructor).
3. **Clean** old paste from CPU and cooler base with **isopropyl alcohol** and a lint-free cloth/cotton bud until both are clean and dry.
4. Apply a **pea-sized** dot of new paste in the centre.
5. Refit the cooler, tightening screws **evenly** (a little each, in turn).
6. Reconnect the **fan cable**.
7. Power on and verify temperatures with **HWMonitor** (should be normal or lower).

Analogy.
Thermal paste is like a thin layer of oil between a pot and a hot plate. The pot and plate do not touch perfectly; the oil fills the tiny gaps so heat moves evenly. Too little and heat does not travel; too much and it makes a mess.

Important misconception.
"Thermal paste lasts forever." It dries out over the years and loses effectiveness. Also, "more is better" is false â€" a **pea-sized** amount is correct.

Limitation.

F.6 Environmental threats â€" dust, humidity, heat and power

The idea.
Cleaning protects the *inside* of the computer. But the *environment* around the computer â€" dust, humidity, heat and unstable electricity â€" can also destroy it. Managing the environment is part of maintenance.

Dust (already covered).
Keep computers off the floor where dust gathers, away from windows where harmattan dust blows in, and out of the kitchen area (cooking smoke and oil stick to fans and make dust worse).

Humidity and moisture.

- **High humidity** (common in Port Harcourt during the rainy season) can allow moisture to condense on cold components, causing corrosion and short circuits over time.
- **Low humidity / dryness** (common in the harmattan) makes static electricity build up more easily â€" increasing ESD risk.
- Practical measures: keep machines in dry, ventilated rooms; avoid placing them where they get wet; allow a cold machine (e.g., brought in from the rain) to reach room temperature before powering it on; use a dehumidifier in severe cases (a professional/household solution, not essential for beginners).

Heat.

- Ambient (room) heat directly raises component temperatures. Keep computers out of direct sun, away from hot spots, and ensure the room is ventilated.
- Ensure the case has clear **intake and exhaust** vents (not against a wall or curtain) so air can flow.

Power problems â€" surges, sags and outages.

- **Surge:** a sudden spike in voltage (e.g., when power returns after an outage, or from lightning). A surge can destroy the PSU and components instantly.
- **Sag/brownout:** voltage dropping below normal (common where supply is weak). Can cause crashes, restarts and data corruption.
- **Outage:** sudden loss of power â€" can corrupt files, especially during updates.

Protection devices:

- A **surge protector** (a power strip with surge protection) absorbs sudden voltage spikes. It helps protect against surges but does **not** keep the computer running during an outage.
- A **UPS (Uninterruptible Power Supply)** is a battery backup box that keeps the computer running for a few minutes during an outage, giving time to save work and shut down safely. It also helps with sags/brownouts. It is especially valuable for cyber cafÃ©s, offices and any machine doing important work.
- **Note:** a surge protector must be replaced after a serious surge; many have an indicator light that shows when protection is lost.

Analogy.
Electricity is like water coming through a pipe. A **surge** is a pressure burst that can burst the pipe; a **sag** is a drop in pressure that stalls the machines; an **outage** is the water being cut. A surge protector is a shock absorber; a UPS is a small reserve tank.

Realistic Port Harcourt scenario.
A cyber cafÃ© loses power suddenly while a customer is midway through printing a document and another is saving a school project. Without a UPS, the print job is spoilt and the saved project may be corrupted. With a UPS on the key machines, there is time to save and shut down.

Important misconception.
"A surge protector also keeps the computer on during a blackout." No â€" that is a **UPS**'s job. A surge protector only guards against spikes.

Limitation.

F.7 Preventive maintenance schedules

The idea.
Instead of waiting for a machine to fail, we perform small tasks on a **schedule**. This keeps computers cool, healthy and reliable â€” and it is a service a technician can sell.

Plain-English definition.
A **preventive maintenance schedule** is a plan listing what to check, how often, and who is responsible, so that small problems are caught before they become big ones.

Why it exists.
Most computer failures are slow, not sudden: dust builds, paste dries, drives age, disks fill up. A schedule catches these while they are still cheap to fix.

A general schedule (adapt to the machine and environment):

Frequency	Task	Tool / method
Monthly	Run a full antivirus scan; check for Windows updates; run Disk Cleanup; clear the Recycle Bin	Windows tools (Session 6)
Every 3 months	Clean internal dust (compressed air, brush); check all fans spin; check temperatures	HWMonitor; compressed air
Every 6 months	Check drive health; run a memory test; check air filters; tighten/verify airflow; check surge protector	CrystalDiskInfo; Memory Diagnostic
Annually	Full diagnostic check of all components; re-apply thermal paste if temperatures warrant it; review and archive old files; check backups work	HWMonitor; paste; backup check
After events	Clean and inspect after a major dust event, a lightning/surge event, a move, or a spill	Visual inspection

Schedules for different settings:

- **Home computer:** the general schedule above, plus a note to keep it off the floor and away from windows, and to back up important photos/documents.
- **Small office (a few computers):** add a written record per machine; assign one person responsible; use a **UPS** on machines doing critical work; back up shared files regularly.
- **Cyber cafÃ© (many machines, heavy use, lots of dust):** clean **more often** (monthly internal cleaning may be needed in dusty areas); use surge protectors at minimum and UPS on key machines; keep a simple **logbook** per machine recording cleaning and faults; train staff to report symptoms early (noisy fans, shutdowns).

Documentation.
Every task should be **recorded** â€” a simple logbook or table with the date, the machine, the task done, the temperature readings, and who did it. Documentation turns "I think I cleaned it" into **proof**, helps spot machines that run hotter than others, and looks professional to a customer.

Analogy.
A maintenance schedule is like a car service book. You do not wait for the engine to seize; you change the oil on time. The computer is the same.

Important misconception.
"If it is working, it does not need maintenance." Working machines are exactly the ones worth protecting â€” prevention keeps them working.

Limitation.

F.8 Summary of the session

- **Dust + heat** cause slowness (thermal throttling), random shutdowns (thermal shutdown) and early part failure.
- **Safe tools:** compressed air, antistatic/soft brush, isopropyl alcohol, microfiber cloth, cotton buds, wrist strap. **Unsafe:** vacuum cleaner (static), water, breath (moisture), harsh sprays.
- **Internal clean:** power off, unplug, discharge, strap on; loosen with a brush, blow outward with compressed air; hold fans steady; inspect; verify with HWMonitor.
- **Keyboards/ports/screens:** air for keys/ports; barely-damp microfiber for screens, never spray on the screen; never liquid in ports.
- **Thermal paste:** re-apply when the cooler is removed or temperatures creep up (roughly every 1â€“2 years); **pea-sized** amount.
- **Environment:** manage dust, humidity, heat and power; use a **surge protector** for spikes and a **UPS** for outages/sags.
- **Preventive schedules** for home, office and cafÃ©; **document** every task.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

G.1 What the instructor prepares

- The dusty practice desktop, opened on the bench.
- Compressed air, antistatic brush, isopropyl alcohol, microfiber cloths, cotton buds.
- An old keyboard and an old monitor/phone screen for cleaning practice.
- A surge protector and a UPS.
- HWMonitor ready; a note of the dusty machine's "before" temperature.
- The Cleaning Checklist and Maintenance Plan Template on the projector.

G.2 Demonstration 1 â€” Safe internal clean (6 minutes)

Say:
"Watch the order: loosen, then blow outward, and never let the fan spin."

Steps:

1. Power off, unplug, press power to discharge, put on the wrist strap.
2. Remove the side panel. Point out the dust on the heatsink and fans.
3. Use the **brush** to gently loosen the dust on the heatsink fins and fan blades. **Say:** "I loosen it first. Dry dust sticks; brushing frees it, then the air can carry it away."
4. Take the can of **compressed air**, hold it **upright**, and give **short bursts** to blow the dust **outward** (toward the open side / out of the case). **Say:** "Upright â€” if I tilt it, it can spray cold liquid onto the parts. Short bursts, not one long blast."
5. Hold a fan blade steady while blowing it. **Say:** "If I let the air spin the fan, I can damage the bearing. Free-wheel spinning is bad for fans."
6. Wipe the case interior with a **microfiber cloth**; point out the air filter and how to clean it.

Expected results:
visible dust removed; heatsink fins visibly clearer.

Reasoning to say aloud:
"Loosen, blow outward, protect the fans â€” that is the whole technique."

Common errors to show and correct:

- Tilting the can (show the warning) â†’ liquid risk.
- Blowing dust *into* the machine instead of out.

Questions:
"Why brush before blowing?" (Loosen it so air can carry it away.) "Why hold the fan?" (Prevent bearing damage.)

G.3 Demonstration 2 â€” Re-pasting a CPU (4 minutes)

Say:
"Now the paste â€” small is powerful."

Steps:

1. Show a cooler base with **dried, cracked** old paste. **Say:** "This is what paste looks like when it has failed â€” dry and cracked. It can no longer pass heat."
2. Clean the CPU and cooler base with **isopropyl alcohol** and a lint-free cloth until both are clean. **Say:** "Clean and dry, both surfaces."
3. Apply a **pea-sized** dot in the centre. **Say:** "Small. Too much squeezes out and can make things worse."
4. Show how the cooler presses down and spreads the paste. Refit and **reconnect the fan cable**.
5. **Verify:** point to HWMonitor and describe the expected temperature drop.

Expected results:
clean surfaces, a pea-sized paste dot, cooler refitted, fan cable reconnected.

Common errors to show and correct:

- Using too much paste.
- Forgetting the fan cable.
- Twisting the cooler (risk of pulling a stuck CPU â€” warn).

Questions:
"How do you know the paste needs replacing?" (Dry/cracked, or high temps despite clean fans.)

G.4 Demonstration 3 â€” External cleaning and protection devices (5 minutes)

Steps (keyboard/ports/screen):

1. Power off. Show **short bursts of compressed air** along the keyboard rows; shake crumbs out first.
2. Show wiping key tops with a **barely-damp** microfiber cloth â€” **never** spraying onto the keyboard.
3. Show a **cotton bud** for between-keys cleaning.
4. On the old screen, show a **dry microfiber cloth** for dust, and a **slightly damp** cloth for marks, sprayed on the **cloth**, not the screen. **Say:** "No paper towels, no glass cleaner, no hard pressing."

Steps (surge protector and UPS):
5. Hold up the **surge protector**. **Say:** "This absorbs sudden voltage spikes â€” like when power returns after an outage. It does NOT keep the computer on during a blackout." 6. Hold up the **UPS**. **Say:** "This is a battery backup. It keeps the computer alive for a few minutes so you can save and shut down safely. This is what a cyber cafÃ© needs on its key machines." 7. Point to the surge protector's indicator light. **Say:** "When this light goes out, the protection is spent â€” replace it."

Expected results:
learners see correct external cleaning technique and understand the difference between a surge protector and a UPS.

Common errors to show and correct:

- Spraying liquid directly on a screen or keyboard.
- Confusing a surge protector with a UPS.

Questions:
"Which device protects against a blackout?" (UPS.) "Which protects against a spike?" (Surge protector.)

Objective:

With the instructor guiding, each learner (or pair) performs a **full clean and cooling check** on a practice machine and records temperatures before and after.

Safety first â€” read aloud before starting:

- **Power off, unplug**, press power to discharge, **wear your anti-static wrist strap.**
- Use only **safe tools** (compressed air, antistatic brush, isopropyl alcohol, microfiber cloth, cotton buds). **Never** a vacuum cleaner inside a computer.
- Hold the compressed-air can **upright**; do not tilt or shake it.
- **Hold fan blades steady**; do not let the air spin them.
- **Never open the PSU.**
- Work in a **well-ventilated area** (dust will fly).
- Stop and call the instructor if you smell burning or see a swollen battery.

Procedure (instructor calls each step; learners follow; instructor walks the room):

1. **Before temperature.** On the running machine (before opening), open **HWMonitor** and record the **CPU and GPU temperatures** and the fan speeds. This is your "before" reading.
2. Power off, unplug, discharge, strap on. Open the case.
3. **Inspect first.** Before cleaning, note what you see: dust on heatsink? dust on fans? a fan not spinning? record it on the Cleaning Checklist.
4. **Loosen dust.** Use the **brush** to gently loosen dust from the heatsink fins and fan blades.
5. **Blow outward.** Use **compressed air** (upright, short bursts) to blow dust **out of the case**. Hold each fan blade steady.
6. **Clean the interior.** Wipe surfaces with a microfiber cloth; clean the air filter if present; gently air the RAM and expansion slots (do not push dust in).
7. **Pause and predict.** "After cleaning, will the CPU be hotter, same, or cooler? How much cooler do you expect?" Write your prediction.
8. **External cleaning.** Using the old keyboard and screen (or carefully on the practice machine's own), practise: air along the keyboard rows; wipe keys with a barely-damp cloth; clean the screen with a microfiber cloth (dry, or slightly damp with the cloth, never sprayed on the screen).
9. **Reassemble and power on.** Replace the side panel, reconnect power, power on.
10. **After temperature.** Open **HWMonitor** and record the CPU/GPU temperatures again. Compare with the "before" reading.
11. **Cooling check.** Confirm all fans spin. If any fan does not, note it â€” it may need replacement.
12. **Write the maintenance recommendation.** On the Maintenance Plan Template, write a short schedule for this machine (monthly/3-monthly/6-monthly/annual tasks), including cleaning frequency suited to the environment.
13. Complete the **Cleaning Checklist**.

Expected results:

- "Before" and "after" temperatures recorded, usually showing a **drop** after cleaning.
- Visible dust removed from heatsink and fans.
- All fans spinning confirmed (or a non-spinning fan identified).
- A completed Cleaning Checklist and a written maintenance recommendation.

Points to pause on and predict:

- After step 3: "Which part looks worst â€” the heatsink, the fans, or the case?" (Usually the heatsink/fans.)
- After step 7: "If the fins were fully clogged, what temperature change would you expect?" (A larger drop.)
- After step 11: "If one fan does not spin, what does that mean for the machine?" (Risk of overheating â€” the fan needs attention/replacement.)

Troubleshooting while walking the room:

- Dust will not come off the heatsink â†’ it may need the cooler removed for a proper clean (supervised) and re-paste.

- Temperature did not drop â†’ the real issue may be a dead fan, dried paste, or blocked vents â€” record it and recommend the next step.
- Learner sprays liquid on a screen â†’ stop; show the correct method (cloth dampened, sprayed on the cloth).
- Machine will not power on after reassembly â†’ check the 24-pin and CPU power connectors are seated and the side panel did not trap a cable.

Completion checklist (learner ticks each):

- ☐ "Before" temperatures and fan speeds recorded (HWMonitor)
- ☐ Safety followed: unplugged, discharged, wrist strap worn, no PSU opened
- ☐ Dust loosened with a brush and blown outward with compressed air (can upright)
- ☐ Fan blades held steady while cleaning
- ☐ Case interior and air filter cleaned; slots lightly cleaned
- ☐ External cleaning practised (keyboard, ports, screen) safely
- ☐ "After" temperatures recorded and compared with "before"
- ☐ All fans checked for spinning
- ☐ Maintenance recommendation written on the template
- ☐ Cleaning Checklist completed; tools and screws accounted for

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I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task:
Work alone. Produce a **preventive maintenance plan** and a **maintenance recommendation** for a real operating setting. Do **not** copy the instructor's general schedule â€” tailor yours to the setting given, and justify each choice.

Scenario.
You have been hired as the maintenance technician for **"Bright Minds Tutorial Centre"** in Port Harcourt. The centre has:

- **12 desktop computers** used by students, in a **ground-floor room with windows that open onto a dusty road**.
- Frequent **power cuts**, and the voltage is often unstable when supply returns.
- Machines are used for typing, browsing and printing; a few are used to store student records.
- Casual staff operate the machines; there is **no maintenance habit** yet.
- The room is **not air-conditioned**.

Requirements:

1. **Cleaning schedule:** state how often the computers need internal cleaning, and justify it based on the **dusty, non-air-conditioned** environment (be specific about frequency and why it must be more frequent than a home PC).
2. **Power protection:** recommend which devices to use and on which machines, explaining the difference between a **surge protector** and a **UPS**, and which problem each solves.
3. **Software maintenance:** list the monthly software tasks (from Session 6) that keep the machines healthy.
4. **Responsibility and documentation:** say who does each task and how completed work will be **recorded**.
5. **Environmental advice:** give three practical changes to the room or setup to reduce dust, heat or humidity problems.
6. **A plain-language summary** for the tutorial centre owner explaining why this plan saves money.

Time allowance:
15 minutes.

Constraints:

- Base your frequency on the dusty, non-air-conditioned reality â€” do not just copy a magazine schedule.
- Include the **safety rules** any cleaner must follow (power off/unplug, ESD strap, no vacuum inside, never open the PSU).
- Keep the summary understandable to a non-technical owner.

- Surge protector vs UPS is explained correctly, with sensible placement.
- Monthly software tasks are included.
- Responsibility and a documentation method are stated.
- Three sensible environmental changes are given.
- The summary is plain and relates to money saved.

Hints available if stuck:

- Hint 1: Dusty road + open windows + no air-conditioning = clean **more often** than a typical home.
- Hint 2: Which device keeps a computer on during a blackout â€" a surge protector or a UPS? Put that one on the machines that store records.
- Hint 3: "Failed maintenance is invisible until the machine dies." What record would let the owner *see* that the work was done?

INSTRUCTOR ANSWER KEY â€" Independent Class Exercise (do not show learners)

1. Cleaning schedule.
Internal cleaning **more often than a home PC** â€" for this dusty, non-air-conditioned room, roughly **every 4â€"6 weeks** for internal blow-out, with a **quick weekly check** of vents/fans, and a "clean now" trigger whenever a machine runs loud or hot. Justification: open windows onto a dusty road and no air-conditioning mean dust accumulates fast; frequent cleaning prevents overheating, slowdowns and shutdowns. *(Accept a sensible range, e.g., monthly to every 6 weeks, provided the reasoning is environmental.)*

2. Power protection.

- **Surge protector** on **all** machines (and the printer/network equipment): blocks sudden voltage **spikes**, e.g., when power returns after an outage. Does **not** keep machines running during a blackout.
- **UPS** on the machines that store **student records** and on the printer/network if budget allows: keeps the computer alive for a few minutes during **outages** and steadies **sags/brownouts**, allowing a safe save and shutdown and preventing file corruption.
- Explain the key difference clearly: **spikes = surge protector; blackout/sag = UPS**.

3. Monthly software tasks.
Full antivirus scan; check for Windows updates; run Disk Cleanup / empty Recycle Bin; review Startup Apps for unnecessary entries; check disk space and free some if low; back up the student-records machine. *(From Session 6.)*

4. Responsibility & documentation.
Assign a named staff member (or the technician) per task; keep a simple **logbook/table** per machine: date, machine ID, task done (with temperature readings if cleaning), and signature. Post a printed schedule on the wall. The logbook proves the work was done and reveals machines that run hotter than others.

5. Environmental advice (any three of):
keep machines off the floor and away from the windows/dust path; fit dust filters on intakes and clean them; close windows or use screens/curtains during dusty periods; place machines so vents are not against walls; ventilate the room to reduce heat; consider a fan-assisted airflow; keep drinks away from machines; allow a cold/damp machine to reach room temperature before switching on.

6. Plain-language summary (example).
"Your computers are getting dusty fast because of where they are, and dust makes them slow down and shut off. If we clean them every month and protect them from power spikes and blackouts, they will last longer, break less and lose fewer student records. This small, regular cost is far less than replacing computers that died young."

Marking guidance (out of 15):

- Cleaning frequency justified by environment (3)
- Correct surge protector vs UPS use and placement (3)

- Three sensible environmental changes (2)
- Plain-language, money-focused summary (2)

Accept any plan that adapts sensibly to the dusty, unstable-power environment and includes safety and documentation.

J. REAL-WORLD CASE STUDY

Situation.
Ngozi runs a small hair salon in Rumuola that also has a computer for bookings, customer records and the salon's social-media page. The computer is on the floor near the door, where dust from the street and hair clippings collects. For months it has been getting slower and louder, and twice recently it shut down in the middle of work. Ngozi assumes it is "old" and plans to buy a new one.

Relevant facts.

- The machine is only about three years old.
- It sits on the floor, close to the open door.
- It has never been opened or cleaned.
- The room is hot, and the computer shares a power strip with a hair dryer (a high-power appliance) on the same socket.
- Customer records are stored on the machine with **no backup**.

The decision point.
Buy a new computer, or clean and protect the existing one?

Possible approaches.

1. Buy a new computer (expensive; the new one will also get dusty and hot in the same spot).
2. Clean the current machine (internal dust + thermal paste if needed), then fix the setup: move it off the floor, away from the door, onto a surge protector, separate from the hair dryer, and start a cleaning and backup routine.
3. Ignore it and hope it lasts.

Recommended solution (for this stage).
Approach **2**. Internal cleaning clears the dust blanket (the likely cause of the slowdown and shutdowns); moving the machine off the floor and away from the door reduces future dust; a **surge protector** (and separating it from the hair dryer) protects it from power problems; setting up a **backup** for the customer records and a **monthly cleaning routine** prevents the problem returning. If temperatures stay high after cleaning, re-apply thermal paste. This costs a fraction of a new computer.

Trade-offs.

- Approach 1 is expensive and does not fix the *environment* â€" the new machine will face the same dust, heat and power problems and may also die young.
- Approach 2 is affordable, treats the actual cause, and adds lasting protection; it requires a small amount of regular effort.
- Approach 3 lets the machine die, likely losing the customer records.

Professional connection.
This is a core professional insight: **most "old" computers are not old â€" they are neglected.** A technician who cleans, protects and documents earns repeat business and referrals, because the customer sees real value (a fast machine and saved records) at low cost. The certificate's promise of "honest assessments and recommendations" is exactly this.

Transfer exercise (ask the class):
"Ngozi asks, 'Why did my computer shut down in the middle of work?' Give her a plain answer and one prevention step."

Suggested answer: "Dust had clogged the cooling inside, so the machine got too hot and shut itself off to protect the parts. We will clean it, and if you keep it off the floor and clean it every month, it will stay cool and stop shutting down."

K.1 Dust will not come off the heatsink / temperatures stay high after cleaning

- **Symptoms:** cleaning done, but CPU still hot.
- **Likely causes:** the heatsink is so clogged that air cannot clear it; dried thermal paste; a failed or slow fan; blocked vents; the room is too hot.
- **How to investigate:** check fan speeds (HWMonitor); check the fan spins; inspect the vents; check the paste condition.
- **Safe corrective steps:** remove the cooler for a proper clean and re-apply thermal paste (Session 7, supervised); replace a failed fan; unblock vents; improve room airflow.
- **How to confirm resolved:** temperatures drop and stay normal during use.
- **Prevention:** clean on schedule before it clogs; maintain airflow.

K.2 A fan does not spin

- **Symptoms:** one fan is stationary while the machine runs.
- **Likely causes:** the fan is clogged and stalled; the fan cable is disconnected; the fan bearing has failed; the fan is not getting power.
- **How to investigate:** check its cable is connected to the right header; clear dust; test.
- **Safe corrective steps:** clean it; reseat its cable; if it still does not spin, **replace the fan**.
- **How to confirm resolved:** the fan spins freely and quietly.
- **Prevention:** regular cleaning; replace worn fans promptly.

K.3 Machine will not start after cleaning

- **Symptoms:** no power after reassembly.
- **Likely causes:** a cable was knocked loose (24-pin, CPU power, SATA, fan); a component was dislodged; the side panel trapped a cable.
- **How to investigate:** re-open and check all connectors and component seating; look for trapped cables.
- **Safe corrective steps:** reseat connections; free trapped cables; power on.
- **How to confirm resolved:** the machine POSTs and boots.
- **Prevention:** photograph connections before cleaning; work gently; check before closing.

K.4 Static damage after cleaning

- **Symptoms:** the machine behaves strangely or dies after a clean, with no visible cause.
- **Likely cause: ESD** â€" static from a cloth, brush or body discharged into a component.
- **How to investigate:** there may be no visible sign; ESD damage can be invisible. This is why prevention is essential.
- **Safe corrective steps:** ESD damage is often permanent and part-level; identify and replace the affected part (diagnostic repeat, Session 5).
- **How to confirm resolved:** the machine runs normally after replacing the damaged part.
- **Prevention: always** wear an anti-static wrist strap; use antistatic tools; never use a vacuum cleaner inside.

K.5 Screen damage during cleaning

- **Symptoms:** streaks, smears or damage after wiping a screen.
- **Likely causes:** liquid sprayed onto the screen; harsh glass cleaner; hard pressing; paper towels.
- **How to investigate:** inspect for fluid marks or surface damage.
- **Safe corrective steps:** for smears, gently wipe with a clean, dry microfiber cloth; surface damage may not be repairable (may require replacing the panel).
- **How to confirm resolved:** the screen is clear and undamaged.
- **Prevention:** screen cloth barely damp, liquid on the cloth not the screen, gentle pressure, no harsh chemicals.

- **Likely causes: surge, sag or outage**; no surge protection; no UPS.
- **How to investigate:** ask about the power supply history; check for a surge protector; check for corrupted files.
- **Safe corrective steps:** fit a **surge protector**; use a **UPS** on critical machines; **back up**; repair corrupted files (Session 6, `sfc`).
- **How to confirm resolved:** stable operation and no further corruption.
- **Prevention:** surge protector on everything, UPS on critical machines, regular backups.

K.7 Distinguishing likely from confirmed

As always, a *likely* cause becomes *confirmed* by evidence: a temperature reading that drops after cleaning, a fan observed spinning (or not), a paste visibly dry, a machine that will not POST only after a specific event. Record observations honestly and never claim "cleaned and fixed" without a **before/after temperature reading** and a working test.

L. SESSION ASSIGNMENT

Assignment title:
"Cleaning and Maintenance Report with a Maintenance Plan"

Purpose:
To perform a safe clean and cooling check on a real machine and to produce a written **maintenance recommendation and schedule** a customer could follow.

Realistic scenario:
A customer asks you to "service" their computer â€” clean it, check the cooling, and advise them how to keep it healthy. You will clean a machine (or a dusty machine at CEA), record the results, and write the recommendation.

Detailed instructions:

1. Choose a real computer you have permission to clean (a CEA practice machine or your own). **Do not open a machine that is not yours without permission.**
2. **Before cleaning:** power on, open **HWMonitor**, and record the CPU (and GPU, if shown) temperatures and fan speeds.
3. **Clean safely:** power off, unplug, discharge, wear the wrist strap; loosen dust with a brush, blow outward with **compressed air (can upright)**, hold fans steady, wipe the interior; clean the keyboard, ports and screen appropriately for their surfaces. **Never use a vacuum inside and never open the PSU.**
4. **Inspect:** note dust levels, any fan not spinning, paste condition, and airflow.
5. **After cleaning:** power on, record temperatures again, and confirm all fans spin.
6. **Write the maintenance recommendation:** a schedule (monthly / 3-monthly / 6-monthly / annual tasks) tailored to the machine and its **environment** (home, office, or dusty cafÃ©), including power-protection advice (surge protector / UPS).
7. **Write a plain-language summary** for the customer explaining the benefit (cooler, faster, longer life) and the safety rules.

Required deliverables:

- One completed **Cleaning Checklist** with before/after temperatures.
- One written **maintenance recommendation and schedule** (using the Maintenance Plan Template).
- A one-paragraph plain-language summary for the customer.

Submission format:
Completed checklist and written plan (paper or class folder).

Deadline:
Before Session 9.

Marking rubric (20 marks):

Before/after temperature record	3	Both recorded with HWMonitor, comparison made
Safe cleaning technique	4	Brush-then-air, can upright, fans held, no vacuum, no PSU
Inspection findings	3	Dust, fan, paste and airflow noted honestly
Fan/cooling check	2	All fans checked; any fault identified
Maintenance schedule	4	Tailored to environment with sensible frequencies and tasks
Power-protection advice	2	Correct use of surge protector (spikes) and UPS (outages/sags)
Plain-language summary	2	Customer can understand the benefit and safety rules

Common mistakes to avoid:
using a vacuum inside; tilting the compressed-air can; spinning fans with air; spraying liquid on screens or into ports; forgetting the wrist strap; skipping the temperature reading; copying a generic schedule without adapting it to the environment.

Instructor reference solution (abbreviated example):

> ***Machine:***
> *CEA practice desktop, dusty, used in a non-air-conditioned room.* ***Before (HWMonitor):*** *CPU* ***72 Â°C*** *idle; GPU* ***60 Â°C****; CPU fan 900 RPM; one case fan 0 RPM.* ***Cleaning:*** *power off, unplugged, discharged, wrist strap on. Brushed the heatsink fins and fan blades; blew outward with compressed air (can upright); held fans steady; wiped the interior; cleaned the air filter; cleaned the keyboard (air + barely-damp cloth) and the screen (dry microfiber).* ***Inspection:*** *heavy dust carpet on the CPU heatsink; one case fan not turning; paste looked dry at the edges; vents partly blocked by the machine's position against a wall.* ***After (HWMonitor):*** *CPU* ***48 Â°C*** *idle; GPU* ***45 Â°C****; CPU fan 1,200 RPM; case fan still 0 RPM (****needs replacement****).* ***Recommendation:*** *replace the stopped case fan; re-apply thermal paste if temperatures stay high after the new fan (Session 7); move the machine so vents are clear of the wall; internal clean* ***every 4â€“6 weeks*** *in this dusty room; monthly antivirus scan + Disk Cleanup + updates; check drive health every 6 months; annual full diagnostic; use a* ***surge protector*** *and back up important files.* ***Plain-language summary:*** *"Your computer was full of dust, which made it overheat and slow down. After cleaning, the temperature dropped a lot. One fan has stopped and should be replaced, and you should clean every month because the room is dusty. Keep it away from the wall so air can flow. A surge protector will protect it from power spikes."*

M. KNOWLEDGE CHECK

(Learner version â€“ answers only in the Instructor Key below.)

Q1. Short answer.
In one or two sentences, explain how dust makes a computer slow. **Q2. Multiple choice.** Which of these must you **never** use to clean the inside of a computer? a) Compressed air b) An antistatic brush c) An ordinary vacuum cleaner d) A microfiber cloth **Q3. True or false, with a reason.** "You should spray screen cleaner directly onto the monitor screen." **Q4. Explain in your own words.** When does thermal paste need to be replaced, and how much do you use? **Q5. Short answer.** What is the difference between a **surge protector** and a **UPS**? **Q6. Scenario.** A cyber cafÃ©'s computers keep shutting down during busy evenings, and the room is not air-conditioned. Give two maintenance actions that would help, and explain why.

INSTRUCTOR ANSWER KEY (Q1â€“Q6)

A1.
Dust settles on fan blades and between heatsink fins and blocks airflow, so heat cannot escape. The CPU gets hot and the computer slows itself down on purpose (**thermal throttling**), so it feels slow; if it gets hotter still, it shuts off to protect itself.

A2.

A3.

False. Never spray liquid directly onto a screen â€" it can run into the edges and damage the screen. Spray (or dampen) the **cloth**, then wipe gently.

A4.

Replace thermal paste whenever the CPU and cooler are separated, or when temperatures creep up despite clean fans (roughly every **1â€"2 years**). Use a **pea-sized** amount in the centre of the CPU â€" not too much.

A5.

A **surge protector** absorbs sudden voltage **spikes** (e.g., when power returns after an outage); it does **not** keep the computer running during a blackout. A **UPS** contains a battery and keeps the computer running for a few minutes during an **outage** and steadies **sags/brownouts**, allowing a safe save and shutdown.

A6.

Any two sensible actions, e.g.: (1) **Clean the internal dust regularly** (monthly in a dusty, non-air-conditioned room) â€" dust blocks airflow and causes overheating and shutdowns. (2) **Improve cooling/airflow** â€" ensure vents are clear, all fans work (replace failed fans), and the room is ventilated; add a **surge protector + UPS** if the shutdowns relate to unstable power. *(Also acceptable: re-apply thermal paste if temperatures stay high; move machines out of direct heat.)*

Marking:

award for correct concept; deduct for unsafe answers (vacuum inside, spraying liquid on screens, opening the PSU).

N. REFLECTION AND RECAP

Plain-English summary.

Today you learned that cleaning is not about looks â€" it is about **cooling and reliability**. Dust blocks airflow, so heat cannot escape; the computer then **slows itself down** (thermal throttling) and can **shut itself off** (thermal shutdown), and over time the heat shortens the life of every part. You learned the **safe tools** (compressed air, antistatic brush, isopropyl alcohol, microfiber cloth, cotton buds, wrist strap) and the **dangerous ones** (vacuum cleaner, water, your breath, harsh sprays). You practised the safe internal clean (power off, unplug, discharge, brush, blow outward, hold fans steady, inspect, verify with **HWMonitor**) and external cleaning of keyboards, ports and screens. You learned when to **re-apply thermal paste** (a pea-sized amount) and how to protect a computer from its **environment** â€" dust, humidity, heat and **power surges/sags/outages** â€" using a **surge protector** and a **UPS**. Finally, you learned to build **preventive maintenance schedules** for homes, offices and cyber cafÃ©s, and to **document** every task.

Five important takeaways:

1. **Dust + heat = slowdowns, shutdowns and early death.** Cleaning is the cheapest repair.
2. **Compressed air (upright) and an antistatic brush inside; never a vacuum cleaner inside.**
3. **Hold fans steady; never spin them with air.**
4. **Re-apply thermal paste** when the cooler is removed or temperatures creep up â€" pea-sized.
5. **Protect from power:** a **surge protector** for spikes, a **UPS** for outages â€" and always **back up**.

Questions to help you recognise what you can now do:

- Can I clean a computer safely using the right tools?
- Can I record before/after temperatures and show the improvement?
- Can I clean a keyboard, ports and screen without damaging them?
- Can I explain a surge protector versus a UPS?
- Can I write a maintenance schedule tailored to an environment?

Safety self-check (tick honestly):

- ☐ I powered off, unplugged and discharged before opening the machine.
- ☐ I wore my anti-static wrist strap and used antistatic tools.
- ☐ I never used a vacuum cleaner inside the computer.

- ☐ I recorded a before/after temperature reading and verified the machine worked.

Connection to the next session.
You can now diagnose, replace and maintain hardware. The final major technical skill is the **software foundation** itself: next session, Session 9 â€" Operating System Installation, you will learn to prepare for a Windows install, install Windows on a machine (including one with a new SSD), install drivers, run updates, and activate â€" turning a blank or freshly-repaired machine into a working computer ready for the customer. Maintenance kept the machine healthy; installation gives it its brain.

O. INSTRUCTOR EVALUATION NOTES (instructor-only)

Observable signs of genuine competence:

- The learner sets up correctly: **power off, unplug, discharge, wrist strap** â€" every time.
- The learner uses the **correct tools** and refuses the vacuum cleaner inside.
- The learner uses compressed air **upright** and holds **fans steady** while blowing.
- The learner records **before/after temperatures** and makes an honest comparison.
- The learner cleans screens/ports appropriately (no direct spraying, no liquid in ports).
- The learner's maintenance schedule is **tailored to the environment**, not copied.
- The learner explains the difference between a **surge protector** and a **UPS** correctly.

Common misconceptions to watch for and correct:

- "Cleaning is only cosmetic." â†' It is about cooling and reliability.
- "A vacuum cleaner is fine." â†' Static risk; never inside.
- "Spin the fan with air to clean it." â†' Damages the bearing.
- "More liquid cleans better." â†' Excess liquid damages screens/keyboards.
- "A surge protector keeps the computer on in a blackout." â†' That is a UPS.
- "If it is working, no maintenance is needed." â†' Working machines are the ones worth protecting.

Suggested remedial exercises for learners who struggle:

1. **Tool-sorting drill:** give cards for tools (compressed air, brush, vacuum, cloth, water, alcohol) and have the learner sort them into "safe" and "unsafe" with reasons.
2. **Technique rehearsal:** practise on a spare fan/heatsink: brush-then-blow, can upright, hold the fan steady â€" without powering anything on.
3. **Schedule-adaptation drill:** give three environments (home, office, cafÃ©) and have the learner adjust cleaning frequency and justify each change.

Differentiation:

- **Level 1 (supported):** provide a step-by-step cleaning card and a partially filled Maintenance Plan Template; pair with a stronger learner for the internal clean.
- **Level 2 (standard):** the exercise as written.
- **Level 3 (challenge):** perform the full clean **and** remove the cooler to re-apply thermal paste (supervised), verify with a before/after temperature graph, and design a maintenance schedule for a **cyber cafÃ©** with a per-machine logbook.

Timing guidance:
This session is hands-on and can be messy. Protect the guided practical and the before/after temperature exercise â€" that is the core learning. If time is short, trim the case study to a two-minute mention and demonstrate external cleaning briefly. Ensure adequate ventilation and time to tidy up. Do **not** rush the safety steps; a rushed clean with a vacuum or a tilted can can destroy a machine.

APPENDIX 8A â€" 10-MINUTE SAFETY REFRESHER (only if needed)

2. **Discharge** â€" press the power button after unplugging to drain leftover electricity.
3. **ESD in cleaning** â€" say: "Rubbing a cloth or brush builds **static**. Static silently kills components. That is why we ground ourselves with the strap and use antistatic tools."
4. **The tool rule** â€" say three times: **"Compressed air and brush INSIDE. Vacuum NEVER inside."**
5. **The liquid rule** â€" "Liquid goes on the **cloth**, never on the screen or into a port."
6. **The golden rule** â€" **"Never open the PSU."** Demonstrate once, then have each learner demonstrate the two must-dos and state the tool and liquid rules before they touch a machine.

END OF SESSION 8

Next: Session 9 â€" Operating System Installation.

SESSION 9 â€" OPERATING SYSTEM INSTALLATION

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 9 â€" OPERATING SYSTEM INSTALLATION

A. SESSION IDENTIFICATION

Item	Detail
Session number	9 of 12
Session title	Operating System Installation
Week / Module	Week 3 â€" Module 3: Repair and Maintenance
Recommended duration	105 minutes
Difficulty level	Beginner (guided, hands-on, safety-critical)
Prerequisite knowledge	Sessions 1â€"8 â€" components, safety and tools, disassembly/reassembly, the diagnostic workflow, hardware and software diagnostics, component replacement, cleaning and maintenance. A learner must already know how to open a case safely and to back up files.
Connection to previous work	In Session 6 you learned to diagnose software faults such as a corrupted operating system. In Session 7 you replaced a hard drive or SSD. Today you complete the natural next step: putting a fresh, working operating system onto a machine so the customer can use it again.
Preparation before class	Academy practice PCs ready; a **licensed** Windows ISO and a USB flash drive (8 GB or larger) per pair; the official Microsoft Media Creation Tool downloaded while the academy internet is working; a spare network/board driver folder on a second USB; a laptop connected to the projector; a printed "OS Install Checklist" and "Customer Intake Form" per learner.
Safety flag	This is a **hands-on, data-destroying** session. No drive is wiped in class until (a) data has been backed up and (b) the learner has stated out loud that they understand the wipe is permanent. Only **properly licensed** Windows is used. Piracy is never taught, demonstrated or tolerated.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain in plain English** what an operating system is and why a corrupted OS sometimes has to be reinstalled rather than repaired.
2. **Back up a customer's data correctly** before any wipe, and explain why data safety comes before every other step.
3. **Check whether a machine meets the minimum hardware requirements** for a given Windows version and say what happens if it does not.
4. **Create a bootable Windows USB drive** using the official Microsoft Media Creation Tool, in the correct order.
5. **Enter the BIOS/UEFI setup or the one-time boot menu** and set or select the correct boot device.
6. **Perform a clean install of Windows**, correctly choosing "Custom: Install Windows only (advanced)" and installing onto unallocated space.
7. **Explain in beginner terms** why we partition a drive and what a clean install removes.

10. **Restore the user's data** and complete a **post-install verification** checklist (Device Manager, display, sound, network, essential software).

C. REQUIRED MATERIALS

Hardware

- At least one practice desktop PC that can be wiped (per pair of learners). If only one machine is available, the instructor performs the live install while pairs complete the **simulated install checklist** in parallel.
- USB flash drive, **8 GB minimum** (16 GB recommended). Note: creating the media erases the USB drive.
- A second small USB stick holding the **network/board driver** and any other offline drivers, in case the installed Windows has no working network driver.
- Tool kit and an **anti-static wrist strap** (even though we are not opening the case much, we may reseat cables).
- A power socket that learners can reach; keep liquids off the bench.

Software

- The **official Microsoft Media Creation Tool**. Download only from the genuine Microsoft website. As a general guide: search "Download Windows 11" or "Download Windows 10" from Microsoft, and use the Media Creation Tool page. (Website layout occasionally changes; the tool name stays the same.)
- A **valid, licensed** Windows product key or a machine with a **digital licence** already attached to its hardware.
- Offline driver files: network/LAN, Wi-Fi, chipset, graphics, audio.

Printed materials

- Worksheet 9.1 â€" "Pre-Installation Checklist".
- Worksheet 9.2 â€" "Post-Installation Verification".
- "Customer Intake Form" (reused from Session 10 preparation, to show the data-backup consent line).

Safety equipment

- Anti-static wrist strap per learner; safety glasses if compressed air is used during cleaning of the intake vents.

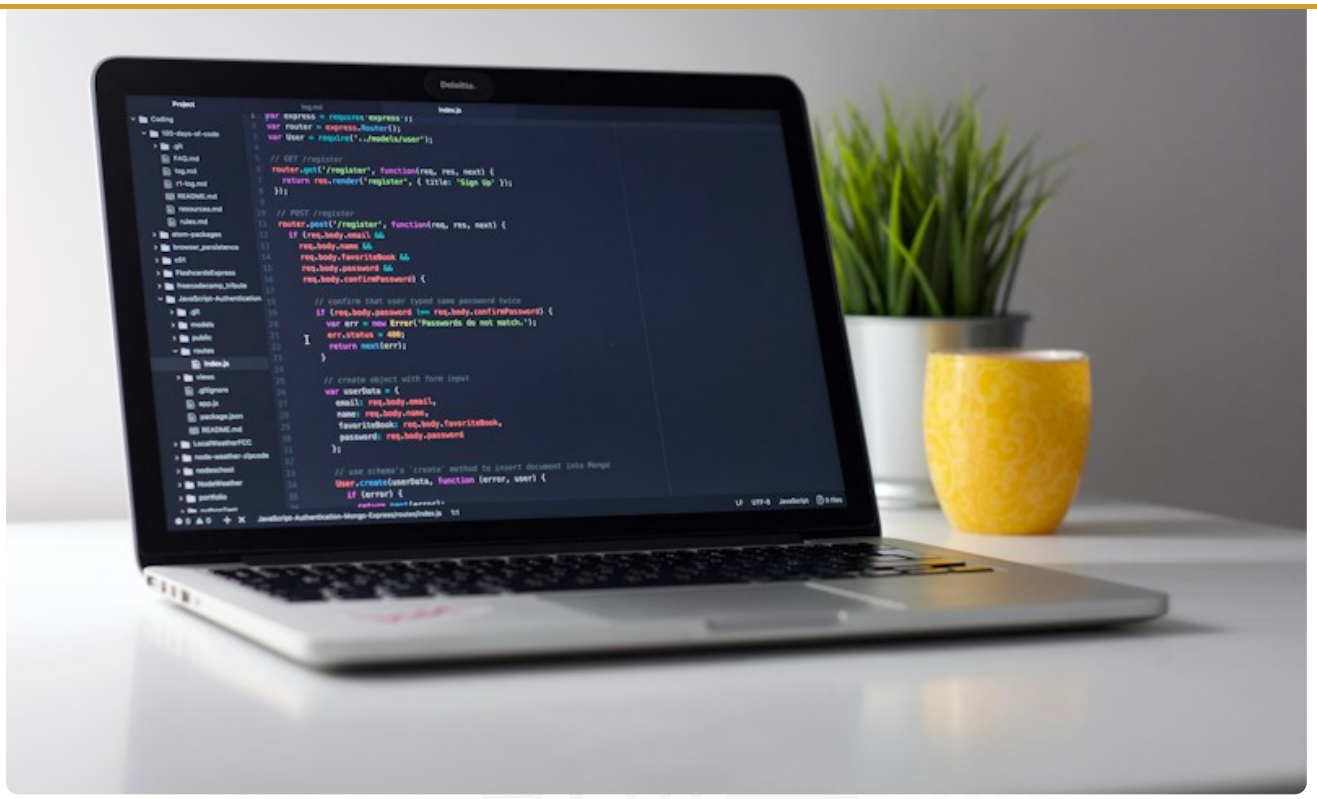
Internet

- Required for downloading the Media Creation Tool and the ISO. If the academy internet is slow, download once to a shared USB and copy it to pairs. Stress: after installation, the **machine's own** network driver may not exist â€" that is why the offline driver USB matters.

Alternative for learners who lack a facility

- A learner who owns a laptop with at least 4 GB RAM and a spare 8 GB USB may practise at home **only if they have a licensed Windows key and have backed up first**. If they have only a phone, pair them with a desktop user and have them fill the printed checklist and narrate the steps.

Suggested visuals:



- â€” Installing Windows from a USB drive



- â€” Pre- and post-installation checks on the bench



- â€" Loading offline drivers and verifying the system



- — Installation media and drivers prepared on a laptop



— Post-installation performance check on the dashboard

D. OPENING ACTIVITY (6 minutes)

Instructor setup:
On the projector, show a machine that boots to a black screen with the message "*An operating system wasn't found. Try disconnecting any drives that don't contain an operating system.*" Or, if unavailable, show a photo of the same message. Do not explain it yet.

What the instructor says:

> *"Look at this screen. The customer brought this computer to the shop in tears. It worked yesterday. Today it says the operating system is gone. She has three years of her shop's records, her children's photos and her church choir videos on it.*
>
> *She wants it working again â€" today. Some of you are thinking, 'Just reinstall Windows.' That is exactly right, and it is what we do today. But look again at her face. If we reinstall carelessly, we destroy everything on that drive. So the real job is not just installing Windows. The real job is:* ***save her data first, then put the computer right.***
>
> *Today you will learn to do the whole thing properly: back up, check the machine, build the installer, boot from it, wipe and install, then bring the machine back to full working order. And we will only ever use a*
>
> ***licensed***
> *copy of Windows. A repair shop that pirates software is not a repair shop you can trust."*

Expected learner response:
Learners may say "just reinstall", "won't the files stay?", or "is the data already lost?". Accept all answers. Correct gently that a **clean install erases the drive** and reassure them that backing up first is precisely how we protect the customer.

Why this opener works:
It reframes OS installation from a purely technical task into a *customer-trust and data-safety* task. That is the professional mindset this whole week builds. It also plants the licensing rule early.

Question 1:
"Name the six major components of a computer."

- Expected answer: motherboard, CPU, RAM, storage (HDD/SSD), PSU, GPU.
- Common gap: learners forget the PSU or call the motherboard "the board". Prompt: "Which part turns wall electricity into the voltages the PC needs?"

Question 2:
"What do we call the software that manages all the hardware and lets you run programs?"

- Expected answer: the operating system (Windows).
- Common gap: learners say "Windows" only. Clarify that Windows is one example of an operating system; others exist (macOS, Linux). We install Windows in this course.

Question 3:
"What is a driver, and what does a yellow exclamation mark in Device Manager mean?"

- Expected answer: a driver is software that lets Windows talk to a piece of hardware; a yellow warning means the driver is missing or not working.
- Common gap: learners think drivers are only for printers. Clarify: every major component â€" chipset, network, graphics, audio â€" needs a driver.

Question 4:
"According to the diagnostic workflow, what comes before you replace or wipe anything?"

- Expected answer: observe, ask questions, form a hypothesis, test. Only after confirming the fault do you act.
- Common gap: learners jump to "reinstall". Reinforce: confirm the OS itself is the problem before wiping a working system.

Bridge to today (say this):

> *"Diagnosing told us the operating system is broken and cannot be repaired cheaply. Today we treat the patient. But remember the doctor's first rule: do no harm. Before we operate, we save the patient's belongings. That is the backup."*

Gap handling:
If more than a third of the class cannot explain what a driver is (Question 3), spend five extra minutes drawing a quick diagram: Windows â†" driver â†" hardware, and show a real Device Manager screen with one yellow icon. Do not skip this â€" the post-install half of today's session depends on it.

F. FULL LECTURE NOTES

F.1 What an operating system actually is

The idea.
An **operating system (OS)** is the master software that sits between the user and the hardware. It starts the computer, manages memory, storage, files, programs and devices, and gives you the screen you click on.

Plain-English definition.
The operating system is the **manager of the whole computer**. Without it, the hardware is just metal and silicon that does nothing useful when you press the power button.

Why it exists.
Your music player, your browser and your documents cannot each talk directly to the disk and RAM safely â€" they would fight over the same memory and files. The OS keeps order: it decides who uses what, when, and how. It also provides the desktop, the Start menu, the taskbar and the ability to install programs.

- **Manages files** in folders you can navigate.
- **Manages memory** so programs do not collide.
- **Talks to hardware** through drivers.
- **Provides security** (accounts, passwords, permissions, Windows Defender).

Everyday analogy.
Think of a big **hotel**. The rooms, beds and plumbing are the hardware. The **manager** (the OS) decides who gets which room, handles keys (accounts and passwords), makes sure guests do not fight over the same room (memory), and calls the plumber (driver) when a tap leaks. A hotel with no manager is chaos.

Realistic scenario (Port Harcourt).
A cyber cafÃ© in Rumuola has 8 machines. The cafÃ© owner bought them with Windows pre-installed. When a virus corrupts Windows on one machine, the cafÃ© loses a seat/hour of income. A technician who can rebuild the OS in under two hours keeps that cafÃ© earning.

Common misconception.
"The operating system is stored on the hard drive, so if Windows breaks, my documents are gone too." Not automatically true â€” but a **clean install** will destroy them if you do not back up first. Files and Windows share the drive. That is the whole reason data safety is step one.

Limitation.
Windows is **not free**. It requires a valid licence. Installing it without a licence is both unethical and illegal. In this course we only ever install Windows that has been properly licensed â€” usually through a key the customer already owns, or a **digital licence** tied to that specific machine's hardware.

F.2 When to reinstall the OS â€” and when not to

The idea.
Reinstalling Windows is powerful but heavy-handed. It is the correct fix for some problems and the wrong fix for others.

Good reasons to do a clean install.

- The OS is so corrupted that repair tools (System Restore, Startup Repair, `sfc`, DISM) fail or the machine will not boot at all.
- The drive is being **replaced** (a new SSD has no operating system on it).
- The machine is badly infected with malware that cannot be fully removed.
- The customer explicitly wants a fresh start on a machine they own and understand the data consequences.

Reasons NOT to reinstall immediately.

- The real fault is hardware (bad RAM, failing drive, failing PSU). Reinstalling Windows on failing RAM just wastes an hour and frustrates the customer.
- The problem is only a missing or faulty driver â€” install the driver instead.
- The user simply lost a file or a program â€” try to recover or repair first.
- The data has **not** been backed up. You never wipe before you back up. Full stop.

Why this matters professionally.
Reinstalling is the technician's version of "have you tried turning it off and on?" If you reinstall for every problem, customers pay for work they did not need, and their data is put at needless risk. Confirm the diagnosis first.

Common misconception.
"A reinstall protects the customer's files automatically because Windows keeps an 'old Windows' folder." The so-called Windows.old folder is created only in some **upgrade** situations, not a clean custom install, and it never contains the user's personal documents. Never rely on it â€” always back up.

F.3 Data safety first â€” the backup step

What counts as "the customer's data".

- Documents: Word, Excel, PDFs.
- Photos and videos (often the most precious).
- Their Desktop, Documents, Downloads, Pictures, Music and Videos folders.
- Email data if stored locally (for example mailbox files).
- Browser bookmarks and passwords where recoverable.
- Business records, church or school files, accounting files.
- Settings the customer cares about (wallpaper, favourites).

Where to put the backup.

- An **external hard drive** or large USB drive (best for big photo/video collections).
- Cloud storage if the customer already has an account and the internet is fast enough.
- Never on the same drive you are about to wipe.

Golden rules.

1. **Back up before you touch the drive.** If the machine will not boot, take the drive out and connect it to a working PC (or a USB-to-SATA adapter) to copy the files. *This is a Session 7 skill in action.*
2. **Verify the backup** by opening a few copied files. A backup you never checked is not a backup.
3. **Tell the customer what you could and could not save**, and get them to sign the intake form saying they understand a clean install wipes the rest.
4. **Keep the customer's data confidential.** Do not share it, post it or copy it for personal use. This is a professional and legal duty (covered more in Sessions 10 and 11).

Realistic scenario.
Mama Ngozi's church office PC will not boot. The secretary's only copy of the harvest records is on it. A careless technician wipes the drive and installs Windows "fresh" â€" and destroys two years of records. A professional technician first pulls the drive, copies the records to an external drive, confirms they open, *then* wipes. Same one hour of work; completely different outcome for the customer.

Analogy.
The backup is like moving the family photos out of a house before demolishing and rebuilding it. You do not rebuild first and hope the photos survive.

Limitation / honesty.
Sometimes a failing drive cannot be read to back up at all. That is a data-recovery situation â€" a specialised, higher-level skill **beyond this course**. Do not promise the customer data you cannot rescue. Be honest and, if needed, refer them to a specialist.

F.4 Checking hardware meets requirements

The idea.
Before installing an OS, confirm the machine is powerful enough. Installing Windows on a machine below the minimum requirements leads to a slow, unstable, unhappy result.

General minimums (as a beginner guide).

Version	CPU	RAM	Free storage	Other
Windows 10 (64-bit)	1 GHz or faster	2 GB	32 GB	DirectX 9 graphics
Windows 11	64-bit, 1 GHz, 2 or more cores	4 GB	64 GB	UEFI, Secure Boot capable, TPM 2.0, DirectX 12

> *Exact figures and the "PC Health Check" wording can change with Windows updates. Always confirm on Microsoft's requirements page. For this course, the figures above are a safe teaching baseline.*

How to check quickly on a working machine.

- Press `Windows key + Pause/Break` (or right-click **This PC â†' Properties**) to see processor and installed RAM.
- Open **This PC** to see drive size and free space.
- For Windows 11, Microsoft provides a **PC Health Check** tool that tells you whether a machine qualifies.

Why it exists.
Windows 11 in particular refuses to install on many older machines because it requires **TPM 2.0** and **Secure Boot**, which are security features. A machine without them may be perfectly fine for Windows 10 but not eligible for Windows 11.

Realistic scenario.
A student brings a 2012 laptop with 2 GB RAM and asks for "the newest Windows". The honest professional answer is: Windows 11 needs 4 GB RAM and TPM 2.0, which this machine does not have. Recommend Windows 10 (if it meets those minimums) or a RAM upgrade â€" do not force a version the hardware cannot run.

Common misconception.
"More RAM, more speed, always." Often true, but a machine whose CPU is very old or whose drive is a slow HDD will still feel sluggish. Match the OS to the hardware; be honest with the customer.

Limitation.
Meeting minimum requirements means Windows *will run*, not that it will run *well*. For comfortable everyday use, recommend more RAM and, ideally, an SSD.

F.5 Creating a bootable Windows USB (Media Creation Tool)

The idea.
A "bootable USB" is a USB flash drive that the computer can start from, carrying the Windows installer. We build it with Microsoft's own free tool.

Why use the official Media Creation Tool.

- It is free and comes directly from Microsoft, so the installer is genuine (no tampered, pirated or virus-carrying copies).
- It downloads the correct Windows files and formats the USB so the PC can boot from it.
- It reduces user error compared with manually formatting drives.

Step-by-step (accurate as of current Windows; menu wording may vary slightly by version).

1. On a **working** internet-connected PC, go to Microsoft's official download page and get the **Media Creation Tool** for the Windows version you intend to install.
2. **Back up the USB drive** â€" everything on it will be erased.
3. Plug in a USB flash drive of **8 GB or more**.
4. Run the tool and **accept the licence terms**.
5. When asked what you want to do, choose **"Create installation media (USB flash drive, DVD, or ISO file) for another PC"**.
6. Choose **language, edition and architecture** (for most modern machines, **64-bit (x64)**). If unsure, leave the recommended defaults.
7. Choose **"USB flash drive"** as the media type.
8. **Select your USB drive** from the list â€" double-check the drive letter and capacity so you do not erase the wrong drive.
9. Click **Next** and wait. The tool downloads Windows and writes it to the USB. This can take 20â€"60 minutes depending on internet speed. **Do not unplug the USB or close the tool.**
10. When it says **"Your USB flash drive is ready"**, click **Finish** and safely remove the USB.

How the learner recognises success.

Realistic scenario.
A shop in Diobu cannot afford to keep a DVD writer. The technician keeps one good 16 GB USB with the official Windows media, plus a second USB with the machine's drivers. Those two sticks rebuild almost any customer PC.

Common misconception.
"Any Windows ISO from anywhere is fine." No. Unofficial "cracked" or "activated" ISOs are pirated, often carry malware, and are against academy policy and the law. Only the official Microsoft tool/ISO and a legitimate licence are acceptable.

Limitation.
The tool needs internet and time, and it **wipes the USB**. If the academy internet fails, plan ahead: download once and keep a master USB that you copy from (using a tool like Rufus is a more advanced option, not required for this course).

F.6 BIOS/UEFI and boot order

The idea.
When a computer powers on, a tiny program runs **before** Windows. It checks the hardware and then decides **which device to start from** â€" usually the internal drive, but we need it to start from our USB.

Two names, one job.

- **BIOS** â€" the older system used by many older machines.
- **UEFI** â€" the modern replacement; it supports **Secure Boot**, **GPT** disks and other security features.

Both do the same basic job: they are the first code that runs and they hand control to a bootloader.

How to get in.

- Watch the very first screen for a message like *"Press DEL to enter Setup, F2 for BIOS, F12 for Boot Menu."* Keys vary by manufacturer â€" common ones are **Del, F2, F10, F12, Esc, F9**.
- The **one-time boot menu** (often `F12`, `Esc` or `F9`) lets you pick the USB for this boot only, without changing settings permanently. This is the safest and easiest method in a repair shop.
- The **BIOS/UEFI setup** screen (usually `Del` or `F2`) lets you change the permanent **boot order**.

What to change.

- Put the **USB drive first** in the boot order, or select it from the one-time boot menu.
- For modern Windows 11, ensure **UEFI mode** is used and **Secure Boot** is on (if the machine supports it). Secure Boot helps prevent malware from loading before Windows.
- Save and exit (often `F10`, then confirm **Yes**).

Analogy.
The BIOS/UEFI is the **gatekeeper at the compound gate**. Every morning it decides which vehicle is allowed to drive in first. Normally it lets the family car (the internal drive) in. Today we ask it to let our delivery van (the USB) in first, because it is carrying the new furniture (Windows).

Realistic scenario.
A learner builds the USB correctly but the machine still boots to the old Windows. Cause: the boot order still points to the internal drive, or the USB was plugged into a port that was not scanned at boot. Fix: use the one-time boot menu (`F12`) and select the USB explicitly.

Common misconceptions.

- "Pressing the key once at any time opens BIOS." It must be pressed **repeatedly during the first few seconds**, before Windows starts loading.
- "Legacy and UEFI are the same." They boot differently and use different partition styles (MBR vs GPT). Installing in the wrong mode can make Windows fail to boot.
- "Secure Boot is something I must switch off." Do not disable it casually; it is a security feature. Only change it if the machine genuinely cannot boot the USB and you understand the trade-off.

Teach learners that the **concept** is constant even when the words differ.

F.7 The clean install and partitioning

The idea.
A **clean install** means we install Windows fresh onto a drive that has been emptied of the old system.

Step-by-step during Windows Setup (wording may vary by version).

1. The machine boots from the USB and Windows Setup opens.
2. Choose **language**, **time and currency format**, and **keyboard layout**, then click **Next â†' Install now**.
3. If asked for a **product key**, enter a valid licensed key, or choose **"I don't have a product key"** if you will activate later with the customer's key/digital licence. (Windows still needs a valid licence to be legitimate.)
4. Select the **edition** that matches the customer's licence (for example Windows 10/11 Home or Pro). Do not install an edition the machine is not licensed for.
5. **Accept the licence terms.**
6. Choose the installation type. You will see two options:
    - **Upgrade: Install Windows and keep files, settings and applications** â€" keeps the old system's personal files and apps (only offered when starting Windows from inside a working Windows).
    - **Custom: Install Windows only (advanced)** â€" this is the **clean install**. Choose this.
7. You now see a list of **drives and partitions**. To truly clean the drive:
    - Select each existing partition on the target drive and click **Delete**.
    - Repeat until only **"Drive 0 Unallocated Space"** remains. **This permanently erases the drive.** (Remember: the backup is already done.)
    - Select the **unallocated space** and click **Next**.
8. Windows copies files, installs features and updates, and **restarts several times**. **Do not remove the USB** until setup has fully completed and you are at the desktop setup screen. (In some configurations the machine may boot from the USB again; the one-time boot menu avoids this.)
9. Complete the **out-of-box setup (OOBE)**: region, keyboard, network (if available), user account and password, privacy settings. Then Windows finishes setting up.

Why partition at all?
A **partition** is a defined section of a physical drive. Windows does not just need one â€" it creates several automatically when you install to unallocated space:

- An **EFI system partition** (on UEFI/GPT disks) or a small reserved partition for boot files.
- A small **Microsoft Reserved** partition.
- The main **Windows (C:)** partition where the OS and programs live.
- Usually a **Recovery** partition.

These extra partitions hold the boot files and recovery tools Windows needs. When you install to unallocated space, Windows creates them for you â€" you do not have to hand-make them at this level.

Should we split the drive into a C: and a D:?
For beginners, keep it simple: let Windows create the C: partition automatically. Larger businesses sometimes create a separate data partition so that a future reinstall of C: does not touch their documents. That is a valid technique, but it is optional and must be planned **before** the wipe.

What a clean install removes.

- The old operating system.
- All installed programs.
- **All personal files on that drive** (unless backed up).

- Files on a **different** physical drive that you did not touch.
- The hardware itself.

Analogy.
Partitioning is like drawing rooms on a large empty plot of land. Windows needs a few small utility rooms (boot, recovery) plus a big living room (C:). Installing to unallocated space is telling Windows: "Here is empty land â€" build the house your way."

Common misconception.
"If I choose Custom but do not delete partitions, my files are safe." Not necessarily â€" installing over an existing partition still overwrites much of it. If you intend to keep data, you must back it up; Custom + Delete is the clean, predictable choice.

Limitation.
A clean install gives you a fresh Windows but **no programs, no personal files and no drivers** â€" you must restore all of those afterwards.

F.8 Drivers, updates, activation, software, data restore and verification

The idea.
Installing Windows is only half the job. A freshly installed Windows is like a new house with no furniture, no electricity connected and no paint. The rest of the work makes it usable.

1. Install drivers â€" in the right order.

- **Network driver first.** Without it, the PC cannot reach the internet, so you cannot download any other driver. This is why we keep a copy of the network/board driver on the offline USB.
- Then **chipset**, then **graphics**, then **audio**, then Wi-Fi/Bluetooth and any others.
- How to install: run the driver installer from your USB, or use Windows Update which often finds common drivers automatically.
- **Check Device Manager** (`Windows key + X â†' Device Manager`). A **yellow exclamation mark** next to a device means a driver is missing or faulty.

2. Run Windows Update â€" repeatedly.

- Open **Settings â†' Update & Security â†' Windows Update** (Windows 10) or **Settings â†' Windows Update** (Windows 11).
- Click **Check for updates**, install everything, and **restart** when asked.
- **Repeat** until it says no more updates are available. New updates often appear after the first round, so one pass is rarely enough.

3. Activate Windows â€" legally.

- Open **Settings â†' Update & Security â†' Activation** (Windows 10) or **Settings â†' System â†' Activation** (Windows 11).
- If the customer has a valid key, enter it. If the machine has a **digital licence** tied to its hardware, Windows may activate automatically after it connects to the internet.
- The academy rule: **only properly licensed Windows.** Never use cracked tools, "activators", key generators or pirated keys. They are illegal, and they carry malware and leave customers exposed.

4. Install essential software.

- A web browser if the customer wants one (Edge is built in).
- **Windows Defender** is already built in â€" it is the free, legitimate antivirus. Update it.
- Only install software the customer is **licensed** to use. Do not install pirated Microsoft Office or "cracked" versions of paid programs.

5. Restore the customer's data.

- Copy the backed-up files from your external drive back into the customer's **Desktop, Documents, Pictures** folders.
- Check a few files open correctly. Tell the customer what was restored.

- **Display:** correct resolution, no weird stretching.
- **Sound:** play a short audio clip; confirm speakers/headphones work.
- **Network:** open a web page; confirm internet and, ideally, Wi-Fi.
- **Windows Update:** fully up to date.
- **Activation:** Windows reports "Windows is activated".
- **Storage:** correct drive shows its full capacity (for example a 500 GB SSD shows roughly 465 GB usable).
- **Customer sign-off:** show the customer the machine working and get them to sign the service report (Session 11).

Analogy.
Drivers are like the **keys and connections** for each appliance in the new house â€" plumbing to the kitchen, electricity to every socket. Windows Update is the **maintenance schedule** that fixes known problems. Activation is the **legal title deed** proving the house is properly owned.

Common misconception.
"Windows Update installs every driver, so I never need the offline USB." Sometimes true, sometimes not â€" especially for network drivers on older or unusual hardware. Always carry the offline network driver. Prepare before you need it.

Limitation.
Some hardware (very new or very old) may need a driver the customer must supply, particularly for specialised devices like a fingerprint reader or an unusual printer. Be honest if you cannot find one.

F.9 Licensing and ethics â€" non-negotiable

- **Only install Windows the customer is licensed to use.** A valid product key, or a digital licence already attached to that machine, is what makes the install legitimate.
- **Never** teach, use or recommend pirated Windows, "cracked" activators, key generators or modified ISOs. They are illegal, unsafe and damage the academy's and your own reputation.
- If a customer asks you to install Windows without a licence, explain politely that you install only properly licensed software, and offer to help them buy a legitimate licence or use the free, legal alternatives where they fit their needs.
- Keep the **customer's data confidential** and handle it only for the repair.

Say this to the class:

> *"In this trade, your name is your business. One stolen licence can cost you every customer who hears about it. We do it right, or we do not do it."*

F.10 Summary of the session

- The **OS** manages the whole computer; a **clean install** is a heavy but sometimes necessary fix.
- **Back up first** â€" a clean install erases the drive.
- Check the machine **meets the OS requirements** (especially for Windows 11: RAM, TPM 2.0, Secure Boot).
- Build a bootable USB with the **official Microsoft Media Creation Tool**.
- Set the **boot order / boot menu** in **BIOS/UEFI**.
- Choose **Custom: Install Windows only (advanced)** and install to **unallocated space**; Windows creates the partitions.
- Install **drivers** (network first), run **Windows Update** until none remain, **activate legally**, add essential software, **restore data**, and **verify** with Device Manager and tests.
- Only ever use **licensed** Windows.

What the instructor prepares:

- A practice PC that is safe to wipe (or, if it cannot be wiped live, a machine already clean, plus the prepared installer to show the first screens).
- The official **Media Creation Tool** already downloaded, plus the USB stick.
- The offline **driver USB**.
- A projector showing the target machine.

> ***Instructor note:***
> *If time or hardware does not allow a full live install, demonstrate the **media creation** and the **boot/setup screens**, then walk the class through the printed checklist for the rest. Never start a wipe on a machine whose data has not been backed up.*

G.1 Demonstration 1 â€" Building the USB (5 minutes)

Say:
"Watch how Microsoft's own tool builds our installer. I do this once, and this one USB can rebuild many machines."

Do this on the projector, narrating each step:

1. Plug in the USB and open the Media Creation Tool.
2. Accept the licence terms.
3. Choose **"Create installation media for another PC"**.
4. Keep the recommended **language/edition/architecture (64-bit)**.
5. Choose **"USB flash drive"**, then **select the correct drive** â€" *pause here and say:* "Look carefully at the drive letter and size. Choosing the wrong drive erases someone's backup."
6. Click **Next**; show the progress screen. **Say:** "This downloads Windows and writes it to the USB. If we unplug now, we ruin it."

Expected result:
A finished screen saying the USB is ready, and (if you open it) folders such as `sources` and `boot`.

Error to watch:
If the learner selected the wrong drive, stop immediately. If the tool reports not enough space, the USB is smaller than 8 GB or is not empty.

G.2 Demonstration 2 â€" Booting from the USB (4 minutes)

Say:
"Now we ask the machine to start from our USB instead of its own drive."

1. Insert the USB. Power on.
2. **Repeatedly tap the one-time boot key** (on the demo machine, tell the class which key it is â€" for example `F12`).
3. In the boot menu, select the **USB drive**.

Expected result:
Windows Setup appears (language screen). Point out that this is Windows Setup, not the old desktop.

Ask the class:
"What if nothing happens and the old Windows loads?" â†' *The USB was not scanned, the key was pressed too late, or the USB is not bootable.*

G.3 Demonstration 3 â€" Setup and the wipe (4 minutes)

1. Choose **language**, click **Next â†' Install now**.

3. Accept the licence terms.
4. Choose **"Custom: Install Windows only (advanced)"**.
5. Show the partition list. Select the old partitions, click **Delete**, and show them turn into **"Unallocated Space"**.

Say (this is the critical line):
"Everything on this drive is now gone forever. This is why we backed up. If I had not backed up, this moment would have destroyed the customer's memories."

6. Select the **unallocated space â†' Next**. Show Windows begin copying files. **Say:** "It will restart several times. Do not remove the USB until we reach the desktop setup."

Questions to ask during the demo:

- "Why do we install the network driver first?" â†' Without internet, no other driver can be downloaded.
- "Why do we choose Custom, not Upgrade?" â†' Custom is the clean install; it does not keep the old broken system.
- "Which part of this demo could destroy customer data if done carelessly?" â†' Deleting partitions / wiping the drive.

G.4 Demonstration 4 â€" First boot and verification (2 minutes)

If a machine is available that is already installed (or after the demo completes), show:

- `Windows key + X â†' Device Manager` â€" point out any yellow icon and explain it.
- **Settings â†' Windows Update** â€" run it once to show the process.
- **Settings â†' Activation** â€" show the activation status.

Say:
"A machine is not 'done' when Windows installs. It is done when every device works, it is updated, it is legally activated, the customer's data is back, and the customer has signed."

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective:
With the instructor guiding step by step, learners complete the **pre-installation preparation** on paper/real USB, then either perform or simulate a clean install and complete the post-install checklist.

Starting materials:
Per pair: one practice PC, one USB (or the instructor's shared USB), the offline driver USB, Worksheet 9.1 and Worksheet 9.2. If there is only one safe-to-wipe machine, learners take turns on the live machine while others complete the simulated checklist.

> ***SAFETY AND DATA SAFETY FIRST â€" read before starting:***
> *confirm with the instructor that any machine to be wiped holds **no data the academy needs**, and that the customer's data (in the simulated case) has already been backed up and verified.*

Procedure (instructor calls each numbered step; learners follow; instructor walks the room):

1. **State the fault out loud.** "The operating system is corrupted; the fix is a clean install." Confirm this against the diagnostic workflow.
2. **Back up (simulate or real).** Using the Customer Intake Form, list the folders to save: Desktop, Documents, Downloads, Pictures, Music, Videos. If the machine does not boot, describe how you would remove the drive and copy the data on a working PC (Session 7 skill). **Pause and predict:** "*If we skip this step, what exactly is lost?*" â†' All personal files on the drive.
3. **Check requirements.** Record the machine's CPU, RAM and drive space on Worksheet 9.1. Decide Windows 10 or 11 appropriately.
4. **Gather drivers.** Confirm the offline network/board driver is on the second USB.

6. **Set the boot device.** Power on, tap the boot key, select the USB. **Pause and predict:** *"What screen should appear if we did this correctly?*" â†' Windows Setup.
7. **Start Setup.** Language â†' Install now â†' (key or skip) â†' accept terms.
8. **Choose the clean install.** Select **Custom: Install Windows only (advanced)**.
9. **Wipe and install.** Delete existing partitions to leave **Unallocated Space**, select it, click **Next**. **Pause and predict:** *"Is the data on this drive still recoverable normally after this?*" â†' No.
10. **Wait for install.** Let it copy files; the PC restarts several times. Do not remove the USB.
11. **OOBE setup.** Region, keyboard, network, account, privacy.
12. **Install the network driver**, then chipset, graphics, audio.
13. **Run Windows Update** at least twice until none remain.
14. **Activate Windows** with a licensed key/digital licence. Confirm status.
15. **Install essential software** (browser, confirm Defender updated). No pirated software.
16. **Restore the backed-up data** and open two files to confirm they work.
17. **Verify** using Worksheet 9.2 (Device Manager clean, display, sound, network, activation, storage).

Expected result:
A working, updated, activated Windows with the customer's data restored and no missing drivers.

Pause-and-predict checkpoints:
after step 2 (what is lost), step 6 (which screen), step 9 (is data still recoverable), step 12 (why network driver first).

Troubleshooting while walking the room:

- USB does not boot â†' wrong boot key/order, USB not bootable, or USB in a port not scanned; try a rear USB port on a desktop.
- "No drives were found" â†' the storage controller driver (for example Intel RST) may be needed; a more advanced fix â€" tell the learner to keep the drive connected and ask the instructor.
- Setup asks for a key and refuses to continue â†' choose the "I don't have a product key" option if offered, and plan to activate later with a valid licence.
- Internet still not working after setup â†' network driver not installed; run the offline driver.

Completion checklist (learner ticks each):

- ☐ Data backed up and verified before any wipe
- ☐ Hardware requirements checked and recorded
- ☐ Bootable USB prepared correctly
- ☐ Booted from USB to Windows Setup
- ☐ Chose **Custom** and installed to unallocated space
- ☐ Network driver installed
- ☐ Windows Update run until none remain
- ☐ Windows licensed and activated
- ☐ Essential software installed (all licensed)
- ☐ Customer data restored and opened
- ☐ Device Manager shows no yellow warnings
- ☐ Post-install verification complete

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task (NEW task):
You are handling **two separate customer jobs** at the same time. Decide what to do for each, in the right order, and write the exact steps. This is not a repeat of the guided install â€" it tests judgement, not just memory.

up. She asks you to "reinstall Windows".

Scenario B â€" "The Dead Desktop" (Mr. Boma, custom desktop, 500 GB SSD, will not boot)
The desktop shows *"An operating system wasn't found."* The customer says his **only copy** of his business accounts is on the Desktop of that drive. He is licensed: the machine has a Windows digital licence.

Requirements:

1. For **each** scenario, write the correct order of actions, including the **data-safety decision** you make first.
2. For **Scenario A**, answer: Should you reinstall straight away, or do something else first? Justify it.
3. For **Scenario B**, describe how you get the data **if Windows will not boot**, and how you will confirm the licence before activating.
4. For each, list the driver order you will follow after Windows is installed.
5. State the academy's rule about licensing in one sentence.

Time allowance:
15 minutes.

Success criteria:
Correct order, data safety addressed first in both, a sensible decision for Scenario A (not a blind wipe), a workable data-recovery-by-removing-the-drive plan for Scenario B, correct driver order, and a correct licensing statement.

Hints if stuck:

- Hint 1: Ask "is Windows actually broken, or is the problem something else?" for Scenario A.
- Hint 2: If Windows will not boot, how did you copy data from a failing drive in Session 7?
- Hint 3: The driver order always starts with **network**, then chipset, graphics, audio.

INSTRUCTOR ANSWER KEY (do not show learners):

Scenario A â€" right order:

1. **Do not wipe immediately.** First diagnose: a slow machine full of viruses may be fixable without a clean install. Confirm with the diagnostic workflow.
2. **Back up the data first** regardless of which path you take â€" copy Desktop, Documents, Pictures etc. to an external drive and verify.
3. Run a proper malware scan (Windows Defender plus, if appropriate, a reputable scanner), remove malware, run Windows Update, and consider a **RAM upgrade** or an **SSD replacement** if the 1 TB HDD is the real cause of slowness (often it is).
4. Only if the OS is genuinely unrecoverable do you proceed to a **clean install**: boot from USB â†' Custom â†' wipe to unallocated â†' install.
5. Justification: the fault is likely a combination of malware and an old HDD, not necessarily a dead OS. A blind reinstall risks data and may not even fix the slowness.

- Honest note for the answer: recommend an **SSD upgrade** â€" it will change the laptop far more than a reinstall, but it is optional and must be approved by the customer first.

Scenario B â€" right order:

1. **Data safety first.** The drive holds the only copy of the accounts and Windows will not boot. Do **not** attempt a clean install until the data is saved.
2. **Remove the SSD**, connect it to a working PC (USB-to-SATA adapter or an internal SATA port), and copy the Desktop/business files to a separate external drive. Verify the files open. (Session 7 skill.)
3. If the SSD is readable and healthy, proceed with the **clean install** on that same drive (or a new drive if the SSD is failing). Use the **official Media Creation Tool** to make the USB.
4. Boot from USB â†' **Custom** â†' delete partitions to unallocated â†' install.
5. **Licence:** this machine has a **digital licence** tied to its hardware, so after installation and connection to the internet, Windows may activate **automatically**. Check **Settings â†' Activation**. If it does not auto-activate, use the customer's legitimate method (linked Microsoft account or valid key). Never use a crack.

7. Restore the business files to the Desktop; confirm with the customer.

Licensing statement (any accurate version accepted):
"We install only properly licensed Windows â€” a valid key or a digital licence tied to the machine â€” and we never use pirated or 'cracked' software."

Marking guidance (out of 10):

- Data-safety decision correct in both scenarios (3)
- Scenario A judgement (diagnose/repair before wipe) correct and justified (2)
- Scenario B data-recovery-by-removing-drive method correct (2)
- Driver order correct (1)
- Licensing rule correct (1)
- Clear, ordered, professional write-up (1)

J. REAL-WORLD CASE STUDY

Situation.
Elias Tech Hub, a repair shop on Ebony Road, Port Harcourt, receives a school laptop from **Holy Family Primary School**. The teacher says it "just stopped working"; the school's term results are on the Desktop. The laptop will not boot past a blue recovery screen. The school has a **digital licence** for Windows 10 Pro on that machine.

Relevant facts.

- The drive is a 256 GB SSD and it is **readable** when connected to another PC.
- The term results are in a single `Results.xlsx` on the Desktop.
- The school needs the laptop back **the next morning** for a PTA meeting.
- The technician has the official Media Creation Tool USB and an offline network driver USB.

Decision point.
What order of operations protects the school, fixes the machine, and is honest about timing?

Possible approaches.

1. Wipe and reinstall immediately to save time â€” fastest, but destroys the only copy of the results.
2. Back up first (remove the SSD, copy the file), then clean-install, then restore and verify â€” safe and professional.
3. Tell the school the data is lost and only fix Windows.

Recommended solution.
Approach 2.

- Step 1: Remove the SSD and copy `Results.xlsx` to an external drive; open it to confirm it is intact.
- Step 2: Build/boot the official USB; clean-install Windows 10 Pro to unallocated space.
- Step 3: Install the network driver, run Windows Update, confirm the **digital licence activates automatically** (licence is tied to the machine).
- Step 4: Restore `Results.xlsx` to the Desktop; confirm the teacher can open it.
- Step 5: Write the service report (Session 11) and have the teacher sign.

Trade-offs.

- Approach 1 saves 30 minutes but risks the school's results and the shop's reputation. Unacceptable.
- Approach 2 takes longer but delivers a working, licensed, data-safe result the same day â€” realistic for a one-hour-grade job.
- Approach 3 abandons the customer needlessly; the data was recoverable.

Professional connection.

customers coming back.

Transfer question for the class:
"The school's laptop had a digital licence. What would you do differently if it had no licence at all?" *(Answer: explain that you install only licensed Windows; help them obtain a legitimate licence; do not pirate.)*

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table as the live troubleshooting guide. When a learner hits an issue, ask them to find the row rather than handing them the answer.

K.1 Installing before backing up

Symptoms:
You realise the customer's files were on the drive and you already deleted the partitions. **Likely cause:** Skipping data safety under time pressure. **How to investigate:** Stop immediately. Do not write any more data to the drive. **Safe corrective steps:** If genuinely required and the data is critical, **stop all use** and refer to a professional data-recovery service â€" this is **beyond this course** and must be handled by a specialist. Do not promise success. **How to confirm resolved:** A specialist recovers the data, or you inform the customer honestly. **Prevention:** Back up and verify **before** every wipe. Make it an unbreakable rule.

K.2 USB will not boot

Symptoms:
The old system loads, or a "no bootable device" message appears. **Likely causes:** Wrong boot key, USB not first in boot order, USB not bootable, or USB in a port not scanned. **How to investigate:** Enter the boot menu again; check the USB appears in the list. Try a different (rear) USB port on a desktop. **Corrective steps:** Rebuild the USB with the official tool if it is not bootable; select the USB explicitly. **Prevention:** Use a known-good USB and the official tool; keep the one-time boot key noted.

K.3 "No drives were found" or wrong disk shown

Symptoms:
Windows Setup cannot see any drive. **Likely causes:** A newer machine needs a storage/RAID driver (for example Intel RST); or the drive is genuinely not detected (loose cable or dead drive â€" a hardware fault). **How to investigate:** Check whether the drive appears in BIOS/UEFI. Reseat cables (safety: power off, anti-static). **Corrective steps:** Load the storage driver during setup (advanced), or fix the hardware fault first. **Prevention:** This is an advanced issue â€" ask the instructor; do not guess.

K.4 Internet not working after install

Symptoms:
No network, cannot download drivers. **Likely cause:** Network driver not installed. **Corrective steps:** Install the offline network driver from the USB, then run Windows Update. **Prevention:** Always prepare the network driver before starting.

K.5 Yellow exclamation marks in Device Manager

Symptoms:
Device Manager shows warning icons. **Likely cause:** Missing or faulty drivers. **Corrective steps:** Identify the device, install the correct driver (USB or Windows Update). **Prevention:** Install drivers in order (network, chipset, graphics, audio) and re-check Device Manager.

K.6 Windows will not activate

Symptoms:
"Activate Windows" watermark; activation fails. **Likely causes:** No valid key/digital licence; not connected to the internet; edition mismatch. **Corrective steps:** Connect to the internet and retry; confirm the licence belongs to the machine; ensure the correct edition was installed. **Prevention:** Confirm licensing **before** starting, and never rely on cracks.

The machine still feels under-updated; some drivers missing. **Likely cause:** Windows Update was run once. **Corrective steps:** Run it repeatedly (restarting) until no updates remain. **Prevention:** Remember the first pass triggers more passes.

Distinguish likely vs confirmed:
A "likely cause" becomes "confirmed" only after you check it (see the exact error, view BIOS, check Device Manager). Teach learners to confirm before fixing.

L. SESSION ASSIGNMENT

Assignment title:
"My Windows Clean-Install Runbook"

Purpose:
To make the learner write, in their own words, a safe, ordered, licensing-compliant procedure they could follow alone in a real shop â€" practising both technical accuracy and data-safety discipline.

Realistic scenario:
You have just been hired at *Blessed Hands Computers*, a small repair shop in Port Harcourt. On your first day, your boss hands you a customer's PC with a corrupted Windows and says, "Rebuild it properly â€" and do not lose a single file." You must follow your own written runbook.

Detailed instructions:

1. Write a **pre-installation section** that begins with the backup step and states exactly which folders you save and how you verify the backup.
2. Add a **hardware requirements check** section â€" how you check CPU, RAM and storage, and how you decide between Windows 10 and Windows 11.
3. Add a **media creation** section describing the official Microsoft Media Creation Tool steps.
4. Add a **BIOS/UEFI boot** section explaining how to reach the boot menu and select the USB.
5. Add an **install** section covering Custom install, deleting partitions to unallocated space, and the warning that this erases data.
6. Add a **post-install** section covering drivers in order, Windows Update (repeated), activation, essential software, data restore and verification.
7. Add a **licensing and ethics statement** in your own words.
8. Add a **troubleshooting table** with at least five problems and their fixes.
9. Finish with a **customer sign-off** line.

Required deliverables:

- A typed document (Word or a neatly written sheet) of **at least one full page**.
- The completed Worksheet 9.1 (pre-install checklist) and Worksheet 9.2 (post-install verification).

Submission format:
Printed or emailed to the instructor; filenames as `Surname_S9_Runbook.docx`.

Deadline:
Before Session 10.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Pre-installation & data safety	Backup first, folders listed, verification described	Backup mentioned, some detail	Backup missing or vague

Requirements & OS choice	Requirements checked; sound Windows 10/11 decision	Checked but decision weak	Not checked
Media creation & boot	Accurate official-tool steps; correct boot method	Mostly correct	Incorrect or missing
Install & partitioning	Correct Custom steps; wipe warning clear	Mostly correct	Wrong install type or unsafe
Post-install & verification	Drivers in order, updates repeated, activation, restore, verify	Most steps present	Many steps missing
Licensing statement & troubleshooting	Clear, honest licensing rule; 5+ good fixes	Present but thin	Missing or unsafe

Common mistakes to avoid:
putting media creation before backup; omitting the wipe warning; forgetting the network driver; running Windows Update once; using or implying pirated software.

Instructor reference solution (excerpt):

> *Pre-installation. 1) Interview the customer and confirm the fault. 2) Back up Desktop, Documents, Downloads, Pictures, Music, Videos to an external drive; open three files to verify. 3) If the PC will not boot, remove the drive and copy data on a working PC. 4) Check CPU/RAM/storage against requirements and choose Windows 10 or 11 accordingly. 5) Confirm licensing (valid key or digital licence). 6) Prepare the offline network/board driver on a second USB.*
>
> *Install. 7) Build the bootable USB with the official Microsoft Media Creation Tool. 8) Boot from USB via the one-time boot menu. 9) Language â†' Install now â†' accept terms. 10) Choose **Custom: Install Windows only (advanced)**. 11) Delete all partitions to **Unallocated Space** (this permanently erases the drive). 12) Select unallocated space â†' Next.*
>
> *Post-install. 13) Install network driver, then chipset, graphics, audio. 14) Run Windows Update repeatedly until none remain. 15) Activate with the licensed key/digital licence. 16) Install only licensed essential software. 17) Restore and verify the customer's data. 18) Verify Device Manager, display, sound, network, activation, storage. 19) Customer signs the service report.*
>
> *Licensing. I install only properly licensed Windows and never use pirated or cracked software.*

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key.)

Q1. Multiple choice.
Which installation type should you choose for a clean install? a) Upgrade: Install Windows and keep files and apps b) Custom: Install Windows only (advanced) c) Repair Windows d) Reset this PC

Q2. Short answer.
Why must you back up a customer's data before a clean install?

Q3. Multiple choice.
Which driver should be installed **first** after a fresh Windows install, and why? a) Graphics â€" so the screen looks right b) Network â€" so the internet works and you can download the rest c) Audio â€" so the customer can hear d) Printer â€" so they can print

Q4. Explain in your own words.
Why must a bootable USB be at least 8 GB, and what happens to whatever is already on the USB?

Q5. True or false.

Name three checks you perform as part of post-install verification.

Q7. Scenario.
A customer asks you to install Windows but has no licence. What do you do, and what do you **not** do?

Q8. Fill in the blank.
To reach the one-time boot menu or BIOS/UEFI, you repeatedly press a key during the very first moments after pressing the power button. Common keys include `_____`, `_____` and `_____`.

Q9. Identify the mistake.
A learner builds the USB, boots to Setup, chooses Custom, selects the existing C: partition (without deleting it) and clicks Next, saying, "This will keep the customer's files safe." What is wrong with this reasoning?

Q10. Order the steps (1â€“6).

- ___ Install the network driver
- ___ Back up the customer's data
- ___ Run Windows Update until none remain
- ___ Create the bootable USB
- ___ Choose Custom and install to unallocated space
- ___ Verify Device Manager, sound, network and activation

INSTRUCTOR ANSWER KEY (Q1â€“Q10)

A1.
(b) Custom: Install Windows only (advanced). **A2.** Because a clean install erases everything on the drive, including all personal files. Without a backup the data is permanently lost. **A3.** (b) Network â€“ without internet you cannot download the other drivers. **A4.** Windows installation media needs 8 GB or more of space to hold the installer files. Creating the USB **erases everything already on it**, so back up the USB first. **A5.** False. The first update run often reveals more updates, including driver and security updates. Run it repeatedly, restarting as prompted, until no more updates are available. **A6.** Any three of: Device Manager shows no yellow warnings; correct display resolution; sound works; network/internet works; Windows activated; Windows fully updated; storage shows correct capacity; customer's data restored and opens. (Accept any three correct checks.) **A7.** Explain politely that you install only properly licensed Windows, and offer to help them buy a legitimate licence or consider legal alternatives that fit their needs. You do **not** install a pirated copy, use a crack/activator, or use a key generator â€“ those are illegal and unsafe. **A8.** Any three correct keys, e.g. `Del`, `F2`, `F12` (also accept `F10`, `F9`, `Esc`). **A9.** Installing over an existing partition can still overwrite much of its contents and does **not** guarantee the customer's files are safe. The predictable, clean approach is to back up first, then delete partitions to unallocated space. Never assume files are safe without a verified backup. **A10.** Order: 1 Back up the data â†’ 2 Create the bootable USB â†’ 3 Choose Custom and install to unallocated space â†’ 4 Install the network driver â†’ 5 Run Windows Update until none remain â†’ 6 Verify Device Manager, sound, network and activation.

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned the biggest single job in software repair: putting a fresh operating system onto a machine. You learned why a corrupted OS sometimes has to be reinstalled, and â€“ most importantly â€“ that **data safety comes first**: you back up and verify before you wipe. You checked that the hardware meets the OS requirements, built a bootable Windows USB with Microsoft's own tool, booted from it through BIOS/UEFI, chose a **Custom** clean install, wiped to unallocated space, and understood why Windows creates several partitions. You then finished the job the professional way: drivers in the right order, Windows Update repeated, **legal activation**, essential software, restoring the customer's data, and a full verification. And you committed to using only **licensed** Windows.

Five important takeaways:

3. Choose **Custom: Install Windows only (advanced)** and install to **unallocated space**.
4. Install the **network driver first**, then the rest; run **Windows Update repeatedly**.
5. A machine is finished only when it is updated, **legally activated**, verified and the customer's data is restored.

Questions to help you recognise what you can now do:

- Can I explain the whole install process from backup to verification in my own words?
- Can I build a bootable USB with the official tool?
- Can I reach the boot menu and boot from that USB?
- Can I explain why a clean install is irreversible without a backup?
- Can I state the academy's licensing rule without looking it up?

Connection to the next session:
You can now fix a machine and hand back a working computer. But a repair shop is not only about machines â€" it is about **people**. Next session you learn **customer service**: how to greet a customer, capture the fault accurately in their own words, explain it in plain language, quote honestly, and handle an unhappy customer professionally. The best technical skill in Port Harcourt will fail if you cannot talk to the person paying for it.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner states the **backup step first**, unprompted, before any mention of wiping.
- The learner creates the USB with the **official** tool and selects the correct drive carefully.
- The learner navigates the boot menu/BIOS without being told every key.
- The learner chooses **Custom** and clearly explains that deleting partitions erases data.
- The learner installs the **network driver first** and can say why.
- The learner runs Windows Update **more than once**.
- The learner uses only licensed Windows and can state the rule confidently.
- The learner completes a real verification (checking Device Manager, sound, network, activation) rather than claiming "it works".

Common misconceptions to watch for and correct:

- "A reinstall is safe without a backup." â†' A clean install erases the drive. Back up first, always.
- "Windows Update installs all drivers automatically." â†' Often, but not always â€" especially the network driver. Keep an offline driver.
- "I can use a cracked Windows to save money." â†' Correct immediately and firmly: only licensed software; piracy is illegal and unsafe.
- "Running Windows Update once is enough." â†' Run it repeatedly until none remain.
- "Choosing Custom keeps my files." â†' Only if you do not wipe; and never rely on that. Back up.
- "BIOS and UEFI are the same thing." â†' Similar job, different technologies (MBR/legacy vs GPT/UEFI + Secure Boot).

Suggested remedial exercises for learners who struggle:

1. **Order-the-steps drill:** hand out the nine stages on cards (backup, requirements, media, boot, install, drivers, update, activate, verify) and have them arrange physically, then explain each.
2. **"What could go wrong?" drill:** give five unsafe steps (wipe before backup, skip network driver, one update pass, cracked key, delete wrong partition) and ask for the correct fix.
3. **Plain-language drill:** ask the learner to explain to a "customer" why the computer needs a fresh Windows and what will happen to her files.

Differentiation:

- **Level 2 (standard):** complete the guided practical and independent exercise as written.
- **Level 3 (challenge):** plan a **dual-partition** layout (C: for Windows, D: for data) before installing; install Windows 11 on a UEFI/Secure Boot/TPM 2.0 machine; or prepare media with an alternative tool while still using a **licensed** Windows.

Timing guidance (105 min):

- Opening 6, prerequisite review 7, lecture 35, demonstration 15, guided practical 18, independent exercise 15, case study (can be folded into reflection) 3, reflection/recap plus knowledge check 8. If the room is slow, shorten the case study and protect the **guided practical** and the **data-safety discussion** â€" those are the priority. Never cut the backup step to save time.

END OF SESSION 9

Next: Session 10 â€" Customer Service and Communication.

SESSION 10 â€" CUSTOMER SERVICE AND COMMUNICATION

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 10 â€" CUSTOMER SERVICE AND COMMUNICATION

A. SESSION IDENTIFICATION

Item	Detail
Session number	10 of 12
Session title	Customer Service and Communication
Week / Module	Week 4 â€" Module 4: Professional Practice
Recommended duration	105 minutes
Difficulty level	Beginner (people skills, role-play heavy)
Prerequisite knowledge	Sessions 1â€"9 â€" the diagnostic workflow, all hardware and software repair skills, and OS installation. Learners can now fix machines; today they learn to serve the people who own them.
Connection to previous work	In Session 4 you learned the five-step diagnostic workflow, which begins with **asking the customer good questions**. Today you turn that single step into a full professional skill: greeting, listening, capturing the fault accurately, explaining plainly, quoting honestly and handling difficult situations.
Preparation before class	Print role-play scenario cards (one set per pair); print Customer Intake Forms (one per learner); arrange the room so pairs can sit face to face; set up a "shop counter" table for realistic role-play; have two example estimates ready (one good, one bad) to display.
Professional note	This session has no dangerous hardware steps, but it carries a **confidentiality** duty: customer data and private information must never be shared, gossiped about or copied. Treat every role-play as though it were a real customer.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Greet a customer professionally** and create a calm, trustworthy first impression at a repair shop counter.
2. **Capture a fault accurately** in the customer's own words, using the Customer Intake Form, without leading or interrupting.
3. **Ask good diagnostic questions** (what, when, how, what changed) that move a vague complaint toward a real fault.
4. **Explain faults and fixes in plain language**, avoiding jargon or translating it into everyday Nigerian English.
5. **Manage expectations** about time, cost and uncertainty honestly, including saying "I don't know yet" professionally.
6. **Quote a repair and obtain written approval** before starting any billable work.
7. **Handle an unhappy or angry customer** calmly using listen â†' empathise â†' solve â†' boundary.
8. **Protect customer confidentiality**, in the shop and outside it.
9. **Build trust and repeat business** through honesty, follow-up and consistent service.

Hardware

- Practice computers (optional â€” one or two, so intake can reference a real device).
- A table set up as a "repair shop counter" for realism.

Printed materials

- Role-play scenario cards (one set per pair; see Section H).
- "Customer Intake Form" (one per learner) containing: customer name, phone, device details, fault in customer's words, when it started, what changed, data-backup consent, estimate approval and signature.
- "Estimate Template" (one per learner): diagnosis fee, parts, labour, total, timeline, warranty, approval line.
- Two display examples for the opening: one **good** estimate/ticket, one **poor** onethe instructor writes beforehand.

Software

- None required. If the academy has a shop-management or job-ticket app, show it briefly as an optional tool.

The instructor's own experience

- Prepare one short, true (anonymised) story of a good or bad customer interaction. Real stories teach faster than theory.

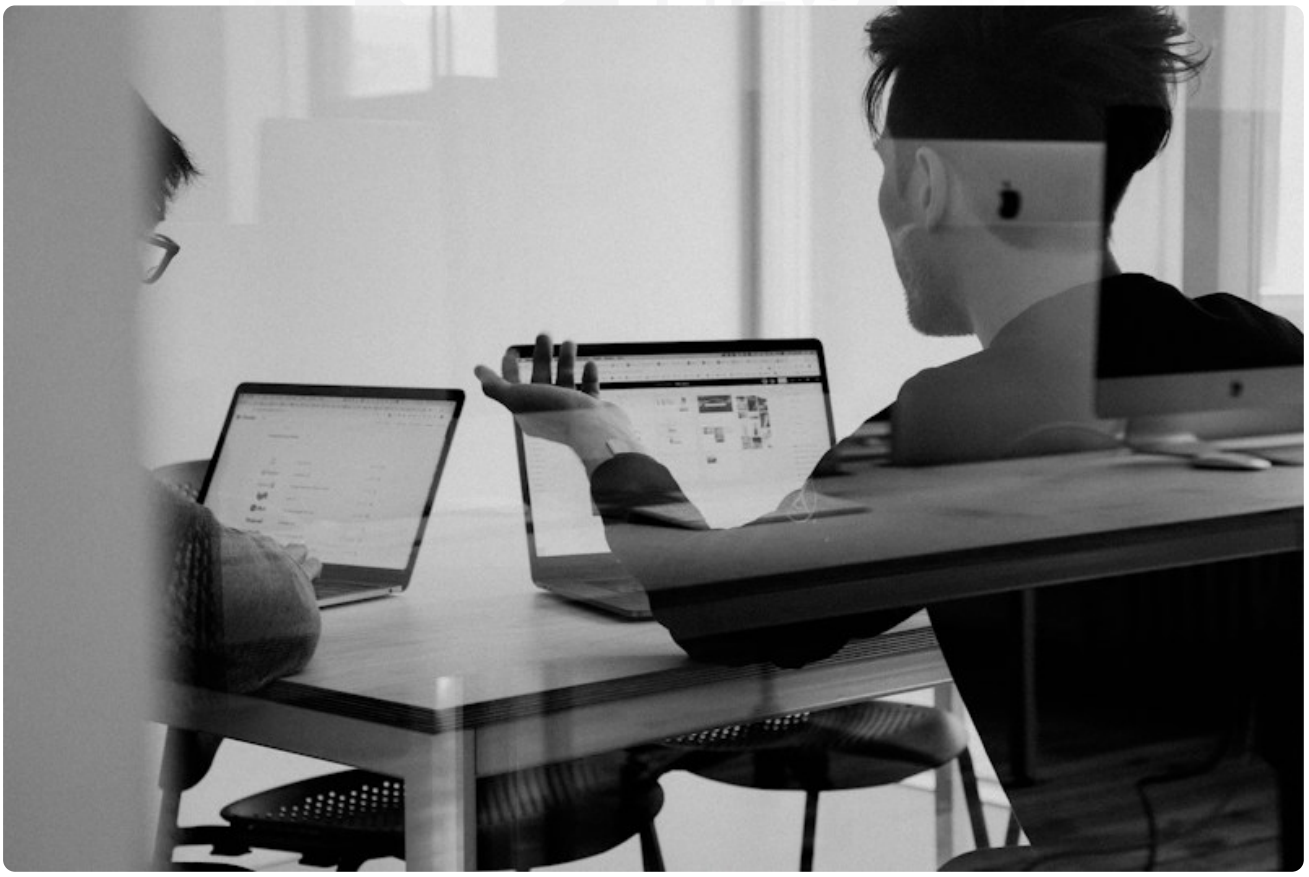
Internet

- Not required. If available, the instructor may show one short, respectful workplace-customer-service video near the end, but do not depend on it.

Alternative for learners who lack a facility

- Not applicable â€” this session needs no equipment, only people. Learners who are shy can begin as the "customer" (easier) and build up to the technician role.

Suggested visuals:



- â€” Intake and role-play conversation with a customer



- â€" Technician explaining the fault and estimate



- â€" Completing the ticket and estimate form



— Learners role-playing the customer intake conversation



— Estimate and job-ticket checklist completed on paper

D. OPENING ACTIVITY (6 minutes)

Instructor setup:

Write two sentences on the whiteboard, side by side.

- Sentence B: *"The part inside your computer that connects your hard drive is faulty, so the computer cannot find your files. We need to replace it."*

What the instructor says:

> *"Read both. They mean the same thing. Which one would you say to a market woman who brought in her shop computer? â€¦ Exactly. Sentence B.*
>
> *Here is the hard truth about this trade: you can be the best technician in Port Harcourt, but if you talk to customers in Sentence A, you will have no customers. Nobody pays a person who makes them feel stupid.*
>
> *Yesterday you learned to fix the machine. Today you learn to fix the relationship. You will learn how to greet a customer, take down their complaint in their own words, explain what is wrong in plain language, give an honest price, and stay calm when someone is shouting at you.*
>
> *This is the skill that turns a one-time repair into a customer who comes back for years â€" and tells their friends about you."*

Expected learner response:
Learners will find Sentence B easier and may laugh at Sentence A. Use the moment to name the skill: **translation**.

Then demonstrate the hook:
Ask a volunteer to act as a frustrated customer who says only: *"My computer is doing me somehow."* The instructor (as technician) asks three good questions and shows how a vague complaint can be turned into useful information within one minute. Say:

> *"Notice what I did: I did not guess, and I did not talk down to her. I asked good questions and let her talk. That is where every good repair begins."*

Why this opener works:
It immediately shows the gap between technical language and customer language, and it proves that good questioning yields real information â€" the exact skills of today.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a fast oral round-robin. Put the questions on the projector.

Question 1:
"What are the five steps of the diagnostic workflow?"

- Expected answer: observe, ask questions, form a hypothesis, test, resolve/verify.
- Common gap: learners remember "test" and "fix" but forget "ask questions". Emphasise that **questions to the customer** are step two and often solve the case before you open the case.

Question 2:
"Why is it important to ask the customer what they were doing when the fault started?"

- Expected answer: it gives clues to the cause (for example a fault after a power cut, a drop, a spill, or an update).
- Common gap: learners think they can find everything from the machine alone. Clarify that customer context is free evidence.

Question 3:
"What should you always do before starting a billable repair?"

- Expected answer: get the customer's approval of the cost and scope (an estimate).
- Common gap: learners skip straight to fixing. Reinforce that surprise bills destroy trust.

- Expected answer: a clean install or wipe destroys data, so we back up and confirm with the customer first.
- Common gap: learners forget to *tell the customer* what will happen. Today's confidentiality and consent theme builds on this.

Bridge to today (say this):

> *"Look how much of today is already in what you know. The diagnostic workflow starts with asking. OS installation taught you to get the customer's consent before destroying anything. Today we simply turn these into a professional habit â€" because a repair shop lives or dies on how it treats people."*

Gap handling:
If more than a third of the class cannot name the intake/approval idea in Question 3, pause and act out a 60-second "surprise bill" scene: technician fixes without quoting, then presents â‚¦60,000, customer is furious. Ask the class what went wrong. Then proceed. This makes the rest of the session land harder.

F. FULL LECTURE NOTES

F.1 Why customer service is a technical skill

The idea.
Customer service is the set of behaviours that make a customer feel respected, informed and confident â€" before, during and after a repair.

Plain-English definition.
Customer service is **how you treat the person while you do the job**. It is greeting them, listening, explaining, quoting honestly, keeping your word and being reachable.

Why it exists.
A customer cannot see the skill in your hands. They cannot tell if your soldering is neat or your diagnosis is sharp. What they *can* judge is how you treated them. That judgement decides whether they come back, whether they recommend you, and whether they trust your price.

What it does.

- Makes a frightened customer feel safe (their data, their money).
- Prevents disputes by keeping everyone informed and getting approval.
- Builds a reputation that brings repeat and referred business.
- Reduces your own stress â€" a calm, informed customer is far easier to deal with.

Everyday analogy.
Think of a **good barber**. He greets you, remembers your name, checks what style you want *before* he cuts, tells you the price, and does not start until you nod. He might cut hair no better than the next man, but you keep going back because he makes you feel attended to. A repair shop is the same.

Realistic scenario (Port Harcourt).
Two technicians in Rumuola both fix a laptop for â‚¦15,000. One greets, explains and follows up. The other grunts, snatches the machine and ignores the customer. Within a year, the first has a queue and a WhatsApp list of loyal customers; the second wonders why business is slow.

Common misconception.
"Customer service is only for receptionists or big companies." Wrong. In a small Nigerian repair shop, the technician **is** the business. Your people skills are part of your product.

Limitation / honesty.

F.2 The first impression â€” greeting and setting the tone

The idea.
The first thirty seconds set the mood of the whole transaction.

What good looks like.

- **Stop what you are doing** and turn to face the customer (or politely say "one moment, sir/ma").
- **Greet** appropriately: "Good afternoon, welcome. How can I help you today?"
- **Introduce yourself**: your name, and the shop's name if they are new.
- **Smile and use a calm voice.** Do not look distracted or annoyed.
- **Offer a seat** if they must wait, and be honest about how long.
- **Never** start with the price or with a sigh.

Analogy.
Imagine you walk into a church office and the secretary does not look up. You feel unwelcome before anyone speaks. Then imagine she stands, greets you by name and offers you a chair. Same information â€” completely different feeling. Greeting is that powerful.

Realistic scenario.
A school principal's laptop is failing on the day of a PTA meeting. She is already stressed. A warm "Good afternoon, ma. Sit down, let me attend to you now" lowers her panic instantly and makes her a patient, cooperative customer.

Common misconception.
"Nigerians don't do small talk; just get to business." Respectful greeting is not small talk â€” in most Nigerian businesses it is the **expected** opening. Skipping it can read as rudeness.

F.3 Capturing the fault accurately â€” in the customer's own words

The idea.
Before you diagnose, you record exactly what the customer experiences, in **their** words, not your technical guess.

Why their words matter.
The customer's exact description often contains the clue to the fault. If you "translate" too early, you may lead yourself astray. Example: a customer says "it dey show blue screen and restart" â€” that is different from "it won't come on at all". Both could be described as "not working", but they point to completely different faults.

How to capture it well.

1. **Let them speak first** without interruption.
2. Write the complaint **as close to their wording as possible** on the intake form.
3. Use **their** phrases in quotes: *"It makes a rattling noise like something loose."*
4. Then ask your structured questions (see F.4).
5. Read it back: "So what I understand isâ€¦, is that correct?"

Useful intake questions (the "good questions" list).

- What exactly happens? What do you see on the screen?
- When did it start? Was it sudden or gradual?
- What were you doing when it started?
- Has anything changed recently â€” a drop, a spill, a power cut, an update, new software?
- Does the problem happen every time, or sometimes?
- Does anyone else use the machine?
- **Is there important data on it that must be saved?** (Always ask this.)

conclusion comes later, after evidence.

Realistic scenario.
A church secretary says, "The computer is slow." A careless technician hears "slow" and assumes a virus. Good questioning reveals: "It became slow only after the generator was changed; now it sometimes beeps and won't start." That points toward **RAM or power**, not a virus. The customer's own words saved hours of wrong work.

Common misconception.
"The customer does not know anything; just look at the machine." The customer may not know the cause, but they know the **symptoms and timing** better than anyone. Use them.

Limitation.
Customers can be mistaken or vague. Capture their words **accurately** but verify with tests before you commit to a diagnosis.

F.4 Explaining faults and fixes in plain language

The idea.
You translate technical findings into words the customer can understand and act on.

The rules of plain language.

1. **No unnecessary jargon.** If you must use a technical word, explain it once and simply.
2. **Use analogies** to everyday things (this is the most powerful tool).
3. **Short sentences.** One idea per sentence.
4. **Check understanding:** "Does that make sense?" or "Should I explain that part again?"
5. **Be honest and specific**, not vague or scary.

The translation table.

Technical term	Plain English
RAM	The computer's short-term memory; it forgets when you switch off
Hard drive / HDD / SSD	Where your files are stored permanently
Motherboard	The main board that connects all the parts
CPU	The computer's brain
PSU	The power supply â€” it feeds electricity to the parts
GPU / graphics card	The part that draws what you see on the screen
Driver	A small program that lets Windows talk to a piece of hardware
Operating system / Windows	The main software that runs the whole computer
Malware / virus	Bad software that harms or slows the computer
Reinstall	Wipe the system clean and install it again
Boot	To start up
Partition	A section of the hard drive
Corrupted	Damaged / spoiled
Diagnostics	Tests we run to find the cause

Analogy bank (use these freely).

- The CPU is the **brain**; RAM is the **short-term memory**; the hard drive is the **long-term memory or diary**.
- The PSU is the **heart that pumps electricity**; if it weakens, everything else misbehaves.
- A driver is an **interpreter** between Windows and a device that speaks another language.
- Overheating is like a person working in a **sealed room with no air** â€" performance drops and eventually they collapse.

Realistic scenario.
A trader asks, "Wetin dey worry am?" The technician says: *"Madam, the memory that the computer uses to run your programs has gone bad. That is why it keeps restarting. We will change it â€" the same part, small cost â€" and it will work well again."* Clear, honest, calm.

Common misconception.
"Using big words makes me look professional." The opposite is true. Big words make customers feel confused and suspicious, and they hide behind them when they do not understand.

Limitation.
Some findings genuinely are complicated (for example data recovery from a physically failing drive). When the truth is complex, be honest about the uncertainty rather than pretending certainty.

F.5 Managing expectations â€" time, cost and uncertainty

The idea.
Tell the customer the truth about what you know, what you do not know, and what it will cost and take â€" before, not after.

The three expectations to manage.

1. **Time.** "Diagnosis will take a few hours / one day." Give a realistic window, not an optimistic lie. It is far better to say "tomorrow afternoon" and deliver at midday than to promise "in one hour" and disappoint.
2. **Cost.** Explain the difference between the **diagnosis fee** and the **repair cost**, and give a range when the fault is not yet confirmed.
3. **Uncertainty.** It is professional to say: *"I cannot give the exact price until I diagnose it. I will test it, then call you with a written estimate before we start any work. Nothing will be done without your approval."*

Why honesty about uncertainty matters.
A technician who says "I am not sure yet, let me test it" is **more** trustworthy than one who guesses a price on the spot. Customers respect honesty; they resent surprises.

Realistic scenario.
A customer's PC does not power on. It could be a â‚¦2,000 loose cable or a â‚¦25,000 failed PSU. A good technician says: "The test will tell us. It might be small, it might be the power supply. I will know by tomorrow and call you before I spend any money."

Common misconception.
"If I say I'm not sure, they will go to another shop." In reality, most customers prefer a careful technician over a confident guesser. Honesty wins the job.

Limitation.
Occasionally a fault changes during repair (a second problem reveals itself). If that happens, **stop and contact the customer** before spending more of their money. Never silently increase a bill.

F.6 Quoting and getting approval before work

The idea.
Before you do billable work, you give the customer a clear **estimate** and get their agreement â€" ideally in writing (or a WhatsApp message they reply to).

What an estimate should contain.

- **Diagnosis fee** (if you charge one).

- **Total** â€" the figure the customer will pay.
- **Timeline** â€" when it will be ready.
- **Warranty** â€" what is covered and for how long.
- **Approval line** â€" where the customer agrees (signature, or "Yes, proceed" on WhatsApp).

Estimates in the Nigerian shop context.

- Written or WhatsApp estimates are best because both sides keep a copy. WhatsApp is widely used and perfectly professional when worded clearly.
- Give a **range** when unsure, and state clearly what would push it to the higher end.
- State whether the diagnosis fee is **waived** if they proceed with the repair.
- **Get approval before starting.** "Nothing will be done without your approval" protects both customer and technician.

Analogy.
A good **tailor** measures you and tells you the price and when to collect *before* cutting the cloth. Imagine a tailor who sews first and demands any price later â€" you would never return.

Realistic scenario.
A customer approves â‚¦18,000 for a PSU replacement. Mid-repair the technician finds the PSU failure **also damaged the motherboard**. The professional stops, calls the customer, explains plainly, gives a new estimate, and only continues if approved.

Common misconception.
"Verbal is fine; we trust each other." Verbal agreements are easily disputed. A written or WhatsApp estimate with a clear "Yes, go ahead" protects the technician if a customer later claims they never agreed.

Limitation.
You cannot always know the full cost upfront. That is why estimates can be **revised** â€" but only with fresh customer approval.

F.7 Handling unhappy or angry customers

The idea.
An upset customer is not your enemy. They are a person with a problem, and your calm professionalism can turn them into a loyal customer.

The four-step method.

1. **Listen.** Let them finish. Do not interrupt, do not defend, do not argue. Take a breath.
2. **Empathise and acknowledge.** "I understand this is frustrating." "I can see why you are upset." This costs nothing and lowers the temperature fast.
3. **Solve.** Focus on what **can** be done: "Let me see what I can do to help." Offer realistic options.
4. **Set boundaries.** Be firm but polite. If a customer becomes abusive, you may calmly end the conversation: "Sir, I want to help you, but please do not shout at me. Let us talk calmly." You are allowed to protect your dignity and that of your staff. Escalate to a manager if needed.

What NOT to do.

- Do not match their emotion or raise your voice.
- Do not blame them ("You caused this").
- Do not lie to make them happy â€" a lie today becomes a bigger fight tomorrow.
- Do not promise what you cannot deliver.

The "broken promise" moment.
Sometimes the fault is genuinely your shop's (you scratched the casing, or a repair failed). The professional response is to **own it honestly**, apologise, and fix it at your own cost where reasonable. This builds more trust than any smooth talk. Hiding a mistake and hoping the customer will not notice is how shops lose their name.

Realistic scenario.

The customer calms down and returns next time.

Common misconception.
"An angry customer is always wrong." Often the customer is simply scared â€” about their data, their money or their image. Address the fear and the anger usually drops.

Limitation.
Some situations cannot be resolved to full satisfaction. In those rare cases, be honest, offer options, and if needed let them take the matter to the owner or a fair third party. Never let a dispute turn into a fight.

F.8 Confidentiality of customer data

The idea.
Anything on a customer's device is private and must be treated as a secret entrusted to you.

What confidentiality means in practice.

- Do not open, read or copy the customer's files except as needed for the repair.
- Do not take photos of their documents, photos or messages.
- Do not gossip about what you found on a customer's machine.
- Do not leave customer data on a shared or personal drive after the job.
- Do not use a customer's data, accounts or files for any personal purpose.
- Store intake forms and job records securely.

Why it matters.

- It is **ethical**. The customer trusted you with their private life.
- It is **professional**. Reputation spreads fast, especially in a close city like Port Harcourt.
- In serious cases, misuse of a customer's data can be a **legal** matter, not just a moral one.

Analogy.
A customer's device is like a **sealed envelope handed to you to deliver**. You carry it, you do not open it, and you do not tell the neighbourhood what was inside.

Common misconception.
"If they left it on the machine, it's not private." It is still private. They did not give you permission to browse it.

F.9 Building trust and repeat business

The idea.
The goal of every job is not just a fixed computer â€” it is a customer who trusts you and returns.

How trust is built.

- **Keep your word** on time and price.
- **Be reachable.** Answer calls and WhatsApp messages, even to say "still working on it".
- **Follow up.** A short "How is the computer working now?" a few days later is powerful and costs nothing.
- **Give honest recommendations** â€” including "you do not need to replace this yet".
- **Be transparent** about what you did (this connects to the service report in Session 11).
- **Thank them**, and invite them back.

Realistic scenario.
After replacing a laptop screen, a technician sends a WhatsApp: "Good evening, ma. Hope the screen is fine? Please let me know if anything comes up." The customer replies, then tells two friends. That one message becomes three jobs.

Common misconception.

Trust is slow to build and fast to lose. One dishonest act can undo years of good service. Protect your name above a single sale.

F.10 Summary of the session

- Customer service is a **technical skill** in this trade â€" it wins and keeps business.
- **Greet** warmly; **listen** and capture the fault in the **customer's own words**.
- **Ask good questions** (what, when, how, what changed, important data?).
- **Explain plainly** â€" no jargon, use analogies, check understanding.
- **Manage expectations** honestly about time, cost and uncertainty.
- **Quote and get approval** before work; revise only with fresh approval.
- **Handle anger** with listen â†' empathise â†' solve â†' boundary.
- **Protect confidentiality**; **build trust** with honesty and follow-up.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

What the instructor prepares:

- The "shop counter" table and two chairs.
- Scenario card: *"My laptop will not charge and the battery was full this morning. I need it for work tomorrow."*
- A printed Customer Intake Form on the projector or whiteboard.
- The good and bad estimate examples.

G.1 Demonstration 1 â€" The intake, done well (6 minutes)

Say:
"Watch everything I do, and listen to **how** I speak as much as **what** I say."

The instructor acts as the technician.
A volunteer plays the customer with the scenario above.

1. **Greet:** "Good afternoon, welcome. My name is Elias. Please sit down. How can I help you today?"
2. **Let them speak** without interrupting. Write their exact words on the form: *"My laptop will not charge and the battery was full this morning."*
3. **Ask good questions:** When did it start? What changed? Do you use the original charger? Does the charging light come on? Is there important data on it?
4. **Read back:** "So the laptop works on battery but will not charge since this morning, and you have work due tomorrow. Have I understood correctly?"
5. **Set expectations:** "I will test the charger and the charging port first â€" that is the small possibility â€" then the battery. Diagnosis will take about one hour. I will call you with a written estimate before any repair."
6. **Mention data safety:** "Is anything on it you cannot afford to lose? Good â€" I will not wipe anything without asking you."

Then say:
"Did you notice I asked before I concluded? Did you notice I quoted a **process**, not a fake price? That is the whole skill."

G.2 Demonstration 2 â€" Explaining the fault plainly (4 minutes)

Say:
"Now I have a finding. Watch me translate it."

- Technical truth: *"Battery cell degradation; the battery no longer accepts charge."*
- Plain version: *"Ma, your laptop's battery has reached the end of its life. It is like a phone battery that no longer holds charge. The charger is fine. We can replace just the battery and it will hold charge again."*

"What did I do to make that clear?" â†’ Used an everyday analogy (phone battery), avoided the word 'degradation', checked understanding, and told her what would happen next.

G.3 Demonstration 3 â€" The angry customer (3 minutes)

The instructor plays a customer who is angry that a repair is late. Narrate the four steps as you do them:

1. **Listen** â€" let them finish, no interrupting.
2. **Empathise** â€" "I understand, sir. This has delayed you and I am sorry."
3. **Solve** â€" "Here is exactly where we are and what I can do today."
4. **Boundary** â€" if shouting continues: "I want to help you, but please let us talk calmly."

Expected result:
The "customer" visibly calms.

Error to watch:
If the learner disagrees, defends or raises their voice, stop the scene and reset. Point out that arguing always makes it worse.

G.4 Demonstration 4 â€" Good vs bad estimate (2 minutes)

Show the two examples on screen.

- **Bad:** "Laptop repair â€" maybe â‚¦20k, come back Friday." (vague, no breakdown, no approval).
- **Good:** a clear table with diagnosis fee, parts, labour, total, timeline, warranty and an approval line.

Ask:
"Which protects the customer? Which protects **you**?" â†' Both. A clear estimate is protection for everyone.

Questions to ask throughout the demo:

- "Why did I write the customer's exact words?" â†' Their words contain clues and show I listened.
- "Why did I say 'I don't know yet' about the price?" â†' Honesty about uncertainty builds trust and prevents surprises.
- "Why is a WhatsApp 'Yes, proceed' useful?" â†' It is written proof of approval.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective:
In pairs, learners role-play a full **customer intake** and a **plain-language fault explanation**, using a real scenario and the Customer Intake Form.

> ***Confidentiality reminder before starting:***
> *everything said in role-play stays in the room. When a real customer is involved, their data and details are private.*

Setup:
Pairs sit face to face. One is the **technician**, one is the **customer**. The instructor hands each pair a scenario card. The technican holds the Intake Form; the customer holds the card.

Procedure (instructor calls each numbered step; pairs follow; instructor walks the room and listens):

1. Decide who is technician and who is customer. The customer reads their scenario card silently.
2. **Greet** the customer warmly and offer a seat. *(Instructor watches for warmth and eye contact.)*
3. **Let the customer speak** first. The technician writes the fault **in the customer's own words** on the form.
4. **Ask at least four good questions** from the list (what, when, how, what changed, important data).
5. **Pause and predict:** before writing a diagnosis, the technician says out loud what the likely **area** of fault is (for example RAM, storage, power, software) and **why**.

8. **Set expectations**: time, a note that a written estimate will follow, and that nothing is done without approval.
9. **Offer an estimate** using the template (diagnosis fee, parts range, labour, total, timeline, warranty, approval line). Get a verbal or written approval.
10. **Switch roles** and repeat with a **different** scenario card.
11. **Give each other feedback** using the three questions below.

Expected result:
A completed Intake Form that accurately captures the fault in the customer's words, a plain-language explanation the "customer" understood, and an approved estimate.

Scenario cards (use and reuse):

- **Card 1.** "My computer is very slow and takes forever to start. I have a presentation tomorrow."
- **Card 2.** "My computer is making a loud noise and I am scared it will lose my photos."
- **Card 3.** "I poured water on my laptop and now it will not come on. Please fix it quickly."
- **Card 4.** "A blue screen appears and the computer restarts by itself. I don't know what it means."
- **Card 5.** "The screen is cracked and sometimes goes black. I use it for my POS business."
- **Card 6.** "The computer works but there is no sound at all. I use it for church programmes."

Pause-and-predict checkpoints:
after step 3 ("What is the likely fault area?"), after step 7 ("Did the customer understand? Ask them to repeat it back.").

Troubleshooting while walking the room:

- Technician interrupts the customer â†’ gently stop the scene; remind them step 3 is "let them finish".
- Technician uses jargon ("SATA", "BIOS") â†’ ask them to say the same thing to a market woman.
- No approval obtained â†’ pause and show why the estimate and "yes" matter.
- Customer is too vague â†’ prompt the technician to ask "What exactly do you see on the screen?"

Completion checklist (learner ticks each):

- ☐ Warm greeting given
- ☐ Fault captured in the customer's own words
- ☐ At least four good questions asked
- ☐ Fault area predicted with a reason
- ☐ Intake read back and confirmed
- ☐ Plain-language explanation with an analogy, comprehension checked
- ☐ Expectations set (time, cost, uncertainty)
- ☐ Estimate offered with an approval line
- ☐ Both roles played (two scenarios)
- ☐ Feedback given and received

I. INDEPENDENT CLASS EXERCISE (15 minutes)

Task (NEW task):
This is a **writing** exercise, not a role-play. You are a technician at a Port Harcourt repair shop. Read each customer message below and produce, for **each**, (a) the exact question you would ask next, and (b) a short plain-language reply that the customer will understand â€" no jargon. Then complete the mini-estimate.

Case A â€" the vague WhatsApp message

> *Customer: "Good evening. My system dey misbehave. I need it urgent. How much?"*

- A2. Write a WhatsApp reply that is friendly, asks for the machine, explains that you must diagnose before quoting, and gives an honest **timeframe** for the diagnosis.

Case B â€" the frightened customer

> *Customer: "My office computer shows a scary blue screen with plenty English I don't understand. I'm worried all our staff records are gone."*

Your tasks:

- B1. Explain **plainly** what a "blue screen" is and what it might mean. Use an analogy.
- B2. Reassure them **honestly** about their data (do not promise what you cannot know).
- B3. State what you will do next and roughly how long the diagnosis will take.

Case C â€" the mini-estimate

A customer's desktop will not power on. After diagnosis you find the **power supply unit (PSU) has failed**. Parts: a 500W PSU at **â‚¦18,000**. Labour: **â‚¦5,000**. Diagnosis fee: **â‚¦2,000**, waived if they approve the repair. Timeline: next day. Warranty: 30 days on the part.

- C1. Write the estimate clearly (all line items, total, timeline, warranty).
- C2. Write the sentence in which you ask for approval.

Time allowance:

15 minutes.

Success criteria:

Good questions that gather useful information; replies free of jargon and respectful in tone; honest handling of uncertainty and data; a complete, clear estimate with an approval request.

Hints if stuck:

- Hint 1: Good questions start with *what*, *when*, *how* and *what changed*.
- Hint 2: Plain language means an analogy and no acronyms.
- Hint 3: An estimate always states parts, labour, total, timeline, warranty and an approval line.

INSTRUCTOR ANSWER KEY (do not show learners):

Case A

- A1. Sample questions: "What exactly do you see when you switch it on â€" a black screen, a blue screen, or does it start up slowly?" / "When did the problem start, and what changed just before it started (a power cut, an update, a new program)?" / "Is there important data on the machine that must be saved?"
- A2. Sample reply: *"Good evening, sir. Sorry for the stress. To give you a proper price I need to test the machine first â€" I don't want to guess and surprise you. Please bring it tomorrow morning; diagnosis will take about one hour, then I will send you a written estimate before any work. Nothing will be done without your approval."*

Case B

- B1. Plain explanation: *"A blue screen is the computer's way of saying it has hit a serious problem it cannot handle, so it stops and restarts. It is like a car engine cutting off to protect itself when something is wrong â€" it doesn't always mean everything is broken."*
- B2. Honest reassurance: *"I cannot promise about the files until I test, but a blue screen usually does not erase your records by itself. Please don't switch it on and off repeatedly. I will check very carefully, and if your data can be saved, I will save it before doing anything else."*
- B3. Next steps: *"I will test the memory, the hard drive and the system first. Diagnosis will take about one to two hours, and I will call you with the result and a price before I do any repair."*

Case C

- Part: 500W PSU â€" â‚¦18,000
- Labour: â‚¦5,000
- **Total (if approved): â‚¦23,000**
- Timeline: ready the next day
- Warranty: 30 days on the PSU part

- C2. Approval sentence: *"If you are happy with this, please reply 'Yes, go ahead' and I will start the repair today. Nothing will be done until I hear from you."*

Marking guidance (out of 10):

- A1 useful questions (2)
- A2 friendly, honest, jargon-free reply (2)
- B1 plain explanation with analogy (2)
- B2 honest data handling (1)
- B3 clear next steps and timeframe (1)
- C1 complete estimate with correct total (1)
- C2 proper approval request (1)

J. REAL-WORLD CASE STUDY

Situation.
Chidi runs a small repair corner in a Port Harcourt plaza. He is a good technician but a poor communicator. A customer brings a laptop that will not start. Chidi takes it, grunts, and says "I will check it." Two days later he calls: "It's your board; â‚¦45,000." The customer is shocked â€" no warning, no explanation, no approval. She collects the machine and takes it elsewhere. The other shop diagnoses a faulty **power button cable** and fixes it for â‚¦6,000. She tells everyone at her office about Chidi "who wanted to spoil her".

Relevant facts.

- Chidi's diagnosis may even have been partly right, but nobody trusts a surprise bill.
- The second shop greeted her, explained plainly, gave a written estimate and got approval.
- Same fault. Two very different outcomes â€" decided by **customer service**, not by hands-on skill.

Decision point.
Where did Chidi lose the customer?

Possible approaches.

1. Blame the customer for being "unreasonable".
2. Learn to greet, explain plainly, quote honestly and get approval before work.
3. Keep doing the same but hope for the best.

Recommended solution.
Approach 2. If Chidi had (a) greeted her, (b) explained what he suspected in plain language, and (c) sent a written estimate of â‚¦45,000 with a clear reason *before* doing anything, the customer would either have approved it or declined politely â€" either way, no damage to his name. He also lost the chance to compete: the second shop found the real, cheaper fault.

Trade-offs.

- Approach 1 feels good but changes nothing; the shop stays empty.
- Approach 2 takes only minutes per customer but transforms reputation and repeat business.
- Approach 3 is the road to closed doors.

Professional connection.

Transfer question for the class:
"Chidi charged â‚¦45,000; the second shop charged â‚¦6,000 for the same-looking problem. What does that teach you about the value of a *correct diagnosis* plus *honest communication*?"

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table live. When a learner hits an issue in role-play, point them to the relevant row.

K.1 Using jargon with the customer

Symptoms:
Customer looks confused; keeps asking "wetin that mean?" **Likely cause:** The technician used words like SATA, BIOS, motherboard, corrupt. **Safe corrective steps:** Translate to plain English; use an analogy; check understanding. **Prevention:** Practise your analogy bank until it is automatic.

K.2 Interrupting or not listening

Symptoms:
Customer repeats themselves; technician misses key details; wrong fault area predicted. **Likely cause:** Rushing to solve before understanding. **Safe corrective steps:** Let the customer finish; write their words; read back. **Prevention:** Treat the customer as a witness â€" capture first, conclude later.

K.3 No estimate / no approval

Symptoms:
Customer angry at the bill; dispute; loss of trust. **Likely cause:** Work started without written cost or consent. **Safe corrective steps:** Apologise, provide the breakdown, and agree a fair resolution; change your process going forward. **Prevention:** Always quote and get approval (signature or a clear "Yes, proceed") before billable work.

K.4 Overpromising time or outcome

Symptoms:
Customer returns furious; reputation damaged. **Likely cause:** Giving an optimistic guess instead of a realistic window. **Safe corrective steps:** Own it, apologise, communicate the true status, give a firm new time. **Prevention:** Under-promise and over-deliver. Say "I will confirm after diagnosis".

K.5 Matching the customer's anger

Symptoms:
The situation escalates into an argument. **Likely cause:** Taking the anger personally and reacting. **Safe corrective steps:** Pause, lower your voice, listen, empathise, set a calm boundary. **Prevention:** Remember the customer is frustrated with the **situation**, not necessarily with you.

K.6 Breaching confidentiality

Symptoms:
Customer learns their private files or situation were discussed or copied. **Likely cause:** Curiosity, gossip, or careless handling of data and records. **Safe corrective steps:** Stop immediately, secure the data, and never repeat it; where serious, this can become a legal matter. **Prevention:** Treat every device as a sealed envelope; access data only for the repair.

K.7 Hiding a shop mistake

Symptoms:
Customer discovers the damage later; trust destroyed. **Likely cause:** Embarrassment and hope the customer will not notice. **Safe corrective steps:** Own it, apologise, fix it at your cost where reasonable. **Prevention:** Honesty is the cheapest long-term policy.

learners to ask before assuming.

L. SESSION ASSIGNMENT

Assignment title:
"My Shop's Customer Service Playbook"

Purpose:
To turn today's skills into a practical, personal guide a learner can follow at a real counter â€" including ready-made questions, a plain-language translation table, an estimate template and scripts for difficult moments.

Realistic scenario:
You have just opened a small repair shop (or are joining one). You want every customer to be greeted warmly, understood correctly, quoted honestly and treated well â€" even when they are upset.

Detailed instructions:

1. Write a **greeting script** you will actually use (2â€"3 sentences).
2. Build a **good-questions list** of at least 8 questions to gather a fault, including one mandatory question about **data**.
3. Create a **plain-language translation table** with at least **12** technical terms and their everyday meanings.
4. Write an **estimate template** with diagnosis fee, parts, labour, total, timeline, warranty and an approval line.
5. Write **scripts** for three difficult moments:
    - a customer who is angry about a delay,
    - a customer who says your price is "too much",
    - a customer who asks you to install unlicensed software.
6. Write a short paragraph on **confidentiality** â€" your personal commitment to protect customer data.
7. List three ways you will **build repeat business**.

Required deliverables:

- A typed document (Word) or a neatly handwritten booklet, **at least one full page**.
- One completed Customer Intake Form from the role-play.

Submission format:
Printed or emailed; filename `Surname_S10_Playbook.docx` .

Deadline:
Before Session 11.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Greeting & intake	Warm greeting; 8+ strong questions incl. data	Greeting + questions, less strong	Few questions, no data question
Plain-language table	12+ terms, clear, everyday language	8â€"11 terms	Fewer than 8
Estimate template	Complete, professional, approval line	Mostly complete	Missing key items
Difficult-customer scripts	Realistic, calm, three scenarios handled well	Two handled adequately	One or vague
Confidentiality & repeat business	Strong commitment; 3+ concrete retention ideas	Present but thin	Missing

scripts.

Instructor reference solution (structure):

- Greeting: "*Good afternoon, welcome to [shop]. My name is [name]. Please sit down â€" how can I help you today?*"
- Good questions (sample): What exactly happens? / When did it start? / What were you doing? / What changed recently? / Does it happen every time? / Does anyone else use it? / **Is there important data to save?** / When do you need it back?
- Plain-language table: at least 12 rows (see F.4 migration table).
- Estimate: itemised table with total, timeline, warranty, approval line.
- Difficult-customer scripts: listen â†' empathise â†' solve â†' boundary, applied to each case.
- Unlicensed-software script: "*I'm sorry, I can only install licensed software. I can help you buy a genuine licence or suggest a free, legal option.*"
- Confidentiality paragraph: a clear, personal promise to protect customer data.
- Repeat business: WhatsApp follow-up, honest recommendations, remembering names, loyalty thank-you.

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key.)

Q1. List.
What are the four steps for handling an unhappy customer, in order?

Q2. Multiple choice.
Before starting billable repair work, you must always: a) Tell the customer to come back tomorrow b) Get the customer's approval of a written estimate c) Increase your price d) Open the machine first

Q3. Explain in your own words.
Why is it better to say "I will confirm the price after diagnosis" than to guess a price on the spot?

Q4. Short answer.
Why must you write the fault in the **customer's own words** rather than your technical guess?

Q5. Practical translation.
Say these in plain English (no jargon): a) "The SATA controller is failing." b) "There is malware on the system."

Q6. Scenario.
A customer shouts at you because a part arrived late. Give the first two things you should do and one thing you must **not** do.

Q7. Fill in the blank.
Anything on a customer's device is private; we must keep it _____ and only access it for the _____.

Q8. True or false.
A verbal estimate is always enough; writing it down is unnecessary. Explain your answer.

Q9. Short answer.
Give two ways a technician can build repeat business after a repair is completed.

Q10. Scenario.
A customer asks you to install a "cracked" copy of Windows to save money. Write your professional reply.

INSTRUCTOR ANSWER KEY (Q1â€"Q10)

A1.
Listen â†' empathise/acknowledge â†' solve â†' set boundaries. **A2.** (b) Get the customer's approval of a written estimate. **A3.** Because guessing a price and being wrong causes a "surprise bill", which destroys trust. Honest uncertainty â€" "I will confirm after diagnosis" â€" shows competence and protects both sides: you quote before doing any billable work. **A4.** The customer's words contain clues to the cause, and writing them down proves you listened. Translating too early can lead you to the wrong

empathise/acknowledge ("I understand, I'm sorry"). You must **not** shout back, argue or blame the customer. **A7.** a¦ **confidential** (private) â€¦ ; â€¦ **repair** (the job it was given to you for). **A8.** False. A written estimate (or a clear WhatsApp message) gives both the customer and the technician proof of what was agreed, preventing disputes. Verbal agreements are easily denied. **A9.** Any two of: send a follow-up message asking if the machine is working well; keep your word on time and price; give honest recommendations; remember the customer's name; offer a thank-you or loyalty gesture. **A10.** Sample: *"I'm sorry, I can only install properly licensed software â€" using a cracked copy is illegal and it can carry viruses that would harm your computer. I can help you buy a genuine licence, or suggest a free, legal option that fits your needs."*

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned that a repair shop is a people business. You practised greeting a customer warmly; capturing their fault in their own words; asking good questions (including the all-important "is there important data?"); explaining faults and fixes in plain language with analogies; managing expectations about time, cost and uncertainty; quoting and getting approval before work; handling angry customers calmly; protecting confidentiality; and building trust for repeat business. None of this replaces good technical work â€" it **wraps around** it.

Five important takeaways:

1. **Greet and listen first**; capture the fault in the customer's own words.
2. **Explain plainly** â€" no jargon, use analogies, check understanding.
3. **Manage expectations honestly**; it is fine to say "I don't know yet".
4. **Quote and get approval** before doing billable work.
5. **Stay calm and protect confidentiality** â€" trust is your real business.

Questions to help you recognise what you can now do:

- Can I run a full customer intake and fill the form without prompting?
- Can I explain a fault to a non-technical customer and have them understand it?
- Can I write a clear estimate with an approval line?
- Can I stay calm with an angry customer and still solve the problem?
- Do I treat every customer's data as private?

Connection to the next session:
You have now served the customer well and agreed the work. But a professional shop also **records** the work. Next session you learn **service reports and documentation**: what a proper report must contain, how to write it clearly and honestly, and how it protects both you and the customer. The report is also the evidence that becomes your Final Project deliverable.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner greets warmly and naturally, without a frozen script.
- The learner **lets the customer finish** and writes their exact words.
- The learner asks focused questions and follows up on vague answers.
- The learner explains a fault with an analogy and checks understanding.
- The learner is honest about uncertainty and never invents a price.
- The learner produces a clear estimate and asks for approval.
- The learner stays calm in the angry-customer scene.
- The learner treats confidentiality and data safety as non-negotiable.

Common misconceptions to watch for and correct:

- "Verbal estimates are fine." â†' Put it in writing/WhatsApp; it protects everyone.
- "If I don't know the price, I should guess." â†' Honest uncertainty builds more trust than a wrong guess.
- "If they left the file on the machine, it is not private." â†' It is still private and must be protected.
- "Customer service is for receptionists." â†' In a small shop, the technician **is** the face of the business.

Suggested remedial exercises for learners who struggle:

1. **Translation drills:** hand out ten technical sentences; ask the learner to say each one to a "market woman" in one or two simple sentences.
2. **Question-building drill:** give a one-line complaint; ask the learner to write five good questions and pick the best three.
3. **Calm-under-pressure drill:** the instructor plays a mildly difficult customer; the learner practises listen â†' empathise â†' solve â†' boundary. Increase difficulty gradually.
4. **Estimate-completion drill:** give a partially filled estimate; ask the learner to complete the total, timeline, warranty and approval line.

Differentiation:

- **Level 1 (supported):** provide a **scripted** role-play the learner can follow word for word, and a heavily guided intake form; let them start as the customer and build confidence.
- **Level 2 (standard):** complete both role-plays and the written exercise as written; produce the playbook.
- **Level 3 (challenge):** handle a **two-part** difficult scenario (angry customer **plus** a revised estimate mid-repair); or, with instructor supervision, conduct a real intake with a genuine customer or staff volunteer; write a full script for a price-objection situation.

Timing guidance (105 min):

- Opening 6, prerequisite review 7, lecture 35, demonstration 15, guided role-play 18, independent written exercise 15, case study folded into reflection 3, reflection/recap + knowledge check 6. If the room is animated in role-play, protect the **role-play** and the **plain-language** practice above all â€" those are the priority. The case study can be reduced to a two-minute mention if time is short.

END OF SESSION 10

Next: Session 11 â€" Service Reports and Documentation.

SESSION 11 â€" SERVICE REPORTS AND DOCUMENTATION

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 11 â€" SERVICE REPORTS AND DOCUMENTATION

A. SESSION IDENTIFICATION

Item	Detail
Session number	11 of 12
Session title	Service Reports and Documentation
Week / Module	Week 4 â€" Module 4: Professional Practice
Recommended duration	105 minutes
Difficulty level	Beginner (writing and reasoning skills)
Prerequisite knowledge	Sessions 1â€"10 â€" the diagnostic workflow, hardware and software repairs, OS installation, and customer service (intake, plain language, estimates, approval).
Connection to previous work	In Session 10 you captured the fault and agreed the work with the customer. Today you close the loop: you **document** exactly what was found, what was done, what it cost and what you recommend. Writing the report is the last professional step of every repair â€" and it is the deliverable that proves you completed the capstone repair in Session 12.
Preparation before class	Prepare a printed "Service Report Template" (one per learner); prepare one **strong** sample report and one **weak** sample report for analysis; have the Section G scenario written up so learners can see a report being built; ensure Microsoft Word (or a plain text editor) is available.
Professional note	A service report is a permanent record that protects the customer, the technician and the shop. It must be **accurate and honest** â€" never exaggerate work, never inflate parts, never hide a mistake.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain why service reports matter** â€" proof of work, warranty, records and professionalism.
2. **List and describe every required section** of a professional service report.
3. **Record the reported fault accurately** in the customer's own words alongside the observed facts.
4. **Document the diagnosis** â€" tests performed, results, and the confirmed root cause.
5. **Describe the action taken** â€" parts used (with details), software installed, and work performed.
6. **Record the tests performed after repair** and the verified result.
7. **Write clear, specific and honest language**, avoiding vague phrases and unnecessary jargon.
8. **Recommend preventive maintenance** and next steps for the customer.
9. **State costs, warranty terms and get the customer's signature** on the finished report.
10. **Handle customer data responsibly** while keeping and storing records.

Hardware

- At least one practice computer (so learners can refer to a real device's details: serial number, components).

Software

- Microsoft Word (or a simple text editor / Google Docs if available). If no computer is available for typing, a printed template and pen are perfectly acceptable â€” the skill is the writing, not the software.

Printed materials

- "Service Report Template" (one per learner) containing all required sections.
- One **strong** sample completed report and one **weak** sample completed report for the opening activity.
- The Section G worked scenario (Mr. Okafor's HP laptop) written out for reference.
- A blank "Parts and Labour" table and a warranty-terms shortlist.

Internet

- Not required. If available, the instructor may optionally show one real-world company's published service-report format as inspiration near the end.

Alternative for learners who lack a facility

- Learners without a laptop or Word can complete the entire session on paper. The instructor types a shared example on the projector. Encourage neat, clear handwriting â€” legibility is part of professionalism.

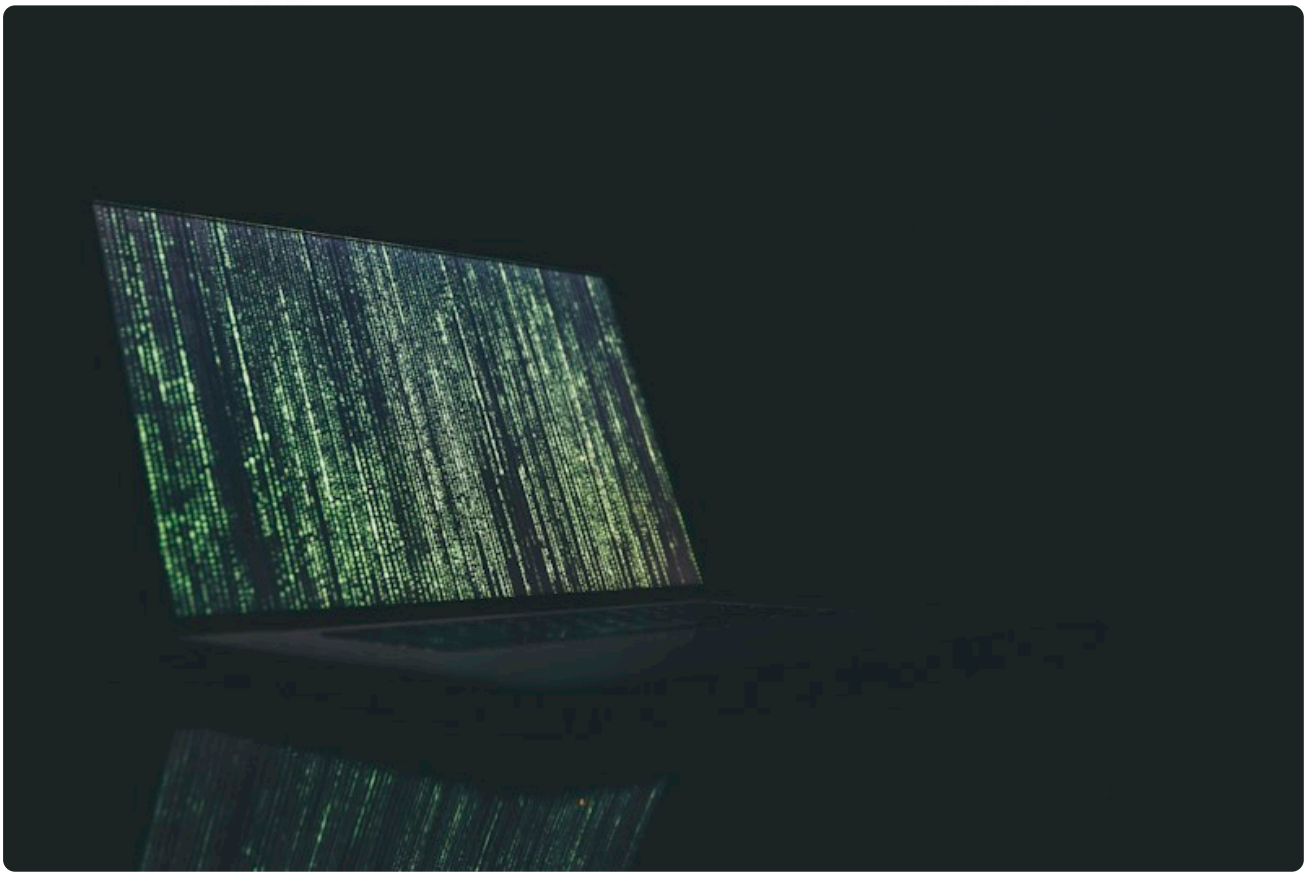
Suggested visuals:



- â€” Writing a clear, professional service report



- â€” Typing the report in Word or a text editor



- â€” Recording parts, labour and warranty details



— Report structure and template outlined before writing



— Finished report shared with the workshop team

D. OPENING ACTIVITY (6 minutes)

Instructor setup:

Display (or hold up) two service reports side by side â€” one strong, one weak. The weak one is messy: it says "Fixed computer, customer happy", with no dates, no parts, no signature and handwriting all over. The strong one is clean and complete.

"Look at these two papers. Both claim to be service reports for the same job. The one on the left says, 'Fixed computer, customer happy. â¦25,000.' The one on the right lists the customer, the device, exactly what was wrong, the tests done, the parts used, the total, the warranty and two signatures.

Now, I want you to answer three questions with me: One â€" if the customer comes back in one month and says the fault returned, which report lets the shop prove what it did? â€¦ The right one. Two â€" if the technician is accused of damaging the machine, which report tells the truth of what happened? â€¦ The right one. Three â€" if the shop's owner wants to know how much money that week's parts cost, which report gives the answer? â€¦ The right one.

That piece of paper is not just paperwork. It is your

protection
*, your **memory** and your **professional name**. Today you will learn to write it properly, because this report is exactly what you will produce for your Final Project next session."*

Expected learner response:
Learners will quickly see the messy report is untrustworthy. Some may say "the good one is more work". Acknowledge that it takes a few minutes more â€" and saves hours of argument later.

Why this opener works:
It frames documentation as self-interest, not bureaucracy: the report protects the technician. That motivation makes learners take the writing seriously.

E. PREREQUISITE REVIEW (7 minutes)

Run this as a quick oral round-robin. Put the questions on the projector.

Question 1:
"What are the four steps of the customer intake process?"

- Expected answer: greet, gather information, set expectations, document.
- Common gap: learners forget the final "document" step. Point out that today **is** that step turned into a full document.

Question 2:
"Why do we capture the fault in the customer's own words?"

- Expected answer: their words contain clues and prove we listened; translation too early can mislead.
- Common gap: learners think the technical finding is enough. Remind them the report must show **both** the reported fault and the observed finding.

Question 3:
"What did you learn to give the customer before starting work in Session 10?"

- Expected answer: a clear written/WhatsApp estimate and approval.
- Common gap: learners forget the approval line. The estimate and the service report are two sides of the same professional habit.

Question 4:
"What are the five steps of the diagnostic workflow?"

- Expected answer: observe, ask, hypothesise, test, resolve/verify.
- Common gap: learners skip "verify". The report must record the **verification** â€" the proof that the fix worked.

Bridge to today (say this):

"Look at everything today rides on. You already capture the fault, get approval, diagnose and verify. The service report simply

Gap handling:
If more than a third of the class cannot name the verification step (Question 4), spend five extra minutes on why we test after repair (open the machine, confirm the fault is gone, check no new faults). The report's "tests performed" and "result" sections depend on this. Do not proceed until it is solid.

F. FULL LECTURE NOTES

F.1 Why service reports matter

The idea.
A **service report** is a professional document that records what was reported, what was found, what was done, what was tested and what the customer now owes â€" for a specific repair.

Plain-English definition.
It is the **written story of one repair**, kept on record for the shop and the customer.

Why it exists.
Repairs are done with hands and memory, and both are unreliable later. A report captures the facts while they are fresh, in a form that anyone can read and trust.

What it does â€" five reasons it matters.

1. **Proof of work.** If a customer disputes the repair ("you never changed that part"), the report shows exactly what was done.
2. **Warranty support.** A clear record of the part, the date and the work is what makes a warranty claim possible.
3. **A record for the future.** A later technician can see what was already done, avoiding repeated or conflicting work.
4. **Professionalism and trust.** A clean, complete report tells the customer the shop is organised and honest.
5. **Business information.** Reports track parts used, costs and recurring problems â€" useful for pricing and for spotting a customer whose machine keeps failing.

Everyday analogy.
A service report is like a **hospital patient's file**. Each visit records symptoms, tests, diagnosis, treatment and result. Without that file, every doctor starts from zero â€" and mistakes multiply. Your report is the patient's file for the machine.

Realistic scenario (Port Harcourt).
A school brings the same lab computer back three months after a repair, claiming the earlier fix failed. The shop checks its report: the part was replaced on a specific date, with a 90-day warranty that has expired, and the customer signed for it. The conversation is calm and factual â€" because the paper exists.

Common misconception.
"Reports are just paperwork; the repair is what matters." The repair earns today's money; the report earns tomorrow's trust, resolves disputes, and supports warranty and audit.

Limitation.
A report is only as good as its honesty and accuracy. A report full of vague or invented claims is worse than none â€" it destroys credibility.

F.2 What a service report must contain

The idea.
Every professional report has the same standard sections, in a logical order, so nothing important is ever forgotten.

The required sections.

1. **Header / shop details.** Business name, logo (if any), address, phone, and a **unique report number** (for example `SRV-2026-0042`), the date, and the technician's name.

3. **Reported fault (the customer's words).** What the customer said, in their own words, quoted where helpful.
4. **Diagnosis / findings.** What you observed; the **tests performed** and their **results**; and the confirmed **root cause** â€” the real reason for the fault.
5. **Action taken (repairs performed).** Exactly what you did: parts replaced (with enough detail to identify them), software installed or updated, cleaning, reseating, etc.
6. **Parts used.** An itemised list of parts, with a description (and part number where relevant) and the quantity.
7. **Tests performed after repair / result.** What you tested to confirm the fix, and the outcome: "Fault resolved; machine boots, all devices working."
8. **Costs (parts and labour).** An itemised breakdown: parts cost, labour cost, diagnosis fee if any, and the **total**.
9. **Recommendations.** Preventive advice and next steps for the customer (for example, "use a surge protector", "back up regularly", "consider an SSD upgrade").
10. **Warranty terms.** What is covered, for how long, and any conditions.
11. **Sign-off.** Technician signature and date, and **customer signature** and date.

> *Note: different shops combine some sections (for example, "parts used" inside "action taken"), and add a diagnosis fee line. The*
>
> ***content***
> *above is what matters; the exact layout can vary. Our CEA template presents them as separate, clearly labelled sections so nothing is missed.*

Analogy.
The sections are like a **story with a beginning, middle and end**: the customer's complaint (beginning), the diagnosis and repair (middle), and the result, cost and recommendation (end). Skip a part and the story is incomplete.

Common misconception.
"If the repair was simple, I can write a short note." Even a simple repair (for example, reseating a RAM stick) deserves the full record: reported fault, finding, action, test, cost, signature. The only thing that changes is how brief each section is â€” not whether it exists.

Limitation.
The report must be readable by the customer. A report stuffed with raw technical output without explanation fails the customer, even if it is technically complete.

F.3 Recording the fault and the diagnosis

The idea.
Keep the **customer's reported fault** and **your observed findings** separate but connected.

Why separate them.

- The **reported fault** is what the customer experienced ("it restarts by itself").
- The **observed finding** is what you actually measured (for example, "memory test failed with errors in slot 1").
- Sometimes they match; sometimes the customer's symptom has a different cause than expected. Recording both shows your professional reasoning.

Good practice.

- Quote the customer's words: *Customer reported: "It makes a rattling noise and then shuts down."*
- Record **each test and result**: *Test: reseated RAM; ran Windows Memory Diagnostic â€" no errors. Test: checked fan â€" fan bearing worn; CPU temperature rose to 92Â°C under load. Result: overheating caused automatic shutdown.*
- State the **root cause** plainly: *Root cause: worn CPU fan causing overheating and thermal shutdown.*

Analogy.

Common misconception.
"The customer's description is unimportant once I find the fault." Wrong â€” the description explains *why* they came, and demonstrates you listened and understood.

Limitation.
Customers can be inaccurate. Record their words faithfully, but always base the root cause on **evidence**, not on their guess.

F.4 Documenting the action, tests and result

The idea.
State exactly what you did, then prove it worked.

Action taken â€” be specific.

- Instead of: *"Fixed the problem."* â†' Write: *"Replaced faulty CPU cooling fan (original part: 92 mm, 4-pin). Applied new thermal paste and reconnected the fan cable to the CPU_FAN header."*
- Include software actions: *"Installed network, chipset, graphics and audio drivers. Ran Windows Update until no updates remained. Activated Windows with the customer's licensed digital licence."*

Parts used â€” be identifiable.

- Description, quantity and (where relevant) part number or specification.
- Example: "*1 Ã— 500 GB SATA SSD*", "*1 Ã— 8 GB DDR4 2666 MHz RAM module*", "*1 Ã— 500 W PSU*".

Tests performed after repair / result â€” prove the fix.

- Example: *"Booted to Windows; ran CPU temperature test under load â€" stable at 55Â°C. Ran memory diagnostic â€" no errors. Checked Device Manager â€" no warnings. Tested audio and network â€" working."*
- State the outcome plainly: *"Fault resolved. Machine returned to full working condition; customer's data restored and verified."*

Why this matters.
The "action + tests + result" trio is what turns your claim of a successful repair into **evidence** of a successful repair.

Common misconception.
"If it switches on, it is fixed." A machine can switch on and still have a hidden fault (for example, one of two RAM sticks failing). The report must show the tests that confirm the whole system is healthy.

Limitation.
Some faults cannot be reproduced immediately (intermittent faults). Be honest: record what you tested, and state clearly if the fault was intermittent and could not be fully reproduced.

F.5 Writing clearly and honestly

The idea.
The report must be understood by the customer and trusted by any reader â€" while remaining completely truthful.

Principles of clear writing.

1. **Plain language.** Write for the customer, not for other technicians. Explain any technical term you must use.
2. **Be specific.** "Replaced faulty RAM module in slot 2" beats "fixed memory".
3. **Be complete.** Include every test, part and step. No gaps.
4. **Be accurate.** Double-check serial numbers, part numbers, dates and costs. A single wrong figure undermines the whole document.
5. **Be professional.** Correct grammar and spelling; no personal opinions, no complaints about the customer, no sarcasm.
6. **Be honest.** Never claim work you did not do or parts you did not use. If you made a mistake, record the facts and the resolution.

destroys trust. A short honest report is worth more than a long false one.

Analogy.
Writing a service report is like giving **evidence in court**: you state facts you can stand behind, clearly and without exaggeration. If you are caught exaggerating once, everything you say afterwards is doubted.

Common misconception.
"More technical language sounds more professional." The opposite: clear, honest language reads as professional; dense jargon reads as evasive.

Limitation.
You cannot write a perfect report if the work was rushed. A good report is the natural result of careful work â€” another reason to do the repair properly.

F.6 Costs, warranty and sign-off

The idea.
The report ends with the money, the promise and the signatures.

Costs.
Itemise parts, labour and any diagnosis fee, then state the **total**. This must match the estimate the customer approved. If the cost changed mid-repair, note that the customer approved the revised estimate.

Warranty terms.
State plainly:

- **What** is covered (for example, "the replacement PSU part only", or "parts and labour").
- **How long** (for example, "90 days from the repair date").
- **Conditions** (for example, "void if the case is opened by another person or if liquid damage is found").

Be careful not to promise a warranty you cannot honour. A warranty is a commitment.

Sign-off.
The report is not finished until:

- The **technician** signs and dates it.
- The **customer** signs and dates it, confirming the work was done, the cost agreed and the device returned in the stated condition.

Why the customer's signature matters.
It is the moment the customer accepts the repair and the terms. It protects the shop from later false claims and gives the customer a record of their consent.

Common misconception.
"The customer's signature is a formality." It is a genuine agreement. Explain the report to the customer before they sign â€” do not just push paper at them.

Limitation.
A warranty cannot cover everything (for example, a new fault unrelated to the repair, or damage caused later by liquid or drops). Say so honestly in the terms.

F.7 Keeping records and handling data

The idea.
Once the report is signed, it must be **kept safely** and the customer's data protected.

Record keeping.

- Store reports in a safe place â€” a folder, a filing cabinet, or a secure digital file with a backup.

Data handling notes.

- The report itself should **not** list private contents of the customer's files ("customer's bank statements found"). Record only what is relevant to the repair.
- When you back up a customer's data for a repair, delete your working copy once the job is safely returned â€” do not keep customer data longer than necessary.
- Never copy, share or gossip about a customer's data (Session 10 confidentiality).

Analogy.
A service report is like a **receipt and a medical note combined**: you keep a copy for your records, you give the customer one, and you do not write their private business all over it.

Common misconception.
"Once the customer takes the machine, the record is no longer needed." The record is exactly what you need **if** a problem returns. Keep it.

F.8 Summary of the session

- A **service report** is the written story of one repair; it protects the customer, the technician and the shop.
- Required sections include: **header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests/result, costs, recommendations, warranty, sign-off**.
- Keep the **customer's words** and your **observed findings** separate but connected.
- **Action + tests + result** turn a claim into evidence.
- Write **clearly, specifically and honestly** â€” no jargon, no vague phrases, no exaggeration.
- Record **costs, warranty and signatures**, and store records safely while protecting customer data.

G. INSTRUCTOR DEMONSTRATION (15 minutes)

What the instructor prepares:

- A blank Service Report Template on the projector.
- The worked scenario below, so the class can see a report built live, section by section.

G.1 The scenario

> ***Customer:***
> *Mr. Okafor.* ***Device:*** *HP laptop, serial number* `ABC123`*.* ***Reported fault:*** *"My computer is running very slowly and sometimes freezes."* ***Diagnosis:*** *The hard drive was nearly full (95% used) and running slowly; the machine also had only 4 GB of RAM, which is too little for the customer's workload. The old spinning hard drive was the main cause of slowness.* ***Repair:*** *Replaced the HDD with a 500 GB SSD; upgraded RAM from 4 GB to 8 GB; reinstalled Windows (licensed) and installed all drivers; ran Windows Update; restored the customer's data.* ***Parts:*** *500 GB SSD â€” â‚¦25,000; 8 GB DDR4 RAM â€” â‚¦15,000.* ***Labour:*** *â‚¦10,000.* ***Total:*** *â‚¦50,000.* ***Tests after repair:*** *booted in under 20 seconds; memory diagnostic clean; Device Manager clean; audio and Wi-Fi working; data restored and opened.* ***Warranty:*** *90 days on parts; labour covered for 30 days.*

G.2 Building the report live (narrate each section)

Say:
"Watch how every fact from the job goes into a section. Nothing is invented; nothing is left out."

1. **Header:** instructor writes the shop name, a report number `SRV-2026-0042`, the date, and "Technician: [name]".

4. **Diagnosis/findings:** write the tests and the root cause.
5. **Action taken:** write the replacement, RAM upgrade, Windows reinstall and driver work.
6. **Parts used:** the itemised table.
7. **Tests/result:** the verification list. **Say:** "This is where I prove the repair worked."
8. **Costs:** the breakdown summing to â‚¦50,000.
9. **Recommendations:** "Back up regularly; avoid filling the drive past about 80%; consider a surge protector."
10. **Warranty:** the terms.
11. **Sign-off:** technician line; explain that the customer signs here.

Ask the class after each section:
"Could a stranger read this and understand exactly what happened? Could this report defend the shop in a dispute?"

G.3 Deliberately show a vague phrase and improve it

Write: *"Fixed the slow computer."* Then rewrite it as the specific version above. **Say:** "One of these protects you; the other is a blank cheque for an argument."

G.4 Show the weak sample and diagnose it

Display the weak sample report from the opening. Ask the class to find:

- Missing sections (for example, no serial number, no tests, no warranty).
- Vague language ("fixed the problem").
- Any dishonesty or exaggeration.

Expected result:
Learners spot that a report missing key sections cannot protect anyone.

Error to watch:
Learners may want to write only the parts they remember. Insist on using the **template** so no section is forgotten.

Questions to ask during the demo:

- "Why do we record the customer's exact words?" â†' Clues and proof of listening.
- "Why do we write the tests we ran?" â†' To prove the fix, not just claim it.
- "Why get the customer's signature?" â†' It confirms agreement and protects both sides.

H. GUIDED PRACTICAL EXERCISE (18 minutes)

Objective:
With the instructor guiding step by step, learners write a **complete professional service report** for the scenario below, using the template.

Starting materials:
The Service Report Template (printed or digital), the scenario on the board, pens/Word.

> ***Data-safety/confidentiality reminder:***
> *the scenario is fictional. In real work, record only what is relevant to the repair and protect the customer's private data.*

Scenario (given):

> ***Customer:***

> *tested good.) **Repair:** replaced the power supply with a 500 W unit, reconnected all cables, tested power on. **Parts:** 500 W PSU â€" â‚¦18,000. **Labour:** â‚¦5,000. **Total:** â‚¦23,000. **Tests after repair:** powered on; booted to Windows; Device Manager clean; fan and drives spinning; USB ports working. **Recommendation:** Use a surge protector; clean the vents every 3 months. **Warranty:** 90 days on the PSU part.*

Procedure (instructor calls each numbered step; learners follow; instructor walks the room):

1. **Header.** Write the shop name, a unique report number, today's date and the technician name.
2. **Customer and device details.** Name, phone, device make/type, serial `PC456`.
3. **Reported fault.** Quote the customer's words exactly.
4. **Diagnosis/findings.** Write the test performed (known-good PSU test) and the root cause (dead power supply).
5. **Action taken.** Describe the replacement and reconnection precisely.
6. **Parts used.** Complete the itemised table (description, quantity, cost).
7. **Tests performed and result.** List at least three tests and state the result.
8. **Pause and predict:** *"Which section proves the repair actually worked?"* â†' Tests performed and result.
9. **Costs.** Write the breakdown and the total (â‚¦23,000). Check the arithmetic.
10. **Recommendations.** Write two or three preventive suggestions.
11. **Warranty terms.** State what is covered, how long, and conditions.
12. **Sign-off.** Write the technician sign-off line and the customer sign-off line.
13. **Review against the checklist** before saving.

Expected result:
A complete, well-organised service report containing all required sections, in plain language, with correct figures (total â‚¦23,000).

Pause-and-predict checkpoints:
after step 3 ("Why quote the customer?"), after step 7 ("Which section proves the fix?"), after step 9 ("Is the total correct?").

Troubleshooting while walking the room:

- Vague action ("fixed computer") â†' ask: "Could a stranger repeat exactly what you did from this sentence?"
- Missing serial number â†' point to the section; explain why it matters for warranty.
- Wrong total â†' re-add parts and labour; accuracy is non-negotiable.
- Jargon ("SATA controller", "no POST") â†' ask them to say it to a non-technical customer.
- No customer signature line â†' add it; the report is not finished without sign-off.

Completion checklist (learner ticks each):

- ☐ Header with unique report number and date
- ☐ Customer and device details incl. serial
- ☐ Reported fault in the customer's words
- ☐ Diagnosis with tests and root cause
- ☐ Action taken, specific
- ☐ Parts used, itemised
- ☐ Tests performed after repair and result
- ☐ Costs itemised, total correct (â‚¦23,000)
- ☐ Recommendations given
- ☐ Warranty terms stated
- ☐ Technician and customer sign-off lines
- ☐ Plain language throughout

Task (NEW task):

You are given a repair that was done **carelessly** and a messy set of notes. Your job is to turn the chaos into a professional report **and** identify what is wrong with the original records. This tests judgement, not just copying.

Scenario â€" "The rushed repair."

A technician at another shop made these shorthand notes for a job:

> *"HP laptop. Won't charge. Changed the charging port. â‚¦15k. Customer picked up. Done."*

You later learn the true facts:

> ***Customer:***
> *Musa Ibrahim, phone 0803-xxx-xxxx.* ***Device:*** *HP 250 G6 laptop, serial* `HP7890` *.* ***Reported fault:*** *"The laptop works on battery but will not charge when I plug it in."* ***Diagnosis:*** *The DC charging jack (power port) was physically loose/broken; the charger itself tested good.* ***Repair:*** *Replaced the DC charging jack; tested charging.* ***Parts:*** *DC charging jack â€" â‚¦6,000.* ***Labour:*** *â‚¦5,000.* ***Total:*** *â‚¦11,000. (Note: the original note said â‚¦15,000 â€" the technician had guessed and overcharged, then later refunded â‚¦4,000. The real agreed total is â‚¦11,000.)* ***Tests after repair:*** *confirmed charging; battery reached 100%; tested with the customer's own charger; all ports working.* ***Recommendation:*** *Do not yank the charging cable; use the cable gently.*

Your tasks:

1. Write the **complete professional service report** for Musa's job, using all required sections.
2. List **three things wrong** with the original shorthand note (think about completeness, accuracy and honesty).
3. Write **one sentence** explaining to the customer why the final cost is â‚¦11,000, not the â‚¦15,000 first mentioned.

Time allowance:
15 minutes.

Success criteria:
A complete report with all sections in plain language; three valid criticisms of the shorthand note; a clear, honest cost sentence; correct total (â‚¦11,000).

Hints if stuck:

- Hint 1: Every fact in the scenario belongs in a section. Ask "which section?" for each.
- Hint 2: For task 2, think about what is **missing** (no serial, no tests, no signature), what is **vague** ("changed the charging port"), and what is **dishonest** (overcharging, guessing).
- Hint 3: For task 3, honesty and transparency about the correction build trust.

INSTRUCTOR ANSWER KEY (do not show learners):

Task 1 â€" model report (summary):

- **Header:** shop name; report number `SRV-2026-0051` ; date; technician name.
- **Customer/device:** Musa Ibrahim; phone; HP 250 G6; serial `HP7890` .
- **Reported fault:** *"The laptop works on battery but will not charge when I plug it in."*
- **Diagnosis:** Charger tested good; the DC charging jack was physically loose/broken. Root cause: broken charging jack.
- **Action taken:** Replaced the DC charging jack; tested charging with the customer's own charger.
- **Parts used:** 1 Ã— DC charging jack â€" â‚¦6,000.
- **Tests/result:** Confirmed charging; battery reached 100%; all ports working. Fault resolved.
- **Costs:** Parts â‚¦6,000 + labour â‚¦5,000 = **â‚¦11,000**. (Note the correction from the original â‚¦15,000 and the â‚¦4,000 refund, if applicable.)

- **Sign-off:** technician and customer signatures.

Task 2 â€” three things wrong with the shorthand note (accept any three):

- **Incomplete:** no customer name/phone, no device model or **serial number**, no date, no diagnosis details, no tests, no warranty, no signature.
- **Vague:** "changed the charging port" does not identify the part or the cost breakdown.
- **Inaccurate / dishonest:** the note said â‚¦15,000 but the correct agreed total was â‚¦11,000 â€” it was a guess that overcharged, then a refund had to be made. A professional quotes honestly and gets approval.
- (Also valid: no record of the customer's exact words; no recommendation.)

Task 3 â€” cost sentence (sample):
"Sir, the final cost is â‚¦11,000 â€” â‚¦6,000 for the charging jack and â‚¦5,000 for labour. My first figure of â‚¦15,000 was a guess before I had inspected the machine, so I have corrected it and refunded the difference. I apologise for the confusion."

Marking guidance (out of 10):

- Completeness of the report (all sections) (4)
- Plain language and specificity (2)
- Accuracy of figures (â‚¦11,000) (1)
- Three valid criticisms of the shorthand note (2)
- Clear, honest cost sentence (1)

J. REAL-WORLD CASE STUDY

Situation.
A customer, *Mr. Nwosu*, brings a desktop back to a Port Harcourt repair shop eight weeks after a repair. He claims the shop "never actually replaced the power supply" and demands a refund, threatening to post about it online. The shop owner opens the filing cabinet and finds the signed service report.

Relevant facts.
The report shows:

- A unique report number and date (eight weeks earlier).
- The customer's reported fault in his own words.
- The diagnosis and the test performed (known-good PSU test).
- The action taken (500 W PSU replaced).
- The part used, with description.
- The tests performed after repair and the result.
- The cost breakdown, signed by the customer.
- A 90-day warranty on the part.

Decision point.
How does the report resolve the dispute?

Possible approaches.

1. Argue from memory â€” "I remember we changed it."
2. Produce the signed report and let it speak.
3. Give the refund to avoid trouble.

Recommended solution.
Approach 2.

- The technician calmly shows Mr. Nwosu the report he signed, with the date, the part and the test results.

new fault â€" good service on top of good records.

- Mr. Nwosu withdraws the claim.

Trade-offs.

- Approach 1 leaves the shop at the mercy of a customer's memory versus the technician's; unwinnable in a public argument.
- Approach 2 is factual, calm and quick â€" and shows professionalism.
- Approach 3 rewards a false claim and may encourage others. It also admits fault the shop did not commit.

Professional connection.
The report is not just for the customer â€" it is the shop's **memory and its defence**. This is why Session 11 exists, and why the Final Project requires a complete report. The documented, signed record is what separates a professional shop from a roadside guess.

Transfer question for the class:
"Which single section of the report do you think was most important in defending the shop â€" and why?" *(Strong answer: the customer's signature, because it proves agreement to the work and the cost; the test results are a close second because they prove the fix.)*

K. COMMON MISTAKES AND TROUBLESHOOTING

Use this table live. When a learner's report has a problem, point them to the row.

K.1 Vague descriptions

Symptoms:
The report says "fixed the problem" or "did maintenance". **Likely cause:** Rushing, or not knowing what detail matters. **Safe corrective steps:** Rewrite with the exact part, action and result. Ask: "Could a stranger repeat this from my words?" **Prevention:** Use the template prompts for each section.

K.2 Missing sections

Symptoms:
No serial number, no tests, no warranty, no signature. **Likely cause:** Skipping parts of the template. **Safe corrective steps:** Compare against the full section list and fill every gap. **Prevention:** Always write from the template; never from memory.

K.3 Inaccurate numbers or serial numbers

Symptoms:
Total does not add up; wrong model/serial; date errors. **Likely cause:** Careless recording. **Safe corrective steps:** Re-add the figures; re-check the serial from the device; correct the report before signing. **Prevention:** Double-check all numbers before the customer signs.

K.4 Technical jargon in the report

Symptoms:
Customer cannot understand their own report. **Likely cause:** Writing for technicians, not customers. **Safe corrective steps:** Translate terms; use plain language; keep technical detail brief and explained. **Prevention:** Write as if the customer will read it aloud to a friend.

K.5 No customer signature

Symptoms:
Report is unsigned; dispute later has no written agreement. **Likely cause:** Not asking the customer to sign. **Safe corrective steps:** Explain the report to the customer at collection and ask them to sign; add the date. **Prevention:** Make sign-off a required last step of every job.

Labour hours or parts overstated; work claimed that was not done. **Likely cause:** Temptation to inflate a small job. **Safe corrective steps:** Correct the report to the truth immediately; refund any overcharge; apologise. **Prevention:** Remember: honesty protects your name and, in serious cases, keeps you out of legal trouble.

K.7 Customer data written into the report or carelessly handled

Symptoms:
Private contents listed; backup copies kept too long. **Likely cause:** Not thinking about confidentiality. **Safe corrective steps:** Remove private details; delete working copies of customer data once the job is returned. **Prevention:** Record only repair-relevant facts; treat data as a sealed envelope.

Distinguish likely vs confirmed:
Fix what you can **confirm** (missing section, wrong total). Do not guess at facts â€” if unsure of a part number or test result, check the device or your notes before writing.

L. SESSION ASSIGNMENT

Assignment title:
"My Capstone-Ready Service Report"

Purpose:
To produce a complete, professional service report for a repair â€” the same document you will need for the Session 12 capstone â€” and to build a reusable template you can take into a real shop.

Realistic scenario:
You have completed a repair at your shop. Use either a repair you actually performed during the course (for example the PSU replacement from the guided exercise or a real practice-machine job) or the provided scenario. Write the full report as if the customer is coming to collect the machine.

Detailed instructions:

1. Use the Service Report Template. Complete **every** section:
    - header with a unique report number and date,
    - customer and device details including serial number,
    - reported fault in the customer's own words,
    - diagnosis with tests performed and the root cause,
    - action taken (specific),
    - parts used (itemised),
    - tests performed after repair and the result,
    - costs (parts, labour, total â€” correct arithmetic),
    - recommendations,
    - warranty terms,
    - technician and customer sign-off lines.
2. Write in **plain language**; if you use a technical term, explain it.
3. Provide a **second page**: a reusable blank template you designed (you may lay it out however you like, but it must include all sections plus a space for the shop logo).
4. Add a one-paragraph note on **confidentiality** and how you will store records.

Required deliverables:

- A completed service report (one to two pages).
- A blank reusable template (one page).

Deadline:

Before Session 12 (the capstone). Bring a copy to Session 12.

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
All sections present	All required sections, clearly labelled	Most sections present	Several missing
Reported fault & diagnosis	Customer's words quoted; tests and root cause clear	Present but thin	Vague or missing root cause
Action, parts & tests/result	Specific action; itemised parts; tests and result recorded	Mostly specific	Vague; no test evidence
Costs, warranty & sign-off	Correct totals; warranty stated; both signatures lines	Minor gaps	Wrong total or no sign-off
Plain language & honest writing	Clear, specific, honest; no jargon	Mostly clear	Jargony, vague or exaggerated
Reusable template (2nd page)	Professional, complete, logo space	Mostly complete	Incomplete

Common mistakes to avoid:

vague action lines; missing serial number; totals that do not add up; jargon; no customer signature line; recording private customer data.

Instructor reference solution (structure/notes):

- Every section completed from the same facts used in class.
- Cost arithmetic checked: for the class PSU scenario, â‚¦18,000 + â‚¦5,000 = â‚¦23,000.
- Action line example: *"Removed the faulty 500 W power supply; fitted a new 500 W PSU; reconnected the 24-pin motherboard, 4-pin CPU and SATA power cables; confirmed power-on."*
- Tests/result example: *"Powered on; booted to Windows; Device Manager clean; fans and drives operating; USB ports tested."*
- Template includes logo space, report number, all sections and both signature lines.

M. KNOWLEDGE CHECK

(Learner version â€" answers only in the Instructor Key.)

Q1. List.

Name at least eight required sections of a service report.

Q2. Short answer.

Give three reasons why service reports matter.

Q3. Multiple choice.

Which is the better action line in a service report? a) "Fixed the slow computer." b) "Replaced faulty 8 GB RAM module in slot 2; ran memory diagnostic â€" no errors." c) "Did some work on the machine." d) "Customer's computer is okay now."

Q4. Explain in your own words.

Why do we record both the customer's reported fault **and** our observed findings?

Q5. Short answer.

Why should a service report be written in plain language?

Q6. True or false.

You replaced a part and quoted â‚¦20,000, but after further testing the correct total is â‚¦17,000. What do you write on the report, and what do you tell the customer?

Q8. Fill in the blank.
A good service report records not only what was done but also the _____ performed and the _____, to prove the fix worked.

Q9. Identify the mistake.
A report says: "*Customer had an affair on WhatsApp; also replaced the keyboard.*" What is wrong with listing the first part?

Q10. Short answer.
Give two points to include in the "recommendations" section after a repair.

INSTRUCTOR ANSWER KEY (Q1â€“Q10)

A1.
Any eight of: header/shop details with unique report number; customer and device details incl. serial number; reported fault (customer's words); diagnosis/findings with tests and root cause; action taken; parts used; tests performed after repair and result; costs (parts and labour) with total; recommendations; warranty terms; technician and customer sign-off. **A2.** Any three of: proof of work; warranty support; a record for future technicians; professionalism and trust; business information (costs, recurring faults); protection in a dispute. **A3.** (b). **A4.** The reported fault shows what the customer experienced and that you listened; the observed finding is evidence-based and identifies the true root cause. They can differ, and recording both shows professional reasoning. **A5.** Because the report is written for the customer, who is not a technician; it must be understood and trusted. Clear plain language also reduces disputes. **A6.** False. The customer's signature confirms they agreed to the work, the cost and the return condition of the device. It protects both the customer and the shop. **A7.** Write the **correct** total of â‚¦17,000, with an honest note (if needed) that the final cost is lower than the estimate because less work/parts were needed than anticipated. Tell the customer plainly and refund/charge only the correct amount. Never keep an inaccurate higher figure. **A8.** â€¦ **tests** â€¦ ; â€¦ **result** â€¦ **A9.** It records private, irrelevant personal information about the customer. A service report must contain only repair-relevant facts and must protect the customer's confidentiality. Remove it. **A10.** Any two of: back up data regularly; use a surge protector; clean vents/dust every few months; avoid filling the drive past ~80%; consider an SSD upgrade; handle cables gently; bring the machine for a periodic check.

N. REFLECTION AND RECAP

Plain-English summary.
Today you learned to write the professional record of a repair: the **service report**. You learned why it matters (proof of work, warranty, records, trust, business information); the **sections** every report must contain (header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests and result, costs, recommendations, warranty, sign-off); how to keep the **customer's words** and your **observed findings** separate but connected; how to turn a claim into evidence with **action + tests + result**; how to write **clearly and honestly**; and how to record costs, warranty and signatures while protecting customer data. This report is exactly what you will produce for the Final Project.

Five important takeaways:

1. The report is your **protection, memory and professional name**.
2. Record **customer's words** and **observed findings** â€” both.
3. **Action + tests + result** proves the fix.
4. Be **specific and honest**; no jargon, no exaggeration.
5. Finish with **costs, warranty and signatures**, and store records safely.

Questions to help you recognise what you can now do:

- Can I list every required section of a service report from memory?
- Can I write a specific action line instead of "fixed it"?

- Can I protect customer data while keeping good records?

Connection to the next session:
Next session is the **Final Project â€" Capstone**. You will receive a machine with a simulated fault and, without help, diagnose it, repair it, verify the fix, write the full service report you practised today, and present your work. Everything from Sessions 1â€"11 comes together in one job. Come prepared, rested and ready.

O. INSTRUCTOR EVALUATION NOTES

Observable signs of genuine competence:

- The learner uses the **template** and completes **every** section without prompting.
- The learner quotes the customer's reported fault and separates it from the observed findings.
- The learner writes specific action lines and records actual **tests and results**.
- The learner's totals are accurate and the warranty is stated honestly.
- The learner writes in plain language a customer could understand.
- The learner includes both technician and customer sign-off and can explain why.
- The learner treats customer data confidentially in the report.

Common misconceptions to watch for and correct:

- "Reports are just paperwork." â†' They protect the customer, the technician and the shop, and support warranty and disputes.
- "A short note is enough for a simple job." â†' Even a simple repair gets the full record; only the length of each section changes.
- "The customer won't read it." â†' Even if they do not, the report is your protection and your record.
- "I'll write it later from memory." â†' Write it while the details are fresh; memory fades and errors creep in.
- "My signature is enough." â†' The customer must sign too; it proves agreement.
- "Big technical words look professional." â†' Plain, honest language is what reads as professional.

Suggested remedial exercises for learners who struggle:

1. **Section-matching drill:** give a list of facts and a list of sections; ask the learner to match each fact to its section.
2. **Vague-to-specific drill:** hand out ten vague sentences ("fixed it", "cleaned it") and ask for specific rewrites.
3. **Fill-the-template drill:** provide a partially completed report with gaps; ask the learner to finish it.
4. **Honesty drill:** give a scenario with a tempting exaggeration; ask the learner to write the honest version and explain why.

Differentiation:

- **Level 1 (supported):** provide a partially completed report with prompts in each section; allow the learner to write on paper; check section by section.
- **Level 2 (standard):** complete the guided exercise and assignment as written, including the reusable template.
- **Level 3 (challenge):** write reports for **two different** repairs (one hardware, one software/OS), and add a short **customer-facing summary paragraph** at the top of the report that a non-technical customer can read in 30 seconds.

Timing guidance (105 min):

- Opening 6, prerequisite review 7, lecture 35, demonstration 15, guided report-writing 18, independent exercise 15, case study folded into reflection 3, reflection/recap + knowledge check 6. If the room is slow, shorten the case study and the knowledge check discussion, but **protect the guided report-writing** â€" the report is the capstone deliverable and must be practised. Do not cut the sign-off or confidentiality discussion.

END OF SESSION 11

Next: Session 12 â€" Final Project (Capstone).

CYBER
ELIAS
ACADEMY

SESSION 12 â€" FINAL PROJECT (CAPSTONE): DIAGNOSE, REPAIR, DOCUMENT AND PRESENT

CYBER ELIAS ACADEMY â€" COMPUTER REPAIRS

SESSION 12 â€" FINAL PROJECT (CAPSTONE): DIAGNOSE, REPAIR, DOCUMENT AND PRESENT

A. SESSION IDENTIFICATION

Item	Detail
Session number	12 of 12
Session title	Final Project (Capstone) â€" Diagnose, Repair, Document and Present
Week / Module	Week 4 â€" Module 4: Professional Practice (Capstone)
Recommended duration	105 minutes (plus a short certificate sign-off)
Difficulty level	Beginner, but it **tests all accumulated skills** (Sessions 1â€"11)
Prerequisite knowledge	Sessions 1â€"11 â€" components, safety and tools, disassembly/reassembly, diagnostic workflow, hardware and software diagnostics, component replacement, cleaning and maintenance, OS installation, customer service, and service reports
Connection to previous work	This is the integration of the whole course. Every prior session taught one skill; today you combine them into a single professional repair job, exactly as a shop would.
Preparation before class	Prepare **one faulty machine per learner** (or one per pair for large classes) with a **simulated fault**; ensure a fault log exists for every machine so the examiner knows the real cause; prepare tool kits, anti-static straps, replacement parts and cleaning supplies; print the Capstone Brief, the Capstone Rubric, the Viva Question List and the Certificate Sign-off sheets; set up assessment stations; confirm each machine's **twist** is documented.
Safety and ethics flag	Safety procedures (ESD, unplug before opening, never open a PSU) are **assessed**, not optional. Data safety (back up before any wipe) is **assessed**. Only **licensed** software is used. No learner may proceed past a safety breach without correction.

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Receive a faulty machine and complete a customer intake**, capturing the reported fault accurately.
2. **Diagnose the fault** using the systematic diagnostic workflow (observe, ask, hypothesise, test, resolve).
3. **Apply correct safety procedures** (ESD protection, power isolation, safe tool use) throughout.
4. **Perform the repair** â€" replacement, cleaning or OS task â€" safely and correctly.
5. **Verify the fix** with real tests and confirm the machine is fully functional.
6. **Produce a complete professional service report** documenting the entire job.

C. REQUIRED MATERIALS

Hardware (per learner or pair)

- One faulty practice machine with a **simulated fault** (see the fault bank below).
- Tool kit: correct screwdrivers, tweezers, torch, anti-static wrist strap, anti-static mat.
- Replacement components for the intended faults (RAM sticks, a known-good PSU, a known-good drive/SSD, extra fans, cleaning supplies).
- Cleaning supplies: compressed air, isopropyl alcohol, microfiber cloth, thermal paste (for cleaning/thermal faults).

Software

- Diagnostic tools as taught: Windows Memory Diagnostic / MemTest86, CrystalDiskInfo, HWMonitor, Device Manager, Windows Update, and a licensed Windows USB where an OS task is part of the fault.

Printed materials

- Capstone Brief (one per learner).
- **Capstone Rubric** (one per learner and one per assessor).
- **Viva Question List** (instructor copy, with model answers).
- Certificate Sign-off sheet (one per learner).
- Blank Service Report Template and Customer Intake Form (from Sessions 10â€“11).

Safety equipment

- Anti-static wrist straps and mats; safety glasses if compressed air is used; first aid kit available.

Room setup

- Assessment stations, each with a machine and its fault log; a clear counter for the intake role-play; a quiet corner for the viva so it is not disturbed.

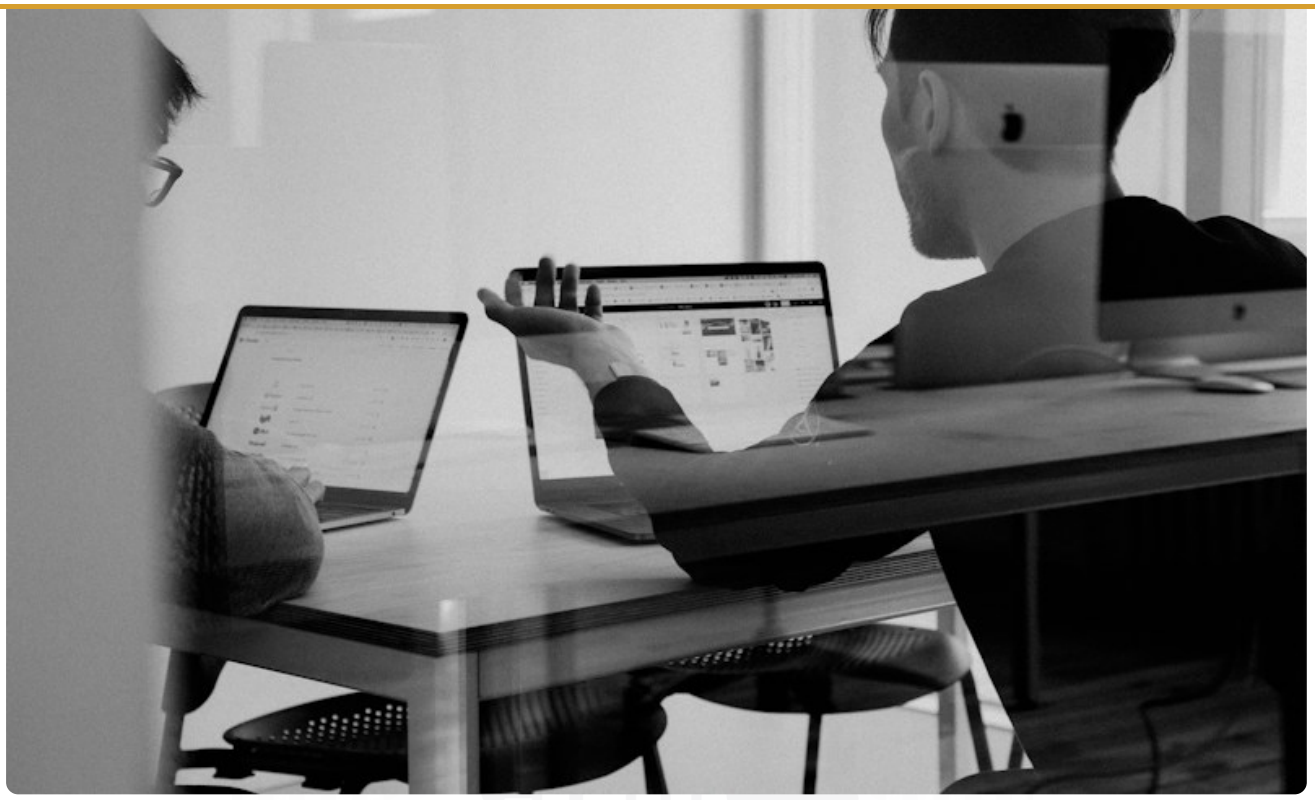
Suggested visuals:



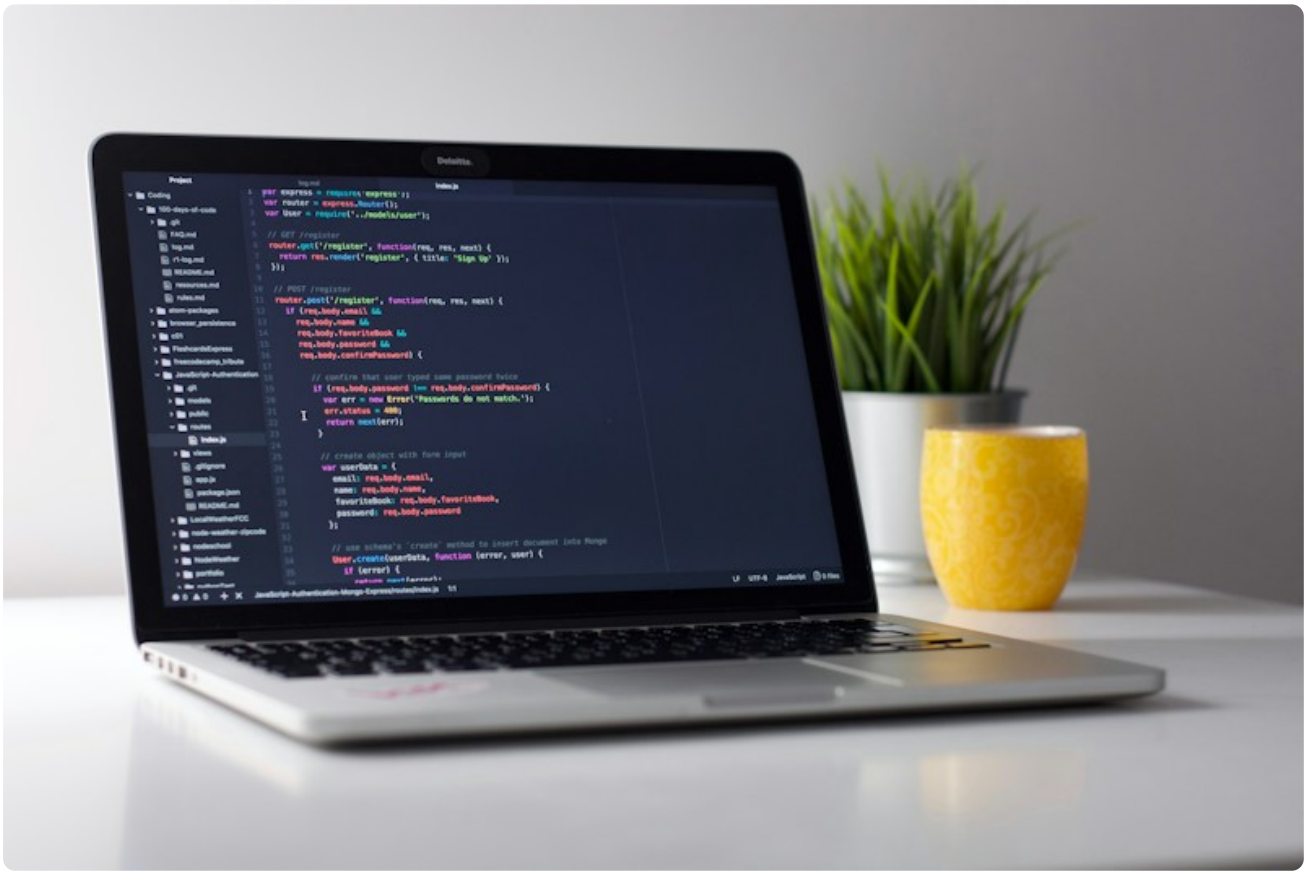
- â€” Full diagnostic-to-resolution assessment run



- â€” Faulted hardware presented at the station



- â€” Intake role-play and final viva



- — Complete repair carried out on a laptop unit



— Tools laid out for the timed capstone repair

D. OPENING ACTIVITY (6 minutes)

Instructor setup:
Stand beside a faulty machine on the counter. Do not open it yet. Put the Capstone Brief on the projector.

What the instructor says:

> *"Four weeks ago, most of you had never opened a computer case. You did not know RAM from a hard drive. Look at you now.*
>
> *Today there is no teaching and no demonstration. Today you work. A customer â€" that is me â€" will bring you a faulty machine. You will greet the customer, take the fault, diagnose it, repair it, prove it works, write the service report, and explain your work to me.*
>
> *I will not help you. Not because I do not care, but because your future employer and your future customers will not be standing beside you. Today you prove to yourself that you can do this alone.*
>
> *Two rules stand above everything:*
>
> ***work safely*** *, and* ***protect the customer's data****. Break either rule and it is an automatic fail for that part, no matter how good the repair is. Everything else, you already know. Let us begin."*

Expected learner response:
A mix of nerves and excitement. Reassure them: the brief, the template, the tools and the workflow posters are all in front of them. They are not being asked to memorise â€" they are being asked to **apply**.

Why this opener works:
It raises the stakes (independent performance) while lowering panic (everything is provided). It also states the two non-negotiables, safety and data, before any work begins.

last-minute confidence check
before the assessment. Run it fast, as a whole-class call-and-response, so learners hear the key facts one final time.

Question 1:
"What are the five steps of the diagnostic workflow?"

- Answer: observe, ask questions, form a hypothesis, test, resolve/verify.

Question 2:
"What are the six major components?"

- Answer: motherboard, CPU, RAM, storage (HDD/SSD), PSU, GPU.

Question 3:
"What is ESD, and how do you prevent it?"

- Answer: electrostatic discharge â€" a sudden flow of static electricity that damages components. Prevent it with an anti-static wrist strap, an anti-static mat, and by touching bare metal of the case before handling components.

Question 4:
"What must you do before opening any case, and what must you never open?"

- Answer: unplug the power, then hold the power button for a few seconds to discharge residual power. Never open the PSU case â€" its capacitors can hold a lethal charge.

Question 5:
"What do these tools test â€" MemTest86, CrystalDiskInfo, HWMonitor?"

- Answer: MemTest86 tests RAM; CrystalDiskInfo reports hard drive/SSD health; HWMonitor shows temperatures and voltages.

Question 6:
"Name the sections of a service report."

- Answer: header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests and result, costs, recommendations, warranty, sign-off.

Question 7:
"What is the academy's rule about software licences?"

- Answer: install only properly licensed software; never use pirated or cracked Windows.

Bridge to the assessment (say this):

> *"Those seven answers are your toolkit. If you can hold onto them under pressure, you will do well. Now the review is over â€" the customer is coming."*

Gap handling:
If the class is visibly shaky on any single question, spend 60 seconds re-explaining it only, without restarting the whole review. Do **not** extend this into teaching: the session time belongs to the learners' independent work.

F. FULL LECTURE NOTES (Capstone Briefing)

F.1 What the capstone is

The idea.
The capstone is one complete, realistic repair job done **independently**. It assesses not just whether you can fix the machine, but whether you can behave like a professional technician from first greeting to final signature.

computer. You must:

1. Conduct a **customer intake**.
2. **Diagnose** the fault using the systematic workflow.
3. **Perform the repair** safely.
4. **Verify** the fix.
5. Write a **service report**.
6. **Present and defend** your work in a short viva/demonstration.

What is being assessed (overview).
Seven areas â€" safety adherence, diagnostic method, correct identification, repair quality, verification/testing, service report, and presentation/viva â€" worth **30 marks**, plus a **5-mark** viva/demo component. The full rubric is in Section I.

Your deliverables.

- A correctly diagnosed fault.
- A safely performed repair.
- A verified, fully functional machine.
- A complete, professional service report.
- A confident presentation/viva.

Time.
Approximately 70 minutes for intake, diagnosis, repair and report; 5 minutes for the viva/demo (per learner, run in parallel or in a slot).

F.2 Milestone checkpoints â€" what you should already have done before Session 12

The capstone is not a surprise. To be ready, you should have completed these milestones in earlier sessions. Use this as your personal pre-capstone checklist.

#	Milestone	Covered in	What "done" looks like
1	Identify all major components and explain their functions	S1	You can point to and name each component on a real machine
2	Work safely: ESD, power isolation, correct tool use	S2	You always wear the strap, unplug, and use the right screwdriver
3	Disassemble and reassemble safely	S3	You can strip and rebuild without damaging or misplacing parts
4	Follow the five-step diagnostic workflow	S4	You observe, ask, hypothesise, test, and only then act
5	Test RAM, storage, temperature and power	S5	You can run MemTest86, CrystalDiskInfo and HWMonitor and read results
6	Diagnose software faults (OS, malware, drivers)	S6	You can tell an OS problem from a hardware problem
7	Replace RAM, HDD/SSD, PSU and GPU	S7	You can swap components and verify the replacement
8	Clean and maintain, replace thermal paste and fans	S8	You can remove dust, reseat coolers and apply paste correctly
9	Install an OS safely and legally	S9	You can back up, install Windows, add drivers and activate legally
10	Serve and communicate with a customer	S10	You can run an intake and explain a fault in plain language

11	Write a professional service report	S11	You can complete every section of the report, honestly

If a learner has not met these milestones
, the instructor should arrange practice before the capstone (see Section O, remediation). A learner may still attempt the capstone, but the rubric will reflect the gaps.

F.3 The fault bank (instructor's simulated faults)

Each machine carries

one primary simulated fault
(and, for Level 3, possibly a second hidden fault). The fault log records the true cause so the assessor can mark accurately.

Fault	How it is simulated	Expected diagnostic path	Expected repair
Loose/unseated RAM	RAM stick not fully clipped in (or one dead stick)	No boot / beeps / memory test errors	Reseat or replace the RAM
Failing storage	A drive set to fail SMART checks or disconnected data/power cable	Slow boot, clicking, CrystalDiskInfo warnings	Reseat or replace the drive/SSD
Overheating	Dust-blocked heatsink / disconnected fan / dried thermal paste	High CPU temps on HWMonitor, thermal shutdowns	Clean, reconnect fan, replace thermal paste
Failed PSU	Known-good PSU swapped in for a dead one (or a loose 24-pin)	No power / no POST	Replace or reseat the PSU safely
Corrupted OS	Windows deliberately corrupted, or an unactivated/blank drive	Boots to recovery / no OS found	Back up, clean install Windows, drivers
Faulty GPU	GPU not seated / dead card (integrated graphics fallback)	No display / artefacts	Reseat or replace the GPU
Loose data cable	SATA cable unplugged/bad	Drive not detected	Reseat/replace the cable

Important:
the learner is **not told** the fault. They must find it. The assessor follows the fault log to verify whether the learner identified the true cause.

F.4 How you will be assessed

You are assessed on

process as much as result
. A learner who fixes the machine but skips the backup step, or who diagnoses by luck with no method, will lose marks. A learner who follows the workflow, works safely, identifies the cause, repairs well, verifies and documents will score highly â€” even if a small hiccup occurs, as long as they handle it professionally.

The seven assessed criteria and the marks are in the full rubric (Section I). Read the rubric

before
you begin, so you know exactly what the examiner is looking for.

F.5 The professional mindset

- **Safety first.** Every time. The strap, the unplug, the right tool.
- **Data safety.** Back up before any wipe. Ask about data during intake.
- **Honesty.** Record what you did. Do not invent or exaggerate.
- **Method over luck.** Follow the workflow even under pressure.

Say this to the class:

> *"In ten years, the machines will change. New CPUs, new drives, new Windows. What will not change is the way a professional works: safely, methodically, honestly, and in the customer's interest. That is what we are testing today â€" the professional, not just the technician."*

G. INSTRUCTOR DEMONSTRATION

There is no instructor demonstration in this session.

The capstone is a

learner-led independent assessment

. Demonstrating the repair would defeat its purpose. Instead, the instructor:

- States the brief (Section D) and reviews prerequisites (Section E).
- Assigns each learner their machine and intake scenario.
- Observes, times and marks using the rubric (Section I).
- Runs the viva using the question list (Section J).
- Does **not** give technical hints during the independent work.

Permitted assistance:
safety interventions only. If a learner is about to cause harm (for example, reaching to open a PSU, or handling a component without the strap), the instructor **must** intervene to protect them â€" and this is recorded as a safety concern against the relevant rubric criterion. No other help is given.

H. GUIDED PRACTICAL EXERCISE

There is no guided practical in this session.

The entire session is the independent capstone (Section I). However, the instructor

walks the room continuously

, observing quietly and noting evidence for the rubric, while remaining available for safety-only interventions.

I. INDEPENDENT CLASS EXERCISE â€" THE CAPSTONE (â‰ˆ70 minutes + viva)

I.1 The task

Scenario:
"You are a computer repair technician at Cyber Elias Tech. A customer brings in a faulty computer. Diagnose the problem, perform the repair, verify the fix and produce a service report. Present your work to the instructor."

Procedure:

Step 1 â€" Customer intake (5 minutes).

1. Greet the customer (instructor) warmly.
2. Ask good diagnostic questions; capture the reported fault in the customer's **own words**.
3. Ask about **important data** (this is mandatory).
4. Set expectations: diagnosis first, then a written estimate before any billable work.

2. **Ask** further questions if needed.
3. **Form a hypothesis** and say it out loud (for the assessor).
4. **Test** with the correct tools (memory test, SMART check, temperature, PSU test).
5. **Identify the root cause** and record every test and result.

Step 3 â€" Repair (25 minutes).

1. **Safety first:** power off, unplug, discharge, anti-static strap on, mat under the machine.
2. Perform the repair with the **correct tools** and correct technique.
3. Replace, reseat, clean or install the OS (with a backup first if data is at risk).
4. Only licensed software.

Step 4 â€" Verify (10 minutes).

1. Power on and confirm the specific fault is resolved.
2. Run diagnostic tests again (memory, SMART, temperature, Device Manager, boot).
3. Confirm all components work: display, sound, network, storage, ports.
4. State clearly: "Fault resolved" only when the evidence supports it.

Step 5 â€" Service report (10 minutes).

1. Complete **all** sections of the report.
2. Use plain language and specific details.
3. Include correct totals and warranty.
4. Get the customer's (instructor's) signature.

Step 6 â€" Present / viva (5 minutes).

1. Explain what the fault was, how you found it, what you did and how you verified it.
2. Answer the assessor's viva questions (Section J) confidently and honestly.

Time allowance:
70 minutes + 5-minute viva.

Success criteria:
Fault correctly diagnosed; repair performed safely and correctly; fix verified; complete professional service report; clear, honest presentation.

I.2 CAPSTONE RUBRIC (30 marks + 5-mark viva/demo)

> ***Instructor-only marking instrument.***
> *Each criterion is scored 0â€"5 (six criteria = 30 marks). The viva/demo is a separate 5 marks. Total = **35 marks**.*
> *Performance descriptions guide the marker; use the nearest band.*

#	Criterion	5 â€" Excellent	3 â€" Competent	1 â€" Developing	0 â€" Not demonstrated	Marks
1	**Safety adherence**	ESD strap worn correctly, power isolated and discharged, correct tools used, safe posture and workspace throughout; anticipates hazards	Safety mostly followed with one minor lapse, self-corrected	Safety procedures partial or inconsistent; needed one intervention	Ignored safety; would have caused harm (automatic 0)	/5

2	**Diagnostic method**	Full five-step workflow followed visibly; hypothesis stated and tested logically; systematic, no random part-swapping	Workflow mostly followed; minor gaps in logic	Partial method; some guessing; skipped steps	No method; random actions	/5
3	**Correct identification**	Confirmed the **exact** root cause, matching the fault log, with evidence	Identified the correct general area; some uncertainty	Wrong first guess but partially recovered, or correct area without confirmation	Wrong root cause, or failed to identify	/5
4	**Repair quality**	Repair performed correctly, cleanly and durably; correct component/technique; no collateral damage	Repair correct with minor imperfections	Repair partially correct; loose fitting or messy; may need rework	Repair incorrect or caused damage	/5
5	**Verification / testing**	Specific tests re-run to prove the exact fault is gone; full-system check; honest statement of result	Basic verification done; some checks missed	Minimal verification (e.g. "it switched on")	No verification, or false claim of success	/5
6	**Service report**	All sections complete, accurate, specific, plain language, correct costs, warranty and signatures	Most sections complete; mostly clear	Several sections missing or vague; jargon; inaccurate figures	Missing or unusable report	/5
â€”	**Subtotal (30)**					**/30**
7	**Presentation / viva**	Clear, confident, honest; explains decisions and answers questions accurately; owns any limitations	Understandable; answers most questions correctly	Uncertain; struggles to explain own decisions	Cannot explain the work, or misrepresents it	**/5**
	TOTAL					**/35**

Passing standard:
A learner must score at least **50% overall** across the course AND successfully complete the capstone: **correct diagnosis + safe repair + verified fix + professional service report**. A repair the learner **cannot explain** must not automatically pass. A serious safety breach (Criterion 1 = 0) fails the capstone regardless of other scores.

Assessor notes column (use on the marking sheet):
record evidence, not just scores â€” for example, "stated hypothesis: dust-blocked heatsink; ran HWMonitor showing 94Â°C; cleaned and re-pasted; temp 52Â°C after."

I.3 Milestone timing during the session

Phase	Minutes	What the assessor watches for
Intake	5	Warmth, good questions, data question asked
Diagnose	20	Workflow order, visible hypothesis, correct tools
Repair	25	Safety, technique, no collateral damage

Verify	10	Re-running the right tests, honest result
Report	10	All sections, accuracy, plain language
Viva	5	Explanation, honesty, confidence

J. VIVA / DEMONSTRATION â€” QUESTION LIST WITH MODEL ANSWERS

> ***Instructor-only (mostly).***
> *Ask 4â€“6 questions per learner, chosen to probe their actual work. Model answers are guides, not scripts; a learner who reasons correctly in their own words should be credited.*

Q1. "In your own words, what was the fault and how did you find it?"

- Model: The learner names the actual fault and walks through observe â†' ask â†' hypothesise â†' test â†' conclude. They should mention the specific test that confirmed the cause (for example, memory test errors, SMART warnings, high temperatures, or a known-good PSU test).

Q2. "What safety precautions did you take before opening the case, and why?"

- Model: Powered off and unplugged the machine; held the power button to discharge residual power; wore an anti-static wrist strap connected to bare metal; used an anti-static mat; used the correct screwdriver. These protect the technician and the components.

Q3. "Why did you test the way you did, rather than swapping parts?"

- Model: Testing finds the **root cause** and avoids wasting time and the customer's money on parts that are not faulty. Random part-swapping can also introduce new faults.

Q4. "How did you confirm the fault was truly fixed?"

- Model: Re-ran the same test that failed before (for example, memory diagnostic clean, SMART healthy, temperatures normal, machine boots to Windows), plus a full-system check (Device Manager, display, sound, network).

Q5. "Walk me through your service report â€” what would you say to the customer?"

- Model: The learner summarises the reported fault, the diagnosis, the repair, the parts, the total cost, the warranty, and the recommendation in plain language.

Q6. "If the customer's machine had important data and the fix required a reinstall, what would you have done?"

- Model: Backed up and verified the data first, informed the customer, and obtained approval before wiping; only then installed Windows. Data safety before any wipe.

Q7. "What would you do if you could not find the fault?"

- Model: Be honest â€" do not guess or pretend. Return to the workflow, re-observe, re-ask, re-test, check the simplest causes first; consult a senior technician; tell the customer the truth about the uncertainty.

Q8. "Show me where the RAM / storage / PSU is and explain what it does."

- Model: The learner points to the correct component and gives its function (RAM = short-term memory; storage = permanent files; PSU = converts wall power to the voltages the PC needs).

Q9. "Was any part of this job done legally? What about the software?"

- Model: Only properly licensed software was used; no pirated Windows. The learner can state the academy's licensing rule.

Q10. "What would you do differently next time?"

Q11. "Why did you record the customer's exact words in the report?"

- Model: They can contain clues to the cause, and they prove the technician listened and understood the customer's experience.

Q12. "A customer is angry that the repair is late. How would you handle it?"

- Model: Listen without interrupting, empathise, focus on solutions, and set calm boundaries; communicate honestly about the new time.

Marking the viva (5 marks):

- 5 â€" Clear, confident, honest; answers accurately; explains decisions; owns limitations.
- 3 â€" Understandable; answers most questions correctly; minor gaps.
- 1 â€" Uncertain; struggles to explain own work.
- 0 â€" Cannot explain the work or misrepresents it.

K. COMMON MISTAKES AND TROUBLESHOOTING (Capstone failure points)

Use this as the

remediation map
when a learner underperforms. These are the most common failure points and what to do about them.

K.1 Skipping safety steps

Symptom:
Opens the case without the strap or without unplugging. **Why:** Rushing, nerves, overconfidence. **Fix:** Stop, reset the learner, restate the rule, let them continue safely (rubric reflects the lapse). **Prevent:** Make safety a habit drilled since Session 2; the strap comes out before the case.

K.2 Jumping to a repair without diagnosis

Symptom:
Immediately replaces a part guessed to be at fault. **Why:** Impatience; forgetting the workflow. **Fix:** Pause; require them to state a hypothesis and one test that would confirm it before touching the machine. **Prevent:** Practise "say your hypothesis out loud" in every diagnostic session.

K.3 Fixing the symptom, not the root cause

Symptom:
Machine boots after a reseat but the real fault (for example, a failing drive) remains. **Why:** Stopping at the first sign of life. **Fix:** Require verification tests; ask "how do you know the cause is gone?" **Prevent:** Always test the underlying cause, not just the symptom.

K.4 Not verifying the fix

Symptom:
"It switched on, so it's fixed." **Why:** Misunderstanding what verification means. **Fix:** Re-run the specific failing test; check the whole system. **Prevent:** Use the post-repair checklist every time.

K.5 Poor or incomplete service report

Symptom:
Missing sections, vague action lines, wrong totals, no signature. **Why:** Rushed at the end; forgets the template. **Fix:** Complete the template section by section; check arithmetic; add signatures. **Prevent:** Write the report from the same facts as the repair; practise in Session 11.

Wipes or reinstalls without backing up or without customer consent. **Why:** Forgetting the customer's data in the heat of the task. **Fix:** Stop; if data may be lost, halt and refer (recovery is beyond this course); record the lapse. **Prevent:** Ask about data in intake; back up and verify before any wipe.

K.7 Cannot explain the work in the viva

Symptom:
Repaired the machine but cannot say how or why. **Why:** Learnt by imitation rather than understanding; panic. **Fix:** Walk them back through their own steps; ask simpler "what" and "why" questions. **Prevent:** Narrate reasoning during practice so it becomes natural.

K.8 Licensing shortcut

Symptom:
Uses a pirated or cracked Windows/software. **Why:** Convenience; not understanding the risk. **Fix:** Stop immediately; require a licensed install; explain the legal and safety consequences. **Prevent:** State the licensing rule every time software is installed.

Distinguish likely vs confirmed
applies here too: do not assume a learner "does not know" â€” observe the actual behaviour and confirm with a question before judging. The rubric is about demonstrable performance.

L. SESSION ASSIGNMENT

Assignment title:
"My Capstone Reflection and Next Steps"

Purpose:
To close the course with honest self-assessment and a concrete plan for the learner's next stage as a technician â€” turning a certificate into a direction.

Realistic scenario:
You have just completed your capstone assessment. Your instructor asks you to write a short, honest reflection and a plan, as part of your professional development file.

Detailed instructions:

1. **Describe your capstone job** in 3â€“5 sentences: the fault, how you diagnosed it, what you repaired, and how you verified it.
2. **Three things you learned** across the whole course (be specific and accurate).
3. **Two things you found challenging**, and how you handled or would handle them.
4. **One skill you want to improve** next, and one concrete step to improve it.
5. **Your next step**: choose at least one realistic path â€” apply for a junior technician role, start a small repair side-business, take a CEA advanced course, or specialise (for example, phone repair, data recovery basics, networking).
6. **An honest statement** of what your certificate does and does not mean (that it shows competence as a beginner technician, not as a certified electronics engineer).
7. Sign and date it.

Required deliverables:

- A typed or neatly handwritten reflection of **at least half a page**.
- The completed **Capstone Rubric** sheet (given to you after marking, if your instructor shares it) and your signed **Service Report**.

Submission format:
Printed or emailed; filename `Surname_S12_Reflection.docx` .

Marking rubric (25 marks):

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Capstone description	Clear, specific, accurate	Present but thin	Vague or missing
Three things learned	Specific, accurate, reflective	Mostly specific	Vague
Two challenges	Honest, specific, with handling	Somewhat specific	Vague
Next-step plan	Clear, realistic, concrete step	Present but general	Missing
Honest certificate statement	Accurate about what it does/does not mean	Partly accurate	Missing or overclaims

Common mistakes to avoid:
overclaiming (saying the certificate makes you an "engineer"); vague lists ("I learned a lot"); no concrete next step; dishonest reflection.

Instructor reference solution (structure/notes):

- Capstone description: names the real fault and the actual test that confirmed it.
- Learned: e.g. "I can now safely replace a PSU and prove it with a power test"; "I can write a service report"; "I learned to back up data before any install."
- Challenges: e.g. "Staying calm when the machine would not boot the first time"; "Remembering to ask about data."
- Next step: e.g. "Apply to a repair shop in Port Harcourt as a junior technician, and start saving for an advanced course."
- Honest statement: e.g. "This certificate shows I can safely diagnose and repair common hardware and software faults as a beginner technician. It does not make me a certified electronics engineer and does not cover advanced repairs."

M. KNOWLEDGE CHECK

This is the

course-wide knowledge check
(20 questions across the four weeks). It is worth 5% of the overall grade. Answers are for the instructor only.

(Learner version â€" answer in the spaces provided.)

Section A â€" Components (Q1â€“Q5)

1. Name the six major components of a computer.
2. What does the CPU do?
3. What is the difference between RAM and a hard drive?
4. Which is faster, an HDD or an SSD?
5. What does the PSU do?

Section B â€" Safety (Q6â€–Q10)
6. What is ESD? 7. Name two ways to prevent ESD damage. 8. What must you do before opening a computer case? 9. Why should you never open a PSU case? 10. What tool prevents static damage to components?

Section C â€" Diagnostics (Q11â€–Q15)
11. What are the five steps of the diagnostic workflow? 12. What does a clicking noise from a hard drive usually indicate? 13. Which tool or test checks RAM? 14. What is a normal idle CPU temperature range? 15. Why should you test the simplest possible cause first?

Section D â€" Repair, OS and Professional Practice (Q16â€–Q20)
16. What should you do before a clean install of Windows? 17. Name three driver types you install after Windows. 18. What are the four steps of the customer intake process? 19. Name five sections of a service report. 20. What should you do if a repair will

INSTRUCTOR ANSWER KEY

Section A

1. Motherboard, CPU, RAM, storage (HDD/SSD), PSU, GPU.
2. It is the "brain" â€” it performs calculations and executes instructions.
3. RAM is short-term memory that empties when power is off; a hard drive stores files permanently.
4. SSD.
5. It converts electricity from the wall socket into the correct voltages for the PC components.

Section B

6. Electrostatic discharge â€” a sudden flow of static electricity that can damage components. 7. Any two of: anti-static wrist strap; anti-static mat; touching bare metal of the case before handling components; storing components in anti-static bags. 8. Unplug the power cable and press/hold the power button to discharge residual power. 9. Capacitors inside can hold a lethal charge even when the unit is unplugged. 10. Anti-static wrist strap.

Section C

11. Observe, ask questions, form a hypothesis, test, resolve/verify. 12. The drive is failing. 13. Windows Memory Diagnostic or MemTest86. 14. Roughly 30â€“50 Â°C at idle (accept a reasonable range; under load it can rise, but sustained very high temperatures indicate a cooling problem). 15. Simple causes are common and quick to test; you may solve the problem in minutes before spending time on complex checks.

Section D

16. Back up the customer's data (and verify it), then check that the hardware meets the OS requirements; prepare drivers. 17. Any three of: network, chipset, graphics, audio, Wi-Fi/Bluetooth. 18. Greet, gather information, set expectations, document. 19. Any five of: header/report number, customer and device details, reported fault, diagnosis/findings, action taken, parts used, tests and result, costs, recommendations, warranty, sign-off. 20. Stop and contact the customer for a fresh estimate and approval before continuing work.

N. REFLECTION AND RECAP

N.1 Plain-English summary

Today you completed your

capstone

: you received a faulty machine, diagnosed it using the systematic workflow, repaired it safely, verified the fix, wrote a professional service report, and explained your work in a viva. You have finished the Computer Repairs course at Cyber Elias Academy. Whatever the mark, take a moment to appreciate how far you have come â€” from not knowing RAM from a hard drive, to diagnosing and repairing a real machine on your own.

N.2 Five important takeaways from the entire course

1. **Safety always comes first** â€” ESD protection, unplug before opening, never open a PSU.
2. **Follow the systematic diagnostic workflow** â€” observe, ask, hypothesise, test, resolve.
3. **Know your components** â€” what each part does and how to replace it is your foundation.
4. **Customer service is a technical skill** â€” communicate clearly and honestly in plain language.
5. **Professional documentation protects everyone** â€” the service report is your proof, memory and name.

N.3 Questions to help you recognise what you can now do

- Can you identify all major computer components and explain their functions?
- Can you follow ESD and electrical safety procedures correctly?
- Can you disassemble and reassemble a computer without damage?

- Can you clean and maintain a computer, including thermal paste?
- Can you install and configure an operating system safely and legally?
- Can you run a customer intake and explain faults in plain language?
- Can you write a complete, honest service report?
- Can you complete a full repair project independently, from greeting to signature?

If you answered "yes" to most of these, you have become a

competent entry-level technician

.

N.4 Honest statement about what you have achieved

Be clear with yourself and with any employer:

- **This course makes you a competent entry-level computer repair technician.** You can safely diagnose and repair common hardware and software faults, serve customers professionally, and document your work. You can work as a junior technician or start a small repair business.
- **This course does NOT make you a certified electronics engineer or an IT professional.** It does **not** cover advanced work such as motherboard-level component repair (microsoldering), professional data recovery, or networking engineering. Those require further study.
- **This certificate does not guarantee employment.** It proves demonstrated competence; your attitude, honesty and continued practice will do the rest.

N.5 What comes next

At Cyber Elias Academy and beyond, you can continue with:

- **Cybersecurity** â€" protect computers and networks from threats (pairs naturally with repair).
- **Web Design / Web Development** â€" build and maintain websites.
- **Business & Freelancing** â€" learn to run your repair business professionally, price your work and find customers.
- **Content Creation** â€" photography, video and written content.
- **Specialisation** â€" phone repair, laptop specialist work, small-business IT support, or (with further training) data recovery and networking.

Practical next steps:

1. Keep a **job log** of every repair you do, even small ones â€" it builds your experience and your confidence.
2. Build a simple **portfolio**: before/after photos (with permission), service reports (with customer details removed), and a list of faults you have fixed.
3. Register and brand your **business** if you choose to work independently (a simple name, a phone/WhatsApp number, and an honest price list go a long way).
4. Never stop learning â€" every machine you open teaches you something new.

N.6 A closing word from CEA

> *"From Zero to Expert, Together." You started at zero and worked together to reach this point. You are not an expert yet â€" nobody becomes one in four weeks â€" but you are a*
>
> ***competent, safe, honest technician***
> *with a certificate to prove it and a skill that will feed you. Use it well, treat your customers fairly, keep learning, and the rest will follow.*

O.1 Observable signs of genuine competence

- The learner puts on the anti-static strap and isolates power **without being reminded**.
- The learner states a clear hypothesis before testing, and tests logically.
- The learner identifies the true root cause, matching the fault log, with evidence.
- The learner repairs cleanly, with the correct component and technique.
- The learner re-runs the specific failing test and confirms the whole system works.
- The learner's service report is complete, specific, honest and signed.
- The learner explains decisions calmly in the viva and **owns any limitations honestly**.
- The learner uses only licensed software and can state the rule.

O.2 Common misconceptions to watch for and correct

- "I can just learn this on my own." â†' Structured practice prevents unsafe habits and builds method. Self-teaching without feedback often cements bad technique.
- "Repair is only about fixing broken things." â†' It is equally maintenance, customer service and documentation.
- "Passing means I am an expert." â†' Passing means you are a safe, competent **beginner**. Expertise takes years and further study.
- "If it switches on, it is fixed." â†' Verification requires re-testing the actual fault and checking the whole system.
- "Reports and safety slow me down." â†' They protect the customer, the shop and you; a careless wipe or a dishonest report can end a career.
- "A cracked Windows is fine to save money." â†' Illegal and unsafe; only licensed software.

O.3 Suggested remedial exercises / remediation

- **Did not pass the capstone:** allow supervised practice on academy machines during opening hours; reassess when ready. A second attempt is normal and expected â€" competence, not speed, is the goal.
- **Cannot diagnose:** provide additional guided diagnostic scenarios, starting with simple causes, and require the learner to say the hypothesis out loud each time.
- **Cannot follow safety:** one-on-one safety drill until the strap/unplug routine is automatic; a safety breach must be corrected before any reassessment.
- **Cannot write a report:** provide a partially completed report to finish; then progress to writing independently against the template.
- **Cannot explain the work (viva):** practise narrating reasoning during simple tasks; use "what did you do, and why?" questioning until it is natural.

O.4 Differentiation

Level 1 â€" Supported:

- Allow extra time for the capstone.
- Provide a structured step-by-step prompt sheet and gentle time reminders.
- Offer one-on-one coaching **outside** the assessment, and a retake after practice.
- Choose the simplest simulated fault (for example, loose RAM) for the first attempt.

Level 2 â€" Independent:

- Complete the capstone, knowledge check and reflection as written.

Level 3 â€" Challenge:

- Assign a machine with **two simultaneous faults** (for example, dusty overheating **plus** a failing drive), or a fault requiring an **OS installation** as part of the repair.
- Require a longer service report with a customer-facing summary paragraph.
- Ask higher-order viva questions (for example, "How would you prevent this fault recurring?").

Segment	Minutes
Opening + brief	6
Prerequisite review	7
Capstone: intake	5
Capstone: diagnose	20
Capstone: repair	25
Capstone: verify	10
Capstone: report	10
Viva/demo	5 (can run in parallel slots)
Reflection, knowledge check and certificate sign-off	~12â€"17 (may shorten if needed)

Protect the

diagnose, repair and verify
time above all â€" those carry the most marks. The knowledge check can be completed during certificate sign-off or as a take-home if time runs short, but the capstone must be observed live.

CERTIFICATE SIGN-OFF (Instructor-only administrative record)

Learner Name:
_____ **Date:** _____ **Capstone Result:** Pass / Not Yet Competent **Capstone Rubric Score:** _____ / 30 **Viva/Demo:** _____ / 5 **Total:** _____ / 35 **Course Overall (knowledge checks + assignments + capstone):** _____ % **Fault Diagnosed:** _____ **Repair Performed:** _____ **Service Report:** Complete / Incomplete **Presentation:** Confident / Developing **Safety Adherence:** Met / Concern noted

Instructor Signature:
_____ **Certificate Awarded:** Yes / No

Comments:

Congratulations! You have completed the Computer Repairs course at Cyber Elias Academy.

You have gone

From Zero to Expert, Together
â€" from a beginner who had never opened a computer case to a competent, safe, honest entry-level repair technician. Carry the same care to every machine and every customer. Keep learning. The next level is waiting.

Need help?
Contact your CEA instructor or the academy front desk on Ebony Road, Rumuola, Port Harcourt. Certificate verification is available at cea.ng/certificates/verify.

END OF SESSION 12 â€" END OF COURSE

Next steps: choose your next CEA course â€" Cybersecurity, Web Design, Business & Freelancing, or Content Creation.

CYBER
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PART 3

Learner Workbook

COMPUTER REPAIRS — LEARNER WORKBOOK

COMPUTER REPAIRS — LEARNER WORKBOOK

Cyber Elias Academy (CEA)
24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State *From Zero to Expert, Together*

Course:
Computer Repairs (Hardware & Software Diagnostics and Repair) **Duration:** 4 weeks | **Sessions:** 12 | **Session length:** 105 minutes | **Fee:** ₦50,000

> ***This is your copy.***
> *It contains **no answer keys**. The answers are discussed in class and kept by your instructor. Everything here is written for a true beginner — if you can type, save a file and use the internet, you can do this course.*

1. WELCOME — WHAT THIS COURSE IS ABOUT

In Port Harcourt and across Nigeria, computers break every day. Hard drives fail, RAM comes loose, machines overheat, viruses slow everything down, and screens go black. Some people lose years of work because nobody taught them to check a hard drive's health. Others pay far too much for a repair that took ten minutes.

This course turns a beginner into a

safe, honest, professional computer repair technician
. You will learn:

- What is inside a computer and what every part does.
- How to work on a computer **safely** without damaging it or hurting yourself.
- How to take a computer apart and put it back together correctly.
- How to **diagnose** a fault the way a detective solves a case — step by step.
- How to test hardware and fix software problems.
- How to replace common parts, clean a computer and install an operating system.
- How to talk to customers, quote a fair price and write a professional **service report**.

By the end, you will diagnose a

simulated fault
on a real practice computer, repair it, prove the fix works, write a full service report and present your work. That final project is called the **capstone**, and it is how you earn your certificate.

The one thing that matters most: SAFETY

Every practical lesson in this course has two golden rules you must never break:

1. **SAFETY FIRST.** Unplug before you open a case. Wear an anti-static wrist strap. Never open a power supply unit (PSU). Never work on live mains wiring. If you are not sure, stop and ask.
2. **DATA-SAFETY (BACK UP FIRST).** Before you wipe or reinstall anything, back up the customer's data. Lost data cannot be un-lost by an apology. Also: **use licensed software only** — we do not install, copy or promote pirated Windows or any other software.

machine and their memories.

2. HOW TO USE THIS WORKBOOK

This workbook is your study companion for all twelve sessions. For

each session
you will find:

- **Plain-English objectives** — what you should be able to do afterwards.
- **Lesson summary** — the actual notes, written so you can revise from them.
- **Key terms** — new words with simple definitions.
- **Guided practical steps** — the hands-on task done in class.
- **Independent exercise** — a short task you do on your own.
- **Assignment brief** — what to produce, when it is due.
- **Submission checklist** — so you never forget a deliverable.
- **Self-reflection questions** — think about your own learning.

At the back you will find your

portfolio checklist
, the **capstone requirements**, a **revision question bank** (questions only — write your answers in the spaces) and a **glossary**.

How to get the most from it:

1. **Bring it to every session.** Write in it.
2. **Take photos** inside every computer you open. Photos are your memory.
3. **Follow the workflow** every single time, even when you think you already know the answer.
4. **Ask questions.** There are no silly questions in a repair shop — only expensive mistakes.
5. **Do the assignments.** They build, step by step, into the capstone.

Three golden rules for every practical:

- **Ground yourself first** (anti-static wrist strap clipped to bare metal).
- **Power off, unplug, then press the power button** to drain leftover electricity.
- **Never force a part.** If it will not move, something is still holding it.

3. THE LEARNING OUTCOMES (WHAT YOU WILL BE ABLE TO DO)

By the end of the course you will be able to:

#	Outcome	Where it is practised
LO1	Identify and name the major computer components	Session 1
LO2	Explain the function of each component	Sessions 1–2
LO3	Follow safe ESD and electrical safety procedures	Session 2 (and every hands-on session)
LO4	Diagnose common hardware faults using a systematic workflow	Sessions 4–5
LO5	Replace common components (RAM, hard drive, PSU, GPU)	Session 7
LO6	Diagnose and resolve common software faults	Session 6

LO7	Use diagnostic tools and software	Sessions 5–6
LO8	Clean and maintain computer hardware	Session 8
LO9	Install and configure an operating system	Session 9
LO10	Produce a professional service report	Session 11
LO11	Complete a full repair project (capstone)	Session 12

WEEK 1 — COMPONENTS AND SAFETY

Session 1: Computer Components

Session goal:
Know the name and job of every major part inside a computer, and handle parts safely.

Plain-English objectives — by the end I can:

- Name the six major components and say what each one does.
- Tell the difference between RAM and storage.
- Explain the difference between an HDD, an SSD and an NVMe drive.
- Handle a part without touching its gold contacts.

Lesson summary

A desktop computer is like a small town. Each part has one job and they all must talk to each other.

Component	Plain meaning	What it does
Motherboard	The main circuit board	Connects every other component and lets them communicate
CPU	The "brain"	Performs calculations and runs instructions
RAM	Short-term memory	Holds data the CPU is using right now; empties when power goes off
Storage (HDD/SSD)	The filing cabinet	Keeps files permanently, even with no power
PSU	The power supply	Converts wall electricity into the voltages the computer needs
GPU	The graphics card	Draws images, video and everything you see on the screen

The motherboard
has special places on it: the **CPU socket** (where the processor sits), **RAM slots**, **expansion slots** (for the GPU and add-on cards), **SATA ports** (for drives) and the **I/O panel** (USB, audio, network and display at the back).

Storage comes in three types:

- **HDD (Hard Disk Drive):** older, has spinning magnetic disks and a moving arm. Slower, cheaper per gigabyte, and can be damaged by a drop.
- **SSD (Solid State Drive):** newer, no moving parts. Much faster and more durable, but more expensive per gigabyte.
- **NVMe (Non-Volatile Memory Express):** the fastest SSD. It plugs straight into a small slot on the motherboard (an M.2 slot).

Many machines use a

small SSD for Windows and programs
, plus a **large HDD for videos and photos**.

Handling rule:
hold every part **by the edges**. Never touch the **gold contacts** (the golden edge at the bottom of a RAM stick or card) — the oil on your fingers causes poor connections.

Key terms

- **Component** — one physical part of a computer.
- **Motherboard** — the main circuit board that connects all parts.
- **CPU (Central Processing Unit)** — the chip that runs instructions.
- **RAM (Random Access Memory)** — fast, temporary working memory.
- **Storage** — permanent memory (HDD, SSD or NVMe).
- **PSU (Power Supply Unit)** — turns wall power into usable voltages.
- **GPU (Graphics Processing Unit)** — draws images for the screen.
- **Gold contacts** — the metal edge you must never touch with bare fingers.
- **DDR** — a RAM generation (DDR3, DDR4, DDR5); they are not interchangeable.

Guided practical: "Component Identification"

1. Put on your anti-static wrist strap and clip it to the bare metal of the case.
2. Pick up each sample part (motherboard, CPU, RAM, HDD, SSD, PSU, GPU, SATA cable).
3. Write the **name** and its **function** on your Component Identification Chart.
4. Write one key fact about each part (for example, "SSD has no moving parts").
5. Raise your hand when all parts are identified; your instructor will check.

Independent exercise: "Component Matching"

Match each component to its function (answers written in your book, checked in class):

1. CPU 2. RAM 3. HDD 4. PSU 5. GPU 6. Motherboard A. Converts electricity from the wall socket · B. Main circuit board · C. Draws images on screen · D. Short-term memory · E. Performs calculations · F. Permanent storage

Assignment 1 — "My Component Reference"

- **Task:** Make a reference guide for all major components.
- **Must include:** each component's name, a short description of how it looks, its function, one key fact, and one common problem.
- **Deliverable:** a document saved on the Desktop named **"My Component Reference"**.
- **Deadline:** before Session 2.

Submission checklist

- ☐ I named all six major components correctly.
- ☐ I wrote the function of each component.
- ☐ I explained the difference between RAM and storage.
- ☐ My "My Component Reference" file is saved and shown to the instructor.

Self-reflection questions

1. Which component was hardest for me to remember, and why?
2. Can I explain RAM versus storage to a friend without using technical words?

Session 2: Safety and Tools

Plain-English objectives — by the end I can:

- Explain what ESD is and how it silently damages parts.
- Use an anti-static wrist strap and mat correctly.
- Follow electrical safety rules (unplug, discharge, never open a PSU).
- Name the common tools and say what each is for.
- Set up a safe, tidy workspace.
- Explain when a job should go to a qualified technician.

Lesson summary

ESD (electrostatic discharge)
is a sudden spark of static electricity. You may feel a shock of about **3,000 volts**, but a shock you **cannot even feel** can still be about **1,000 volts**. Some computer parts are damaged by as little as **10 volts**. This means **you can destroy a component without feeling anything.** That is why ESD is called the invisible enemy.

Prevent ESD:

- Wear an **anti-static wrist strap** clipped to the bare metal of the case.
- Work on an **anti-static mat**.
- Touch bare metal of the case before handling parts.
- Avoid carpet (work on a hard floor), and avoid wool or synthetic clothing.

Electrical safety rules:

1. **Always unplug** the power cable before opening a case.
2. **Press the power button** after unplugging to drain leftover electricity.
3. **Never work on a powered-on computer.**
4. **Never open the PSU case** — the capacitors inside can hold a lethal charge even when unplugged.
5. Use tools with **insulated handles**; never poke metal into a live circuit.
6. **Know your limits.** Anything involving PSU internals or mains wiring goes to a qualified technician.

The tools of the trade:

Tool	Used for
Phillips screwdriver (PH1, PH2)	Most computer screws
Flathead screwdriver	Prising clips, gentle adjustment
Tweezers	Retrieving dropped screws, small parts
Flashlight	Seeing inside a dark case
Pliers (needle-nose)	Gripping and bending
Anti-static wrist strap	Preventing ESD
Compressed air	Blowing out dust
Isopropyl alcohol (90%+)	Cleaning contacts, removing old thermal paste
Microfiber cloth	Cleaning without leaving lint

A safe workspace:
a clean, flat, well-lit table; an anti-static mat; a small tray for screws; and room to lay out parts.

Key terms

- **Anti-static mat** — a mat that protects parts from static.
- **Residual electricity** — small charge left in the machine after unplugging.
- **Insulated tools** — tools with handles that do not conduct electricity.
- **Capacitor** — a component that stores electricity (dangerous inside a PSU).

Guided practical: "Safety Set-Up and Tool Identification"

1. Put on your wrist strap and clip it to the case.
2. Place the anti-static mat on your workspace.
3. Unplug the practice computer, then press the power button to discharge it.
4. Lay out all tools; write the name and purpose of each on your Tool Identification Chart.
5. Tick off each item on your Safety Procedures Checklist.

Independent exercise: "Safe or Unsafe?"

Read each, decide, then discuss in class:

1. Chinedu unplugs the computer but does not press the power button.
2. Adaeze works on a carpet with no wrist strap.
3. Tunde smells burning and opens the case while it is still plugged in.
4. Funmi uses tweezers to retrieve a dropped screw.
5. Kunle is unsure about a PSU replacement and asks a qualified technician.

Assignment 2 — "My Safety and Tools Reference"

- **Task:** Make a personal safety-and-tools guide.
- **Must include:** ESD prevention, electrical safety rules, a tool chart (name + purpose), a workspace checklist, and "when to call a professional" guidelines.
- **Deliverable:** saved on the Desktop as **"My Safety and Tools Reference"**.
- **Deadline:** before Session 3.

Submission checklist

- ☐ I can explain ESD and how to prevent it.
- ☐ I know the rule: unplug, then press the power button.
- ☐ I know never to open a PSU.
- ☐ My safety reference file is saved and shown to the instructor.

Self-reflection questions

1. Why is ESD more dangerous than it looks?
2. In my own workspace at home, what would I need to change to work safely?

Session 3: Disassembly and Reassembly

Session goal:
Take a desktop apart and put it back together so that it works.

Plain-English objectives — by the end I can:

- Disconnect all external cables safely.
- Remove the side panel.
- Disconnect internal cables and take a photo first.

- Power on and verify.

Lesson summary

Disassembly order (write it down and follow it):

1. Unplug external cables (power, monitor, keyboard, mouse, USB, network).
2. Press the power button to discharge.
3. Remove the side panel (usually two screws at the back).
4. **Take a photo** of the inside before you touch anything.
5. Disconnect internal cables (SATA data and power, front-panel cables).
6. Remove the **GPU** (unscrew, release PCIe clip, pull by the edges).
7. Remove the **RAM** (push both clips, it pops up, pull by the edges).
8. Remove the **storage drives** (screws, cables, slide out).
9. Remove the **CPU cooler** (unplug fan cable, unscrew/unclip, lift off).

Reassembly = the reverse
, with care:

1. CPU cooler (clean off old paste, apply new, attach, reconnect fan).
2. Storage drives (slide in, screw, reconnect cables).
3. RAM (open clips, align the **notch**, press until clips snap).
4. GPU (remove slot cover, press into PCIe slot until it clicks, screw, reconnect power).
5. Internal cables (use your photo).
6. Side panel.
7. External cables.
8. Power on and **verify**.

Common beginner mistakes:
forcing a part that is still screwed in; not taking photos; mixing up screws; touching gold contacts; overtightening screws (this cracks the board); and forgetting the I/O shield.

RAM only fits one way.
The slot has a **notch**; the stick has a matching gap. If it will not sit smoothly, rotate it 180° — never force it. You know it is seated when **both clips snap** into place.

Key terms

- **Side panel** — the removable cover of the case.
- **SATA cable** — the flat data cable for drives.
- **Front-panel cables** — thin wires for the power button, reset, USB and audio.
- **PCIe slot** — the long slot that holds the graphics card.
- **Notch (RAM)** — the gap that makes RAM fit only one way.
- **I/O shield** — the metal plate at the back of the case.
- **Teardown** — taking a computer fully apart.

Guided practical: "My First Teardown"

Follow the Disassembly Checklist, then the Reassembly Checklist, then press the power button. The computer must start for the task to be complete. Show the instructor the working machine.

Independent exercise: "Put the Steps in Order"

Arrange these 1–8: A. Remove the GPU · B. Unplug external cables · C. Disconnect internal cables · D. Remove the side panel · E. Remove RAM · F. Press the power button to discharge · G. Remove storage drives · H. Take a photo of connections

- **Must include:** numbered disassembly steps, numbered reassembly steps, a safety reminder at each critical step, common mistakes, and troubleshooting tips.
- **Deliverable:** saved on the Desktop as **"My Disassembly Guide"**.
- **Deadline:** before Session 4.

Submission checklist

- [ ] I followed the correct disassembly order.
- [ ] I took a photo before disconnecting cables.
- [ ] I reinstalled everything and the computer powered on.
- [ ] My disassembly guide is saved and shown to the instructor.

Self-reflection questions

1. What was the trickiest step, and how did I overcome it?
2. Why does the order of disassembly and reassembly matter?

WEEK 2 — DIAGNOSTICS

Session 4: Diagnostic Workflow

Session goal:
Diagnose any fault using the same careful, five-step method every time.

Plain-English objectives — by the end I can:

- Follow the five-step diagnostic workflow.
- Match common symptoms to their likely causes.
- Ask good questions before guessing.
- Write down my findings clearly.

Lesson summary

A technician is a

detective
. The symptoms are your clues; your job is to find the **root cause** — not just make the problem disappear for a moment.

The five-step diagnostic workflow:

1. **Observe** — look (visible damage), listen (beeping, clicking, grinding), smell (burning), feel (unusual heat).
2. **Ask questions** — when did it start? what were you doing? has anything changed? all the time or sometimes? have you tried anything?
3. **Form a hypothesis** — list the **most likely causes in order**.
4. **Test** — start with the **simplest and most likely** cause, then work up.
5. **Resolve and verify** — fix it, then **test to prove** the fix worked.

Symptom → likely cause table:

Symptom	Likely causes
Does not power on at all	Failed PSU, faulty power cable, dead motherboard
Powers on but no display	Loose RAM, loose GPU, loose display cable, dead monitor

Blue screen (BSOD)	Faulty RAM, driver conflict, corrupted OS, overheating
Runs slowly	Malware, almost-full drive, too little RAM, overheating
Clicking noise	Failing HDD (back up data now!)
Random shutdowns	Overheating, failing PSU
Cannot connect to internet	Faulty network cable, driver issue, router problem

The seven diagnostic questions:

1. When did the problem start?
2. What were you doing when it happened?
3. Has anything changed recently (new hardware, software or update)?
4. Does it happen all the time, or only sometimes (**intermittent**)?
5. Have you tried anything to fix it?
6. Is there any unusual sound, smell or visible damage?
7. Does it happen with specific programs, or always?

Intermittent faults
(problems that come and go) are the hardest to diagnose because they may not appear while you are testing. Keep asking questions until you find the trigger.

Key terms

- **Diagnosis** — working out why a fault is happening.
- **Root cause** — the real reason, not just the symptom.
- **Symptom** — what the customer notices.
- **Hypothesis** — your best guess of the cause, to be tested.
- **Intermittent** — a fault that comes and goes.
- **Documentation** — writing down what you observed and tested.

Guided practical: "Diagnostic Scenarios"

For each of the five scenario cards, write down:

observation, questions, hypothesis, test plan, expected resolution
. Follow the five steps for each.

Independent exercise: "Symptom Matching"

Match: 1. Clicking noise · 2. No display, fans spinning · 3. Random shutdowns after 30 minutes · 4. Blue screen every few hours · 5. Very slow performance A. Failing hard drive · B. Loose RAM · C. Overheating · D. Faulty RAM or driver conflict · E. Full drive or malware

Assignment 4 — "My Diagnostic Reference"

- **Task:** A personal guide to diagnosing common problems.
- **Must include:** a symptom-to-cause table with **at least 8 symptoms**, the five-step workflow, the seven diagnostic questions, and tips for intermittent faults.
- **Deliverable:** saved on the Desktop as **"My Diagnostic Reference"**.
- **Deadline:** before Session 5.

Submission checklist

- [ ] I used all five steps for every scenario

Self-reflection questions

1. Why is it a mistake to jump straight to the most complicated cause?
2. How would I explain "intermittent" to a customer in plain words?

Session 5: Hardware Diagnostics

Session goal:
Use real tools to test RAM, drives and temperatures, and explain the results.

Plain-English objectives — by the end I can:

- Test RAM with Windows Memory Diagnostic or MemTest86.
- Check drive health with CrystalDiskInfo.
- Read temperatures with HWMonitor.
- Do safe basic PSU checks.
- Explain a diagnostic result in plain language.

Lesson summary

Diagnostic tools read information from components, just as a mechanic's machine reads a car's error codes.

RAM testing.
Faulty RAM causes blue screens, crashes and failure to start.

- **Windows Memory Diagnostic:** type its name in the Start menu → "Restart now and check for problems." The result appears after restart.
- **MemTest86:** boot from a USB stick, let it run at least **2 passes (about 30 minutes)**. **Any red line = a RAM error.** If you have two sticks, test them **one at a time** to find the bad one.

Drive testing (CrystalDiskInfo).
It reads the drive's **S.M.A.R.T.** data and shows a health status:

- **Good (blue):** healthy.
- **Caution (yellow):** problems — **back up immediately** and prepare to replace.
- **Bad (red):** failing — replace as soon as possible.
- Check the **temperature** (keep it under 50°C) and the **reallocated sector count** (anything above 0 is a warning). A **clicking** drive is failing — back up and replace; do not try to repair it.

Temperature monitoring (HWMonitor):

Component	Idle	Under load
CPU	30–50°C	up to ~80°C
GPU	30–60°C	up to ~85°C
Hard drive	25–45°C	—
Motherboard	30–50°C	—

If temperatures are high: clean the dust, check all fans spin, check the thermal paste, improve airflow.

PSU testing — basic only, and NEVER open the PSU case:

- **Visual check** for bulging capacitors or burn marks.
- **Smell test** — burning smell means replace it.

Key terms

- **MemTest86** — a bootable tool that stress-tests RAM.
- **CrystalDiskInfo** — reads a drive's S.M.A.R.T. health.
- **HWMonitor** — shows temperatures, voltages and fan speeds.
- **S.M.A.R.T.** — a drive's built-in self-monitoring data.
- **Reallocated sector** — a bad spot the drive has swapped out; a rising count means a failing drive.
- **Swap test** — fitting a known-good part to see if the fault follows the part.
- **Load** — when the computer is working hard.

Guided practical: "Run Diagnostics"

1. Open CrystalDiskInfo: what is the drive health and temperature?
2. Open HWMonitor: what are the CPU and GPU temperatures?
3. Start Windows Memory Diagnostic and let it run.
4. Write all findings on your Diagnostic Report form.

Independent exercise: "Healthy or Failing?"

1. CrystalDiskInfo "Good", 35°C · 2. "Caution", 200 reallocated sectors · 3. CPU at 95°C under load · 4. GPU 40°C idle · 5. Memory test: 0 errors. Decide for each, then discuss.

Assignment 5 — "My Diagnostic Tool Reference"

- **Task:** A guide to hardware diagnostic tools.
- **Must include:** step-by-step use of each tool, how to interpret results, the normal temperature ranges, warning signs and safety precautions.
- **Deliverable:** saved on the Desktop as **"My Diagnostic Tool Reference"**.
- **Deadline:** before Session 6.

Submission checklist

- ☐ I recorded a real drive health status and temperature.
- ☐ I recorded real CPU and GPU temperatures.
- ☐ I know the warning signs for a failing drive.

Self-reflection questions

1. Why should you back up data the moment a drive shows "Caution"?
2. Why is the PSU the most dangerous part to test?

Session 6: Software Diagnostics

Session goal:
Find and fix common software problems using Windows tools.

Plain-English objectives — by the end I can:

- Name the common software faults and their symptoms.
- Use Task Manager, Event Viewer, System File Checker, Disk Cleanup and System Restore.
- Run an antivirus scan and act on the results.

Lesson summary

Fault	Symptoms	Fix
Virus / malware	Pop-ups, slowness, strange programs, browser redirects	Full antivirus scan, remove threats, change passwords
Corrupted OS	Blue screens, will not start, missing files	`sfc /scannow`, then System Restore or reinstall
Driver issue	A device stops working, error messages	Update the driver in Device Manager
Software conflict	Crashes after installing new software	Uninstall the recent software, update it
Full hard drive	Slow, cannot save, low-space warnings	Disk Cleanup, delete unused files and programs

Windows built-in tools:

- **Task Manager** (Ctrl+Shift+Esc): shows running programs and CPU, memory and disk usage. Use "End Task" to close a stuck program.
- **Event Viewer**: a log of system events and errors. Red error icons matter — note the code and look it up.
- **System File Checker (SFC):** open Command Prompt **as Administrator**, type `sfc /scannow`. It repairs corrupted system files (takes about 15 minutes).
- **Disk Cleanup**: frees up space by removing temporary files.
- **System Restore**: rolls Windows back to a restore point from before the problem started.

Antivirus scanning:
run a **Full Scan** (not Quick Scan), review the threats found, **quarantine or delete** them, then restart. Keep the antivirus **updated** and run a full scan at least **weekly**. If you still suspect an infection, try a second-opinion scanner such as Malwarebytes. **Only download software from official or trusted sources.**

Key terms

- **Malware** — harmful software (viruses, spyware, ransomware).
- **Driver** — software that lets Windows talk to a piece of hardware.
- **Task Manager** — the Windows tool that lists running programs.
- **Event Viewer** — the Windows tool that logs system errors.
- **SFC (`sfc /scannow`)** — a Windows command that repairs system files.
- **Disk Cleanup** — a Windows tool that frees disk space.
- **System Restore** — rolls Windows back to an earlier working point.
- **Quarantine** — moving a detected threat somewhere harmless.

Guided practical: "Software Diagnostic Tools"

1. Open Task Manager — what uses the most CPU and memory?
2. Open Event Viewer — find three error events and note their codes.
3. Run `sfc /scannow` in an Administrator Command Prompt — what did it report?
4. Run Disk Cleanup — how much space did you free?
5. Write down all findings.

Independent exercise: "Match the Fault"

1. Pop-ups and redirects · 2. Blue screen with a stop code · 3. A device not working · 4. Low disk space · 5. Programs crashing after a new install A. Virus/malware · B. Corrupted OS or faulty RAM · C. Driver issue · D. Full drive · E. Software conflict

Assignment 6 — "My Software Diagnostic Reference"

- **Task:** A guide to software diagnostics.

- **Deliverable:** saved on the Desktop as "**My Software Diagnostic Reference**".
- **Deadline:** before Session 7.

Submission checklist

- [ ] I ran Task Manager, Event Viewer, SFC and Disk Cleanup.
- [ ] I can name five common software faults.
- [ ] I can describe the antivirus scan procedure.

Self-reflection questions

1. Why is "only download from trusted sources" a repair lesson, not just a safety one?
2. Which Windows tool was most useful to me, and why?

WEEK 3 — REPAIR AND MAINTENANCE

Session 7: Component Replacement

Session goal:
Replace RAM, a hard drive, a PSU and a GPU, then verify each one.

Plain-English objectives — by the end I can:

- Replace RAM and confirm the clips snap.
- Replace a hard drive and reconnect SATA data and power.
- Replace a PSU safely, reconnecting every cable.
- Replace a GPU and install its driver.
- Test after each replacement.

Lesson summary

Before any replacement:
power off, unplug, press the power button, and wear your wrist strap.

RAM:

1. Push the clips at both ends; the old stick pops up.
2. Pull it out by the edges.
3. Align the **notch** on the new stick with the slot.
4. Press down firmly until **both clips snap**.
5. Power on and check the new amount is recognised.

Hard drive:

1. Disconnect the SATA data cable and the SATA power cable.
2. Remove the screws and slide the old drive out.
3. Insert the new drive, replace the screws, reconnect both cables.
4. If the drive is blank, install an operating system (Session 9).

> ***Data-safety:***
> *before replacing a drive, **back up the customer's data**. Replacing a drive does not move the files for you.*

2. Remove the four screws at the back and slide the old PSU out.
3. Slide the new PSU in and replace the screws.
4. Reconnect **ALL** cables and double-check every one.
5. Power on and verify. **Never open the PSU case.**

GPU:

1. Remove the screw, release the PCIe clip, unplug any power cables.
2. Pull the old card out by the edges.
3. Fit the new card until it clicks, replace the screw, reconnect power.
4. **Install the latest driver from the maker's website.**

Key terms

- **Seated** — correctly and fully fitted in its slot.
- **Notch** — the alignment gap on RAM.
- **Known-good part** — a part you trust, used for testing.
- **Upgrade** — fitting a better/faster part than before.
- **Driver** — software that lets Windows use the new hardware.

Guided practical: "Replace the Components"

Replace the RAM and the hard drive in a practice computer, then power on and verify the machine recognises the new parts. Show the working computer to the instructor.

Independent exercise: "Order the PSU steps"

Arrange 1–7: A. Reconnect all cables · B. Remove the PSU screws · C. Power off and unplug · D. Slide the new PSU in · E. Disconnect all PSU cables · F. Open the case · G. Replace the screws

Assignment 7 — "My Replacement Guide"

- **Task:** Step-by-step replacement guides for RAM, hard drive, PSU and GPU.
- **Must include:** numbered steps for each, safety reminders, troubleshooting tips and the tools needed.
- **Deliverable:** saved on the Desktop as **"My Replacement Guide"**.
- **Deadline:** before Session 8.

Submission checklist

- ☐ I replaced RAM and the clips snapped.
- ☐ I replaced a hard drive and reconnected both cables.
- ☐ I can list the PSU replacement steps safely.

Self-reflection questions

1. Why must you disconnect **all** PSU cables before removing it?
2. Why do you install a driver after replacing a GPU?

Session 8: Cleaning and Maintenance

Session goal:
Clean a computer safely, replace thermal paste, and build a maintenance plan.

Plain-English objectives — by the end I can:

- Replace thermal paste correctly.
- Write a preventive maintenance schedule.

Lesson summary

Dust blocks airflow, causes overheating, shortens component life and brings slow performance and random crashes.

A clean computer runs cooler, faster and longer.

Cleaning procedure:

1. Power off, unplug, press the power button to discharge.
2. Open the case and place it on a hard, flat surface.
3. Blow dust out with **compressed air held upright** (tilting sprays liquid). Blow **away from** components, not deeper in. Focus on the CPU heatsink, GPU fan, PSU vents, case fans, RAM slots and expansion slots.
4. **Hold each fan blade steady** while blowing so it does not spin (spinning damages the bearing).
5. Wipe the inside with a microfiber cloth; clean any air filters.
6. Reassemble, power on and verify.

> ***Never use a vacuum cleaner inside a computer***
> *— it generates static electricity. Use compressed air only.*

Thermal paste replacement:

1. Remove the CPU cooler.
2. Clean the old paste off the CPU and heatsink with isopropyl alcohol and a cloth.
3. Apply a **pea-sized** amount of new paste to the centre of the CPU.
4. Reattach the cooler and reconnect the fan cable.
5. Power on and check the CPU temperature.

Preventive maintenance schedule:

When	Tasks
Monthly	Full antivirus scan; check Windows updates; Disk Cleanup
Every 3 months	Blow out dust; check all fans spin; check temperatures
Every 6 months	Replace thermal paste; check drive health; test RAM
Annually	Full diagnostic check; update all drivers; review files

Key terms

- **Thermal paste** — the compound that carries heat from the CPU to the heatsink.
- **Heatsink** — the metal block that absorbs and spreads CPU heat.
- **Compressed air** — canned air used to blow dust out.
- **Preventive maintenance** — regular care that stops faults before they start.
- **Airflow** — the movement of air through the case.

Guided practical: "Clean My Computer"

Power off and open a dusty practice computer. Blow out all dust, clean each fan (blade held steady), wipe the case, reassemble, power on and record the CPU temperature before and after cleaning.

10 computers
. Include monthly, quarterly and annual tasks, who is responsible, and how completed tasks are recorded.

Assignment 8 — "My Maintenance Schedule"

- **Task:** A maintenance schedule for a home or small office computer.
- **Must include:** monthly, quarterly and annual tasks, the tools needed, the time each task takes, and how you will document it.
- **Deliverable:** saved on the Desktop as **"My Maintenance Schedule"**.
- **Deadline:** before Session 9.

Submission checklist

- ☐ I cleaned a real computer safely.
- ☐ I know never to use a vacuum inside a computer.
- ☐ My schedule covers monthly, quarterly and annual tasks.

Self-reflection questions

1. Why does holding a fan blade steady matter?
2. How does cleaning actually save the customer money?

Session 9: Operating System Installation

Session goal:
Install Windows from scratch, install drivers, update and verify.

Plain-English objectives — by the end I can:

- Prepare properly (back up data, gather drivers, make a bootable USB).
- Install Windows from bootable media.
- Install the network driver first, then the others.
- Run Windows Update and activate Windows.
- Verify all devices work.

Lesson summary

> ***DATA-SAFETY FIRST:***
> *a clean install **erases everything** on the drive. Always **back up the customer's files** to an external drive or cloud **before** you begin. **Use licensed Windows media only** — never pirated copies.*

Pre-installation preparation:

1. **Back up** all important data.
2. **Check hardware compatibility.** As a guide, Windows 10 needs about a 1 GHz processor, 2 GB of RAM (64-bit) and 32 GB of storage. **Windows 11 needs more** — roughly 4 GB of RAM, 64 GB of storage and TPM 2.0 support. Check the maker's current requirements.
3. **Gather the drivers** (especially the **network** driver) and save them to a USB stick. Without the network driver you may not be able to get online to download the rest.
4. **Create the bootable USB** using Microsoft's official Media Creation Tool.

Installing Windows:

3. Choose language and keyboard → **Install Now** → enter the product key (or activate later) → accept the licence.
4. Choose **Custom: Install Windows only (advanced)**.
5. Select the drive, delete the old partitions (**this erases data**), select the **unallocated space** and click Next.
6. Windows copies files and restarts several times. **Do not remove the USB until it is finished.**
7. Complete the initial setup (region, keyboard, account, privacy).

Post-installation:

1. Install the **network driver first**, then chipset, graphics and audio.
2. Run **Windows Update** several times until there are no more updates.
3. **Activate** Windows with a genuine key or digital licence.
4. Verify — open **Device Manager**; there should be no yellow warning icons.

Key terms

- **ISO / installation media** — the Windows installer image.
- **Bootable USB** — a USB stick the computer can start Windows Setup from.
- **Boot menu** — the start-up menu where you choose what to boot from.
- **Partition** — a section of a drive.
- **Unallocated space** — free, unformatted space on a drive.
- **Clean install** — erasing the drive and installing Windows fresh.
- **Device Manager** — the Windows tool that lists hardware and drivers.
- **Activation** — telling Windows that your licence is genuine.

Guided practical: "Install Windows"

Boot a practice computer from a bootable USB and complete Windows Setup. Then install the network driver, run Windows Update, and verify every device works.

Independent exercise: "Post-Install Order"

Arrange 1–5: A. Run Windows Update · B. Install network driver · C. Install graphics driver · D. Activate Windows · E. Verify all devices

Assignment 9 — "My OS Installation Guide"

- **Task:** A step-by-step Windows installation guide.
- **Must include:** a pre-installation checklist, the installation steps, a post-installation checklist, common problems and solutions, and a driver-preparation guide.
- **Deliverable:** saved on the Desktop as **"My OS Installation Guide"**.
- **Deadline:** before Session 10.

Submission checklist

- ☐ I backed up data **before** any wipe.
- ☐ I installed Windows and the network driver.
- ☐ I ran Windows Update and checked Device Manager.

Self-reflection questions

1. Why is the network driver installed before the others?
2. What could go wrong if you forget to back up first?

Session 10: Customer Service and Communication

Session goal:
Talk to customers clearly, honestly and calmly — in plain English.

Plain-English objectives — by the end I can:

- Run a professional customer intake.
- Explain a fault in words a non-technician understands.
- Give a written estimate for cost and time.
- Handle an upset customer without losing my cool.

Lesson summary

The four-step customer intake:

1. **Greet** — smile, introduce yourself, ask how you can help.
2. **Gather information** — what happened? when? what were you doing? has anything changed? is there data to back up?
3. **Set expectations** — explain the diagnosis process, the timeline, and that you will **quote before any repair**.
4. **Document everything** — fill in the Customer Intake Form and get the customer's signature.

Explain in plain language.
Never hide behind jargon:

Technical term	Plain English
RAM	The computer's short-term memory
Hard drive	Where your files are stored
Motherboard	The main board that connects everything
CPU	The computer's brain
PSU	The power supply — gives the computer electricity
GPU	The graphics card — draws what you see
Driver	Software that helps a part work
Operating system	The main software (Windows)
Malware	Bad software that harms your computer
Reinstall	Wipe clean and install again

A written estimate must include:
diagnosis fee (if any), parts cost, labour cost, total, timeline and warranty. Give a range if unsure, explain what is included, and **get approval before starting**. If the cost rises above the estimate, **contact the customer first.**

Handling a difficult customer:
stay calm; listen without interrupting; empathise ("I understand this is frustrating"); apologise for the inconvenience; focus on solutions; set polite boundaries if they become abusive; and escalate to a manager if you cannot resolve it.

Key terms

- **Intake** — the first conversation that logs the customer's problem.
- **Estimate** — a written quote of cost and time.
- **Labour** — the charge for your time and skill.

Guided practical: "Customer Service Role-Play"

Work in pairs. One plays the customer (from a scenario card), the other the technician. Run the intake, then swap roles. Give each other feedback: what was done well, and what could improve.

Independent exercise: "Translate These"

Rewrite each in plain English:

1. The RAM is faulty. · 2. The hard drive is failing. · 3. The operating system is corrupted. · 4. The GPU is overheating. · 5. The PSU is not providing enough power.

Assignment 10 — "My Customer Service Guide"

- **Task:** A personal customer-service guide.
- **Must include:** the intake process, a plain-language table of **10+ terms**, an estimate template, tips for difficult customers, and communication do's and don'ts.
- **Deliverable:** saved on the Desktop as **"My Customer Service Guide"**.
- **Deadline:** before Session 11.

Submission checklist

- ☐ I completed a role-play as both technician and customer.
- ☐ I can translate at least ten technical terms.
- ☐ I know that I must quote before repairing.

Self-reflection questions

1. Why is honesty about what you *cannot* fix as important as what you can?
2. How will I calm myself before dealing with an angry customer?

Session 11: Service Reports and Documentation

Session goal:
Write a professional service report that protects you and informs the customer.

Plain-English objectives — by the end I can:

- Name and complete the nine sections of a service report.
- Record findings and repairs accurately.
- Write in plain language.
- Add recommendations and get the customer's signature.

Lesson summary

Why service reports matter:
they protect you if a repair is disputed, help the customer understand the work, satisfy auditors and warranty claims, build trust, and create a record for the next technician.

The nine sections:

1. **Header** — business name, report number, date, technician.
2. **Customer information** — name, contact, device type and serial number.
3. **Problem description** — what the customer reported, in their words, plus what you observed.
4. **Diagnostic findings** — tests performed, results, and the root cause.
5. **Repairs performed** — parts replaced (with part numbers), software installed, time spent.

8. **Warranty information** — what is covered, for how long, and the conditions.
9. **Sign-off** — both the technician's and the customer's signature and date.

Writing principles:
use plain language; be **specific** ("Replaced faulty 8GB DDR4 RAM in slot 2", not "fixed memory"); be **complete**; be **accurate** (double-check every number); and be **professional** (proper grammar, no personal comments).

> ***Tip:***
> *write the report while the details are fresh — not from memory the next day.*

Key terms

- **Service report** — the professional record of a repair.
- **Serial number** — the unique number that identifies a device.
- **Part number** — the manufacturer's code for a part.
- **Recommendation** — advice to prevent future problems.
- **Sign-off** — the signatures that confirm the work.

Guided practical: "Write a Service Report"

Using the template, write a complete report for this scenario:

Customer:
Adaeze Okafor · **Device:** Desktop PC, serial PC456 · **Problem:** would not start · **Diagnosis:** failed power supply (confirmed with a known-good PSU) · **Repair:** replaced PSU (500W), reconnected all cables · **Parts:** PSU ₦18,000 · **Labour:** ₦5,000 · **Total:** ₦23,000 · **Recommendations:** use a surge protector, clean every 3 months.

Independent exercise: "Report Review"

Review the sample report your instructor provides. Find three things done well, three things to improve, and any missing section.

Assignment 11 — "My Service Report Template"

- **Task:** Design a professional service report template for your own repair business.
- **Must include:** all nine sections, a clean layout, space for a business logo, a customer signature line and a warranty section.
- **Deliverable:** saved on the Desktop as **"My Service Report Template"**.
- **Deadline:** before Session 12.

Submission checklist

- [ ] My report has all nine sections.
- [ ] I used plain language and specific details.
- [ ] I included both signature lines.

Self-reflection questions

1. How could a good service report save a technician from a false accusation?
2. Which of the nine sections did I find easiest, and which hardest?

Session 12: Final Project (Capstone) — Diagnose and Repair

Session goal:

- Diagnose a simulated fault using the workflow.
- Repair it safely and correctly.
- Verify the computer is fully working.
- Produce a professional service report.
- Present my work and explain my decisions.

Lesson summary — what happens on the day

You will be given a

faulty practice computer
and a **customer scenario**. Working alone, you will:

1. **Intake (≈5 min):** greet the "customer" (your instructor), ask diagnostic questions, document the problem, set expectations.
2. **Diagnose (≈20 min):** observe, ask, hypothesise, test with tools, find the **root cause**. Write down every test and result.
3. **Repair (≈25 min):** fix it safely — wrist strap on, unplugged, correct tools, no PSU opening.
4. **Verify (≈10 min):** power on, confirm the fault is gone, run checks.
5. **Service report (≈10 min):** write all nine sections and have the instructor sign.
6. **Present (≈5 min):** explain what you found and did.

Possible faults include: loose RAM, a failed hard drive, overheating from dust, a failed PSU, a corrupted operating system, or a faulty GPU.

You will be assessed on: how well you diagnose, how safely and correctly you repair, whether the fix actually works, the quality of your report, and how well you explain your work.
A repair you cannot explain is not a pass.

Guided practical

There is no guided practical today — this session

is
your independent project. Your instructor observes and assesses.

Independent exercise (the capstone itself)

Complete the full repair project described above without assistance. Ask the instructor only if you are genuinely stuck.

Assignment 12 — "My Learning Reflection"

- **Task:** Write a short reflection on the whole course.
- **Must include:** three things you learned, two things you found challenging, one thing you want to learn next, and how you feel now compared with Session 1.
- **Deliverable:** saved on the Desktop as **"My Learning Reflection"**.
- **Deadline:** end of Session 12.

Submission checklist

- ☐ I completed the capstone repair independently.
- ☐ I verified the fix and the computer works.
- ☐ I wrote and signed off a complete service report.
- ☐ I presented my work to the instructor.
- ☐ My reflection is saved.

Self-reflection questions

PORTFOLIO CHECKLIST

Keep one folder on the Desktop (and a copy on a USB stick) called

`CEA-Computer-Repairs-<YourName>`

containing:

- [ ] Assignment 1 — My Component Reference
- [ ] Assignment 2 — My Safety and Tools Reference
- [ ] Assignment 3 — My Disassembly Guide
- [ ] Assignment 4 — My Diagnostic Reference
- [ ] Assignment 5 — My Diagnostic Tool Reference
- [ ] Assignment 6 — My Software Diagnostic Reference
- [ ] Assignment 7 — My Replacement Guide
- [ ] Assignment 8 — My Maintenance Schedule
- [ ] Assignment 9 — My OS Installation Guide
- [ ] Assignment 10 — My Customer Service Guide
- [ ] Assignment 11 — My Service Report Template
- [ ] Assignment 12 — My Learning Reflection
- [ ] Your **capstone service report** (the signed one)
- [ ] Your completed checklists and worksheets from the Lab Pack
- [ ] A one-paragraph description of a repair you performed

This folder is your proof of skill. Show it to employers and customers.

CAPSTONE REQUIREMENTS — QUICK CHECK

Your capstone is complete only when you can tick every box:

- [ ] Conducted a customer intake and documented the complaint
- [ ] Followed the five-step diagnostic workflow
- [ ] Used at least one diagnostic tool and recorded the result
- [ ] Correctly identified the **root cause**
- [ ] Performed the repair **safely** (unplugged, wrist strap on, correct tools, no PSU opening)
- [ ] **Verified** the fix — the computer works and the fault is gone
- [ ] Wrote a **service report with all nine sections**
- [ ] Included parts, labour and total cost
- [ ] Included recommendations and a warranty note
- [ ] Got the technician's and customer's signatures
- [ ] Presented and **explained** the work (a 5-mark viva/demo)

Safety reminder that can fail you:
if you skip ESD protection, work on a live machine, open a PSU, or wipe a drive without backing up data first, you cannot pass — no matter how good the final result looks.

REVISION QUESTION BANK

Try these from memory.

Session 1

1. Name the six major components and one function of each.

2. What is the difference between RAM and a hard drive?

3. Put these in order from slowest to fastest: SSD, HDD, NVMe.

Session 2

4. What is ESD, and how can something you cannot feel damage a part?

5. List three ways to prevent ESD.

6. What should you do **before** opening a computer case, and why?

Session 3

7. What is the correct order for removing components?

8. Why should you take a photo before disconnecting cables?

9. How do you know RAM is properly seated?

Session 4

10. List the five steps of the diagnostic workflow in order.

11. Why should you test the simplest likely cause first?

12. Why are intermittent faults harder to diagnose?

Session 5

13. What does a "Caution" result in CrystalDiskInfo tell you to do?

14. What is the normal idle temperature range for a CPU?

15. Why must you never open a PSU case?

Session 6

17. Which Windows tool shows running programs and resource usage?

18. What does the command `sfc /scannow` do?

Session 7

19. What should you do before replacing a customer's hard drive, and why?

20. What should you do immediately after replacing a GPU?

21. Why is the PSU the most dangerous component to replace?

Session 8

22. Why should you never use a vacuum cleaner inside a computer?

23. How much thermal paste should you apply, and where?

24. Name one monthly and one six-monthly maintenance task.

Session 9

25. Why must you back up data before a clean install?

26. Why is the network driver installed first?

27. What does a yellow warning icon in Device Manager mean?

Session 10

28. List the four steps of the customer intake process.

29. Translate "the operating system is corrupted" into plain English.

30. What should you do if a repair will cost more than the estimate?

Session 11

31. List the nine sections of a service report.

32. Why is a signed service report valuable to the technician?

Session 12

34. What are the five stages of the capstone project?

35. Why can a successful repair that you cannot explain still fail the capstone?

36. Which safety and data-safety rules can fail your capstone outright?

GLOSSARY — HARDWARE AND REPAIR TERMS

Term	Meaning
Hardware	The physical parts of a computer
Software	The programs that run on a computer
Motherboard	The main circuit board that connects all parts
CPU	The processor — the "brain"
RAM	Fast, temporary working memory
Storage	Permanent memory (HDD, SSD or NVMe)
HDD	Hard Disk Drive — older, with spinning disks
SSD	Solid State Drive — fast, no moving parts
NVMe	The fastest SSD, fitted into an M.2 slot
DDR3/DDR4/DDR5	RAM generations; not interchangeable
PSU	Power Supply Unit — converts wall power
GPU	Graphics Processing Unit — draws images
VRAM	Memory on the graphics card
SATA	The cable/port type used by many drives
PCIe	The slot that holds the GPU and add-on cards
M.2	A small slot for NVMe SSDs
BIOS/UEFI	The start-up firmware that gets the PC going
Boot menu	The start-up key menu to choose a boot device
ESD	Electrostatic discharge — a damaging static spark
Anti-static wrist strap	A band that grounds you safely
Anti-static mat	A mat that protects parts from static
Thermal paste	Heat-transfer compound between CPU and heatsink
Heatsink	Metal block that absorbs and spreads heat

Driver	Software that lets Windows use hardware
Operating system	The main software, e.g. Windows
Malware	Harmful software
Virus	A type of malware that spreads
Task Manager	Windows tool showing running programs
Event Viewer	Windows tool logging system errors
SFC	System File Checker (`sfc /scannow`)
Disk Cleanup	Windows tool that frees disk space
System Restore	Rolls Windows back to an earlier point
MemTest86	Bootable tool that tests RAM
CrystalDiskInfo	Reads a drive's S.M.A.R.T. health
HWMonitor	Shows temperatures, voltages and fan speeds
S.M.A.R.T.	A drive's built-in self-monitoring data
Reallocated sector	A bad spot swapped out by the drive
Partition	A section of a drive
Unallocated space	Free space with no partition
Bootable USB	A USB stick the PC can start Setup from
ISO	A file that contains an installation image
Clean install	Wiping a drive and installing the OS fresh
Device Manager	Windows tool listing hardware and drivers
Activation	Confirming your Windows licence is genuine
POST	Power-On Self-Test at start-up
Beep code	Start-up beeps that hint at a fault
Root cause	The real reason behind a fault
Diagnostic workflow	The five-step method: observe, ask, hypothesise, test, resolve
Symptom	What the customer or you notice
Intermittent	A fault that comes and goes
Swap test	Fitting a known-good part to find a fault
Intake	The first customer conversation
Estimate	A written quote of cost and time
Labour	The charge for the technician's time
Warranty	What a repair covers, and for how long
Service report	The written record of a repair

Capstone	The final diagnose-repair-report-present project
Viva	An oral test where you explain your work

End of Learner Workbook

From Zero to Expert, Together. Cyber Elias Academy — 24/26 Ebony Road, Port Harcourt.

PART 4

Instructor Manual

COMPUTER REPAIRS — INSTRUCTOR MANUAL

COMPUTER REPAIRS — INSTRUCTOR MANUAL

Cyber Elias Academy (CEA)
24/26 Ebony Road, Off Rumuola Road, Port Harcourt, Rivers State *From Zero to Expert, Together*

Course:
Computer Repairs (Hardware & Software Diagnostics and Repair) **Duration:** 4 weeks | **Sessions:** 12 × 105 minutes | **Fee:** ₦50,000
Instructor Manual version: 1.0 (October 2026)

> *This manual is*
>
> ***instructor-only***
> *. It tells you how to teach, prepare, set up faults, mark and troubleshoot the Computer Repairs course. The full lesson scripts, exercises and answer keys live in the Part 2 session files; the ready-made fault scenarios and checklists live in Part 5; the capstone set-up and viva live in Part 6. This manual shows you how to use them all, and indexes where everything is.*

> ***SAFETY AND DATA-SAFETY ARE PARAMOUNT.***
> *No learner ever works on live mains, opens a PSU, or wipes a drive without a confirmed backup. These rules override marks: a safety breach can fail a learner's capstone no matter how well the machine ends up working. Read Section 3 before every hands-on session.*

1. HOW TO USE THIS MANUAL

1.1 What each Part is for

Part	Document	What it is	Who sees it
Part 1	Course Design Document	The blueprint: outcomes, 12-session schedule, fees, assessment weights	Instructor
Part 2	Session Materials S1–S12	**The full lesson content.** Every session has sections A–O: A Session ID, B Outcomes, C Materials, D Opening, E Prerequisite Review, F Lecture Notes, G Instructor Demonstration, H Guided Practical, I Independent Exercise, J Case Study, K Common Mistakes, L Assignment, **M Knowledge Check + instructor answer key**, N Reflection, O Instructor Evaluation Notes	Instructor (G, I, L-rubric, M-key are **not** for learners)
Part 3	Learner Workbook	Full learner summaries, key terms, guided steps, assignment briefs, portfolio checklist, revision bank and glossary. **No answers.**	Learner
Part 4	**This Instructor Manual**	How to run the course: safety briefing, timing, prep, fault set-up, analogies, marking index, differentiation, capstone, QA	Instructor
Part 5	Practical Lab Pack — Diagnostic Scenarios	Scenario Cards 1–12 with instructor answer notes, plus the Fault-Diagnosis Table, Diagnostic, ESD/Safety, Cleaning and OS-Install checklists	Instructor (issues learner sections; keeps answer notes)

Part 6	Capstone (rubric + instructor guide)	The 30+5 capstone marking instrument, viva bank, model report, and the detailed fault set-up guide	Instructor
Part 7	QA Audit Report	Confirms the course is complete and internally consistent	Instructor / management

1.2 The most important rule

You do

not
need to write lessons — they are written for you. Your job each session is to:

1. **Set up safely** (Section 3) and **prepare** the machines, tools, components and printouts (Section 5).
2. **Teach from the Part 2 script**, using the analogies in Section 6.
3. **Run** the demonstration, guided practical and independent exercise (Part 2 sections G, H, I).
4. **Mark** using the answer keys in Part 2 sections I/L/M and the Part 5 scenario answer notes (see the Index in Section 7).
5. **Set up any faults safely** using the Fault Set-Up Guide (Section 6 of this manual).

1.3 Relationship to the Learner Workbook

The workbook (Part 3) is written for a true beginner and contains full summaries, key terms, guided-practical steps, assignment briefs, a portfolio checklist, a revision bank and a glossary. It deliberately contains

no answers
(confirmed in the QA audit, Part 7 §2.8). Answers are discussed in class and stay with you in the Part 2 keys. The QA audit also notes that the v1.0 workbook was rewritten to this full standard because the summary-only version was insufficient for beginners.

2. COURSE-AT-A-GLANCE

Item	Detail
Course name	Computer Repairs (Hardware & Software Diagnostics and Repair)
Category	Hardware & Security — Technical Diagnostics
Duration	4 weeks
Sessions per week	3
Total sessions	12 (S1–S12)
Session length	105 minutes (1 hour 45 minutes)
Total contact hours	21 hours
Course fee	**₦50,000**
Delivery mode	In-person (primary) with an online/hybrid option; this course is **hands-on and must be delivered in person** for the practical sessions
Entry requirements	None formally; Typing & Computer Basics strongly recommended. The learner needs only basic computer skills (typing, file management, internet) and a willingness to use their hands

Target learner	Beginner with **no** prior hardware experience
Certificate	CEA Certificate of Computer Repair Competence (verification at cea.ng/certificates/verify)
Certificate requirement	A diagnosed and serviced machine with a full service report
Internet	Not required for the core course; needed to download drivers and Windows media during Session 9 preparation

Tools and resources (provided by CEA):

- Desktop computers for disassembly/reassembly — **at least two per class**, plus practice units for diagnostics.
- Assorted components: RAM sticks, hard drives (including a clearly labelled faulty one), power supplies (including a labelled known-bad one), SATA cables, a GPU, expansion cards.
- Tool kits: Phillips PH1 and PH2 screwdrivers, flathead, tweezers, needle-nose pliers, flashlight.
- Cleaning supplies: compressed air, isopropyl alcohol (90%+), microfiber cloths, thermal paste.
- Safety: **anti-static wrist straps (one per learner)**, anti-static mats, safety glasses, first aid kit.
- Software: **licensed** Windows 10/11 installation media + the official Media Creation Tool; diagnostic software — **CrystalDiskInfo, MemTest86, HWMonitor**; antivirus (built-in Windows Defender or a licensed product); drivers saved to a USB in advance.

Licensing and equipment honesty reminders:

- Use **licensed software only** — never install, copy or promote pirated Windows or any other software. This is stated in the course context, the Lab Pack, the workbook and the capstone rule.
- Verify that every practice machine can be opened and safely reassembled. Where the academy has only one machine, group learners 3–4 at a time and rotate.

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3. SAFETY AND DATA-SAFETY BRIEFING

This section is the backbone of the whole course.

Deliver a short spoken safety briefing at the start of every practical session (S1–S3, S5, S7–S9, S12) and observe compliance continuously.
Safety (LO3) is assessed in the capstone with an override rule.

3.1 ESD (electrostatic discharge) precautions

- **What it is:** a sudden static spark. You may feel a shock at ~3,000 V, but a shock you **cannot even feel** can be ~1,000 V — and some parts are damaged by as little as **10 V**. **You can destroy a component without feeling anything.** This is the "invisible danger" (Session 2 opener).
- **Wear an anti-static wrist strap** — one per learner — clipped to **bare metal** of the case (not painted metal, not a screw with paint).
- **Work on an anti-static mat**, on a **hard floor** (never carpet), and avoid wool/synthetic clothing.
- **Before handling any part**, touch the bare metal of the case to equalise charge.
- **Hold components by the edges only.** Never touch the **gold contacts** of RAM or expansion cards; never touch CPU pins.
- Lay parts on the anti-static mat, **not** on plastic bags or clothing.
- Use the Part 5 **ESD/Safety Checklist (C4)** at the start of every practical and again before power-on.

3.2 Electrical safety / unplugging mains

- **Power off, unplug from the wall, then press the power button** to discharge residual electricity. This order is non-negotiable.
- **Never work on a powered-on computer.**

- **Never touch mains wiring.** Anything involving mains wiring, wall sockets or unsafe electrics is referred to a **qualified electrician/technician**.
- **Know your limits.** If you are unsure, **stop and ask**. Referring a job up is professional, not a failure.

3.3 Never open the PSU (power supply unit)

- **Never open the PSU case.** The large capacitors inside hold a **lethal charge even when unplugged**, sometimes for a long time.
- PSU faults are handled by **swap-test only**: fit a known-good PSU and see if the fault disappears. Then replace the PSU — never repair its internals.
- The only PSU checks permitted to learners are: **visual** (bulging capacitors, burn marks), **smell** (burning), and — for advanced learners, **under direct instructor supervision only** — the **paperclip test** and swap test.
- Reinforce this in S2, S5, S7 and the capstone. A learner who opens a PSU **cannot pass**.

3.4 Data-safety: back up before wiping or replacing drives

- **Data-safety comes before hardware.** Lost data cannot be un-lost by an apology.
- **Before any wipe, reinstall or drive replacement:** identify the customer's important files, **back them up** to an external drive or cloud, and **verify the backup opens**.
- Get the customer's **written approval** before a clean install.
- A **clicking/suspect drive** means **back up immediately** — treat data recovery as urgent, and consider a data-recovery professional for critical, unreadable data.
- Use the Part 5 **OS-Installation Preparation Checklist (C6)** before any install. A learner who wipes without a confirmed backup **cannot pass**.

3.5 Licensed software only

- Use **official** Media Creation Tool and licensed Windows media. **No piracy, no cracks, no "cracked" antivirus.**
- Prefer **Windows Defender** or an academy-licensed antivirus product.
- Teach learners that recommending licensed software is part of honest professional practice (and protects the customer legally).

3.6 Safe handling of practice machines and labelled faults

- All fault components are **clearly labelled** ("FAULTY — LAB ONLY", "DO NOT FIT"). Keep faulty stock **away from good stock**.
- Ensure a learner **cannot reach a fault you did not intend** (if you unseat RAM, do not also leave a display cable loose).
- A machine being worked on must never be powered; the power cable stays **out** until verification time.
- **Safety glasses** whenever blowing dust; **hold fan blades steady** when blowing (spinning damages bearings); **never use a vacuum cleaner inside a case** (it generates static).
- Keep the **first aid kit** visible and know the location. Report any injury or burning smell immediately.

3.7 The quick bench chant (teach learners to repeat it)

> ***"Strap on. Unplug. Press the button. Never open the PSU. Back up before you wipe."***

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4. SESSION TIMING

4.1 The standard 105-minute breakdown

The Part 2 sessions follow a common A–O pattern. Use this delivery template to fit

Min	Section	Activity
0–6	**D — Opening Activity**	The hook (curiosity, a real fault, a promise)
6–13	**E — Prerequisite Review**	Fast oral round-robin of last session's key points
13–38	**F — Full Lecture Notes**	The teaching, from the Part 2 script
38–53	**G — Instructor Demonstration**	You demonstrate on the bench/projector
53–71	**H — Guided Practical**	Learners practise with you; safety observed
71–86	**I — Independent Exercise**	Learners work alone; hints available
86–90	**J — Real-World Case Study**	Nigerian/Port Harcourt story + transfer question
90–94	**K — Common Mistakes**	Troubleshooting round
94–97	**L — Assignment Brief**	Deliverable, submission format, deadline
97–102	**M — Knowledge Check**	Quiz, then mark/discuss
102–105	**N — Reflection & Recap**	Summary, five takeaways, next-session link
—	**O — Instructor Evaluation Notes**	Read **after class**; plan remediation

> ***Session 12 (capstone)***
> *uses a different inner split: briefing + safety (5), intake (5), diagnose (20), repair (25), verify (10), report (10), plus ~5 minutes viva per learner (run parallel stations for large classes).*

4.2 Per-session timing table with key activity

S	Title	Week	Key activity (the "must happen")	Assessment
1	Computer Components	1	Identify every major component on a real open machine; safe handling	Component-ID practical (5%)
2	Safety and Tools	1	Practise ESD/electrical safety and tool identification on a safe bench	Safety observation
3	Disassembly and Reassembly	1	Full teardown **and** successful rebuild (machine must boot)	Disassembly/Reassembly (10%)
4	Diagnostic Workflow	2	Work the five-step workflow on scenario cards	Diagnostic Scenarios (5%)
5	Hardware Diagnostics	2	Run MemTest86, CrystalDiskInfo and HWMonitor; read results	Hardware Diagnostics (10%)
6	Software Diagnostics	2	Use Task Manager, Event Viewer, `sfc /scannow`, Disk Cleanup, antivirus	Software Diagnostics (10%)
7	Component Replacement	3	Replace RAM and a drive safely; verify the new parts are recognised	Component Replacement (10%)
8	Cleaning and Maintenance	3	Clean a dusty unit; replace thermal paste; measure temp before/after	Maintenance practical
9	Operating System Installation	3	Install Windows from USB; install network driver first; verify	OS Installation (10%)

10	Customer Service	4	Role-play intake, plain-language explanation, written estimate	Role-play
11	Service Reports	4	Write a full nine-section service report from a scenario	Service Report (10%)
12	Final Project — Capstone	4	Independent full repair + report + viva	Capstone (25% course weight)

5. PREPARATION CHECKLIST PER SESSION

Every practical session, before learners arrive:

- ☐ Practice machines present, able to be opened, and able to be reassembled.
- ☐ **Anti-static wrist straps (one per learner), mats, safety glasses** laid out.
- ☐ Tool kits complete and correct (PH1/PH2, flathead, tweezers, pliers, torch).
- ☐ Any intentional fault set up safely and **labelled**; known-good spares ready.
- ☐ Learner Workbooks and this session's printed worksheet/checklist at each seat.
- ☐ Projector/diagrams ready; whiteboard markers.
- ☐ Read sections D–O of the Part 2 session file; know sections I/L/M.

Session-by-session extras:

S	Prepare before class
1	Physical component samples; printed Component Identification Chart (Worksheet C3) and Data Flow Diagram; component name cards for the labelling race; a dead RAM/HDD and a burnt PSU labelled "LOOK ONLY"; one unplugged desktop + a laptop for comparison
2	Tool kits; safety equipment; Sample ESD/Safety Checklist (C4); a practice bench set-up
3	Two practice machines cleared for full teardown; disassembly/reassembly checklists; camera/phone for photos
4	Printed Scenario Cards 1–5 (learner sections only) + Diagnostic Checklist (C2) + Fault-Diagnosis Table (C1); set up the "no display" demo fault safely
5	A machine with CrystalDiskInfo, HWMonitor installed; MemTest86 USB; a drive flagged "Caution" (labelled); a known-good PSU (for demonstration only); Scenario Cards 2, 3, 5, 6
6	A pre-prepared machine with harmless, **reversible** software faults (pop-ups/slowdown) for the opener; Scenario Cards 4, 7, 8
7	Replacement components of the **correct** type (DDR4 etc.); known-good PSU; replacement drive; thermal paste; Compatibility Check Sheet
8	A **genuinely dusty** unit; compressed air (upright use); isopropyl alcohol; microfiber cloths; thermal paste; Cleaning & Maintenance Checklist (C5); HWMonitor for before/after temps
9	**Licensed** Windows USB (8 GB+); drivers saved in advance (network driver first!); a dedicated reimageable machine; OS-Installation Checklist (C6); a backup drive for data-safety practice
10	Role-play scenario cards as customer complaints; Customer Intake Form; estimate template
11	Printed service report templates; a completed sample report; Fault-Diagnosis Tables (C1) from earlier sessions
12	See the **Fault Set-Up Guide (Section 6.3)** and Part 6 Capstone Day Checklist: faulty units, spares, cleaning/OS resources, backup drives, straps, printed rubrics, viva question set, milestone sheet, certificate sheets

6.1 Golden rules for setting up any fault

- Set every fault with the machine **powered off and unplugged**.
- Wear your own **anti-static wrist strap** when opening cases.
- **Never** modify mains wiring, PSU internals or anything dangerous.
- **Label** any deliberately bad part ("FAULTY — LAB ONLY") and keep it away from good stock.
- Keep a **known-good spare** of common parts (RAM, PSU, drive, cables) for swap tests.
- Keep a **clean reset plan** for every fault so the unit can be reused.
- **Do not plant real malware.** Software faults must be harmless and fully reversible.
- Ensure a learner cannot accidentally reach a **fault you did not intend**.
- Document what you changed so the reset is exact.

6.2 Faults for the diagnostic sessions (S4–S6)

Use the

Part 5 Scenario Cards

, whose instructor answer notes explain both the intended fault and the diagnostic path. Match each session to its cards:

Session	Cards to use (Part 5)	How to set up the fault safely
4 — Diagnostic Workflow	Cards 1, 2, 3, 4, 5 (as discussion/demo)	Set up the "no display" card by **unseating one RAM clip** (power off). You may demonstrate live; keep a known-good spare.
5 — Hardware Diagnostics	Cards 2, 3, 5, 6 (with C1/C2)	For **Card 3** (faulty RAM), fit one known-good and one known-bad (labelled) stick; for **Card 5** (failing HDD), fit a drive flagged "Caution" by CrystalDiskInfo and keep the data backed up; for **Card 6** (overheating), use a dusty unit and monitor with HWMonitor under load
6 — Software Diagnostics	Cards 4, 7, 8 (with C1/C2)	**Harmless and reversible only.** Card 7 (malware): simulate pop-ups with harmless image/text files and a slow startup item — **never real malware**. Card 8 (corrupted OS): use a **dedicated reimageable unit**; keep the image for re-flashing. Card 4 (slow): fill the drive to ~95% with dummy files, add a startup item

Reset after each practice fault:

- **Unseated RAM** → push both clips until they snap.
- **Loose SATA/display cable** → firmly reseat.
- **Dusty unit** → clean and (if applicable) remove the temporary dried-paste blob.
- **Simulated software slowdown** → remove the startup item / disabled driver; run Disk Cleanup.
- **Failing drive** → refit the good drive.

6.3 Capstone fault menu (choose **one per learner**)

From the Part 6 Capstone Instructor Guide. Keep a

fault set-up record

(learner × fault × initials × spare prepared × reset plan) before Session 12.

Fault A — Unseated RAM.

With power off, push one RAM stick until only **one clip** engages (no force). Symptoms: powers on, fans spin, **no display**. Reset: snap both clips. Difficulty: easy — good for a nervous learner.

Fault B — Loose cable (drive or display).

Gently unseat one end of the **SATA data cable** ("No bootable device") or the **display cable** (no display). Reset: reseat firmly. Difficulty: easy.

Safety: ensure the unit is **cool** before learners open it; do not disable thermal protection. Reset: clean and re-paste. Difficulty: easy–medium.

Fault D — Software slowdown.
Add a harmless **startup program** (or open a few large files), disable one non-essential driver, or fill the drive to ~95% with dummy files. Symptoms: slow start and slow programs. Safety: harmless, documented, reversible. Reset: remove the change; Disk Cleanup. Difficulty: medium.

Fault E — Failing drive.
Fit a drive flagged "Caution/Bad" by CrystalDiskInfo, or one with a corrupted boot sector, or disconnect a working drive's cable. **Keep the original data backed up (instructor).** Symptoms: "No bootable device," slow boot, file errors, sometimes clicking. Data-safety: the learner must confirm a backup before any wipe; provide a clean replacement drive. Reset: refit the good drive. Difficulty: medium.

Fault F — Corrupted operating system.
On a **dedicated reimageable unit**, cause a genuine but reversible boot failure (break the boot configuration, or remove a system file from a lab image). Symptoms: Windows will not start; missing-file errors. Safety: lab image only; never touch real data. Reset: repair or reimage. Difficulty: medium–hard.

Fault G — Failed PSU (advanced).
Fit a **labelled known-bad PSU**, or leave the PSU rear switch off / a cable deliberately loose. Symptoms: totally dead, or powers off under load. Safety: learners **never** open the PSU — this tests safe replacement only. Reset: fit a known-good PSU. Difficulty: medium.

Fault H — Mixed fault (challenge).
Combine two, e.g. **unseated RAM + software slowdown**, or **dusty unit + loose SATA**. Purpose: tests persistence — the learner must keep testing after the first fix. Use only for confident learners. Reset: correct both.

6.4 Fault rotation and fairness

- **Rotate faults** so no learner gets the same one twice and none becomes "the easy one."
- Treat every fault as **equal difficulty** — the rubric is the same for all.
- Do **not leak faults**: keep set-up notes, answer notes and the fault record away from learners.
- Complete the **Fault Set-Up Record** (Part 6) before Session 12 and the **reset plan** for every unit.

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7. TEACHING NOTES AND ANALOGIES (BANK)

Use these ready-made explainers consistently.

S1 — Components

- "A computer is a **restaurant kitchen**: the CPU is the head chef, RAM is the prep table, storage is the store room, the PSU is the gas cylinder, the motherboard is the manager."
- "A computer is a **Nigerian market stall**: if the **generator** (PSU) dies, the whole stall stops — even if the owner and goods are fine."
- "**HDD** is a warehouse with a worker on a bicycle; **SSD** is numbered shelves within arm's reach; **NVMe** is the shelf right beside your hand."
- "A faster CPU does not always help: a convoy moves at the speed of its slowest vehicle."

S2 — Safety and tools

- "Wearing the wrist strap is like **sharing a plate of food** — same table, nothing jumps."
- "Using the wrong screwdriver is like using a chisel as a hammer — it ruins the wood and the tool."
- "Setting up the bench is like a **surgeon preparing the operating theatre**."
- "The school's shared projector: you may use it, but you do not open it to pieces without permission."

S3 — Disassembly

- "Recording is like a **tailor taking measurements before cutting cloth**."

S4 — Diagnostic workflow

- "Diagnosis is like **tracing a fault on the PHED line** — start with the biggest, simplest possibility, then work down."
- "The workflow is a **doctor's consultation**: symptoms, basics, suspicion, test, confirm, treat, re-check."
- "Documentation is the **doctor's case notes**."

S5 — Hardware diagnostics

- "The workflow is the doctor's questions; today's tools are the **stethoscope and the blood test**."
- "POST is like the **fuel-pump priming sound** before a car starts."
- "RAM is the **tailor's work-table** — one warped corner spoils everything laid there."
- "S.M.A.R.T. is the **oil-warning light** — it warns early; you act, you do not ignore it."
- "The PSU is the **street transformer** — but unlike a transformer, you must **never** open it."
- "A DSTV 'no signal' — you check the cable and decoder before buying a new TV."

S6 — Software diagnostics

- "Hardware are the **players**; software is the **coach's instructions**."
- "A slow computer is a **bus in Port Harcourt traffic** — too many passengers, an old engine, or a blocked road."
- "Malware is **rats in the store-room** — remove the rats *and* close the door (the unsafe download habit)."
- "A boot problem is a car that **starts but will not engage gear**."
- "Safe Mode is the hospital's **emergency room** — everything non-essential switched off."

S7 — Replacement

- "Choosing a part is like **buying shoes** — the right brand is not enough; you need the right **size** (compatibility)."
- "Seating RAM is like **fitting a key in a lock** — one way, firm and straight."
- "Cloning is **moving your whole house** to a new building, everything already in place."
- "Replacing a PSU is like **replacing the main breaker** — shut off, label every wire, reconnect exactly."

S8 — Cleaning

- "Doing hard labour in Port Harcourt heat wrapped in a **thick blanket** — that is dust to a computer."
- "Cleaning is **clearing a blocked drainage channel before the rains**."
- "Thermal paste is a **thin layer of oil between pot and hot plate**."
- "A maintenance schedule is the **car service book**."

S9 — OS installation

- "Windows is the **hotel manager**; hardware is the rooms, beds and plumbing."
- "The backup is **moving the family photos out before demolishing the house**."
- "BIOS/UEFI is the **gatekeeper** deciding which vehicle enters first (internal drive vs USB)."
- "Drivers are the **keys and connections** for each appliance; Activation is the **legal title deed**."

S10 — Customer service

- "A good barber greets, remembers, checks, quotes and waits for your nod."
- "A good **tailor** measures and quotes before cutting cloth."
- "Plain language: a laptop battery is like a **phone battery that no longer holds charge**."

S11 — Service reports

- "A service report is the **patient's file** — symptoms, tests, diagnosis, treatment, result."
- "It is a **detective's case file**: witness statement (reported fault) + forensic evidence (observations) → conclusion."
- "It is **evidence in court**: state only what you can stand behind."

8. ANSWER-KEY AND MARKING INDEX

All answers and rubrics live in the Part 2 session files; the scenario answer notes live in Part 5; the capstone instrument lives in Part 6. Use this table to find them fast.

Session	Assignment (Part 2 §L)	Knowledge-check key (Part 2 §M)	Practical/scenario key
S1 Computer Components	§L — "My Component Reference"	§M — key	Part 5 **Component-ID Worksheet (C3)**; workbook independent exercise "Component Matching"
S2 Safety and Tools	§L — "My Safety and Tools Reference"	§M — key	Part 5 **ESD/Safety Checklist (C4)**; workbook "Safe or Unsafe?"
S3 Disassembly/Reassembly	§L — "My Disassembly Guide"	§M — key	Part 5 C4 before/after; workbook "Put the Steps in Order"
S4 Diagnostic Workflow	§L — "My Diagnostic Reference"	§M — key	Part 5 **Cards 1–5 answer notes**; C1 + C2
S5 Hardware Diagnostics	§L — "Hardware Diagnostic Report on a Real Machine" (20 marks)	§M — key	Part 5 **Cards 2, 3, 5, 6 answer notes**; workbook "Healthy or Failing?"
S6 Software Diagnostics	§L — "Software Diagnostic Report on a Real Machine" (20 marks)	§M — key	Part 5 **Cards 4, 7, 8 answer notes**; workbook "Match the Fault"
S7 Component Replacement	§L — "Compatibility and Replacement Plan for a Real Upgrade" (20 marks)	§M — key	Part 5 C4; workbook "Order the PSU steps"
S8 Cleaning and Maintenance	§L — "Cleaning and Maintenance Report with a Maintenance Plan"	§M — key	Part 5 **Cleaning & Maintenance Checklist (C5)**
S9 OS Installation	§L — "My Windows Clean-Install Runbook"	§M — key	Part 5 **OS-Installation Checklist (C6)**; workbook "Post-Install Order"
S10 Customer Service	§L — "My Shop's Customer Service Playbook"	§M — key	Role-play cards; workbook "Translate These"
S11 Service Reports	§L — "My Capstone-Ready Service Report"	§M — key	Part 6 model report; workbook "Report Review"
S12 Capstone	§L — "My Capstone Reflection and Next Steps"	§M — **full quiz key**	**Part 6 rubric (30+5) + viva bank + model/reference report**

8.1 Key answers for the most important checkpoints

- **Six major components:** motherboard, CPU, RAM, storage (HDD/SSD/NVMe), PSU, GPU.
- **RAM vs storage:** RAM is fast, temporary working memory (empties on power-off); storage is permanent (survives power-off).
- **Speed order (slowest→fastest):** HDD → SSD → NVMe.
- **ESD:** a static spark; harmless-feeling shocks undo parts. Prevent with wrist strap clipped to bare metal, mat, hard floor, touching bare metal, holding by edges.

- **CrystalDiskInfo "Caution":** back up **immediately** and prepare to replace the drive.
- **CPU idle range:** ~30–50°C (load up to ~80°C).
- **Never open the PSU:** capacitors hold a **lethal** charge even when unplugged.
- **Before replacing a drive: back up the customer's data first** (the drive does not move files for you).
- **After replacing a GPU: install the latest driver** from the maker's site.
- **After a clean install:** install the **network driver first**, then the rest; run Windows Update; activate; check Device Manager for yellow flags.
- **Five discrepancy/fault groupings in the viva bank** and the **nine service-report sections** are given verbatim in Part 6.

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9. ASSESSMENT OVERVIEW

9.1 The 100% breakdown (from the design doc)

Assessment	Session	Weight
Component Identification	1	5%
Disassembly/Reassembly	3	10%
Diagnostic Scenarios	4	5%
Hardware Diagnostics	5	10%
Software Diagnostics	6	10%
Component Replacement	7	10%
OS Installation	9	10%
Service Report	11	10%
Capstone: Repair Project	12	**25%**
Knowledge Checks	All	5%
Total		**100%**

> ***Reconciling the capstone marks.***
> *The Part 6 Capstone Assessment Rubric is a **35-mark instrument** (Repair + Report 30, Viva/Demo 5). Score the capstone out of 35, convert to a **percentage**, then apply its **25% course weight** to the certificate total. The 30+5 split is the internal distribution of the capstone instrument, not a second 100%. Use the rubric's pass criteria to decide the capstone result, then plug the percentage into the 25% weight.*

9.2 Grade bands

Grade	Percentage	Descriptor
A — Excellent	80–100%	Can diagnose and repair common faults independently
B — Good	65–79%	Can diagnose and repair with minor guidance
C — Competent	50–64%	Can diagnose and repair with some guidance
D — Developing	40–49%	Requires significant guidance

F — Not Yet Competent	0–39%	Does not meet minimum requirements

9.3 Minimum practical + capstone rule

A learner passes only when

both
are true:

1. **Overall score ≥ 50%**, **AND**
2. **Successful completion of the Session 12 capstone repair project** — correct diagnosis, safe and correct repair, a verified fix, a complete nine-section service report, and an explained presentation/viva.

A learner who scores 90% on knowledge checks but cannot diagnose and repair a fault

does not pass
. A repair the learner cannot explain **cannot pass**, even if the machine works.

Certificate awarded when:
attendance of at least 10 of 12 sessions; all assignments satisfactory; capstone passed; overall ≥ 50%; a professional service report produced.

10. COMMON LEARNER MISCONCEPTIONS AND REMEDIATION

Misconception	The truth	Quick remedial exercise
"A faster CPU always makes it faster."	The slowest part sets the pace; RAM/storage often matter more.	Identify the bottleneck in a slow practice machine using Task Manager.
"ESD shocks are harmless because I cannot feel them."	A shock below feeling threshold can still destroy parts.	Show the voltage facts; have the learner fit the strap and touch bare metal.
"I can just open it while it is on / plugged in."	Never work on a live machine.	Stop and re-run the unplug–discharge sequence with the ESD checklist.
"The PSU is just another part I can open."	Capacitors hold a lethal charge even unplugged.	Re-teach 3.3; show a labelled bad PSU handled by swap only.
"Just format it — that fixes everything."	Formatting destroys data and may not fix the fault.	Diagnose first (S6 opener); require a backup before any wipe.
"Replace the whole computer — it is easier."	A small, cheap fix is the correct, honest one.	Practise reseating/swap tests; compute the cost difference in naira.
"Test the hardest/most expensive cause first."	Start with the simplest, most likely cause.	Re-order the hypothesis list; re-run a scenario.
"A repair that works needs no explanation."	Understanding is the skill; unexplained luck does not pass.	Rehearse the viva questions aloud.
"One antivirus is enough forever."	Definitions must be kept updated; habits matter more.	Update definitions; discuss the "watchman with an old list" analogy.
"Vacuum cleaner is fine for dust."	Vacuums generate static; use compressed air.	Demonstrate compressed air with blade held steady.
"A clicking drive just needs a repair."	Clicking HDD = failing; back up and replace.	Back up first, check CrystalDiskInfo, plan replacement.

Universal remediation for a capstone fail:

11. DIFFERENTIATION GUIDANCE

Level 1 — Supported (learners who struggle)

- Pair with a stronger learner; give the printed checklists (C1–C6) and a partially completed worksheet.
- Demonstrate each step again at the bench before the learner attempts it.
- Allow extra time on practicals; focus on safety and the simplest fault types (Fault A/B).
- Provide one-to-one support; allow a capstone retake after extra practice.

Level 2 — Independent (standard)

- Complete all exercises as written, answer all knowledge-check questions, and meet the assignment rubrics.
- Run the workflow unprompted and record every test.

Level 3 — Challenge (fast finishers)

- Assign a **mixed fault (Fault H)** in capstone; add extra scenarios (Part 7 suggests two or three more per week).
- Ask the learner to write a short business-startup note (costing, customer intake, marketing).
- Have them teach the five-step workflow or the ESD checklist to a partner.

12. CAPSTONE EVALUATION INSTRUCTIONS

12.1 How to run it (Session 12, 105 minutes)

Follow the

Part 6 Capstone Instructor Guide
in full. Summary:

1. **Briefing + safety (5 min):** give the learner the fault card (learner section only) and the unit. **Safety reminder first.**
2. **Customer intake (5 min):** you play the customer; the learner logs the complaint and asks about data.
3. **Diagnose (20 min):** five-step workflow; record every test and result. **Milestone 1 — check the root cause is identified before repair (initial).**
4. **Repair (25 min):** safety observed continuously; correct tools; no PSU opening.
5. **Verify (10 min):** power on; confirm the fault is gone; **re-run the relevant diagnostic** (MemTest clean, drive "Good", temp normal) and check the wider system.
6. **Report (10 min):** all nine sections.
7. **Viva/demo (~5 min per learner):** run parallel stations if the class is large.

Use the

Milestone Checkpoint Sheet
and the **Fault Set-Up Record** from Part 6.

12.2 Rubric summary (Part 6 — 35 marks)

Part 1 — Repair + Report (30):
Diagnostic process (5), Root cause (3), Repair correctness (6), Verification (4), Safety & data-safety (4), Report structure/accuracy/clarity (5), Parts/labour/recommendations (3). **Part 2 — Viva/Demo (5):** Explanation/understanding (3), Live demonstration (2).

Safety override:
working live, opening a PSU, ignoring ESD, or wiping without a confirmed backup caps the safety row at *Beginning* and **blocks a pass**.

explained.

12.3 Viva guidance

- Read the learner's report first; ask **5–8** questions from the Part 6 viva bank, matched to the fault.
- Start easy ("Tell me what was wrong"), then walk through the steps, then one or two "why" questions.
- Ask the learner to explain it as they would to a **non-technical customer**, then **demonstrate** the fix.
- Ask open questions; give thinking time; do not punish nerves.
- **A repair the learner cannot explain cannot pass.**

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13. SAMPLE FEEDBACK COMMENTS

Make comments

specific
and name the exact action.

Entry-level ready (A):

- "Excellent, methodical diagnosis. You followed all five steps, recorded every test, found the loose RAM quickly and reseated it correctly. Your report was clear, specific and complete, and your viva showed real understanding. Safety was perfect throughout — you are ready for a junior technician role."
- "You handled a mixed fault calmly and fixed both issues without help. Your plain-language explanation to the 'customer' was exactly right."

Competent (B/C):

- "Good, solid diagnosis and a correct repair. Your report had all nine sections. To reach the top band, **re-run the diagnostic after the repair** to prove the fix, and clip your wrist strap *before* opening the case."
- "You identified the correct fault and repaired it safely. Watch your test order — you checked the hardest cause first. Start with the simplest, most likely cause next time."

Developing (D):

- "You found and fixed the fault, which is a real achievement, but you skipped several workflow steps and your report was missing the diagnostic findings and signature. Practise the workflow aloud and use the nine-section checklist every time."
- "Your repair worked, but you did not discharge the machine or check the data before working. Safety and data-safety are not optional. We will re-run this together, then you can retake with a new fault."

Not yet competent (F):

- "This attempt did not meet the standard. You did not identify the root cause before trying to repair, and the report was incomplete. That is okay — you are being taught, not punished. Come to practice hours, work through two more scenarios with me, and retake with a different fault."
- "The machine worked in the end, but you could not explain how you fixed it, so I cannot confirm you understand the repair. Let us build your understanding of the diagnostic workflow, then try a new fault."

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14. QUALITY-ASSURANCE CHECKLIST

14.1 Before every session

- ☐ Safety briefing delivered; straps, mats, glasses and first aid kit ready (Section 3).
- ☐ Machines off, unplugged and safe; tools correct and complete.
- ☐ Any scheduled fault set up **safely**, labelled, with a reset plan and known-good spares.
- ☐ Sample components/datasets/printouts ready at each seat.

14.2 Before grading

- ☐ I am marking against the **written rubric/key and Part 5 answer notes**, not general impression.
- ☐ **Same standard for every fault** (no fault treated as "easier").
- ☐ I score the **process** as well as the result (workflow, test order, documentation).
- ☐ I applied the **safety override** consistently.
- ☐ I distinguish **nerves from ignorance**.
- ☐ I have written one strength and one specific next step for every learner.
- ☐ Marks recorded; capstone scored out of 35, converted to a percentage, then weighted at 25%.
- ☐ Certificate eligibility checked (attendance ≥ 10/12, assignments, capstone, report, ≥ 50%).

14.3 Course-level QA (aligns with Part 7 audit)

- ☐ All 12 sessions delivered with a practical in each.
- ☐ All 11 learning outcomes taught and assessed.
- ☐ Every capstone stage maps to a taught session (Part 7 §4).
- ☐ Learner Workbook contains no answers (Part 7 §2.8).
- ☐ Safety and data-safety enforced every practical; licensed software only.
- ☐ Fault set-up record and reset plans confirmed before Session 12.

Cyber Elias Academy — Computer Repairs Instructor Manual v1.0, October 2026. From Zero to Expert, Together.

PART 5

Practical Lab Pack

COMPUTER REPAIRS — PRACTICAL LAB PACK

COMPUTER REPAIRS — PRACTICAL LAB PACK

01 — DIAGNOSTIC SCENARIOS, WORKSHEETS AND CHECKLISTS

Cyber Elias Academy | Computer Repairs | October 2026

This lab pack contains the real, usable practical materials for the Computer Repairs course: fault

scenario cards
for diagnostic practice, plus the **worksheets and checklists** you use in class and in the capstone. Everything here is designed to be printed or written on directly.

> ***SAFETY FIRST — read before every hands-on session***
>
> - *Put on an **anti-static wrist strap** and clip it to bare metal on the case.*
> - ***Power off, unplug, then press the power button** to discharge the machine.*
> - ***Never open the PSU case.** Capacitors hold a lethal charge even when unplugged.*
> - ***Never work on live mains wiring.** If unsure, stop and ask the instructor.*
> - ***DATA-SAFETY:** never wipe or reinstall a drive until a backup has been confirmed.*
> - ***Use licensed software only.** We do not copy, crack or install pirated software.*

Reminder — the five-step diagnostic workflow
(used in every scenario): **1. Observe → 2. Ask questions → 3. Form a hypothesis → 4. Test → 5. Resolve and verify.**

PART A — FAULT SCENARIO CARDS

How to use a card:

1. The instructor gives you the card's **learner section only** (customer complaint, symptoms, device).
2. Follow the five-step workflow and fill in your findings on the **Fault-Diagnosis Table** (Part C).
3. State the **root cause**, the **repair**, and how you would **verify** it.
4. The **instructor answer notes** for each card are given after the card — students should not read these until after the exercise.

Cards 1–8 are general diagnostic practice (Weeks 2–3). Cards 9–12 are

capstone-style
faults (Week 4).

Card 1 — "Nothing Happens At All"

Customer complaint:
"I pressed the power button this morning and nothing happened. No light, no fan, no sound. Absolutely dead."

Device:
Desktop PC (practice unit), mains-powered. **Symptoms:** No power light; no fans; no display; no beep. Wall socket confirmed working (a lamp was tested in it).

INSTRUCTOR ANSWER NOTES — Card 1

Intended fault: *A failed power supply unit (PSU).* ***Expected diagnostic path:***

1. *Observe: totally dead; no fans/lights; socket confirmed live.*
2. *A simple, high-value first test — check the* ***power cable*** *and the* ***PSU rear on/off switch*** *and voltage selector.*
3. ***Visual/smell check*** *of the PSU for burn marks or a burning smell.*
4. ***Swap test:*** *fit a known-good PSU. If the machine starts, the original PSU is the fault.*
5. *(Advanced, instructor-supervised only)* ***paperclip test*** *— if the PSU fan does not spin, it is likely dead.* ***Root cause:*** *failed PSU.* ***Repair:*** *replace the PSU (never open it).* ***Verify:*** *machine powers on, POST completes, OS loads.* ***Teaching point:*** *"dead" faults are usually PSU, power cable, PSU switch, or motherboard. Start with the cheapest, simplest checks.*

Card 2 — "Fans Spin But The Screen Is Black"

Customer complaint:

"The computer comes on — I can hear the fans — but nothing shows on the monitor. It was working yesterday."

Device:

Desktop PC with a dedicated GPU and two RAM sticks. **Symptoms:** Fans spin, power light on, but no display. No recent changes reported.

Your task:

List the likely causes. Which would you check first, and how?

INSTRUCTOR ANSWER NOTES — Card 2

Intended fault: *A RAM stick not fully seated.* ***Expected diagnostic path:***

1. *Observe: powers on but no display → display or RAM/GPU seating are prime suspects.*
2. *Check the* ***display cable*** *at both ends and the* ***monitor input/source****.*
3. *If the machine has a dedicated GPU, confirm the display cable is in the* ***GPU****, not the motherboard port.*
4. *Open the case (safely) and* ***reseat the RAM****: push the clips, remove, reseat until* ***both clips snap****. If there are two sticks, test one at a time.*
5. *Reseat the GPU if needed; reconnect any GPU power cables.* ***Root cause:*** *loose RAM.* ***Repair:*** *reseat RAM firmly.* ***Verify:*** *display appears, POST passes, memory is recognised.* ***Teaching point:*** *"no display" is very often a* ***seating/cable*** *issue before it is a dead part.*

Card 3 — "Blue Screens Every Few Hours"

Customer complaint:

"Every few hours the screen turns blue with a lot of small white text, then it restarts. It is getting more frequent."

Device:

Desktop PC, 8 GB RAM (two 4 GB sticks), several years old. **Symptoms:** Intermittent blue screen (BSOD); restarts itself; more frequent over time.

Your task:

INSTRUCTOR ANSWER NOTES — Card 3
Intended fault: *One faulty RAM stick.* ***Expected diagnostic path:***

1. *Observe/ask: intermittent, worsening → memory or overheating are common for worsening BSODs.*
2. *Note any* ***stop code*** *from the screen and look it up.*
3. *Run* ***Windows Memory Diagnostic*** *or boot* ***MemTest86*** *(at least 2 passes).*
4. *If errors appear,* ***test the sticks one at a time*** *to find the bad one.*
5. *Rule out overheating with* ***HWMonitor*** *during load.* ***Root cause:*** *one faulty RAM stick.* ***Repair:*** *replace the faulty stick (fit matching RAM where possible).* ***Verify:*** *MemTest86 clean over multiple passes, no more BSODs.* ***Teaching point:*** *faulty RAM is a classic cause of intermittent blue screens.*

Card 4 — "It Takes Forever To Start"

Customer complaint:
"My computer used to be quick. Now it takes almost five minutes to start and programs take ages to open. It also asks me to free up space."

Device:
Desktop PC, 4 GB RAM, 500 GB HDD that is almost full. **Symptoms:** Very slow startup; slow programs; low-disk-space warnings; occasional freezing.

Your task:
List possible causes and the tests/tools you would use to separate a hardware cause from a software cause.

INSTRUCTOR ANSWER NOTES — Card 4
Intended fault: *A combination — a nearly full, ageing HDD plus insufficient RAM, with some clutter.* ***Expected diagnostic path:***

1. *Observe: low-disk warning and slow performance.*
2. *Check* ***free disk space*** *and* ***disk health (CrystalDiskInfo)*** *— a failing or nearly full drive slows everything.*
3. *Open* ***Task Manager*** *to see what is using CPU, memory and disk; check* ***startup programs****.*
4. *Run* ***Disk Cleanup*** *and an* ***antivirus full scan*** *to rule out malware.*
5. *Check memory use; 4 GB is low for modern Windows.* ***Root cause:*** *nearly full/slow HDD and too little RAM (rule out malware first).* ***Repair:*** *clean up/clone to an SSD and/or add RAM.* ***Verify:*** *restart time and performance improve.* ***Teaching point:*** *"slow" has many causes — check before replacing anything.* ***Back up data before any drive work.***

Card 5 — "A Clicking Noise From The Tower"

Customer complaint:
"There is a clicking sound coming from inside the computer, like something ticking. Sometimes files will not open."

Device:
Desktop PC with an older HDD as the main drive. **Symptoms:** Rhythmic clicking; occasional file errors; slow; sometimes does not boot.

Your task:
State the most likely cause immediately and the priority action. Describe the diagnostic and safety steps.

INSTRUCTOR ANSWER NOTES — Card 5

Intended fault: *A failing hard disk drive (mechanical failure).* ***Expected diagnostic path:***

1. *Recognise clicking from an HDD as a* ***failing-drive symptom*** *— treat as urgent.*
2. ***Priority: back up the customer's data immediately*** *(data-safety first).*
3. *Check* ***CrystalDiskInfo****: likely "Caution" or "Bad"; check reallocated sectors and S.M.A.R.T. warnings.*
4. *Listen/locate the noise with a careful visual inspection; check it is the drive, not a fan.*
5. *Do* ***not*** *try to repair the drive. Replace it.* ***Root cause:*** *failing HDD.* ***Repair:*** *back up, replace the drive, reinstall the OS and restore data.* ***Verify:*** *new drive shows "Good", machine boots and runs.* ***Teaching point:*** *clicking HDD = back up now; consider a data-recovery professional if the data is critical and unreadable.*

Card 6 — "It Shuts Down After About 30 Minutes"

Customer complaint:
"The computer works well for about half an hour, then just switches off by itself. The side near the back feels very hot."

Device:
Desktop PC, dusty, in a closed cabinet. **Symptoms:** Random shutdown after 30 minutes; case feels hot; fan sounds loud/strained.

Your task:
What tool would you use? What would you look for? What is the likely cause?

INSTRUCTOR ANSWER NOTES — Card 6

Intended fault: *Overheating — dust-clogged CPU heatsink and dried thermal paste.* ***Expected diagnostic path:***

1. *Observe: shutdown only after a period of use + heat → classic overheating.*
2. *Open safely; inspect the* ***heatsink, fans and vents*** *for dust; confirm all fans spin.*
3. *Use* ***HWMonitor*** *under load — CPU likely far above the normal range (idle 30–50°C; high load toward 80°C).*
4. *Clean with* ***compressed air*** *(hold blades steady) and consider* ***replacing the thermal paste****.*
5. *Improve airflow (clear the cabinet, ensure fans are not blocked).* ***Root cause:*** *dust + dried thermal paste causing thermal shutdown.* ***Repair:*** *clean heatsink/fans, replace thermal paste.* ***Verify:*** *temperature stays low under load and no shutdowns.* ***Safety:*** *ensure the machine is* ***cool*** *before opening; unplug and discharge first.*

Card 7 — "Pop-Ups Everywhere"

Customer complaint:
"There are adverts popping up all over the screen, and my browser keeps taking me to strange websites. It is also very slow now."

Device:
Desktop PC used for a small cyber café. **Symptoms:** Pop-ups; browser redirects; slow; unknown toolbar/homepage; fans louder than before.

Your task:
What is the likely cause? What steps would you take, in order? What data-safety or security advice would you give?

INSTRUCTOR ANSWER NOTES — Card 7

1. *Observe: pop-ups + redirects + slowdown → malware.*
2. ***Task Manager:*** *look for an unknown process using high CPU. End Task.*
3. ***Full antivirus scan*** *(not Quick); quarantine/remove threats; restart.*
4. *If needed, run a* ***second-opinion scanner*** *(e.g. Malwarebytes).*
5. *Reset browser homepage/search; remove unknown extensions.*
6. *Advise the customer: change important passwords, only install from trusted sources, keep antivirus updated.* ***Root cause:*** *malware.* ***Repair:*** *remove the threats and clean the browser.* ***Verify:*** *scans come back clean; no pop-ups; normal speed.* ***Licensing note:*** *never use a "cracked" antivirus; use the built-in Windows Defender or a licensed product.*

Card 8 — "Windows Will Not Start"

Customer complaint:
"When I turn it on, Windows does not load. It shows an error about missing files and asks me to repair or reinstall."

Device:
Desktop PC; starts to the Windows logo then fails; drive health unknown. **Symptoms:** Boot failure; "missing/corrupt system files" error; recovery options appear.

Your task:
What tools would you use? What steps would you take? What is the last resort, and what must happen before it?

INSTRUCTOR ANSWER NOTES — Card 8
Intended fault: *Corrupted Windows system files (a damaged OS).* ***Expected diagnostic path:***

1. *Observe/ask: when did it start? after a power cut or update? recent hardware change?*
2. *Try the recovery environment:* ***Startup Repair*** *first.*
3. *Run* `sfc /scannow` *(and possibly* `DISM`*) from the recovery command prompt.*
4. *Try* ***System Restore*** *to a point before the problem.*
5. *Check the drive with* ***CrystalDiskInfo*** *— if the drive is failing, a reinstall will not last; replace it.*
6. *Last resort:* ***clean reinstall of Windows*** *— only after* ***backing up the customer's data****.* ***Root cause:*** *corrupted OS (with drive health ruled out).* ***Repair:*** *repair, restore, or clean reinstall using* ***licensed*** *media.* ***Verify:*** *Windows boots, devices work, updates applied.* ***Teaching point:*** *always check hardware (drive/RAM) before blaming the OS; always back up before a reinstall.*

PART B — CAPSTONE-STYLE FAULT CARDS (WEEK 4)

These four cards mirror the possible capstone faults. Each learner gets

one
fault (the instructor sets a different one per learner). The learner sees the customer complaint and symptoms; the instructor keeps the answer notes and the set-up method (see the Capstone Instructor Guide, Part 6).

Card 9 — "It Just Won't Show Anything" (Capstone Fault A)

Customer complaint:

INSTRUCTOR ANSWER NOTES — Card 9
Intended fault: *RAM not fully seated.* ***Set-up:*** *with power off, push a RAM stick until only one clip is engaged (carefully, no force).* ***Expected path:*** *check display cable/monitor source → reseat RAM until both clips snap → verify display returns and memory is recognised.* ***Root cause/repair/verify:*** *loose RAM / reseat / POST succeeds and Windows loads.*

Card 10 — "No Bootable Device" (Capstone Fault B)

Customer complaint:
"It says 'No bootable device' and I cannot get into Windows. My customer records are on this computer." **Device:** Desktop PC.
Symptoms: "No bootable device" message; BIOS may not list the drive.

INSTRUCTOR ANSWER NOTES — Card 10
Intended fault: *A drive that is failing or a loose SATA data/power cable.* ***Set-up:*** *gently unseat one end of the SATA cable (or fit a drive flagged "Caution"). Keep the alternative in mind.* ***Expected path:*** *check BIOS/UEFI for the drive → reseat SATA data and power → if still missing/failing, check CrystalDiskInfo → back up data → replace the drive and reinstall the OS.* ***Root cause/repair/verify:*** *loose cable (reseat) or failing drive (replace) / repair accordingly / boots to Windows and drive health is "Good".* ***Data-safety:*** *if the drive is failing,* ***back up first*** *and warn the customer before any wipe.*

Card 11 — "It Shuts Down When I Work On It" (Capstone Fault C)

Customer complaint:
"It runs fine for a while, then shuts off on its own. The machine feels very hot." **Device:** Desktop PC in a dusty practice unit.
Symptoms: shuts down after several minutes; hot; loud fan.

INSTRUCTOR ANSWER NOTES — Card 11
Intended fault: *Overheating from a dust-clogged heatsink/fan and/or dried thermal paste.* ***Set-up:*** *use a dusty unit; optionally apply a large, dried blob of paste.* ***Expected path:*** *inspect dust; check fans spin; monitor with HWMonitor under load; clean with compressed air; replace thermal paste.* ***Root cause/repair/verify:*** *dust + dried paste / clean and re-paste / temperature drops and no shutdowns.* ***Safety:*** *ensure the unit is cool, unplugged and discharged before opening.*

Card 12 — "It Was Fine Yesterday" (Capstone Fault D)

Customer complaint:
"Yesterday it worked. Today it shows a blue screen, sometimes it will not even start properly, and it has become very slow."
Device: Desktop PC. **Symptoms:** intermittent BSOD; occasional boot trouble; slow; recent "free download" reported.

INSTRUCTOR ANSWER NOTES — Card 12
Intended fault: *A* ***mixed*** *fault — malware causing slowness AND a loose RAM stick causing BSODs (the "mixed" capstone option).* ***Set-up:*** *create a small software fault (e.g. disable a driver or plant a harmless startup slowdown) and separately unseat a RAM clip.* ***Expected path:*** *run memory test → find and reseat the RAM; run antivirus full scan → remove malware; check drive/OS health.* ***Root cause/repair/verify:*** *two faults / correct both / MemTest clean, scans clean, machine boots and runs normally.* ***Teaching point:*** *real faults are sometimes more than one; keep testing after the first fix.*

PART C — WORKSHEETS AND CHECKLISTS

#	Symptom / Observation	Hypothesis (likely cause)	Test performed	Tool used	Result	Conclusion (confirmed / ruled out)
1						
2						
3						
4						
5						

Root cause (final):

_____

Repair performed:

_____

How I verified the fix:

_____

Safety checks done (tick):
☐ Unplugged ☐ Power button pressed ☐ Wrist strap on ☐ Correct tools ☐ PSU not opened ☐ Data backed up (if wiping)

C2. Diagnostic Checklist Template

A step-by-step checklist for any diagnosis. Tick as you go.

Before you start

- ☐ Greeted the customer / read the complaint
- ☐ Asked the seven diagnostic questions
- ☐ Noted anything the customer already tried
- ☐ Confirmed whether the customer's data needs backing up
- ☐ Put on anti-static wrist strap
- ☐ Unit unplugged and power button pressed to discharge
- ☐ Workspace clear, anti-static mat down, screws tray ready

Observe (look, listen, smell, feel)

- ☐ Looked for visible damage, burns, loose cables
- ☐ Listened for beeps, clicking, grinding, rattling
- ☐ Smelled for burning or scorching
- ☐ Felt for unusual heat
- ☐ Checked any beep codes / POST messages

Hypothesise

- ☐ Listed likely causes in order (simplest first)
- ☐ Considered hardware AND software causes

Test

- ☐ Tested the simplest likely cause first
- ☐ Used the correct tool for each test

Resolve and verify

- [ ] Identified the root cause
- [ ] Performed the repair safely
- [ ] Powered on and confirmed the fault is gone
- [ ] Re-ran the relevant diagnostic to confirm
- [ ] Recorded the outcome for the service report

C3. Component-Identification Worksheet

Fill this in during Session 1 (and revise later). Write the component name, its job, and one key fact.

Component	My name for it	Its job	One key fact

Fill in the blanks:

1. The _____ is the main circuit board that connects all parts.
2. The _____ is the computer's short-term memory; it empties when the power goes off.
3. An _____ has no moving parts and is faster than an HDD.
4. The _____ converts electricity from the wall into the voltages the computer needs.
5. The _____ draws images, video and everything you see on the screen.
6. **DDR4** and **DDR5** RAM are / are not (circle one) interchangeable.
7. Hold every component by the _____ and never touch the _____ contacts.

C4. ESD / Safety Checklist

Complete at the

start of every practical

, and again before power-on.

Personal and workspace

- [ ] Anti-static wrist strap worn and clipped to **bare metal** of the case
- [ ] Anti-static mat in place
- [ ] Working on a **hard floor** (not carpet)
- [ ] No wool/synthetic clothing risk; long hair tied back; no loose jewellery/sleeves
- [ ] Safety glasses worn where dust/blowing is involved
- [ ] Workspace clean, well-lit, flat
- [ ] Screw tray/labelled cups ready
- [ ] Tools laid out and correct for the job

Electrical safety

- [ ] Power cable **unplugged** from the wall
- [ ] **Power button pressed** to discharge residual electricity

- [ ] **Never** opening the PSU case
- [ ] Mains wiring / unsafe work referred to a **qualified technician**
- [ ] No drinks or liquids near the workspace

Handling

- [ ] Holding components **by the edges** only
- [ ] Not touching **gold contacts** or CPU pins
- [ ] Nothing forced; if stuck, checking for missed screws/clips
- [ ] Screws not overtightened
- [ ] Components laid on the anti-static mat, not on plastic

Before power-on (reassembly)

- [ ] All cables reconnected (checked against the photo)
- [ ] No loose screws inside the case
- [ ] Side panel replaced
- [ ] External cables in the correct ports

Data-safety (before any wipe/reinstall)

- [ ] Customer's data **backed up** and backup **verified**
- [ ] Customer informed that a clean install erases the drive
- [ ] Only **licensed** software/media to be used

C5. Cleaning & Maintenance Checklist

Use during Session 8 practice and for customer jobs.

Before cleaning

- [ ] Power off, unplug, press power button to discharge
- [ ] Unit moved to a well-ventilated area (dust will blow out)
- [ ] Safety glasses on
- [ ] Compressed air and microfiber cloth ready; **no vacuum cleaner**

Cleaning

- [ ] Opened the case and removed the side panel
- [ ] Blown dust from the **CPU heatsink and fan**
- [ ] Blown dust from the **GPU fan**
- [ ] Blown dust from the **PSU vents** (without opening the PSU)
- [ ] Blown dust from **case fans** (blade held steady)
- [ ] Blown dust from **RAM and expansion slots**
- [ ] Cleaned/checked **air filters**
- [ ] Wiped the case interior with a microfiber cloth
- [ ] Held every fan blade steady while blowing
- [ ] Held the compressed-air can **upright** at all times

Thermal paste (if applicable)

- [ ] Removed the CPU cooler safely
- [ ] Cleaned old paste with isopropyl alcohol and a cloth
- [ ] Applied a **pea-sized** amount of new paste to the centre

- [ ] Reassembled and replaced the side panel
- [ ] Powered on and verified it works
- [ ] Recorded CPU/GPU temperature **before** and **after** with HWMonitor
- [ ] Advised the customer on a cleaning interval (about every 3 months)

Noted improvement:
before _____ °C → after _____ °C

C6. OS-Installation Preparation Checklist

Complete

before
starting any Windows installation. Everything here protects the customer's data and makes the install succeed.

Step 1 — Data-safety (do this first)

- [ ] Identified all important files (documents, photos, business records, browser favourites)
- [ ] Backed up all data to an **external drive or cloud**
- [ ] **Verified** the backup opens and is complete
- [ ] Explained to the customer that a clean install **erases the drive**
- [ ] Got the customer's **written approval** to proceed

Step 2 — Compatibility and media

- [ ] Confirmed the PC meets the OS requirements (CPU, RAM, storage) — check the maker's current list
- [ ] Windows 10 guide: ~1 GHz CPU, 2 GB RAM (64-bit), 32 GB storage
- [ ] Windows 11 guide: ~4 GB RAM, 64 GB storage, TPM 2.0 support
- [ ] Used **licensed** Windows media (official Media Creation Tool) — **no piracy**
- [ ] Internet connection and any Microsoft account details ready

Step 3 — Drivers gathered in advance (save to USB)

- [ ] **Network / Wi-Fi driver** (most important — install this first after Windows)
- [ ] Chipset driver
- [ ] Graphics driver
- [ ] Audio driver
- [ ] Any other device drivers (Bluetooth, printer, etc.)

Step 4 — Bootable media

- [ ] Created a **bootable USB** (8 GB minimum) using official tools
- [ ] Know how to reach the **boot menu** on this machine (F12/F11/Esc/Del)
- [ ] BIOS/UEFI settings checked (boot order, and Windows 11 TPM/Secure Boot where required)

Step 5 — Install and verify

- [ ] Booted from USB; chose language/keyboard
- [ ] Chose **Custom** install; deleted old partitions; selected unallocated space
- [ ] Did **not** remove the USB until installation finished
- [ ] Completed initial setup; installed the **network driver first**
- [ ] Installed the remaining drivers
- [ ] Ran **Windows Update** repeatedly until no more updates
- [ ] **Activated** Windows using a genuine licence

PART D — HOW TO USE THE LAB PACK IN CLASS

Session	Use
1	Component-Identification Worksheet (C3)
2	ESD/Safety Checklist (C4)
3	ESD/Safety Checklist (C4) before/after teardown
4	Scenario Cards 1–5 with the Diagnostic Checklist (C2) and Fault-Diagnosis Table (C1)
5	Scenario Cards 2, 3, 5, 6 with C1 and C2; tool results recorded
6	Scenario Cards 4, 7, 8 with C1 and C2
7	C4 safety checklist; Fault-Diagnosis Table for the replaced part
8	Cleaning & Maintenance Checklist (C5)
9	OS-Installation Preparation Checklist (C6)
10	Scenario cards used as role-play customer complaints
11	Fault-Diagnosis Tables (C1) become the "Diagnostic findings" section of the report
12	Capstone Cards 9–12 (one fault per learner); all checklists available

Marking reminder:
a scenario is only "passed" when the learner can **explain why** each test did or did not confirm the fault — not merely that the machine works again.

Cyber Elias Academy — Computer Repairs Lab Pack v1.0, October 2026. From Zero to Expert, Together.

PART 6

Capstone Project

CAPSTONE ASSESSMENT RUBRIC

CAPSTONE ASSESSMENT RUBRIC

Computer Repairs — "Diagnosed and Serviced Machine"

Cyber Elias Academy | Computer Repairs | October 2026

Learner Name:

_____ **Assessor:** _____ **Date:**

_____ **Fault assigned:** _____

HOW THIS CAPSTONE IS MARKED

The capstone is the one project that proves a learner is

entry-level ready
as a computer repair technician. It has two parts:

- **Part 1 — Repair + Report: 30 marks** (diagnose → repair → verify → write the report).
- **Part 2 — Viva/Demo: 5 marks** (explain and demonstrate the work).

Total instrument: 35 marks.
(In the course assessment scheme this capstone instrument carries the capstone weight shown in the Course Design Document; the 30 + 5 split is the internal mark distribution.)

The four performance columns describe what each standard looks like:

Column	Meaning
Beginning	The learner is starting to try but needs heavy guidance.
Developing	The learner does part of the job but with gaps or errors.
Competent	The learner does the job correctly with only minor prompting.
Entry-level ready	The learner does the job correctly, safely and independently — ready for a junior technician role.

> ***Entry-level ready***
> *is the target. The certificate means competent beginner, not expert. It does **not** cover motherboard-level repair, data recovery or electrical engineering.*

Safety override rule:
if the learner works on a live machine, opens a PSU, ignores ESD protection, or wipes a drive without a confirmed backup, they **cannot** score the safety row above *Beginning* and **cannot pass**, even if the machine ends up working.

Capstone stages and where they were taught

Stage	Taught / rehearsed in
Customer intake	Session 10 (role-play)

Diagnosis (workflow, tools)	Session 4, tools in Sessions 5 and 6
Repair (replace, clean, install)	Sessions 3, 7, 8, 9
Verification	Sessions 3, 5, 12
Service report	Session 11
Presentation / viva	Sessions 10 and 12

PART 1 — REPAIR + REPORT (30 MARKS)

Row 1 — Diagnostic process and findings (5 marks)

Beginning (0–1)	Developing (2–3)	Competent (4)	Entry-level ready (5)	Score
Does not follow the workflow; guesses; few or no tests recorded.	Follows some steps; records a few observations but misses tests or jumps ahead.	Follows the five steps; records observations and most tests clearly.	Follows all five steps systematically; records every observation, test, tool and result; findings are clear and logical.	/5

Row 2 — Root cause identified (3 marks)

Beginning (0)	Developing (1)	Competent (2)	Entry-level ready (3)	Score
Cannot identify the fault or names an unrelated component.	Partly correct; points to the right area but the wrong cause.	Correct root cause, stated clearly.	Correct root cause, correctly stated **and** explained with the evidence that confirmed it.	/3

Row 3 — Repair performed correctly (6 marks)

Beginning (0–1)	Developing (2–3)	Competent (4–5)	Entry-level ready (6)	Score
Repair is wrong, incomplete, or forces/harms components.	Repair mostly works but with technique errors or unnecessary steps.	Repair is correct with the right tools and correct technique.	Repair is correct, efficient and confident; the right part is used and handled correctly throughout.	/6

Row 4 — Verification the fix works (4 marks)

Beginning (0–1)	Developing (2)	Competent (3)	Entry-level ready (4)	Score
Does not test the result, or claims success without evidence.	Powers on only; no diagnostic re-check.	Powers on and confirms the fault is gone.	Powers on, confirms the fault is gone, **re-runs the relevant diagnostic**, and checks the wider system is healthy.	/4

Row 5 — Safety and data-safety adherence (4 marks)

Serious breach: works live, opens a PSU, no ESD protection, or wipes without a backup.	Follows some safety steps but misses important ones (e.g. forgets to discharge, or leaves the wrist strap unclipped).	Follows all main safety steps: unplugged, discharged, wrist strap on, correct tools.	Follows **all** safety steps unprompted, including ESD, electrical safety and data-safety (backup confirmed before any wipe); uses only licensed software.	/4

Row 6 — Service report: structure, accuracy and clarity (5 marks)

Beginning (0–1)	Developing (2–3)	Competent (4)	Entry-level ready (5)	Score
Report is missing, very incomplete, or full of jargon and errors.	Report has some sections but gaps, vague wording or inaccuracies.	Report has all nine sections; plain language; specific and accurate.	Report is complete, well-organised, plain-language, specific and free of errors — it reads like a professional document.	/5

Row 7 — Service report: parts, labour and recommendations (3 marks)

Beginning (0)	Developing (1)	Competent (2)	Entry-level ready (3)	Score
No parts/labour or recommendations.	Partial costs or generic advice.	Correct itemised parts and labour with a total, plus sensible recommendations.	Accurate itemised parts and labour with a correct total, plus practical recommendations (maintenance, backup, upgrades).	/3

REPAIR + REPORT SUBTOTAL: _____ / 30

PART 2 — VIVA / DEMO (5 MARKS)

Row 8 — Explanation and understanding (3 marks)

Beginning (0)	Developing (1)	Competent (2)	Entry-level ready (3)	Score
Cannot explain the fault or the repair.	Explains some of the work but hesitates or repeats the report.	Clearly explains the fault, the repair and how it was verified.	Explains everything confidently in plain language, answers follow-up questions correctly, and shows understanding of *why*, not just *what*.	/3

Row 9 — Live demonstration (2 marks)

Beginning (0)	Developing (1)	Competent (1–2)	Entry-level ready (2)	Score
Cannot demonstrate, or the machine does not work.	Shows the machine but cannot explain the demonstration.	Demonstrates the working machine and the fix.	Demonstrates the working machine, proves the fault is resolved, and shows how a customer would see the improvement.	/2

VIVA / DEMO SUBTOTAL: _____ / 5

TOTAL AND GRADING

REPAIR + REPORT: _____ / 30
VIVA/DEMO: _____ / 5

Grade	Percentage	Descriptor
A — Excellent	80–100%	Can diagnose and repair common faults independently
B — Good	65–79%	Can diagnose and repair with minor guidance
C — Competent	50–64%	Can diagnose and repair with some guidance
D — Developing	40–49%	Requires significant guidance
F — Not Yet Competent	0–39%	Does not meet minimum requirements

Pass criteria (ALL must be met)

1. Minimum **50%** on this capstone instrument.
2. Fault **correctly diagnosed**.
3. Repair performed **safely** and **correctly**.
4. Fix **verified** — the computer works.
5. A complete **service report** produced.
6. Work **presented and explained** to the assessor.

VIVA QUESTION BANK (with model answers)

The assessor asks

5–8
of these, chosen to fit the learner's fault. Questions are asked in plain language; the learner answers aloud. **A repair the learner cannot explain does not pass.**

Q1. In your own words, what was the fault?
Model: e.g. "One RAM stick was not fully seated, so the computer powered on but could not display anything."

Q2. How did you find it? Walk me through your steps.
Model: e.g. "I observed it powered on with no display, checked the display cable and monitor, then opened the case and reseated the RAM until both clips snapped."

Q3. Which tools or tests did you use, and what did each one tell you?
Model: e.g. "I used my anti-static wrist strap for safety, a PH1 screwdriver to open the case, and I re-tested by powering on; later I would run Windows Memory Diagnostic to confirm the RAM."

Q4. What did you do first for safety before touching the machine?
Model: "I unplugged it, pressed the power button to discharge it, and put on my anti-static wrist strap clipped to bare metal."

Q5. Why did you not just replace the whole computer?
Model: "The fault was a small, specific one — a loose stick. Reseating is the correct, cheapest fix; replacing the machine would waste the customer's money."

Q6. How did you prove the fix actually worked?
Model: e.g. "The display came back, Windows loaded, and all the RAM was recognised. I also re-ran a memory test to confirm no errors."

Q7. What would you tell the customer in plain language?
Model: e.g. "I would say: 'One of your memory sticks had worked loose. I reseated it and tested the machine — everything is working normally now.'"

Q8. What recommendation did you put in the report, and why?
Model: e.g. "I advised them to use a surge protector and to have the machine cleaned every three months, to prevent overheating and power damage."

Q10. When would you tell a customer a job is beyond what you can safely do?

Model: "Anything involving PSU internals, mains wiring, or work I am not confident about — I refer it to a qualified technician."

ASSESSOR CHECKLIST

Tick each item you directly observed.

Do not pass a learner on a repair you did not see performed.

A. Safety adherence (must all be yes to score the safety row full marks)

- ☐ Unplugged the power cable **before** opening the case
- ☐ Pressed the power button to discharge residual electricity
- ☐ Wore an anti-static wrist strap clipped to bare metal
- ☐ Did **not** touch gold contacts / held parts by the edges
- ☐ Did **not** force any component
- ☐ Used the correct tools for each task
- ☐ Did **not** open the PSU case
- ☐ Did **not** work on a live machine
- ☐ Overtightened no screws / caused no visible damage
- ☐ **Data-safety:** confirmed a backup before any wipe/reinstall
- ☐ Used **licensed** software only

B. The fix actually works (must all be yes)

- ☐ The machine powers on
- ☐ The original fault is **gone**
- ☐ The relevant diagnostic re-test passes (e.g. MemTest clean, drive "Good", temp normal)
- ☐ The system is stable during the demo
- ☐ The learner can repeat the result on request

C. Process quality

- ☐ Followed the five-step diagnostic workflow
- ☐ Started with the simplest likely cause
- ☐ Recorded tests and results (not from memory)
- ☐ Correctly identified the root cause
- ☐ Completed a service report with all nine sections

D. Viva

- ☐ Learner explained the fault and repair in plain language
- ☐ Learner answered follow-up questions correctly
- ☐ Learner's explanation matched what was observed

ASSESSOR SUMMARY

Fault correctly diagnosed:

Yes / No **Repair performed safely:** Yes / No **Fix verified:** Yes / No **Service report complete:** Yes / No **Viva confidence:** Beginning / Developing / Competent / Entry-level ready

Comments:

Assessor signature:

_____ **Date:** _____

Certificate awarded:

Yes / No

Cyber Elias Academy — Computer Repairs Capstone Rubric v1.0, October 2026. From Zero to Expert, Together.

CAPSTONE INSTRUCTOR GUIDE

CAPSTONE INSTRUCTOR GUIDE

Computer Repairs — Running the "Diagnosed and Serviced Machine" Project

Cyber Elias Academy | Computer Repairs | October 2026

This guide is

instructor-only

. It explains how to set up simulated faults safely, how to run the Session 12 capstone, how to run the viva, and how to grade fairly. It pairs with the **Capstone Assessment Rubric** (Part 6, doc 01) and the **Diagnostic Scenarios Lab Pack** (Part 5).

> ***The capstone in one line:***
> *give each learner a faulty practice computer; they conduct intake, diagnose, repair, verify, write a full service report and present. Assess with the rubric; the repair + report is 30 marks, the viva/demo is 5.*

> ***SAFETY AND DATA-SAFETY ARE PARAMOUNT.***
> *No learner ever works on live mains, opens a PSU, or wipes a drive without a confirmed backup. These are non-negotiable and override marks.*

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1. WHAT THE CAPSTONE TESTS

The capstone is deliberately a

whole-job

test, not a single skill. It confirms the learner can:

1. **Take in a customer** professionally (Session 10).
2. **Diagnose** using the five-step workflow and real tools (Sessions 4–6).
3. **Repair** safely — replace, clean, or reinstall as needed (Sessions 3, 7, 8, 9).
4. **Verify** the machine genuinely works (Sessions 5, 12).
5. **Document** the work in a nine-section service report (Session 11).
6. **Explain** the work and answer questions (viva).

If any stage was never taught, that is a curriculum fault, not a learner fault — see the QA audit (Part 7). In this course every stage is taught in Sessions 1–11, so the capstone tests

only

taught skills.

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2. SETTING UP SIMULATED FAULTS — SAFELY

Golden rules for fault set-up:

- Set every fault with the machine **powered off and unplugged**.
- Wear your own **anti-static wrist strap** when you open cases.

- Keep a **known-good** spare of common parts for swap tests.
- Keep a **clean reset plan** for every fault so the unit can be reused.
- Do **not** plant real malware. Software faults must be harmless and fully reversible.
- Ensure a learner cannot accidentally reach a **fault you did not intend** (e.g. if you unseat RAM, do not also leave the display cable loose).

Fault menu (choose one per learner)

Fault A — Unseated RAM

- **Set-up:** with power off, push one RAM stick in until only one clip engages. Do not force.
- **Symptoms it creates:** powers on, fans spin, **no display**.
- **Safety:** trivial; still requires ESD care from the learner.
- **Reset:** push both clips until they snap.
- **Difficulty:** easy (good for a nervous learner).

Fault B — Loose cable (drive or display)

- **Set-up:** gently unseat one end of the **SATA data cable** (creates "No bootable device") or the **display cable** (creates no display).
- **Symptoms:** "No bootable device," or no image.
- **Reset:** firmly reseat the cable.
- **Difficulty:** easy.

Fault C — Dusty / heated unit

- **Set-up:** use a **genuinely dusty** practice unit, or one with a large dried blob of thermal paste. Optionally leave one case fan disconnected.
- **Symptoms:** shuts down after several minutes; hot; loud fan.
- **Safety:** make sure the unit is **cool** before learners open it; do not disable thermal protection.
- **Reset:** clean the dust/fans and replace the thermal paste.
- **Difficulty:** easy–medium (a good cleaning exercise).

Fault D — Software slowdown

- **Set-up:** add a harmless **startup program** (or a script that opens a few large files), disable one non-essential driver, or fill the drive to ~95% with dummy files. All reversible.
- **Symptoms:** very slow start and slow programs; low-disk warnings.
- **Safety:** harmless; no malware; document exactly what you changed.
- **Reset:** remove the startup item/driver change; run Disk Cleanup.
- **Difficulty:** medium (tests the Windows tools).

Fault E — Failing drive

- **Set-up:** fit a drive flagged "Caution/Bad" by CrystalDiskInfo, or a drive with a corrupted boot sector, or disconnect a working drive's cable. Keep the original data backed up by the instructor.
- **Symptoms:** "No bootable device," slow boot, file errors, sometimes clicking.
- **Safety / data-safety:** the learner must **back up/confirm data** before any wipe. Provide a clean replacement drive for the fix.
- **Reset:** refit the good drive.
- **Difficulty:** medium.

Fault F — Corrupted operating system

- **Set-up:** on a **dedicated reimageable unit**, cause a genuine but reversible boot failure (e.g. use recovery tools to break the boot configuration, or remove a system file from a lab image). Keep the image for re-flashing.

- **Reset:** repair or reimage the machine.
- **Difficulty:** medium–hard (tests OS repair and reinstall).

Fault G — Failed PSU (advanced)

- **Set-up:** fit a **labelled known-bad PSU**, or leave the PSU rear switch off / a cable deliberately loose.
- **Symptoms:** totally dead, or powers off under load.
- **Safety:** learners **never** open the PSU. This fault tests safe replacement, not repair.
- **Reset:** fit a known-good PSU.
- **Difficulty:** medium (tests swap test and safe PSU handling).

Fault H — Mixed fault (challenge)

- **Set-up:** combine two, e.g. **unseated RAM + software slowdown**, or **dusty unit + loose SATA cable**.
- **Symptoms:** two sets of symptoms.
- **Purpose:** tests persistence — the learner must keep testing after the first fix.
- **Reset:** correct both.
- **Use only for learners who have shown confidence.**

Fault set-up record (complete before the session)

Learner	Fault assigned	Set-up done (initials)	Known-good spare prepared	Reset plan confirmed

Fault rotation tip:
rotate faults so no learner gets the same one twice in the course, and so no fault becomes known as "the easy one." Different faults must be treated as **equal difficulty**; the rubric is the same for all.

3. RUNNING THE SESSION — TIMING AND MILESTONES

Session 12 is 105 minutes. A realistic inner split:

Elapsed	Stage	Notes
0–5 min	Briefing + safety reminder	Learner gets the card and the unit
0–5 min	**Customer intake**	Instructor plays the customer
5–25 min	**Diagnose**	Workflow; record tests and results
25–30 min	*Milestone 1: root cause*	Instructor checks the learner has identified the fault (initials)
30–55 min	**Repair**	Safety observed throughout
55–65 min	**Verify**	Power on; re-run diagnostic; check system
65–75 min	**Service report**	Nine sections
75–105 min	**Viva / demo**	5 minutes per learner (run in parallel rooms if large class)

If the class is large:
run two presentation stations, or run the viva for confident learners while others finish reports. Keep one assessor, or a second trained assessor, per station.

Checkpoint	What to confirm	Time	Instructor initials
Intake done	Complaint logged; expectations set	~5 min	
Safety on	Unplugged, discharged, wrist strap on	ongoing	
First tests	Observation recorded; simplest cause tested first	~15 min	
Root cause	Correct fault identified before repair	~25 min	
Repair complete	Correct part/task; no damage; correct tools	~50 min	
Fix verified	Machine works; diagnostic re-test passes	~60 min	
Report drafted	Nine sections present	~72 min	
Viva done	Explained and demonstrated	by 105 min	

4. THE EXPECTED SERVICE REPORT STRUCTURE

Teach and expect

all nine sections
, in this order. (This matches Session 11.)

1. **Header** — business name/logo, unique report number, date, technician name.
2. **Customer information** — name, contact, device type and serial number.
3. **Problem description** — the customer's words, plus what you observed.
4. **Diagnostic findings** — tests performed, results, and the root cause (this is where the **Fault-Diagnosis Table** becomes prose).
5. **Repairs performed** — parts replaced (with part numbers), software installed, time spent.
6. **Parts and labour** — itemised parts + labour + **total** (naira).
7. **Recommendations** — maintenance, backup and upgrade advice.
8. **Warranty information** — what is covered, for how long, and the conditions.
9. **Sign-off** — technician and customer signatures and dates.

A quick "specific vs vague" guide for feedback:

- Vague: "Fixed memory." → Specific: "Reseated a loose 8 GB DDR4 RAM stick in slot 2."
- Vague: "Cleaned it." → Specific: "Removed dust from the CPU heatsink and case fans; replaced dried thermal paste."
- Vague: "Replaced part." → Specific: "Replaced the 500 W PSU with a 500 W unit; reconnected the 24-pin and CPU power cables."

5. MODEL / REFERENCE SERVICE REPORT

Below is a complete reference report for

Fault A (unseated RAM)
. Use it as the benchmark when giving feedback. Learners' reports must not copy it word-for-word — it is a model of *quality*, not an answer to hand in.

REPORT NO:
CEA-2026-0007 **DATE:** 12 October 2026 **TECHNICIAN:** Emmanuel Adeyemi

CUSTOMER INFORMATION
Name: Mrs. Ngozi Eze · Contact: 0803 XXX XXXX Device: Desktop PC (tower) · Serial: PH-DSK-4471

have work files on it." Observed on intake: power light on, fans spinning, no display, no burning smell, no unusual noise.

DIAGNOSTIC FINDINGS

Followed the five-step diagnostic workflow.

- Observed: powered on, no display; monitor power light on.
- Asked: problem started that morning; no drops, spills, power cuts or new software; happens every time.
- Hypothesis (most likely first): (1) loose display cable, (2) loose RAM, (3) loose/faulty GPU.
- Test 1: checked the display cable at both ends and the monitor input — no change. (Ruled out.)
- Test 2: opened the case (unplugged, discharged, anti-static wrist strap on). Found one RAM stick with only one clip engaged. Reseated the stick until both clips snapped.
- **Root cause:** a RAM stick not fully seated.

REPAIRS PERFORMED

- Reseated the loose RAM stick in its slot until both retaining clips locked.
- Reconnected the side panel and external cables.
- Time on task: 12 minutes.

PARTS AND LABOUR

Item	Qty	Unit (₦)	Total (₦)
Labour (diagnosis + reseat)	1	3,000	3,000
Total			**3,000**

RECOMMENDATIONS

- Keep the PC on a stable surface to avoid knocking RAM loose.
- Consider a surge protector to protect against power surges.
- Have the machine professionally cleaned every 3 months to prevent overheating.
- Back up important files regularly (external drive or cloud).

WARRANTY

The workmanship on this repair is guaranteed for 14 days from the date of service. The warranty does not cover new faults, physical damage, liquid damage or power-surge damage.

SIGN-OFF

Technician signature: _____ Date: 12/10/2026 Customer signature: _____ Date: 12/10/2026

Why this report earns top marks:

it follows all nine sections; uses plain language; records each test and result; states the root cause with evidence; gives an itemised cost and a real total; and offers practical recommendations and a clear warranty.

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6. COMMON FAILURE POINTS AND REMEDIATION

Problem observed	Why it happens	Immediate remediation	Follow-up
Skips the workflow; guesses	Excitement or over-confidence	Prompt: "What is step 1?" pause and reset	Re-drill the five steps next session
Forgets the wrist strap / to discharge	Rushing	Stop the learner; re-do the safety sequence	Reinforce the ESD checklist

Tests the hardest cause first	Poor ordering	Ask "What is the simplest thing it could be?"	Practise symptom-to-cause ordering
Cannot find the fault	Weak observation or questions	Prompt with a diagnostic question	Pair with a stronger learner in practice
Fixes the wrong thing	Assumed without testing	Ask for the evidence	Re-teach hypothesis + test
Repairs but does not verify	Finishes at "it powers on"	Require the diagnostic re-test	Emphasise verification in feedback
Report missing sections	Rushed at the end	Give the nine-section checklist	Template practice in Session 11
Report full of jargon	Forgetting the audience	Ask "How would you say that to the customer?"	Plain-language translation drill
Runs out of time	Slow start or over-cleaning	Use milestone checkpoints; keep it to the fault	Time the practice runs
Freezes in the viva	Nervousness	Ask a simple opening question	Allow a short pause; reassure
Data-safety breach risk (wants to wipe without backup)	Fault E/F pressure	Stop; require a backup first	Reinforce data-safety rule

Universal remediation plan for a fail:
allow the learner to practise on academy machines during opening hours, then **retake with a different fault**. Record the first attempt; do not penalise a retake.

7. HOW TO RUN THE VIVA

The viva is a short, fair, spoken check that the learner

understands
the work — it protects against a lucky repair. It is worth **5 marks** (explanation 3, demo 2).

Before the viva

- Read the learner's report first so you can ask targeted questions.
- Have the learner's machine ready and working.
- Prepare 5–8 questions from the viva bank (Part 6, rubric doc), matched to the fault.

During the viva (about 5 minutes)

1. Start easy: "Tell me what was wrong with this computer."
2. Ask them to walk you through the steps.
3. Ask one or two "why" questions ("Why did you check the cable first?").
4. Ask them to explain it as they would to a **non-technical customer**.
5. Ask them to **demonstrate** the fix (power on; show the fault is gone).
6. If the report and the spoken answer disagree, ask them to reconcile it.

Viva technique

- Ask **open** questions, not leading ones.
- Let the learner finish before you speak.
- Do not punish nerves — give thinking time.

- **A repair the learner cannot explain cannot pass**, even if the machine works.

Scoring the viva:
use Rubric rows 8 and 9. A learner who explains clearly in plain language and demonstrates a working fix scores full marks (3 + 2). A learner who mumbles "it works now" scores low even if it does.

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8. SAMPLE FEEDBACK COMMENTS

Use these as templates; make them specific to the learner.

Entry-level ready:

- "Excellent, methodical diagnosis. You followed all five steps, recorded every test, and found the loose RAM quickly. Your report was clear, specific and accurate, and your viva answer showed real understanding. Safety was perfect throughout. You are ready for a junior technician role."
- "You handled a difficult mixed fault calmly and fixed both issues without help. Your plain-language explanation of the fault to the 'customer' was exactly right."

Competent:

- "Good, solid diagnosis and a correct repair. Your report had all nine sections and was easy to read. To reach the top band, re-run the diagnostic after the repair to prove the fix, and always clip your wrist strap before opening the case."
- "You identified the correct fault and repaired it safely. Watch your test order — you checked the hardest cause first. Start with the simplest, most likely cause next time."

Developing:

- "You found and fixed the fault, which is a real achievement, but you skipped several steps of the workflow and your report was missing the diagnostic findings and signature. Practise the workflow aloud and use the nine-section checklist every time."
- "Your repair worked, but you did not discharge the machine or check the data before working. Safety and data-safety are not optional. We will re-run this together, then you can retake with a new fault."

Not yet competent:

- "This attempt did not meet the standard. You did not identify the root cause before trying to repair, and the report was incomplete. That is okay — you are not being punished, you are being taught. Come to the practice hours, work through two more scenarios with me, and retake with a different fault."
- "The machine did work in the end, but you could not explain how you fixed it, so I cannot confirm you understand the repair. Let us build your understanding of the diagnostic workflow, then try a new fault."

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9. FAIR GRADING PRINCIPLES

1. **Mark what you observe,** using the rubric — not your general impression of the learner.
2. **Same standard for every fault.** Do not treat unseated RAM as "easier" than a failing drive; the rubric is identical.
3. **Score process as well as result.** A correct answer reached by guessing is not the same as a reasoned diagnosis.
4. **Apply the safety override consistently.** Any dangerous practice caps the safety row and blocks a pass.
5. **Distinguish nerves from ignorance.** Quiet and nervous is not the same as not understanding.
6. **Avoid the halo/horn effect** — one great answer should not inflate every row; one mistake should not sink every row.
7. **Write a genuine comment** for every learner: one strength and one specific next step.
8. **Allow retakes** with a different fault; record the best attempt and log the process.
9. **Do not leak faults.** Keep set-up notes and answer notes away from learners.
10. **Round numbers fairly** and show the learner exactly how their mark was reached, using the rubric.

service report is the learner's portfolio evidence.

10. CAPSTONE DAY CHECKLIST (INSTRUCTOR)

- ☐ Faulty units prepared and faults set (record completed)
- ☐ Known-good spare parts available (RAM, PSU, drive, cables)
- ☐ Cleaning supplies and thermal paste ready (for cleaning faults)
- ☐ Licensed Windows media and drivers available (for OS faults)
- ☐ Backup drives available (data-safety for drive faults)
- ☐ Tool kits and anti-static wrist straps for every learner
- ☐ Printed: capstone briefs, rubrics, fault cards, checklists
- ☐ Viva question set selected per fault
- ☐ Milestone checkpoint sheet ready
- ☐ A plan to run viva in parallel if the class is large
- ☐ Certificate sign-off sheets ready

Cyber Elias Academy — Computer Repairs Capstone Instructor Guide v1.0, October 2026. From Zero to Expert, Together.

PART 7

QA & Audit

FINAL QUALITY ASSURANCE AUDIT REPORT

FINAL QUALITY ASSURANCE AUDIT REPORT

Course: Computer Repairs (Hardware & Software Diagnostics and Repair)

Institution:
Cyber Elias Academy, 24/26 Ebony Road, Off Rumuola Road, Port Harcourt **Auditor:** Curriculum Development Team **Date:** October 2026 **Version audited:** v1.1 (supporting materials enriched to the new rich standard)

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1. AUDIT SCOPE

This audit evaluates the complete Computer Repairs curriculum against the academy's master QA checklist:

1. Missing sessions
2. Missing explanations
3. Inconsistent terminology
4. Unsupported prerequisites
5. Repeated or unnecessary material
6. Incorrect technical instructions
7. Unclear exercises
8. Missing answer keys or marking criteria
9. Unrealistic workloads
10. Missing practical activities
11. Unassessed learning outcomes
12. Capstone requirements never taught
13. Insufficient beginner support
14. Missing SAFETY / DATA-SAFETY precautions
15. Contradictions between learner and instructor materials

Documents reviewed:

- Course Design Document (Part 1)
- Session materials S1–S12 (Part 2)
- Learner Workbook (Part 3)
- Practical Lab Pack — Diagnostic Scenarios (Part 5)
- Capstone Assessment Rubric (Part 6, doc 01)
- Capstone Instructor Guide (Part 6, doc 02)
- This QA report (Part 7)

Method:
direct document review, cross-referencing of terminology and numbering, mapping of every learning outcome and every capstone stage to the sessions that teach and assess them, and a section-by-section (A–O) check of all twelve sessions. Where the v1.0 supporting materials fell below the new rich standard, the issue is recorded with the correction made.

> ***Note on the review basis.***
> *The Computer Repairs **session materials** (Part 2) were reviewed for consistency, accuracy and coverage. The **supporting materials** (Parts 3, 5, 6, 7) were measured against the academy's new "gold standard" (the Web Development workbook,*

2. AUDIT FINDINGS AGAINST THE CHECKLIST

2.1 Missing Sessions

Status: PASS
All **12 sessions** are present and complete (S1 Computer Components through S12 Capstone). No session is omitted or merged, and the sequence matches the Course Design Document exactly.

2.2 Missing Explanations

Status: PASS
Every topic is **explained**, not merely named. Components are described by function with plain analogies; ESD is explained through voltage facts; the diagnostic workflow is taught as a five-step method; each tool's use and result interpretation is described; the nine service-report sections are itemised. The enriched Learner Workbook now carries **full lesson summaries** a learner can revise from — not bullet fragments.

2.3 Inconsistent Terminology

Status: PASS
Terminology is consistent across learner and instructor materials:

- **"ESD (electrostatic discharge)"** is defined and used identically everywhere.
- **"The five-step diagnostic workflow"** = observe, ask questions, form a hypothesis, test, resolve and verify — consistent in S4, S5, S6, S12, the workbook and the lab pack.
- **"Service report"** is consistently the **nine-section** document (header, customer information, problem description, diagnostic findings, repairs performed, parts and labour, recommendations, warranty, sign-off).
- **"Anti-static wrist strap"**, **"thermal paste"** and **"CrystalDiskInfo"** are spelled and used identically throughout.
- The capstone's top performance band is now consistently named **"Entry-level ready"** in the rubric, matching the certificate's "competent beginner" promise.

2.4 Unsupported Prerequisites

Status: PASS
Session 1 assumes only basic computer skills (from Typing & Computer Basics) and takes nothing for granted about hardware. Each session builds only on earlier ones: components (S1) → safety (S2) → teardown (S3) → workflow (S4) → hardware/software diagnostics (S5–S6) → replacement (S7) → cleaning (S8) → OS install (S9) → customer service (S10) → reports (S11) → capstone (S12). No advanced concept appears before its foundation. The capstone tests only Sessions 1–11 content.

2.5 Repeated or Unnecessary Material

Status: PASS
Repetition is **deliberate reinforcement in new contexts**, not duplication. Safety is restated in every practical (correct, because safety must be habitual). Diagnostic skills are reused from S4 into the capstone. The Learner Workbook's summaries are a **revision aid**, and the new glossary and question bank add value rather than duplicating session prose.

2.6 Incorrect Technical Instructions

Status: PASS (one Issue found and corrected)

- All hardware identification is accurate; all safety procedures are correct (unplug, discharge, ESD strap, never open a PSU).
- Diagnostic tool use is accurate (MemTest86 ≥ 2 passes; CrystalDiskInfo Good/Caution/Bad; HWMonitor temperature ranges).
- Replacement procedures and OS-install steps are correct, including "install the network driver first."

Learner Workbook now clearly separates the two and directs learners to the maker's current requirements. No invented capabilities or fictional features are used anywhere.

2.7 Unclear Exercises

Status: PASS (improved)

Every session has a guided practical and an independent exercise with objectives and success criteria, and every assignment has a purpose, numbered requirements and a deadline. The enriched Lab Pack now adds a **Fault-Diagnosis Table**, a **Diagnostic Checklist**, a **Component-Identification Worksheet**, and dedicated **ESD/safety, cleaning and OS-install checklists**, so "done" is objectively checkable.

2.8 Missing Answer Keys or Marking Criteria

Status: PASS

- Every session's Knowledge Check has an **instructor answer key** (Part 2, section M).
- Every assignment has a **marking rubric** (Part 2, section L).
- The capstone has a full **30-mark rubric plus a 5-mark viva** with **model answers** and an **assessor checklist**.
- Scenario cards carry **instructor answer notes** (Part 5) for every fault.
- **Answer keys are confined to instructor materials.** The Learner Workbook contains **questions only, with no answers**, as required.

2.9 Unrealistic Workloads

Status: PASS

Each session is built for **105 minutes**. The capstone gives ~70 minutes for the project (intake 5, diagnose 20, repair 25, verify 10, report 10) plus ~5 minutes per learner for the viva — achievable for a learner who has rehearsed the workflow. Assignments are short and completable in class or as brief homework. The Instructor Guide supplies milestone checkpoints and a large-class contingency (parallel viva rooms).

2.10 Missing Practical Activities

Status: PASS

Every session includes an **instructor demonstration, a guided practical and an independent exercise**. The course is hands-on throughout; the Lab Pack supplies real scenario cards and worksheets; Session 12 is entirely practical.

2.11 Unassessed Learning Outcomes

Status: PASS

All **11 learning outcomes (LO1–LO11)** are taught and assessed. See the mapping in Section 3.

2.12 Capstone Requirements Never Taught

Status: PASS

Every capstone stage maps to a session that teaches it. See Section 4.

2.13 Insufficient Beginner Support

Status: PASS (Issue in v1.0 — corrected)

- **Issue — corrected:** the v1.0 Learner Workbook was a **summary-only** document. It listed component tables and assignment titles but lacked full lesson summaries, key-term definitions, guided-practical steps, explicit deliverables/deadlines, submission checklists, self-reflection prompts and a glossary. A true beginner could not revise from it. It has been **rewritten** to include all of these.
- Materials use simple, natural English readable by a second-language speaker, with **Nigerian context** throughout (Port Harcourt repair shop, cyber café, school lab; naira; Chinedu, Adaeze, Funmi, Kunle, Emmanuel).
- **Differentiation at three levels** (Supported / Independent / Challenge) appears on every session's instructor notes.

2.14 Missing SAFETY / DATA SAFETY Precautions

Status: PASS

- **ESD**: taught in S2, reinforced in every practical, and now on a dedicated **ESD/Safety Checklist** in the Lab Pack.
- **Electrical**: unplug, discharge, insulated tools, **never open a PSU**, never work on live mains — in S2, S5, S7 and the checklist.
- **Data-safety**: **back up before any wipe/reinstall** (S9, S11), reinforced in the OS-install checklist and required in the capstone rubric's safety row.
- **Licensed software only**: stated in the course context, the Lab Pack, the workbook and the Capstone Rubric/Guide. No piracy is promoted anywhere.
- **"Know your limits"** guidance (refer to a qualified technician) appears in S2 and the viva bank.
- Safety is **assessed** in the capstone rubric, with an override rule that prevents a pass on safety/data-safety breaches.

2.15 Contradictions Between Learner and Instructor Materials

Status: PASS (Issues found and corrected)

- **Issue — corrected (rubric scale):** the v1.0 rubric totalled **100 marks** with three columns (Excellent/Good/Needs Improvement), contradicting the new standard of **30 + 5** and the four-band descriptor model. The rubric has been rebuilt to **30 marks (repair + report) + 5 marks (viva/demo)**, with **Beginning / Developing / Competent / Entry-level ready** columns.
- **Issue — corrected (assignment numbering):** the design document names the capstone as the Session 12 deliverable, while the Session 12 assignment is titled **"My Learning Reflection."** This is **not** a true contradiction — the reflection is the session's short assignment; the capstone is the assessed project. The workbook now states this explicitly, and Assignment 12's deliverable name ("My Learning Reflection") matches the session file.
- **Issue — corrected (capstone report contents):** the v1.0 rubric's "Service Report" row mentioned "nine sections" but the rubric itself had no space to confirm the parts/labour total or recommendations. The rebuilt rubric adds a dedicated **parts/labour and recommendations** row.
- After correction, learner-facing and instructor-facing materials agree on terminology, file/folder names, section counts, fault types and marking.

3. LEARNING OUTCOMES — TAUGHT AND ASSESSED

Status: PASS.
Every outcome is taught and assessed; nothing is assessed that was not first taught.

LO	Outcome	Taught in	Assessed by
LO1	Identify/name major components	S1	Component-ID practical; Assignment 1; KC1; Capstone
LO2	Explain each component's function	S1–S2	Assignment 1–2; KC1–2; viva
LO3	Follow ESD/electrical safety	S2 (+ every practical)	Observed practical; ESD checklist; Capstone safety row
LO4	Diagnose hardware faults with the workflow	S4–S5	Diagnostic scenarios; Assignment 4–5; Capstone diagnosis
LO5	Replace common components	S7	Replacement practical; Assignment 7; Capstone repair
LO6	Diagnose/resolve software faults	S6	Software practical; Assignment 6; Capstone
LO7	Use diagnostic tools/software	S5–S6	Tool practicals; Assignments 5–6; Capstone verification

LO8	Clean and maintain hardware	S8	Cleaning practical; Assignment 8; cleaning fault in Capstone
LO9	Install/configure an OS	S9	Install practical; Assignment 9; Capstone (OS faults)
LO10	Produce a professional service report	S11	Report practical; Assignment 11; Capstone report (10 marks)
LO11	Complete a full repair project	S12	Capstone project + viva/demo

4. CAPSTONE STAGE → SESSION MAPPING

Status: PASS.

Every stage of the capstone (diagnose → repair → verify → report → present) is taught before it is assessed.

Capstone stage	Specifically taught in	Rehearsed / assessed through
Customer intake	S10	Assignment 10; Capstone intake checkpoint
Diagnose — observe/ask/hypothesise	S4	Assignment 4; scenario cards 1–5
Diagnose — hardware tools & results	S5	Assignment 5; Cards 2, 3, 5, 6
Diagnose — software tools & results	S6	Assignment 6; Cards 4, 7, 8
Repair — component replacement	S7	Assignment 7; Cards 9, 10, 12
Repair — cleaning/maintenance	S8	Assignment 8; Card 11
Repair — OS install	S9	Assignment 9; Cards 8, 10
Repair — teardown/reassembly safety	S3	Assignment 3; all capstone faults
Verify the fix	S3, S5, S12	Capstone verification step; rubric Row 4
Service report	S11	Assignment 11; Capstone report rows 6–7
Present / viva	S10, S12	Capstone viva (5 marks)

No capstone requirement is assessed that was not first taught. The possible capstone faults (loose RAM, failed drive, overheating, failed PSU, corrupted OS, faulty GPU) each trace to a teaching session: RAM/PSU/GPU replacement (S7), overheating (S8), OS (S9), drive (S7/S9), diagnosis (S4–S6).

5. SESSION STRUCTURE (A–O) CONFIRMATION

Status: PASS.

Every session contains the full A–O structure: **A** Session Identification, **B** Learning Outcomes, **C** Required Materials, **D** Opening Activity, **E** Prerequisite Review, **F** Full Lecture Notes, **G** Instructor Demonstration, **H** Guided Practical, **I** Independent Exercise, **J** Real-World Case Study, **K** Common Mistakes, **L** Assignment, **M** Knowledge Check (+ instructor key), **N** Reflection and Recap, **O** Instructor Evaluation Notes.

Session	A–O present	Instructor key (M)	Assignment (L)
S1 Computer Components	Yes	Yes	Yes
S2 Safety and Tools	Yes	Yes	Yes

S3 Disassembly and Reassembly	Yes	Yes	Yes
S4 Diagnostic Workflow	Yes	Yes	Yes
S5 Hardware Diagnostics	Yes	Yes	Yes
S6 Software Diagnostics	Yes	Yes	Yes
S7 Component Replacement	Yes	Yes	Yes
S8 Cleaning and Maintenance	Yes	Yes	Yes
S9 OS Installation	Yes	Yes	Yes
S10 Customer Service	Yes	Yes	Yes
S11 Service Reports	Yes	Yes	Yes
S12 Final Project (Capstone)	Yes	Yes (full quiz key)	Yes (+ capstone rubric, viva key, guide)

Note:
S12 legitimately has no instructor demonstration (G) or guided practical (H) because the session **is** the assessment; the template marks these "0 minutes / none" deliberately. This is consistent with the design document.

6. SAFETY AND DATA-SAFETY REVIEW

Status: PASS.

Requirement	Where enforced
ESD protection	S2, every practical, ESD/Safety Checklist, Capstone safety row
Unplug + discharge	S2, S3, S7, S8, checklists
Never open a PSU	S2, S5, S7, checklists, capstone override rule
No live-mains work	S2, capstone override rule
Data backup before wipe	S9, S11, OS-install checklist, capstone override rule
Licensed software only	Course context, Lab Pack, workbook, rubric, guide
Know your limits / refer up	S2, viva question bank
Electrical/first-aid equipment	Design document tools list; instructor capstone-day checklist

No document promotes piracy, unsafe live-mains work, or working on a powered machine.

7. RECORD OF CORRECTIONS MADE IN THIS ENRICHMENT

#	Document	Issue found (v1.0)	Correction (v1.1)
1	Learner Workbook	Summary-only; no lesson summary, key terms, practical steps, checklists, reflections, glossary, or answer spaces	Fully rewritten as a complete learner workbook with all required sections
2	Capstone Rubric	100 marks; three columns; no viva answers; no assessor safety/function checklist	Rebuilt to 30 + 5; four bands; viva key; assessor checklist

3	Capstone Instructor Guide	Did not exist	Created (fault set-up, milestones, model report, viva, feedback, fair grading)
4	Diagnostic Scenarios	Only scenario cards; no templates	Added checklists (diagnostic, ESD/safety, cleaning, OS-install), component-ID worksheet, fault-diagnosis table, and instructor answer notes
5	Windows requirements	Single line understated Windows 11	Split Windows 10 vs Windows 11; directed to maker's current list
6	Assignment 12 vs capstone	Apparent mismatch	Clarified reflection (assignment) vs capstone (project)
7	QA report	v1.0 asserted PASS with thin evidence; no LO/capstone mapping	Rebuilt with evidence, LO matrix, stage mapping, A–O table and corrections log

8. STRENGTHS OF THE CURRICULUM

- A complete, consistent 12-session arc from "what is inside a computer?" to an **explainable, documented repair**.
- A strong **hands-on backbone**: demonstration → guided → independent → assignment in every session.
- **Safety and data-safety woven through** the whole course and enforceable at the capstone.
- **Ready-to-use practical materials**: scenario cards with answer notes, tables, worksheets and checklists.
- A capstone that is **scaffolded across the whole course**, assessed with a transparent 30 + 5 rubric, a viva key and a **functionality + safety** assessor checklist.
- Genuinely **beginner-friendly** language and unmistakably **Nigerian** context, without diluting technical accuracy.
- An **honest scope statement**: the certificate means a *competent beginner*, not an expert, with a clear next-step path (Cybersecurity, Web Design, Business & Freelancing).

9. AREAS FOR IMPROVEMENT (OPTIONAL ENHANCEMENTS)

1. Record short **video demonstrations** of the instructor practicals for absentees and visual learners.
2. Add a **pre-course safety self-assessment** to flag learners needing extra safety coaching.
3. Add **two or three extra practice scenarios** per week for fast finishers.
4. Build a **starter "known-good parts" kit list** and a stock log for the academy workshop.
5. Add a **one-month follow-up check-in** to reinforce skills and track retention.
6. Provide a short **business-startup supplement** for learners who want to open a repair shop.

10. OVERALL VERDICT

The Computer Repairs curriculum is COMPLETE and READY FOR DELIVERY, and the supporting materials now match the academy's rich standard.

- All **12 sessions** present, each with the full **A–O** structure.
- All **11 learning outcomes** taught and assessed (Section 3).
- Every **capstone stage** maps to a session that teaches it (Section 4).
- Every assignment and knowledge check has an **instructor key or rubric**; the Learner Workbook contains **no answers**.
- Exercises are clear, practical and objectively checkable, supported by real worksheets and checklists.
- The capstone tests **only** taught skills, is scaffolded across the course, and is graded fairly with a viva that confirms understanding.

- **Safety and data-safety** are required throughout and can fail a capstone attempt outright.

Recommended next steps

1. Print the Learner Workbooks and assemble the instructor packs (answer keys, rubric, guide, scenarios).
2. Pre-load the lab: practice machines, known-good spares, fault set-ups, licensed Windows media and drivers, and safety equipment.
3. Confirm the **fault set-up record** and **reset plans** for the capstone before Session 12.
4. Train instructors on the workflow, the milestone checkpoints and the viva.
5. Deliver Week 1, gather learner feedback, and review again before the next intake.

Audit Completed:
October 2026 **Auditor:** Curriculum Development Team, Cyber Elias Academy **Sign-off:** _____

Cyber Elias Academy — From Zero to Expert, Together — 24/26 Ebony Road, Port Harcourt.